Real Analysis

MIT 18.100A

Professor Casey Rodriguez

Course-note convention

The page references to theorems in this note pack could be off by fewer than five pages, because pages were added in between. In this typeset version, such references point to the typeset page on which the cited handwritten page begins.

Definitions Results
Examples Notes

Contents

1 Sets, Set Operations, and Mathematical Induction 1
1.1 Sets . . . 1
1.2 Set operations . . . 2
1.3 Induction . . . 3

2 Cantor's Theory of Cardinality 5
2.1 Functions . . . 6
2.2 Cardinality . . . 7

3 Cantor's Cardinality Theorem and the Incompleteness of the Rational Numbers 14
3.1 Cardinality . . . 14
3.2 Real numbers . . . 15
3.3 Ordered sets and the least upper bound property . . . 15

4 The Characterization of the Real Numbers 21
4.1 Ordered fields . . . 21
4.2 The real numbers . . . 24

5 The Archimedean Property, Density of the Rationals, and Absolute Value 25
5.1 The Archimedean property and the density of the rationals in $\mathbb{R}$ . . . 25
5.2 Suprema and infima . . . 27
5.3 Absolute values . . . 30

6 The Uncountability of the Real Numbers 31
6.1 Decimal representations . . . 31
6.2 Cantor's theorem . . . 33

7 Convergent Sequences of Real Numbers 35
7.1 Sequences and series: first definitions . . . 35
7.2 Bounded sequences . . . 35
7.3 Convergence of a sequence . . . 36
7.4 Proving convergence and divergence . . . 38
7.5 Monotone sequences . . . 40
7.6 Subsequences . . . 41

8 The Squeeze Theorem and Operations Involving Limits 43
8.1 Operations involving limits . . . 43
8.2 Limits and the order of the real numbers . . . 44
8.3 Limits and algebraic operations . . . 45
8.4 Square roots and absolute values . . . 47

9 Limsup, Liminf, and the Bolzano–Weierstrass Theorem 48
9.1 Recollections and some standard limits . . . 48
9.2 Limsup and liminf . . . 50
9.3 The Bolzano–Weierstrass theorem . . . 52

10 The Completeness of the Real Numbers and Basic Properties of Infinite Series 55

10.1 Cauchy sequences . . . 55
10.2 Series . . . 57

11 Absolute Convergence and the Comparison Test 61
11.1 The harmonic series and the converse of the nth-term test . . . 61
11.2 Linearity and series with non-negative terms . . . 63
11.3 Absolute convergence . . . 64
11.4 Convergence tests . . . 66

12 The Ratio, Root, and Alternating Series Tests 68
12.1 Using the comparison and p-series tests . . . 68
12.2 The ratio test . . . 69
12.3 The root test . . . 71
12.4 The alternating series test . . . 72
12.5 Rearrangements of absolutely convergent series . . . 73

13 Introduction to Topology in Metric Spaces 76
13.1 Countability theorems . . . 76
13.2 Metric spaces . . . 78
13.3 Compact sets . . . 81

14 Limits of Functions 83
14.1 Continuous functions . . . 83

15 Limits of Functions in Terms of Sequences and Continuity 89
15.1 Left and Right Limits of Functions . . . 91
15.2 Continuity . . . 93

16 The Continuity of Sine and Cosine, and the Discontinuity of Dirichlet's Function 94

17 The Min/Max Theorem and Bolzano's Intermediate Value Theorem 100
17.1 Bounded functions . . . 100
17.2 The extreme value theorem . . . 101
17.3 The bisection method and the intermediate value theorem . . . 104

18 Uniform Continuity and the Definition of the Derivative 106
18.1 Uniform continuity . . . 106
18.2 The derivative . . . 111

19 Weierstrass's Example of a Continuous and Nowhere Differentiable Function 112

20 Differentiation Rules, Rolle's Theorem, and the Mean Value Theorem 119
20.1 Mean Value Theorem . . . 121
20.2 Important Remark on Continuity and Differentiability . . . 124

21 Taylor's Theorem and the Definition of Riemann Sums 124
21.1 Taylor's theorem . . . 124
21.2 The second derivative test and L'Hôpital's rule . . . 127
21.3 The Riemann integrals . . . 132

22 The Riemann Integral of a Continuous Function 133
22.1 Existence and uniqueness of the Riemann integral . . . 133
22.2 Properties of Riemann integrals . . . 138

23 The Fundamental Theorem of Calculus, Integration by Parts, and the Change of Variable Formula 139
23.1 Basic properties of the Riemann integral . . . 139
23.2 The Fundamental Theorem of Calculus . . . 141
23.3 Integration by parts and change of variables . . . 143

24 Pointwise and Uniform Convergence of Sequences of Functions 144
24.1 Fourier series: an application of integration by parts . . . 144
24.2 Sequences of functions and power series . . . 146
24.3 Pointwise and uniform convergence of sequences of functions . . . 147

25 Uniform Convergence, the Weierstrass M-Test, and Interchanging Limits 151
25.1 Interchange of limits . . . 152
25.2 The Weierstrass M-test . . . 155

26 Power Series and the Weierstrass Approximation Theorem 156
26.1 Back to power series . . . 157
26.2 The Weierstrass approximation theorem . . . 159

27 The Riemann Series Theorem (continued) 164

1 Sets, Set Operations, and Mathematical Induction

1.1 Sets

Definition

A *set* is a collection of objects called *elements* or *members*.

Definition

The *empty set* is the set with no elements, denoted by "$\emptyset$".

Notation

- $a \in S$ means that a is an element of S; $a \notin S$ means that a is *not* an element of S.
- $\forall$ reads "for all"; $\exists$ reads "there exists".
- $\Longrightarrow$ reads "implies"; $\Longleftrightarrow$ means that the two statements imply each other, "if and only if".

Definition

1) A set A is a *subset* of B, denoted by $A \subset B$, if $a \in A \Longrightarrow a \in B$.
2) Two sets are *equal* if $A \subset B$ and $B \subset A$.
3) A is a *proper subset* of B if $A \subset B$ and $A \neq B$. This is sometimes denoted by $A \subsetneq B$ (for the purpose of this course).

Notation: set-builder notation

- {a list of elements};
- $\{x \in A : x \text{ satisfies property } P(x)\}$;
- $\{x : x \text{ satisfies } P(x)\}$. The colon ":" is equivalent to a vertical bar "|".

Examples

1) $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$, the natural numbers.
2) $\mathbb{Z} = \{0, 1, -1, 2, -2, \ldots\}$, the integers.
3) $\mathbb{Q} = \{\frac{m}{n} : m, n \in \mathbb{Z},\ n \neq 0\}$, the rationals.
4) $\mathbb{R}$, the real numbers: "$\mathbb{Q}$ along with the irrationals $\pi, \sqrt{2}, \ldots$". Our first goal is to describe $\mathbb{R}$.
5) The odd numbers $\{2m - 1 \mid m \in \mathbb{N}\} = \{1, 3, 5, \ldots\}$.

Remark

It is trivial that

$$\text{odd numbers} \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \ (\subset \mathbb{C}).$$

1.2 Set operations

Definition

1) The *union* of A, B is the set $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

2) The *intersection* of A, B is the set $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

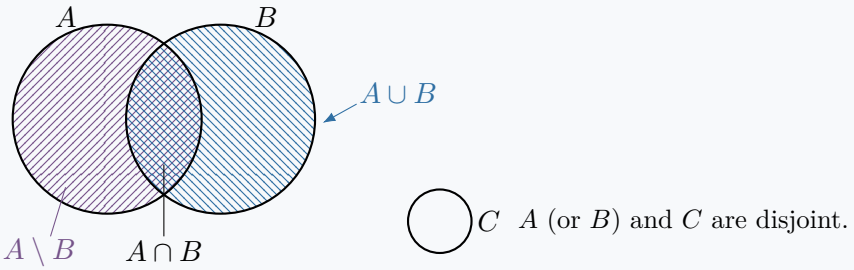
3) The *set difference* of A with respect to B is the set

$$A - B \quad \text{or} \quad A \setminus B = \{x : x \in A \text{ and } x \notin B\},$$

read as "A take away B".

4) The *complement* of A is the set $A^c = \{x : x \notin A\}$. Note that the complement mostly depends on a bigger set of which A is a subset; for example, $\mathbb{Q}^c$ in $\mathbb{R}$ is the irrationals, but $\mathbb{Q}^c$ in $\mathbb{Q}$ is $\emptyset$.

5) A and B are *disjoint* if $A \cap B = \emptyset$.



Theorem (De Morgan's laws)

If A, B, C are sets, then

1) $(B \cup C)^c = B^c \cap C^c$;

2) $(B \cap C)^c = B^c \cup C^c$;

3) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$;

4) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

Proof

1) Let B, C be sets. We want to prove that $(B \cup C)^c \subset B^c \cap C^c$ and $(B \cup C)^c \supset B^c \cap C^c$.
We want to show that $(B \cup C)^c \subset B^c \cap C^c$. Let $x \in (B \cup C)^c$. Then

$$x \notin B \cup C \implies x \notin B \text{ and } x \notin C \implies x \in B^c \text{ and } x \in C^c \implies x \in B^c \cap C^c.$$

Thus $(B \cup C)^c \subset B^c \cap C^c$.
We want to show that $B^c \cap C^c \subset (B \cup C)^c$. Let $x \in B^c \cap C^c$. Then $x \in B^c$ and $x \in C^c$, so

$$x \notin B \text{ and } x \notin C \implies x \notin B \cup C \implies x \in (B \cup C)^c.$$

Thus $B^c \cap C^c \subset (B \cup C)^c$. Therefore $B^c \cap C^c = (B \cup C)^c$.

2) Let $x \in (B \cap C)^c$. Then

$$x \notin B \cap C \iff x \notin B \text{ or } x \notin C \iff x \in B^c \text{ or } x \in C^c \iff x \in B^c \cup C^c.$$

Thus $x \in (B \cap C)^c \iff x \in B^c \cup C^c$, so $(B \cap C)^c \subset B^c \cup C^c$ and $(B \cap C)^c \supset B^c \cup C^c$. Therefore $(B \cap C)^c = B^c \cup C^c$.
3) Let $x \in A \setminus (B \cup C)$. Then

$$\begin{aligned} x \in A \text{ and } x \notin B \cup C &\iff x \in A,\ x \notin B \text{ and } x \notin C \\ &\iff x \in A \text{ and } x \notin B,\quad x \in A \text{ and } x \notin C \\ &\iff x \in A \setminus B \text{ and } x \in A \setminus C. \end{aligned}$$

Thus $x \in (A \setminus B) \cap (A \setminus C)$; that is,

$$x \in A \setminus (B \cup C) \iff x \in (A \setminus B) \cap (A \setminus C).$$

Thus $A \setminus (B \cup C)$ is both a subset and a superset of $(A \setminus B) \cap (A \setminus C)$, so the two sets are equal.
4) Let $x \in A \setminus (B \cap C)$. Then

$$\begin{aligned} x \in A \text{ and } x \notin B \cap C &\iff x \in A \text{ and } (x \notin B \text{ or } x \notin C) \\ &\iff x \in A \text{ and } x \notin B,\quad \text{or}\quad x \in A \text{ and } x \notin C \\ &\iff x \in A \setminus B \text{ or } x \in A \setminus C. \end{aligned}$$

Thus $x \in (A \setminus B) \cup (A \setminus C) \iff x \in A \setminus (B \cap C)$. Therefore $(A \setminus B) \cup (A \setminus C) = A \setminus (B \cap C)$.

1.3 Induction

Induction is a tool to prove theorems related to the natural numbers.

Recall

$\mathbb{N} = \{1, 2, \ldots\}$ has an ordering $1 < 2 < 3 < \cdots$.

Axiom (well-ordering property of $\mathbb{N}$)

If $S \subset \mathbb{N}$ and $S \neq \emptyset$, then S has a *least element*; i.e. there exists $x \in S$ such that $x \leq y$ for all $y \in S$.

Theorem (Induction; Pascal, 1665)

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. Assume that $P(n)$ satisfies

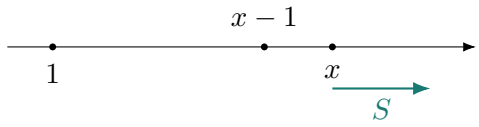
1) (Base case) $P(1)$ is true;

2) (Inductive step) if $P(m)$ is true, then $P(m+1)$ is true.

Then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof

Let $S = \{n \in \mathbb{N} : P(n) \text{ is not true for } n\}$. Then we want to show that $S = \emptyset$. We will prove this by contradiction. Suppose 1) and 2) are true for P.
[By contradiction: assume that the negation of the statement is true, namely $S \neq \emptyset$; we want to arrive at some false statement. Then, by the rules of logic, we can conclude that the assumption $S \neq \emptyset$ is false.]
Suppose $S \neq \emptyset$. By the well-ordering property of $\mathbb{N}$, S has a least element $x \in S$. Since $P(1)$ is true by assumption, $1 \notin S$, so $x \neq 1$. Since $S \subset \mathbb{N}$, $x > 1$. Since x is the least element in S, and $x - 1 < x$, we have $x - 1 \notin S$.



Therefore $P(x-1)$ is true. Then, by property 2) of P, $P[(x-1)+1] = P(x)$ is true, which means $x \notin S$.
Thus, if $S \neq \emptyset$, we showed that there exists x such that $x \in S$ and $x \notin S$, a contradiction. Thus the assumption $S \neq \emptyset$ is false, and $S = \emptyset$.

Using induction

If we want to show that $P(n)$ is true for all $n \in \mathbb{N}$, then we prove:

1) $P(1)$ is true (base case);

2) assuming $P(m)$ is true, $P(m+1)$ is true (inductive step).

Theorem (geometric sum)

For all $c \neq 1$ and all $n \in \mathbb{N}$,

$$1 + c + c^2 + \cdots + c^n = \frac{1 - c^{n+1}}{1 - c}. \qquad (*)$$

We call this statement $P(n)$.

Proof

We prove $(*)$ by induction.
1) Base case. For $P(1)$,

$$1 + c^1 = \frac{1 - c^2}{1 - c} = \frac{(c-1)(c+1)}{(c-1)} = c + 1.$$

2) Inductive step. Assume $P(m)$ is true:

$$1 + c + \cdots + c^m = \frac{1 - c^{m+1}}{1 - c}.$$

Then

$$\begin{aligned} 1 + c + \cdots + c^m + c^{m+1} &= \frac{1 - c^{m+1}}{1 - c} + c^{m+1} \\ &= \frac{1 - c^{m+1} + c^{m+1} - c^{m+2}}{1 - c} \\ &= \frac{1 - c^{m+2}}{1 - c}. \end{aligned}$$

Thus $P(m+1)$ is true (if $P(m)$ is true). By induction, $(*)$ is true for all $n \in \mathbb{N}$.

Theorem

If $c \geq -1$, then for all $n \in \mathbb{N}$,

$$(1 + c)^n \geq 1 + nc.$$

Proof (by induction)

1) (Base case) For $P(1)$, $(1 + c)^1 = 1 + c$; thus $P(1)$ is true.
2) (Inductive step) Assume $P(m)$ is true: $(1 + c)^m \geq 1 + mc$. We have

$$(1 + c)^{m+1} = (1 + c)(1 + c)^m \geq (1 + c)(1 + mc) = 1 + (1 + m)c + \underbrace{mc^2}_{\geq 0}.$$

[We can do this since $1 + c \geq 0$.] Thus

$$(1 + c)^{m+1} \geq 1 + (1 + m)c + mc^2 \geq 1 + (1 + m)c.$$

Thus $P(m+1)$ is true. By induction, $P(n)$ is true for all $n \in \mathbb{N}$.

2 Cantor's Theory of Cardinality

Question

When do two sets A and B have the same size?
Answer (Cantor). When the elements of the sets can be paired off. (This is the theory of cardinality.)

Example

For $\{a, b, c\}$ and $\{1, 2, 3\}$:

$$a \longrightarrow 1, \qquad b \longrightarrow 2, \qquad c \longrightarrow 3.$$

2.1 Functions

Definition

If A, B are sets, then a *function* $f : A \to B$ is a mapping that assigns to each $x \in A$ a unique element $f(x) \in B$.

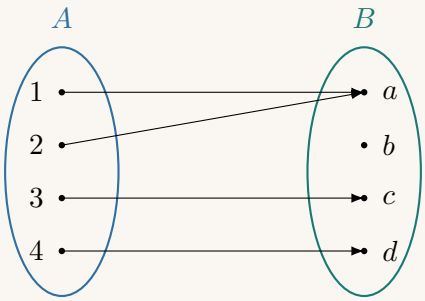
Definition

Let $f : A \to B$.

1) If $C \subset A$, we define $f(C) = \{f(x) : x \in C\}$.
2) If $D \subset B$, we define $f^{-1}(D) = \{x \in A : f(x) \in D\}$.

We call $f^{-1}(D)$ the *inverse image* of D; it is not the inverse of the function f. The inverse image always exists; the inverse of f does not.

Example



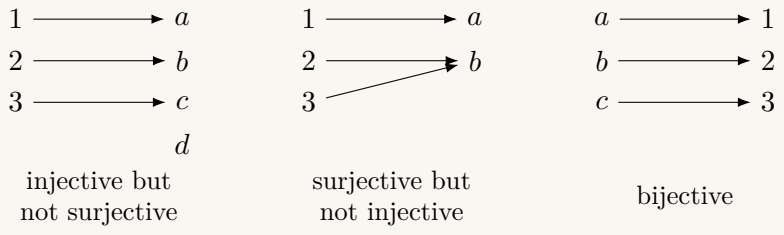
$f(\{1,2\}) = \{a\}$, $f(\{1,3\}) = \{a,c\}$, $f^{-1}(\{a\}) = \{1,2\}$, $f^{-1}(\{a,c,d\}) = \{1,2,3,4\}$.

Definition

Let $f : A \to B$.

1) f is *injective*, or "one-to-one", if $f(x_1) = f(x_2) \implies x_1 = x_2$. [Equivalently, $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$.]
2) f is *surjective*, or "onto", if $f(A) = B$. [Equivalently, for every $y \in B$ there exists $x \in A$ such that $f(x) = y$.]
3) f is *bijective* if f is injective and surjective.

Examples



In the first case the mapping is injective but not surjective. In the second case it is surjective but not injective, and in the third case it is bijective.

Definition

Let $f : A \to B$ and $g : B \to C$.

1) The *composition* $g \circ f : A \to C$ is defined by

$$g \circ f : \ x \longmapsto g(f(x)), \qquad x \in A.$$

2) If f is bijective, the *inverse function* $f^{-1} : B \to A$ is defined by

$$f^{-1} : \ B \longrightarrow A, \qquad y \longmapsto f^{-1}(y) \in A, \qquad \text{such that } f(f^{-1}(y)) = y.$$

Note

The inverse only exists for bijective functions.

- If f^{-1} is a function (a mapping), then it is an *inverse function*, written $f^{-1}(a)$.
- If $f^{-1}(a)$ is a set, then it is an *inverse image*.

2.2 Cardinality

Definition

Two sets A and B have the same *cardinality* if there exists a bijection $f : A \to B$.

Notation (shorthands)

1) If A, B have the same cardinality, we write $|A| = |B|$ (if they do not, we write $|A| \neq |B|$).
2) If $|A| = |\{1, 2, \ldots, n\}|$, we write $|A| = n$, and we say A is *finite*; otherwise A is *infinite*.
3) If there exists an injective map $f : A \to B$, we write $|A| \leq |B|$.
4) If $|A| \leq |B|$ but $|A| \neq |B|$ (i.e. there exists an injective map $f : A \to B$, but there is no surjective map from A to B), we write $|A| < |B|$.

Theorem (Cantor–Schröder–Bernstein; no proof)

If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.

Note

This is a statement about *cardinality*, *not* about the real numbers. Although this seems obvious with the ordering property in $\mathbb{R}$, when it comes to cardinality (especially with infinite sets), things get different from $\mathbb{R}$. Essentially we cannot treat these symbols as real numbers; rather, this is only a notation for cardinality (essentially, for mappings).

Theorem

If $|A| = |B|$, then $|B| = |A|$.

Proof

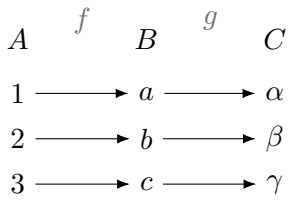
Suppose $|A| = |B|$. Then there exists a bijection $f : A \to B$. Then $f^{-1} : B \to A$ is a bijection, so $|B| = |A|$.

Theorem

If $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$.

Proof

Since $|A| = |B|$ and $|B| = |C|$, there exist two bijections $f : A \to B$ and $g : B \to C$. We want to show that $g \circ f : A \to C$ is a bijection; then $|A| = |C|$. Let $h : A \to C$ be the function $h(x) = g \circ f(x)$, $x \in A$.



$g \circ f$ is still a bijection.

h is one-to-one (injective). Suppose $h(x_1) = h(x_2)$. Then

$$(g \circ f)(x_1) = (g \circ f)(x_2), \qquad g(f(x_1)) = g(f(x_2)).$$

Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$. Hence $h(x_1) = h(x_2) \implies x_1 = x_2$; therefore h is injective (one-to-one).
h is onto (surjective). Let z be any element in C. Since g is surjective, there exists $y \in B$ such that $g(y) = z$. Since f is surjective, there exists $x \in A$ such that $f(x) = y$. Then for all $z \in C$ there exists $x \in A$ such that

$$g(f(x)) = (g \circ f)(x) = h(x) = z.$$

Thus h is surjective (onto).
There exists a bijection $h : A \to C$; hence $|A| = |C|$.

Definition

If $|A| = |\mathbb{N}|$, then A is *countably infinite*. If A is a finite or countably infinite set, then we say A is *countable*. If A is not countable (neither finite nor countably infinite), we say A is *uncountable*.

Theorem

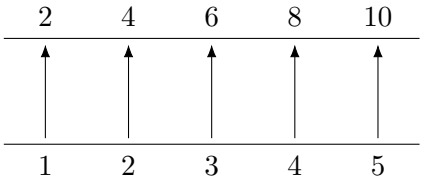
1) $|\{2n : n \in \mathbb{N}\}| = |\mathbb{N}|$.

2) $|\{2n-1 : n \in \mathbb{N}\}| = |\mathbb{N}|$.

Proof

1) We want to show that $|\mathbb{N}| = |\{2n : n \in \mathbb{N}\}|$. Let

$$f : \mathbb{N} \longrightarrow \{2n : n \in \mathbb{N}\}, \qquad n \longmapsto 2n, \quad n \in \mathbb{N}.$$



f *is one-to-one*, i.e. if $f(n_1) = f(n_2)$, then $n_1 = n_2$. Suppose $f(n_1) = f(n_2)$. Then $2n_1 = 2n_2$, so $n_1 = n_2$. Thus f is one-to-one (injective).
f *is onto*: for all $m \in \{2n : n \in \mathbb{N}\}$ there exists $n_0 \in \mathbb{N}$ such that $f(n_0) = m$. Since $m \in \{2n : n \in \mathbb{N}\}$, there exists $k \in \mathbb{N}$ such that $2k = m$. Let $k = n_0$; then $f(n_0) = 2n_0 = 2k = m$. Thus f is onto (surjective).
So $f : \mathbb{N} \to \{2n : n \in \mathbb{N}\}$ is a bijection. Therefore $|\mathbb{N}| = |\{2n : n \in \mathbb{N}\}|$.
2) Similarly, $g : \mathbb{N} \to \{2n-1 : n \in \mathbb{N}\}$, $n \longmapsto 2n-1$, $n \in \mathbb{N}$, is also a bijection.

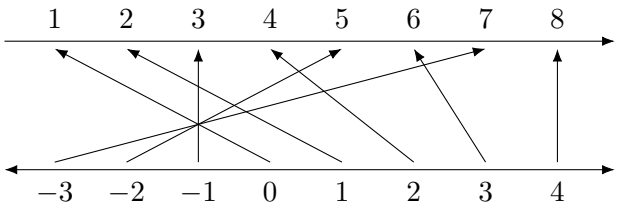
Theorem

$|\mathbb{Z}| = |\mathbb{N}|$.

Proof

Let $f : \mathbb{Z} \to \mathbb{N}$ be defined by

$$\begin{cases} a \longmapsto 2a, & a \in \mathbb{Z} \text{ and } a > 0, \\ 0 \longmapsto 1, & \\ b \longmapsto -2b+1, & b \in \mathbb{Z} \text{ and } b < 0. \end{cases}$$



f *is onto.* Let x be a natural number.

(1) If $x = 1$, then $f(0) = 1$, with $0 \in \mathbb{Z}$.

(2) If x is an odd number and $x > 1$: then there exists $n \in \mathbb{N}$ such that $2n - 1 = x$. Thus $\frac{x-1}{2} = \frac{2n-2}{2} = n - 1$ is an integer. Since $x > 1$, $\frac{x-1}{2} > 0$. Then $\frac{x-1}{-2}$ is a negative integer.

Thus

$$f(-n+1) = f\Big(\frac{x-1}{-2}\Big) = x. \tag{*}$$

(3) If x is an even number and $x > 1$: then there exists $n \in \mathbb{N}$ such that $2n = x$. Thus $\frac{x}{2} = n$ is a positive integer; therefore $f(n) = f\big(\frac{x}{2}\big) = x$.

For all $x \in \mathbb{N}$ there exists $n \in \mathbb{Z}$ such that $f(n) = x$; thus $f : \mathbb{Z} \to \mathbb{N}$ is onto (surjective).
f is one-to-one. Let $f(n_1) = f(n_2) = x$, where $x \in \mathbb{N}$ and $n_1, n_2 \in \mathbb{Z}$. Suppose $a \in \mathbb{Z}$ is such that $f : a \longmapsto x$.

(1) $x = 1$: if $2a = 1$, then $a = \frac{1}{2} \notin \mathbb{Z}$ (so n_1, n_2 cannot be positive integers). If $-2a + 1 = 1$, then $a = 0 \in \mathbb{Z}$, but $a = 0$ is not a negative integer. Thus $f(n_1) = f(n_2) = 1 \Longrightarrow n_1 = n_2 = 0$.

(2) x is odd and $x > 1$: if $2a = x$, then $a = \frac{x}{2}$; since x is odd, $a \notin \mathbb{Z}$ (so n_1, n_2 cannot be positive integers). If $-2a + 1 = x$, then $a = \frac{x-1}{-2}$. Thus, if $f(n_1) = f(n_2) = x$, then $n_1 = n_2 = \frac{x-1}{-2}$, a negative integer by $(*)$.

(3) x is even and $x > 1$: if $-2a+1 = x$, then $a = \frac{x-1}{-2}$; since x is even, $a = \frac{x-1}{-2} \notin \mathbb{Z}$ (so n_1, n_2 cannot be negative integers). If $2a = x$, then $a = \frac{x}{2}$; since x is even, $a = \frac{x}{2} \in \mathbb{Z}_{>0}$. Thus $f(n_1) = f(n_2) = x \Longrightarrow n_1 = n_2 = \frac{x}{2}$.

For all $x \in \mathbb{N}$, $f(n_1) = f(n_2) \Longrightarrow n_1 = n_2$. Therefore f is one-to-one (injective).
Hence $f : \mathbb{Z} \to \mathbb{N}$ is a bijection. Thus $|\mathbb{Z}| = |\mathbb{N}|$.

Theorem

Let $\mathbb{Q}_{>0} := \{m \in \mathbb{Q} : m > 0\}$. Then $|\mathbb{Q}_{>0}| = |\mathbb{N}|$.

Note

Be aware that the proof below does not take $m = 1$ into consideration, because of its triviality.

Proof

Since $m \in \mathbb{Q}_{>0}$, there exist $p, q \in \mathbb{N}$ such that $(p, q) = 1$ (coprime) and $m = \frac{p}{q}$. By the Fundamental Theorem of Arithmetic, there exists a unique expression

$$m = \frac{p}{q} = \frac{p_1^{r_1} p_2^{r_2} \cdots p_s^{r_s}}{q_1^{l_1} q_2^{l_2} \cdots q_t^{l_t}},$$

where $r_1, \ldots, r_s, l_1, \ldots, l_t$ are natural numbers and $p_1, \ldots, p_s, q_1, \ldots, q_t$ are distinct prime numbers.
Note. It seems that $p_1^{r_1} p_2^{r_2} \cdots p_s^{r_s} q_1^{l_1} \cdots q_t^{l_t}$ is a good idea, yet the problem is that if we use this, $\frac{9}{7}$ and $\frac{7}{9}$ will both be mapped to 63. Then this map is not injective. So we need to figure out a way to separate the cases of numerators and denominators (reciprocals). Besides this, the map is unique because of the coprimeness of p, q and the uniqueness given by the Fundamental Theorem of Arithmetic. Another way to see this: if we take the map to $p_1^{r_1} \cdots p_s^{r_s} q_1^{l_1} \cdots q_t^{l_t}$, then, given an n, we cannot determine which primes are $q_1, \ldots, q_t$ and which are $p_1, \ldots, p_s$. For

example, $n = 315 = 3^2 \times 7 \times 5$; then is

$$m = \frac{3^2}{7\times 5}? \quad \frac{7}{3^2\times 5}? \quad \frac{7\times 5}{3^2}? \quad 7\times 3^2\times 5? \quad \frac{1}{7\times 3^2\times 5}? \quad \dots$$

So we cannot determine which of $p_1, \dots, q_t$ are in the denominator and which are in the numerator (but $m \neq \frac{3}{3\times 7\times 5}$ for sure, since $(p, q) = 1$ are coprime, i.e. $p_1, \dots, p_s, q_1, \dots, q_t$ are distinct primes).
Let

$$f : \mathbb{Q}_{>0} \longrightarrow \mathbb{N}, \qquad m = \frac{p}{q} = \frac{p_1^{r_1}\cdots p_s^{r_s}}{q_1^{l_1}\cdots q_t^{l_t}} \longmapsto p_1^{2r_1}\cdots p_s^{2r_s}\, q_1^{2l_1-1}\cdots q_t^{2l_t-1} =: n.$$

f is onto (surjective). Let x be a natural number. Then, by the Fundamental Theorem of Arithmetic, there exists a unique prime factorization

$$x = x_1^{j_1} x_2^{j_2} \cdots x_i^{j_i},$$

where $x_1, \dots, x_i$ are distinct prime numbers and $j_1, \dots, j_i$ are natural numbers. Among $j_1, \dots, j_i$, suppose $j_{y_1}, \dots, j_{y_\alpha}$ are the odd exponents and $j_{z_1}, \dots, j_{z_\beta}$ are the even exponents. Then

$$m = \frac{p}{q} = \frac{x_{z_1}^{j_{z_1}/2}\cdots x_{z_\beta}^{j_{z_\beta}/2}}{x_{y_1}^{(j_{y_1}+1)/2}\cdots x_{y_\alpha}^{(j_{y_\alpha}+1)/2}} \in \mathbb{Q}_{>0},$$

since all the exponents will be positive integers, and $x_{z_1}, \dots, x_{z_\beta}, x_{y_1}, \dots, x_{y_\alpha}$ are primes. Then $f(m) = x$, by our construction of m. For all $x \in \mathbb{N}$ there exists $m \in \mathbb{Q}_{>0}$ such that $f(m) = x$; f is surjective.
f is one-to-one. By the Fundamental Theorem of Arithmetic, the prime factorization is unique. Furthermore, the odd exponents and the even exponents, and their bases (the prime numbers), are all fixed, respectively. Thus, if $f(m_1) = f(m_2) = x$ with $x \in \mathbb{N}$, then the m constructed above such that $f(m) = x$ is unique, since, again, namely

$$\underbrace{j_{z_1}, \dots, j_{z_\beta}}_{\text{even exponents}}, \quad \underbrace{p_{z_1}, \dots, p_{z_\beta}}_{\text{bases}}, \quad \underbrace{j_{y_1}, \dots, j_{y_\alpha}}_{\text{odd exponents}}, \quad \underbrace{q_{y_1}, \dots, q_{y_\alpha}}_{\text{bases}}$$

are *all fixed*. Therefore $f(m_1) = f(m_2) = x \Longrightarrow m_1 = m_2\ (= m)$; f is one-to-one.
Thus f is a bijection, and $|\mathbb{Q}_{>0}| = |\mathbb{N}|$.

Example

Consider

$$f\Big(\frac{3^2}{5\times 7}\Big), \qquad f\Big(\frac{7\times 5}{3^2}\Big), \qquad f(7\times 5\times 3^2), \qquad f\Big(\frac{1}{7\times 5\times 3^2}\Big).$$

They are now mapped to different natural numbers in $\mathbb{N}$.

Theorem

$|\mathbb{Q}| = |\mathbb{N}|$.

Proof

Since $|\mathbb{Q}_{>0}| = |\mathbb{N}|$, there exists a bijection $f : \mathbb{Q}_{>0} \to \mathbb{N}$. Let

$$h : \mathbb{Q}_{<0} \longrightarrow \mathbb{Q}_{>0}, \qquad m \longmapsto -m.$$

It is easy to verify that h is a bijection; thus $|\mathbb{Q}_{<0}| = |\mathbb{Q}_{>0}|$. Since $|\mathbb{N}| = |\mathbb{Z}_{<0}|$ and $|\mathbb{Q}_{>0}| = |\mathbb{N}|$,

$$|\mathbb{Q}_{<0}| = |\mathbb{Q}_{>0}| = |\mathbb{N}| = |\mathbb{Z}_{<0}|.$$

Therefore there exists a bijection

$$g : \mathbb{Q}_{<0} \longrightarrow \mathbb{Z}_{<0}, \qquad m \longmapsto n.$$

Let $\sigma : \mathbb{Q} \to \mathbb{Z}$ be given by

$$\sigma(m) = \begin{cases} f(m), & \text{if } m \in \mathbb{Q}_{>0}, \\ 0, & \text{if } m = 0, \\ g(m), & \text{if } m \in \mathbb{Q}_{<0}. \end{cases}$$

Since f, g are bijections, it is easy to check that σ is also a bijection. Thus $|\mathbb{Q}| = |\mathbb{Z}|$. Since $|\mathbb{Z}| = |\mathbb{N}|$, $|\mathbb{Q}| = |\mathbb{N}|$.

Question

Till now, all the sets have had size smaller than or equal to that of $\mathbb{N}$, i.e. $|S| \leq |\mathbb{N}|$. Does there exist a set A such that $|\mathbb{N}| < |A|$?
Answer. Yes. $\mathbb{R}$ is an uncountable set, yet we cannot define it for now. But there is another type of set that satisfies this.

Definition

If A is a set, we define the *power set* of A, denoted by $\mathcal{P}(A)$ or $P(A)$, by

$$\mathcal{P}(A) := \{B : B \subset A\}.$$

Examples

- $A = \emptyset$: $\mathcal{P}(A) = \{\emptyset\}$; or, if "$\emptyset$" is confusing, $\mathcal{P}(A) = \{\{\}\}$. Even if A is empty, i.e. it has 0 elements, its power set has one element, which is the empty set $\emptyset$, now an element of $\mathcal{P}(A)$.
- $A = \{1\}$: $\mathcal{P}(A) = \{\emptyset, \{1\}\}$, or $\{\{\}, \{1\}\}$.
- $A = \{1, 2\}$: $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Theorem

If $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

Proof (by induction)

1) Base case: $n = 1$. When $|A| = 1$, $A = \{a_1\}$ and $\mathcal{P}(A) = \{\emptyset, \{a_1\}\}$. So $|A| = 1$ and $|\mathcal{P}(A)| = 2 = 2^1$.
2) Inductive step: let the elements in A be a_i. Assume $P(m)$ is true: if $|A| = m$, then $|\mathcal{P}(A)| = 2^m$. Let $A_m = \{a_1, \ldots, a_m\}$. Then, for $P(m+1)$, $A_{m+1} = \{a_1, \ldots, a_m, a_{m+1}\}$. Add a_{m+1} to every subset of A_m in $\mathcal{P}(A_m)$. Then we obtain *all the subsets of* A_{m+1} *that contain* a_{m+1}. Therefore,

$$\begin{aligned}\text{number of subsets of } A_{m+1} \text{ containing } a_{m+1} &= \text{number of elements in } \mathcal{P}(A_m) \\ &= \text{number of subsets of } A_m.\end{aligned}$$

Hence

$$|\mathcal{P}(A_{m+1})| = \underbrace{|\mathcal{P}(A_m)|}_{\substack{\text{number of subsets of } A_m \\ \text{not containing } a_{m+1}}} + \underbrace{|\mathcal{P}(A_m)|}_{\substack{\text{number of subsets of } A_{m+1} \text{ containing } a_{m+1}, \\ \text{obtained by adding } a_{m+1} \text{ to the subsets of } A_m \\ \text{(the elements of } \mathcal{P}(A_m))}}.$$

Since $|\mathcal{P}(A_m)| = 2^m$ by assumption,

$$|\mathcal{P}(A_{m+1})| = 2^m + 2^m = 2^m \times 2 = 2^{m+1}.$$

So $P(m)$ is true $\implies P(m+1)$ is true. Thus, by induction, $P(n)$ is true for all $n \in \mathbb{N} \cup \{0\}$ (the case 0 is trivial).

Remark

We can prove by induction that $2^n > n$ for all $n \in \mathbb{N}$.

Theorem (Cantor)

If A is a set, then $|A| < |\mathcal{P}(A)|$.

The significance of this theorem is that A does not need to be finite. (The proof is given at the beginning of the next lecture.)

Corollary

$$|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < |\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathbb{N})))| < \cdots.$$

Remark

The theorem says that there is an infinity of infinities (a higher order of ∞).

Question (Continuum Hypothesis)

Does there exist a set A such that $|\mathbb{N}| < |A| < |\mathcal{P}(\mathbb{N})|$?

3 Cantor's Cardinality Theorem and the Incompleteness of the Rational Numbers

3.1 Cardinality

Definition

If A is a set, the *power set* of A is

$$\mathcal{P}(A) = \{B : B \subset A\}.$$

Example

$|\emptyset| = 0$, while $|\mathcal{P}(\emptyset)| = |\{\emptyset\}| = 1$.

Theorem (Cantor)

If A is a set (it can be infinite), then $|A| < |\mathcal{P}(A)|$.

Remark

Consequence:

$$|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \cdots.$$

Proof

Let A be a set. Define

$$f : A \longrightarrow \mathcal{P}(A), \qquad x \longmapsto f(x) = \{x\} \in \mathcal{P}(A).$$

Apparently f is one-to-one (injective): if $f(x) = f(y)$, then $\{x\} = \{y\}$, so $x = y$. Since $f : A \to \mathcal{P}(A)$ is injective, by the definition on page 7, we have $|A| \leq |\mathcal{P}(A)|$.
Now we want to show that $|A| \neq |\mathcal{P}(A)|$; this is proved by contradiction. Suppose $|A| = |\mathcal{P}(A)|$. Then there exists a bijection $g : A \to \mathcal{P}(A)$. Let $B \subset A$ be the set

$$B := \{x \in A : x \notin g(x)\}.$$

Remember that $g(x) \in \mathcal{P}(A)$ is a subset of A, so an element of B is not in its image in $\mathcal{P}(A)$. Since $B \subset A$, we have $B \in \mathcal{P}(A)$. Since g is surjective, there exists $b \in A$ such that $g(b) = B$.
Case 1. If $b \in g(b)$, then $b \in B$; but by the definition of B, $b \notin g(b)$.
Case 2. If $b \notin g(b)$, then $b \notin B$; but by the definition of B, since $b \notin g(b)$, we have $b \in B$.
Thus there exists $b \in A$ such that

$$b \in g(b) \quad \Longleftrightarrow \quad b \notin g(b),$$

a contradiction. Thus the assumption $|A| = |\mathcal{P}(A)|$ is false. Therefore $|A| < |\mathcal{P}(A)|$.

The idea of the proof

The idea of this proof is some sort of: we are giving a statement of B, yet the validity of the statement still needs reference to B. Analogue: "I am a liar."

$$\text{True} \Longrightarrow \text{Liar} \Longrightarrow \text{Lying} \Longrightarrow \text{Not a liar} \Longrightarrow \text{Contradiction},$$
$$\text{False} \Longrightarrow \text{Not a liar} \Longrightarrow \text{Not lying} \Longrightarrow \text{Is a liar} \Longrightarrow \text{Contradiction}.$$

3.2 Real numbers

Theorem

There exists a unique ordered field containing $\mathbb{Q}$ with the *least upper bound property* (l.u.b.p.), denoted by $\mathbb{R}$.

Note

This theorem will not be proved. The proof requires the construction of $\mathbb{R}$ from $\mathbb{Q}$; see *Principles of Mathematical Analysis*, pp. 17–21 and the Appendix. The construction of $\mathbb{R}$ can be achieved through *Dedekind cuts* or *equivalence classes of Cauchy sequences*.

Goal

Describe $\mathbb{R}$, and understand this definition of $\mathbb{R}$.

3.3 Ordered sets and the least upper bound property

Definition

An *ordered set* is a set S with a relation, denoted by "$<$", such that

1) for all $x, y \in S$, either $x = y$, $x < y$, or $y < x$;

2) if $x < y$ and $y < z$, then $x < z$.

Example

$\mathbb{Z}$ (or $\mathbb{N}$): $m < n$ if and only if $n - m \in \mathbb{N}$.

Example

$\mathbb{Q}$: $q < r$ if and only if there exist $m, n \in \mathbb{N}$ such that $r - q = \dfrac{m}{n}$.

Non-example

Let $S = \mathcal{P}(\mathbb{N})$. We define a relation $A \prec B$ if and only if $A \subset B$. Then the order $\prec$ satisfies 2):

$$A \prec B \implies A \subset B, \qquad B \prec C \implies B \subset C, \qquad \implies \quad A \subset C \implies A \prec C.$$

However, $\prec$ doesn't satisfy 1): consider any two disjoint sets, for example $\{0\}$ and $\{1\}$. Then

$$\{0\} \not\subset \{1\}, \qquad \{0\} \neq \{1\}, \qquad \{0\} \not\supset \{1\},$$

that is,

$$\{0\} \not< \{1\}, \qquad \{0\} \neq \{1\}, \qquad \{1\} \not< \{0\}.$$

Example: dictionary ordering of $\mathbb{Q} \times \mathbb{Q}$

Recall that $A \times B := \{(a, b) : a \in A,\ b \in B\}$. We say $(a, b) < (q, r)$ if either $a < q$, or $a = q$ and $b < r$. Then we can check that "$<$" is an order on $\mathbb{Q} \times \mathbb{Q}$:

1) For all $(a, b), (q, r) \in \mathbb{Q} \times \mathbb{Q}$:

   (a) if $a < q$, then $(a, b) < (q, r)$;

   (b) if $q < a$, then $(q, r) < (a, b)$;

   (c) if $q = a$, then

   $$\begin{cases} r = b, & \text{then } (a, b) = (q, r); \\ r < b, & \text{then } (q, r) < (a, b); \\ b < r, & \text{then } (a, b) < (q, r). \end{cases}$$

2) Assume $(a, b) < (q, r)$ and $(q, r) < (x, y)$. Without loss of generality (WLOG), let $a = q$ and $q < x$; thus $b < r$. Thus $a < x$, and therefore $(a, b) < (x, y)$. If $a = q = x$, then

   $$(a, b) < (q, r) \implies b < r, \qquad (q, r) < (x, y) \implies r < y, \qquad \implies \quad b < y.$$

   Thus $a = x$ and $b < y$, so $(a, b) < (x, y)$.

Thus $<$ is an order on $\mathbb{Q} \times \mathbb{Q}$.

Definition

Let S be an ordered set and let $E \subset S$.

1) If there exists $b \in S$ such that $x \leq b$ for all $x \in E$, then we say E is *bounded above* and b is an *upper bound* for E.

2) If there exists $b \in S$ such that $b \leq x$ for all $x \in E$, then we say E is *bounded below* and b is a *lower bound* for E.

3) We say $b_0 \in S$ is the *least upper bound* for the subset E if

   (a) b_0 is an upper bound for E;

   (b) if b is any upper bound for E, then $b_0 \leq b$.

   We can also say b_0 is the *supremum* of E, denoted by $b_0 = \sup E$.

4) $b_0 \in S$ is the *greatest lower bound* for E if

   (a) b_0 is a lower bound for E;

   (b) if b is any lower bound for E, then $b \leq b_0$.

   We also say b_0 is the *infimum* of E, denoted by $b_0 = \inf E$.

Example

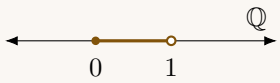
Let $S = \mathbb{Z}$ and $E = \{-1, 0, 2\}$.



Upper bounds: $2, 3, 4, 5, \ldots$ (not 2.5, since $2.5 \notin S = \mathbb{Z}$). So $\sup E = 2$. (This is not a strict proof.) Lower bounds: $-1, -2, -3, \ldots$, so $\inf E = -1$.

Example

Let $S = \mathbb{Q}$ and $E = \{q \in \mathbb{Q} : 0 \leq q < 1\}$.



Upper bounds: $1, 2, 3, \frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \ldots$, so $\sup E = 1$; note that this $\sup E \notin E$. Lower bounds: $0, -\frac{1}{2}, -\frac{1}{3}, \ldots, -1, -2, \ldots$, so $\inf E = 0$.

Remark

Neither $\sup E$ nor $\inf E$ needs to be in E, or to exist.

Example

Let $S = \mathbb{Z}$ and $E = \mathbb{N}$. Then $\sup E$ does not exist (DNE), while $\inf E = 1$.

Definition

An ordered set S has the *least upper bound property* if every subset $E \subset S$ which is non-empty and bounded above has a supremum in S.

Example

$S = \{0\}$. The only choice of E will be $\{0\}$, and $\sup E = 0 \in S$.

Example

$S = \{0, 1\}$. Then

$$\begin{aligned} E = \{0\} &\implies \sup E = 0 \in S, \\ E = \{1\} &\implies \sup E = 1 \in S, \\ E = \{0, 1\} &\implies \sup E = 1 \in S. \end{aligned}$$

Example

$S = \{-1, -2, -3, \ldots\}$. If $E \subset S$ and E is nonempty, then

$$-E = \{-x : x \in E\} \subset \mathbb{N}.$$

Then by the well-ordering property, since $-E$ is a nonempty subset of $\mathbb{N}$, there exists $m \in -E$ such that $m \leq -x$ for all x. Hence $x \leq -m$ for all $x \in E$, and $-m \in E$; so $-m$ is the supremum

of E.
Proof by contradiction. Assume $-m$ is not $\sup E$, and there exists $p \in S$ such that $p = \sup E$. Then $-m$ is not $\sup E$, so there exists an upper bound b of E such that $b < -m$. By definition, $\sup E = p \leq b < -m$, so $p < -m$. But $-m$ is an element of E, thus p is not an upper bound of E, a contradiction.

Example

$\mathbb{Z}$ has the l.u.b.p.

Note

Remember that the l.u.b.p. is talking about *non-empty* subsets that are *bounded above*!

Claim

$\mathbb{Q}$ does not have the l.u.b.p. A counterexample: if

$$E = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\},$$

then $\sup E$ does not exist in $\mathbb{Q}$.

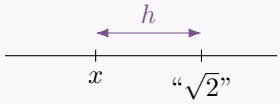
Theorem

If there exists $x \in \mathbb{Q}$ with $x = \sup\{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$, then $x \geq 1$ and $x^2 = 2$.

Proof

Let $E = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$. Since $1 \in \mathbb{Q}$, $1 > 0$, and $1^2 = 1 < 2$, we have $1 \in E$. Then $x = \sup E$ implies $x \geq 1$.
We prove $x^2 = 2$ by contradiction: we prove $x^2 \geq 2$ and $x^2 \leq 2$, which together give $x^2 = 2$.
$x^2 \geq 2$. Assume $x^2 < 2$.

Scratch work: what we want to show



If $x^2 < 2$ (that is, $x <$“$\sqrt{2}$”), then x is not an upper bound of E. So there exists some number in E such that the number is greater than x; thus $x^2 < \text{number}^2 < 2$. So there exists $h \in \mathbb{Q}_{>0}$ such that $(x+h)^2 < 2$, i.e.

$$x^2 + 2xh + h^2 < 2, \qquad 2xh + h^2 < 2 - x^2.$$

If $h < 1$, then $h^2 < h$, so $2xh + h^2 < 2xh + h$. Thus, as long as there exists $h \in \mathbb{Q}$ such that $2xh + h \leq 2 - x^2$, we get

$$2xh + h^2 < 2xh + h \leq 2 - x^2,$$

i.e. this h works for us. (We'll simply choose h such that $2xh + h = 2 - x^2$.) We can't deal with $2xh + h^2$ directly, because we can't explicitly express h in terms of x unless we use square roots, and we shouldn't be using them before we have defined $\mathbb{R}$.

Proof, continued

Let

$$h = \min\Big\{\frac{1}{2},\ \frac{2-x^2}{2x+1}\Big\},$$

thus $h < 1$, and then $h^2 < h$. Thus

$$(x+h)^2 = x^2 + 2xh + h^2 < x^2 + 2xh + h = x^2 + (2x+1)h$$
$$\leq x^2 + (2x+1)\frac{2-x^2}{2x+1} = x^2 + 2 - x^2 = 2.$$

Since $x \geq 1$, we have $2x+1 \neq 0$. Since $x \in \mathbb{Q}$, $\frac{2-x^2}{2x+1} \in \mathbb{Q}$. Since $x \in \mathbb{Q}$ and $h \in \mathbb{Q}$, we have $x+h \in \mathbb{Q}$. Since $x^2 < 2$, $2 - x^2 > 0$; then $\frac{2-x^2}{2x+1} > 0$, thus $x + h > 0$. Therefore $x + h > 0$, $(x+h)^2 < 2$, and $x + h \in \mathbb{Q}$, so $x + h \in E$. Since $h > 0$, $x + h > x$, thus x is not an upper bound of E, a contradiction. The assumption $x^2 < 2$ is false, so $x^2 \geq 2$.
$x^2 \leq 2$. Assume $x^2 > 2$. We want to show that x is not the supremum of E (intuitively).

Scratch work

x is not $\sup E$ means: if $x^2 > 2$, there exists an upper bound of E such that this upper bound is less than x. So there exists $h > 0$ with $h \in \mathbb{Q}$ such that $x - h$ is an upper bound of E, i.e. $(x-h)^2 > 2 > q^2$. We want $x^2 - 2xh + h^2 > 2$. If $h^2 > 0$, then $x^2 - 2xh + h^2 > x^2 - 2xh$. If $x^2 - 2xh \geq 2$, we have $x^2 - 2xh + h^2 > x^2 - 2xh \geq 2$. Thus h can be

$$h = \frac{x^2-2}{2x},$$

since $x > 0$ and $x^2 > 2$. In this case $h^2 > 0$, and $(x-h)^2 > 2 > q^2$.
We want to get rid of h^2. Last time we used $h^2 < h$. However, this time $h^2 < h$ doesn't work, because we need $x^2 - 2xh + h^2 > \cdots > 2$. If we use $h^2 > h$ for $h > 1$, then the problem is that $(x-h)^2$ might be too small, so that $(x-h)^2 < 2$. So this time we use $(x-h)^2 > x^2 - 2xh > 2$: as long as $x^2 - 2xh \geq 2$, we get $(x-h)^2 > 2$ for the chosen h.

Proof, concluded

Let $h = \frac{x^2-2}{2x}$. Thus

$$(x-h)^2 = x^2 - 2xh + h^2 > x^2 - 2xh = x^2 - x^2 + 2 = 2.$$

Since $2 > q^2$, we get $(x-h)^2 > 2 > q^2$, so

$$(x-h)^2 - q^2 > 0, \qquad (x-h-q)(x-h+q) > 0.$$

Since $x - h > 0$ and $q > 0$, we have $x - h + q > 0$. Thus $x - h - q > 0$, that is, $x - h > q$ for all $q \in E$. Apparently $h \in \mathbb{Q}$, thus $x - h \in \mathbb{Q}$. Therefore $x - h$ is an upper bound of E. Since $x - h < x$, x is not the supremum of E, a contradiction. Our assumption that $x^2 > 2$ is false, so $x^2 \leq 2$.
Since $x^2 \leq 2$ and $x^2 \geq 2$, we conclude $x^2 = 2$.

Theorem

The set $E = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$ does not have a supremum in $\mathbb{Q}$.

Proof 1

We want to show that the x such that $x^2 = 2$ does not exist in $\mathbb{Q}$. By contradiction, assume there exists $x \in \mathbb{Q}$ such that $x^2 = 2$. Then there exist $p, q \in \mathbb{N}$ such that $(p, q) = 1$ and $x = \frac{p}{q}$. Thus

$$\frac{p^2}{q^2} = 2 \implies p^2 = 2q^2.$$

Since $(p, q) = 1$ and $p^2 = 2q^2$, we get $2 \mid p^2$, hence $2 \mid p$, so there exists $h \in \mathbb{N}$ such that $2h = p$. Then $(2h)^2 = 2q^2$, so $2h^2 = q^2$. Since $(p, q) = 1$ and $2h = p$, we have $(q, h) = 1$. Then $(q, h) = 1$ and $2h^2 = q^2$ give $2 \mid q^2$, hence $2 \mid q$. Thus $2 \mid p$ and $2 \mid q$, so $(p, q) = 2$, a contradiction. Therefore the assumption that there exists $x \in \mathbb{Q}$ with $x^2 = 2$ is false. Thus $\sup E$ does not exist in $\mathbb{Q}$.

Proof 2

By contradiction, assume $\sup E \in \mathbb{Q}$. Then there exist $p, q \in \mathbb{N}$ such that $x = \frac{p}{q}$. Let

$$S = \{m \in \mathbb{N} : n = mx \text{ for some } n \in \mathbb{N}\}.$$

Since p, q exist, $S \neq \emptyset$. Then $S \neq \emptyset$ and $S \subset \mathbb{N}$, so by the well-ordering property of $\mathbb{N}$, S has a least element k_0.
We want to show that there exists $k_1 \in S$ such that $k_1 < k_0$, which is a contradiction.
Let $k_1 = k_0 x - k_0$. Since $x^2 = 2$, we have $x^2 < 4$, so $x < 2$. By the previous theorem, $x \geq 1$; similarly, we can show that $x > 1$. Thus

$$k_0 - k_0 < k_1 < k_0 \cdot 2 - k_0 \implies 0 < k_1 < k_0.$$

Since $k_0 \in S$, we have $k_0 x \in \mathbb{N}$ and $k_0 \in \mathbb{N}$. Therefore $k_1 = k_0 x - k_0 \in \mathbb{N}$. Moreover,

$$k_1 x = \underbrace{(k_0 x - k_0)}_{>0} \underbrace{x}_{>1} = k_0 x^2 - k_0 x = k_0 \cdot 2 - k_0 x = \underbrace{2k_0}_{\in\mathbb{N}} - \underbrace{k_0 x}_{\in\mathbb{N}}.$$

Thus $k_1 x > 0$ and $k_1 x \in \mathbb{Z}$, hence $k_1 x \in \mathbb{N}$. Therefore $k_1 \in S$. Yet $k_1 < k_0$. This contradicts that k_0 is the least element in S. Thus the assumption that this x exists is false. Therefore $\sup E$ does not exist in $\mathbb{Q}$.

Logic for Proof 2

This x shouldn't exist, so the p, q shouldn't exist, so the set S should be empty, so there shouldn't be a least element in S. If $S \neq \emptyset$, then somehow $x^2 = 2$ and $n = mx$ will always allow me to define an element in S that is smaller than the least element.

Proof 3

Since $1 < 2 < 4$ and $x > 1$, we have $1 < x < 2$. Assume there exist $p, q \in \mathbb{N}$ such that $x = \frac{p}{q}$. Then $1 < \frac{p}{q} < 2$, so $0 < p - q < q$. Since $x^2 = \frac{p^2}{q^2}$, we have $p^2 = 2q^2$. Subtract pq on both sides; then

$$p^2 - pq = 2q^2 - pq \implies p(p-q) = q(2q-p) \implies \frac{p}{q} = \frac{2q-p}{p-q}.$$

Since $p - q > 0$ and $p, q \in \mathbb{N}$, we have $p - q \in \mathbb{N}$. Since $\frac{p}{q} = x > 1$, we have $2q - p > 0$; since $2q - p \in \mathbb{Z}$, we get $2q - p \in \mathbb{N}$. Since $p - q < q$ and $\frac{p}{q} = \frac{2q-p}{p-q}$, we have $2q - p < p$. Let $p_1 = 2q - p$ and $q_1 = p - q$. Then

$$\frac{p}{q} = \frac{p_1}{q_1}, \qquad p > p_1, \quad q > q_1, \quad p_1, q_1 \in \mathbb{N}.$$

By iterating this process, we will have that

$$\frac{p}{q} = \frac{p_1}{q_1} = \frac{p_2}{q_2} = \cdots, \qquad \text{where } p > p_1 > p_2 > \cdots \text{ and } p_i \in \mathbb{N}.$$

Since $\mathbb{N}$ has lower bound 1, we know p_i can't keep decreasing. Yet the process claims the p_i should be able to exist, a contradiction. Thus again, the assumption that x exists in $\mathbb{Q}$ is false.

4 The Characterization of the Real Numbers

4.1 Ordered fields

Definition

A set F is a *field* if it has two operations, addition $(+)$ and multiplication $(\cdot)$, such that:

A1) if $x, y \in F$, then $x + y \in F$ (closure under addition);

A2) for all $x, y \in F$, $x + y = y + x$ (commutativity);

A3) for all $x, y, z \in F$, $(x + y) + z = x + (y + z)$ (associativity);

A4) there exists an element $0 \in F$ such that $0 + x = x$ (additive identity);

A5) for all $x \in F$, there exists $y \in F$ such that $x + y = 0$; denote y as $-x$;

M1) if $x, y \in F$, then $x \cdot y \in F$ (closure under multiplication);

M2) for all $x, y \in F$, $x \cdot y = y \cdot x$ (commutativity);

M3) for all $x, y, z \in F$, $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ (associativity);

M4) there exists $1 \in F$ such that $1 \cdot x = x$ for all $x \in F$ (multiplicative identity);

M5) for all $x \in F \setminus \{0\}$, there exists y such that $x \cdot y = 1$; denote y as x^{-1};

D) for all $x, y, z \in F$, $(x + y) \cdot z = x \cdot z + y \cdot z$ (distributivity).

Example

$\mathbb{Q}$ is a field.

Non-example

$\mathbb{Z}$ is not a field (it is a commutative ring that has a unit element). Reason: there is no multiplicative inverse; $3 \cdot \frac{1}{3} = 1$, but $\frac{1}{3} \notin \mathbb{Z}$.

Example: finite fields

$\mathbb{Z}_2 = \{0, 1\}$ with $1 + 1 = 0 \pmod 2$ arithmetic).
$\mathbb{Z}_3 = \{0, 1, 2\}$, where $a + b := c \bmod 3$ and $a \cdot b := d \bmod 3$; for example, $1 + 2 = 3 = 0 \bmod 3$ and $2 \cdot 2 = 4 = 1 \bmod 3$. In this case, we say the field $\mathbb{Z}_3$ has *characteristic* 3, since $\underbrace{1 + 1 + 1}_{3} = 3 = 0$.

Non-example

$\mathbb{Z}_4 = \{0, 1, 2, 3\} \pmod 4$. Here 2 doesn't have any multiplicative inverse in $\mathbb{Z}_4$:

$$2 \cdot 2 = 4 = 0 \bmod 4, \qquad 2 \cdot 1 = 2 \bmod 4, \qquad 2 \cdot 0 = 0 \bmod 4, \qquad 2 \cdot 3 = 6 = 2 \bmod 4.$$

Theorem

If F is a field, then for all $x \in F$ we have $0x = 0$.

Proof

$$0x = (0 + 0)x = 0x + 0x \implies 0x + (-0x) = 0x + 0x + (-0x) \implies 0 = 0x.$$

Note

Similarly, we can prove trivial facts like: uniqueness of 1 and 0, of the additive inverse $-x$, and of the multiplicative inverse x^{-1}; $(-1)x = -x$; etc.

Definition

An *ordered field* is a field F which is also an ordered set and also satisfies the following conditions:

1) for all $x, y, z \in F$, if $x < y$, then $x + z < y + z$;

2) if $x > 0$ and $y > 0$, then $xy > 0$.

If $x > 0$, then x is *positive*; if $x \geq 0$, we say x is *non-negative*.

Non-example

$\mathbb{Z}_2 = \{0, 1\}$. If $<$ is an order on $\mathbb{Z}_2$, then either $0 < 1$ or $1 < 0$.
Case $0 < 1$. If $0 < 1$, then $1 + 0 = 1$ and $1 + 1 = 2 = 0$. Since $0 < 1$, we have $1 + 0 \not< 1 + 1$, so 1) doesn't hold.
Case $1 < 0$. If $1 < 0$, then $0 + 1 = 1$ and $1 + 1 = 2 = 0$. Since $1 < 0$, we have $1 + 1 \not< 0 + 1$, so 1) again fails.

Example

$\mathbb{Q}$ is an ordered field.

Remark

In general, we can show that *no finite ordered field exists.*
Like the axioms for fields, we can also use the axioms for inequalities to prove some trivial facts.

Theorem

If F is an ordered field and $x > 0$, then $-x < 0$.

Proof

$$x > 0 \implies x + (-x) > 0 + (-x) \implies 0 > -x \implies -x < 0.$$

Theorem

Let $x, y \in F$, where F is an ordered field. If $x > 0$ and $y < 0$ (or $x < 0$ and $y > 0$), then $xy < 0$.

Proof

$$\begin{aligned} x > 0 \text{ and } y < 0 &\implies x > 0 \text{ and } -y > 0 \implies x(-y) > 0 \\ &\implies -xy > 0 \implies -(-xy) < 0 \implies xy < 0. \end{aligned}$$

Question

We are talking about the l.u.b.p. of $\mathbb{R}$ all the time. Is there a greatest lower bound property in $\mathbb{R}$? Answer: yes.

Theorem

Let F be an ordered field with the l.u.b.p. Then if $A \subset F$, $A \neq \emptyset$, and A is bounded below, then $\inf A$ exists in F.

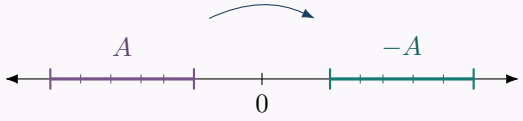
Proof

Since A is bounded below, there exists $b \in F$ such that $b \leq a$ for all $a \in A$. Denote the set

$$-A = \{-a : a \in A\}.$$

Then for all $a \in A$, $-a \leq -b$; thus $-b$ is an upper bound for $-A$. Since F has the l.u.b.p., $-A$ is bounded above, $-A \subset F$, and $-A \neq \emptyset$, the supremum $\sup(-A)$ exists in F, i.e. there exists $x \in F$ such that $-a \leq x$ for all $a \in A$, and $x \leq c$ for any upper bound c of $-A$.
Apparently, if c is an upper bound of $-A$, then $-c$ is a lower bound for A. Thus $a \geq -x$ for all $a \in A$, so $-x$ is a lower bound; and for any lower bound $-c$ for A, we have $-x \geq -c$. Then $-x$ is the greatest lower bound for A, so $\inf A$ exists in F.

Intuition



A bounded below $\implies -A$ bounded above $\implies -\sup(-A) = \inf(A)$.

4.2 The real numbers

Theorem

There exists a unique ordered field with the l.u.b.p. containing $\mathbb{Q}$. The field is denoted by $\mathbb{R}$. (The uniqueness means that any other ordered field that satisfies these conditions has to be *isomorphic* to $\mathbb{R}$.)

Theorem

There exists a unique $r \in \mathbb{R}$ such that $r > 0$ and $r^2 = 2$.

Proof

Let $E = \{x \in \mathbb{R} : x > 0 \text{ and } x^2 < 2\}$. Then E is bounded above by 2. Thus $\sup E$ exists in $\mathbb{R}$; let $r = \sup E$. Then the same proof as before applies to $\mathbb{R}$. Thus $r \geq 1$ and $r^2 = 2$, where $r = \sup E \in \mathbb{R}$.
Now we prove the uniqueness. Suppose $\tilde{r} \in \mathbb{R}$ satisfies $\tilde{r} > 0$ and $\tilde{r}^{\,2} = 2$. We want to show that $r = \tilde{r}$. Then

$$0 = \tilde{r}^{\,2} - r^2 = (\tilde{r} + r)(\tilde{r} - r).$$

Since $\tilde{r} > 0$ and $r > 0$, we have $\tilde{r} + r > 0$. Then, as $\mathbb{R}$ is a field, $(\tilde{r} + r)^{-1}$ exists in $\mathbb{R}$, and

$$0 \cdot (\tilde{r} + r)^{-1} = (\tilde{r} + r)(\tilde{r} - r)(\tilde{r} + r)^{-1} = (\tilde{r} - r)(\tilde{r} + r)(\tilde{r} + r)^{-1} = \tilde{r} - r.$$

So $\tilde{r} - r = 0$, that is, $\tilde{r} = r$.

Remark

In the same way, we can show that if $x \in \mathbb{R}_{>0}$, then $x^{1/n}$ exists in $\mathbb{R}$ for all $n \in \mathbb{N}$. That is: the set $E = \{a \in \mathbb{R} : a > 0 \text{ and } a^n < x\}$ is shown to be bounded above; by the l.u.b.p. of $\mathbb{R}$, $\sup E$ exists in $\mathbb{R}$; show that $(\sup E)^n = x$ (namely $(\sup E)^n \geq x$ and $(\sup E)^n \leq x$, by contradiction); then prove uniqueness: assume $\tilde{a} \in \mathbb{R}$ is another such number, and show that $\tilde{a} = \sup E$.

5 The Archimedean Property, Density of the Rationals, and Absolute Value

5.1 The Archimedean property and the density of the rationals in $\mathbb{R}$

Fact

For all $x, y \in \mathbb{R}$ with $x < y$, there exists $a \in \mathbb{R}$ such that $x < a < y$; take $a = \frac{x+y}{2}$. The same thing works for $\mathbb{Q}$: for all $p, q \in \mathbb{Q}$ with $p < q$, there exists $r \in \mathbb{Q}$ such that $p < r < q$; take $r = \frac{p+q}{2}$.

Question

If $x, y \in \mathbb{R}$ and $x < y$, does there exist $r \in \mathbb{Q}$ such that $x < r < y$? For example, if $x = \sqrt{2}$ and $y = 2\sqrt{2}$, then

$$\frac{x+y}{2} = \frac{3}{2}\sqrt{2} \notin \mathbb{Q}.$$

Can we find $r \in \mathbb{Q}$ such that $\sqrt{2} < r < 2\sqrt{2}$? The answer is yes.

Theorem (Archimedean property and density of $\mathbb{Q}$)

1) (Archimedean property) If $x, y \in \mathbb{R}$ and $x > 0$, then there exists $n \in \mathbb{N}$ such that

$$nx > y.$$

2) (Density of $\mathbb{Q}$) If $x, y \in \mathbb{R}$ and $x < y$, then there exists $r \in \mathbb{Q}$ such that

$$x < r < y.$$

Proof of 1) (by contradiction)

Let $x, y \in \mathbb{R}$ with $x > 0$. Assume that $n \leq \frac{y}{x}$ for every $n \in \mathbb{N}$. Then $\frac{y}{x}$ is an upper bound for $\mathbb{N}$, which $\mathbb{N}$ should not have, since $\mathbb{N}$ should not be bounded. Since $\mathbb{N} \subset \mathbb{R}$, $\mathbb{N} \neq \emptyset$, and $\mathbb{N}$ is bounded above, by the least upper bound property of $\mathbb{R}$ the supremum $\sup \mathbb{N}$ exists in $\mathbb{R}$. Let $\sup \mathbb{N} = a$. Then $a - 1$ cannot be an upper bound for $\mathbb{N}$; that is, there exists $m \in \mathbb{N}$ such that

$$m > a - 1 \quad \Longrightarrow \quad m + 1 > a.$$

Since $m \in \mathbb{N}$, we have $m + 1 \in \mathbb{N}$. Then a is not an upper bound of $\mathbb{N}$, a contradiction. Hence

the assumption that $n \leq \frac{y}{x}$ for every $n \in \mathbb{N}$ is false. Therefore there exists $n \in \mathbb{N}$ such that $n > \frac{y}{x}$, that is, $nx > y$.

Proof of 2)

If $x < y$, then

$$x < y \leq 0 \quad \text{or} \quad x < 0 < y \quad \text{or} \quad 0 \leq x < y.$$

The cases $x < y \leq 0$ and $0 \leq x < y$ are the same, since $\mathbb{R}$ is an ordered field.
Case 1: $x < 0 < y$. Choose $r = 0 \in \mathbb{Q}$. Then $x < 0 < y$.
Case 2: $0 \leq x < y$ (and likewise $x < y \leq 0$). We want to show that there exists $\frac{m}{n} \in \mathbb{Q}$ such that $x < \frac{m}{n} < y$.
Intuition: if x and y get really close, then n has to be big enough. Therefore we need to do something (with respect to x and y) to make sure that n is big enough. Then we control m to get $nx < m < ny$. That is, n needs to be chosen depending on x and y; after that, we can choose m depending on our choice of n and on x, y. By the Archimedean property, there exists $n \in \mathbb{N}$ such that

$$n(y - x) > 1.$$

(This n will be big enough for us.) By the Archimedean property again, there exists $\ell \in \mathbb{N}$ such that $\ell > nx$: indeed, $\ell \cdot \frac{1}{x} > n$ gives $\ell > nx$, and if $x = 0$, let ℓ be an arbitrary natural number, so that $\ell > 0$. Let

$$E = \{s \in \mathbb{N} : s > nx\}.$$

Then $E \subset \mathbb{N}$ and $E \neq \emptyset$. Thus, by the well-ordering property of $\mathbb{N}$, there exists a least element in E. Let that element be m. (Here m is the smallest natural number such that $m > nx$, so $\frac{m}{n}$ is closest to x, and hence $\frac{m}{n}$ is likely less than y.) Then

$$m - 1 \notin E \quad \Longrightarrow \quad m - 1 \leq nx \quad \Longrightarrow \quad m \leq nx + 1.$$

Thus $m \in E$ gives

$$m > nx \quad \Longrightarrow \quad x < \frac{m}{n},$$

and $m - 1 \notin E$ gives $m \leq nx + 1$. Since $n(y - x) > 1$, we have $ny - nx > 1$, so

$$ny > nx + 1.$$

Thus $m \leq nx + 1 < ny$. Therefore this m satisfies

$$nx < m < ny \quad \Longrightarrow \quad x < \frac{m}{n} < y.$$

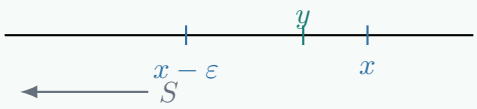
Since $m, n \in \mathbb{N}$, we have $\frac{m}{n} \in \mathbb{Q}$. Take $r = \frac{m}{n}$.

5.2 Suprema and infima

Theorem (characterization of the supremum)

Assume $S \subset \mathbb{R}$, $S \neq \emptyset$, and S is bounded above. Then $x = \sup S$ if and only if x is an upper bound for S and, for every $\varepsilon > 0$, there exists $y \in S$ such that

$$x - \varepsilon < y \leq x.$$



Proof

($\Longrightarrow$) Assume $x = \sup S$. Let $\varepsilon > 0$. Then $x - \varepsilon < x$. Thus $x - \varepsilon$ cannot be an upper bound for S, since x is the supremum. Because $x - \varepsilon$ is not an upper bound, there exists $y \in S$ such that $y > x - \varepsilon$. Since $x = \sup S$, we have $y \leq x$ for every $y \in S$. Therefore, for every $\varepsilon > 0$, there exists $y \in S$ such that $x - \varepsilon < y \leq x$.
($\Longleftarrow$) Since $S \subset \mathbb{R}$ and S is bounded above, $\sup S$ exists in $\mathbb{R}$. Let $\sup S = a$. We want to show that $a = x$. Since x is an upper bound for S, we have $\sup S = a \leq x$. Suppose $a \neq x$, that is, $a < x$. Choose $\varepsilon = \frac{x-a}{2}$. Then there exists $y \in S$ such that $x - \varepsilon < y \leq x$, that is,

$$\frac{x+a}{2} < y \leq x.$$

Since $a < x$, we get

$$\frac{a+a}{2} < \frac{x+a}{2} < y \leq x.$$

Thus $a < y$, where $y \in S$. Then a is not an upper bound for S, a contradiction. Hence the assumption $x \neq a$ is false, and $x = a = \sup S$.

Theorem (characterization of the infimum)

Let $S \subset \mathbb{R}$, $S \neq \emptyset$, and S bounded below. Then $x = \inf S$ if and only if x is a lower bound for S and, for every $\varepsilon > 0$, there exists $y \in S$ such that

$$x \leq y < x + \varepsilon.$$

Proof

Similar to the one above.

Theorem

$$\sup\left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\} = 1.$$

Proof

We will use the Archimedean property (AP) to prove this. Denote

$$E = \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\}.$$

Since $n \in \mathbb{N}$, we have $\frac{1}{n} > 0$, so $1 - \frac{1}{n} < 1$ for every $n \in \mathbb{N}$. Hence 1 is an upper bound for E. We want to show that upper bounds for E cannot be smaller than 1. Suppose x is an upper bound for E such that $x < 1$. Then $1 - x > 0$. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $n(1-x) > 1$. Thus

$$n(1-x) > 1 \implies 1 - x > \frac{1}{n} \implies 1 - \frac{1}{n} > x \implies x \text{ is not an upper bound for } E,$$

a contradiction. Thus $x \not< 1$: any upper bound for E has to be ≥ 1. Since 1 is an upper bound, therefore $1 = \sup E$.
This can also be proven with the theorem above. Let $n = \left[\frac{2}{\varepsilon}\right] + 1$ (integer part plus one); then $n \in \mathbb{N}$, and

$$n > \frac{2}{\varepsilon} > \frac{1}{\varepsilon} \implies \varepsilon > \frac{1}{n} \implies 1 - \varepsilon < 1 - \frac{1}{n} < 1, \qquad n \in \mathbb{N}.$$

Thus there exists $n \in \mathbb{N}$, with $1 - \frac{1}{n} \in E$, such that $1 - \varepsilon < 1 - \frac{1}{n} < 1$. Thus $1 = \sup E$.

Definition

For $x \in \mathbb{R}$ and $A \subset \mathbb{R}$, define

$$x + A := \{x + a : a \in A\}, \qquad xA := \{xa : a \in A\}.$$

Theorem (translates and dilates of a bounded set)

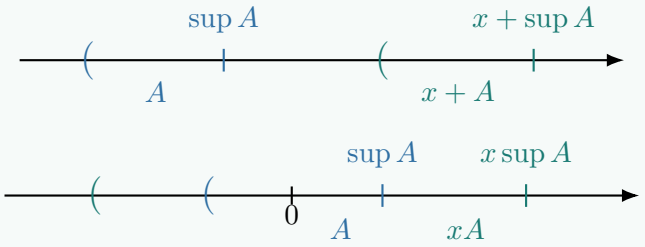
Let $A \subset \mathbb{R}$ be bounded above.

1) If $x \in \mathbb{R}$, then $x + A$ is bounded above and

$$\sup(x + A) = x + \sup A.$$

2) If $x \in \mathbb{R}_{>0}$, then xA is bounded above and

$$\sup(xA) = x \sup A.$$



Proof

This theorem will be easier to prove with the theorem characterizing the supremum by $x - \varepsilon < y \leq x$.
1) Let $a = \sup A$. Let $\varepsilon > 0$. Then there exists $y \in A$ such that

$$a - \varepsilon < y \leq a.$$

Then

$$(a + x) - \varepsilon < y + x \leq a + x.$$

Thus, for every $\varepsilon > 0$, there exists $x + y \in x + A$ such that $(a + x) - \varepsilon < y + x \leq a + x$. Since $a = \sup A$, we have $z \leq a$ for every $z \in A$. Then $x + z \leq x + a$ for every $x + z \in x + A$. Thus, by the previous theorem,

$$x + a = x + \sup A = \sup(x + A).$$

2) Can be proven in a similar way.

Theorem

If $A, B \subset \mathbb{R}$ are such that $x \leq y$ for all $x \in A$ and all $y \in B$, then

$$\sup A \leq \inf B.$$

Proof

Since there exists $y \in B$ such that $x \leq y$ for all $x \in A$, the set A is bounded above. Since there exists $x \in A$ such that $x \leq y$ for all $y \in B$, the set B is bounded below. Since $A, B \subset \mathbb{R}$, both $\sup A$ and $\inf B$ exist in $\mathbb{R}$.
By contradiction, suppose $\sup A > \inf B$. Then let

$$\varepsilon = \frac{\sup A - \inf B}{2} > 0.$$

There exists $\tilde{x} \in A$ such that

$$\tilde{x} > \sup A - \varepsilon = \frac{\sup A + \inf B}{2},$$

and there exists $\tilde{y} \in B$ such that

$$\tilde{y} < \inf B + \varepsilon = \frac{\sup A + \inf B}{2}.$$

Then $\tilde{x} > \tilde{y}$, a contradiction. Thus $\sup A \not> \inf B$, that is, $\sup A \leq \inf B$.

5.3 Absolute values

Definition

If $x \in \mathbb{R}$, then the *absolute value* of x is

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x \leq 0. \end{cases}$$

Remark

$|x|$ should represent the "distance" from x to the origin.

Theorem (properties of the absolute value)

1) For all $x \in \mathbb{R}$, $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$.
2) For all $x \in \mathbb{R}$, $|x| = |-x|$.
3) For all $x, y \in \mathbb{R}$, $|xy| = |x|\,|y|$.
4) For all $x \in \mathbb{R}$, $|x^2| = |x|^2$.
5) If $x, y \in \mathbb{R}$, then $|x| \leq y$ if and only if $-y \leq x \leq y$.
6) For all $x \in \mathbb{R}$, $x \leq |x|$.

Proof

1) If $x \geq 0$, then $|x| = x \geq 0$. If $x \leq 0$, then $-x \geq 0$, so $|x| = -x \geq 0$. ($\Longleftarrow$) If $x = 0$, then $|x| = x = 0$. ($\Longrightarrow$) Suppose $|x| = 0$. Then, if $x \geq 0$, we get $x = |x| = 0$; if $x \leq 0$, we get $-x = |x| = 0$, so $x = 0$. Thus $|x| = 0$ implies $x = 0$.
2) If $x \geq 0$, then $-x \leq 0$, so $|-x| = -(-x) = x = |x|$. If $x \leq 0$, then $-x \geq 0$, so $|-x| = -x = |x|$. Thus $|x| = |-x|$.
3) Trivial if we consider the three cases separately.
4) Trivial from 3).
5) ($\Longrightarrow$) Suppose $|x| \leq y$. If $x \geq 0$, then $-y \leq 0 \leq x = |x| \leq y$. If $x \leq 0$, then $-y \leq x \leq 0 \leq |x| \leq y$. ($\Longleftarrow$) Suppose $-y \leq x \leq y$. If $x \geq 0$, then $|x| = x \leq y$. If $x \leq 0$, then $-x \geq 0$. Since $-y \leq x \leq y$, we have $-y \leq -x \leq y$. Thus, by the case where $x \geq 0$, $|-x| \leq y$. By 2), $|-x| = |x| \leq y$. 6) Let $y = |x|$. Then by 5), since $|x| \leq |x|$, we get $-y \leq x \leq y$, so $x \leq |x|$.

⋆ Theorem (triangle inequality)

For all $x, y \in \mathbb{R}$,

$$||x| - |y|| \leq |x \pm y| \leq |x| + |y|.$$

Proof

First part: $|x + y| \leq |x| + |y|$ (the triangle inequality). We want to show that

$$-(|x| + |y|) \leq x + y \leq |x| + |y|,$$

and then apply 5). By 6), $x \leq |x|$ and $y \leq |y|$. Then

$$-x \leq |-x| = |x|, \qquad -y \leq |-y| = |y| \quad \Longrightarrow \quad x \geq -|x|, \quad y \geq -|y|.$$

Thus

$$-(|x| + |y|) \leq x + y \leq |x| + |y| \quad \overset{5)}{\Longrightarrow} \quad |x + y| \leq |x| + |y|.$$

Second part: $||x| - |y|| \leq |x - y|$ (the reverse triangle inequality). We want to show that

$$-|x - y| \leq |x| - |y| \leq |x - y|.$$

By the first part (the triangle inequality), we have

$$|x| = |(x - y) + y| \leq |x - y| + |y| \quad \Longrightarrow \quad |x| - |y| \leq |x - y|,$$

and

$$|y| = |(y - x) + x| \leq |y - x| + |x| \overset{2)}{=} |x - y| + |x| \quad \Longrightarrow \quad |x| - |y| \geq -|x - y|.$$

Therefore

$$-|x - y| \leq |x| - |y| \leq |x - y| \quad \overset{5)}{\Longrightarrow} \quad ||x| - |y|| \leq |x - y|.$$

Third part: $||x| - |y|| \leq |x + y|$ and $|x - y| \leq |x| + |y|$. Replace every y with $-y$ in the two parts above.

6 The Uncountability of the Real Numbers

Question

$\mathbb{Q}$ is countable. Is $\mathbb{R}$ countable? The answer is no.

Remark

$\mathbb{R}$ is uncountable and $\mathbb{Q}$ is countable, so (by contradiction) $\mathbb{R} \setminus \mathbb{Q}$, the irrationals, is uncountable. Otherwise there would be a bijection $\mathbb{R} \setminus \mathbb{Q} \to -\mathbb{N}$ (the negative integers) and a bijection $\mathbb{Q} \to \mathbb{N}$, since $\mathbb{Q}$ is countable; together these would give a bijection $\mathbb{R} \to \mathbb{Z}$, so that $|\mathbb{R}| = |\mathbb{Z}| = |\mathbb{N}|$, a contradiction.

6.1 Decimal representations

Remark

We can think of $\mathbb{Q}$ and $\mathbb{R}$ as decimal representations.

Example

$$\frac{111}{10} = 11.1 = 10^1 \cdot 1 + 10^0 \cdot 1 + 10^{-1} \cdot 1,$$

$$3.287 = \frac{3287}{1000} = 10^0 \cdot 3 + 10^{-1} \cdot 2 + 10^{-2} \cdot 8 + 10^{-3} \cdot 7.$$

We will focus on $(0, 1] \subset \mathbb{R}$ for now.

Definition

Let $x \in (0, 1]$ and let $d_{-j} \in \{0, 1, 2, \ldots, 9\}$, $j \in \mathbb{N}$. We say that x is *represented* by the digits $\{d_{-j} : j \in \mathbb{N}\}$, that is, $x = 0.d_{-1}d_{-2}\cdots$, if and only if

$$x = \sup\left\{10^{-1}d_{-1} + 10^{-2}d_{-2} + 10^{-3}d_{-3} + \cdots + 10^{-n}d_{-n} : n \in \mathbb{N}\right\}.$$

Example

$$0.25000 = \sup\left\{\frac{2}{10}, \frac{2}{10} + \frac{5}{100}, \frac{2}{10} + \frac{5}{100} + \frac{0}{1000}\right\} = \sup\left\{\frac{2}{10}, \frac{25}{100}\right\} = \frac{25}{100} = \frac{1}{4}.$$

Why a supremum

Essentially, the reason we can use the supremum is that for every additional digit we add, we are essentially adding something to it: $10^{-j}d_{-j} \geq 0$. Thus the supremum above is essentially the supremum of a monotone increasing sequence that is bounded above. Therefore the supremum exists and equals x.

Theorem (decimal expansions)

1) For every set of digits $\{d_{-j} : j \in \mathbb{N}\}$ with $d_{-j} \in \{0, 1, \ldots, 9\}$, there exists a unique $x \in [0, 1]$ such that $x = 0.d_{-1}d_{-2}\cdots$.

2) For every $x \in (0, 1]$, there exist unique digits $\{d_{-j} : j \in \mathbb{N}\}$ such that $x = 0.d_{-1}d_{-2}\cdots$ and

$$0.d_{-1}d_{-2}\cdots d_{-n} < x \leq 0.d_{-1}d_{-2}\cdots d_{-n} + 10^{-n}.$$

Note: the converse of 1) does not hold

Given $x = 0.5$, I can write it as

$$0.5 = 0.5000\cdots = 0.4999\cdots.$$

Thus every set of digits gives a unique real number, yet a unique real number does not give a unique set of digits. However, the second part of this theorem limits our choice of digits, and then gives us a unique set of digits.

Remarks on part 2)

Here requiring $x \neq 0.d_{-1}d_{-2}\cdots d_{-n}$ rules out the case $0.5 = 0.500\cdots$. The proof of this theorem uses the least upper bound property, but it will not be required.

6.2 Cantor's theorem

Theorem (Cantor)

$(0,1]$ is uncountable.

Consequences: $|\mathbb{N}| < |(0,1]| = |\mathbb{R}|$

Then $|\mathbb{N}| < |(0,1]| \leq |\mathbb{R}|$. Thus $\mathbb{R}$ is uncountable. In fact, $|(0,1]| = |\mathbb{R}|$. To see that $|(0,1]| = |\mathbb{R}|$, use the Cantor–Schröder–Bernstein theorem from page 7: if $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$. That is, we find injections $A \to B$ and $B \to A$.

- $(0,1] \to \mathbb{R}$: the identity map is injective, so $|(0,1]| \leq |\mathbb{R}|$.
- $\mathbb{R} \to (0,1]$: let
$$f(x) = \frac{\arctan x + \frac{\pi}{2}}{\pi}.$$
Since $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $f : \mathbb{R} \to (0,1)$, and f is injective, so $|\mathbb{R}| \leq |(0,1]|$.

Therefore $|(0,1]| = |\mathbb{R}|$.

Proof (by contradiction)

Assume $(0,1]$ is countable. Then there exists a bijection

$$f : \mathbb{N} \longrightarrow (0,1], \qquad n \longmapsto 0.d_{-1}^{(n)} d_{-2}^{(n)} \cdots .$$

In other words, f can be seen as a bijection from an $n \in \mathbb{N}$ to a sequence of digits

$$\{d_{-1}^{(n)}, d_{-2}^{(n)}, d_{-3}^{(n)}, \ldots\} = \{d_{-j}^{(n)} : j \in \mathbb{N}\},$$

where $d_{-j}^{(n)} \in \{0,1,2,\ldots,9\}$ for a given j. (The superscript (n) means that the sequence $\{d_{-j}^{(n)} : j \in \mathbb{N}\}$ is the image of n under f, that is, $f(n) = 0.d_{-1}^{(n)} d_{-2}^{(n)} \cdots$, and we assume this map is a bijection.)
We want to show that there exists $y \in (0,1]$ such that $y \notin f(\mathbb{N})$; equivalently, there exists $y \in (0,1]$ such that there is no $n \in \mathbb{N}$ with $f(n) = y$.
Simplification of the idea (an example). If $f : \{1,2,3\} \to \mathbb{R}$ with

$$f(1) = 0.\underline{7}49, \qquad f(2) = 0.4\underline{5}1, \qquad f(3) = 0.91\underline{1},$$

then we choose $y = 0.abc$ such that $a \neq 7$, $b \neq 5$, and $c \neq 1$ (for example, $y = 0.462$). In this case we have shown that, since

- the first decimal place of y is different from that of $f(1)$,
- the second decimal place of y is different from that of $f(2)$,
- the third decimal place of y is different from the third decimal place of $f(3)$,

we get $y \neq f(1), f(2), f(3)$, so $y \notin f(\{1,2,3\})$.
In clear words: we will choose the n-th decimal place of y different from the n-th decimal place of $f(n)$. (This is just for simplicity; if you want, you can use the $(n+1)$-th decimal place of

$f(n)$, or whatever works.) In this case we have constructed a $y \in (0,1]$ such that $y \neq f(n)$ for every $n \in \mathbb{N}$. Thus $y \notin f(\mathbb{N})$, and so f is not a bijection.
Consider the following sequence of digits, which will be the decimal places of y. Let $\{e_{-j} : j \in \mathbb{N}\}$ be given by

$$e_{-j} = \begin{cases} 1, & \text{if } d_{-j}^{(j)} \neq 1, \\ 2, & \text{if } d_{-j}^{(j)} = 1. \end{cases}$$

Then $e_{-j} \neq d_{-j}^{(j)}$, and $d_{-j}^{(j)}$ is the j-th decimal place of $f(j)$ (by the definition of $d_{-j}^{(n)}$); thus the j-th decimal place e_{-j} of y will be different from the j-th decimal place of $f(j)$, so $y \neq f(j)$ for every $j \in \mathbb{N}$.
Visually, we consider the array formed by $f(\mathbb{N})$; we want to show that $f(\mathbb{N}) \subsetneq (0,1]$.

$$f(\mathbb{N}) \left\{ \begin{array}{llllll} f(1) = 0. & d_{-1}^{(1)} & d_{-2}^{(1)} & d_{-3}^{(1)} & d_{-4}^{(1)} & \cdots \quad e_{-1} \neq d_{-1}^{(1)} \implies y \neq f(1) \\ f(2) = 0. & d_{-1}^{(2)} & d_{-2}^{(2)} & d_{-3}^{(2)} & d_{-4}^{(2)} & \cdots \quad e_{-2} \neq d_{-2}^{(2)} \implies y \neq f(2) \\ f(3) = 0. & d_{-1}^{(3)} & d_{-2}^{(3)} & d_{-3}^{(3)} & d_{-4}^{(3)} & \cdots \quad e_{-3} \neq d_{-3}^{(3)} \implies y \neq f(3) \\ f(4) = 0. & d_{-1}^{(4)} & d_{-2}^{(4)} & d_{-3}^{(4)} & d_{-4}^{(4)} & \cdots \quad e_{-4} \neq d_{-4}^{(4)} \implies y \neq f(4) \\ \vdots & & & & & \qquad \vdots \end{array} \right.$$

Let

$$y = 0.e_{-1}e_{-2}e_{-3}\cdots e_{-j}\cdots .$$

Since $e_{-j} > 0$ for all $j \in \mathbb{N}$, we have

$$0.e_{-1}e_{-2}\cdots e_{-n} < y \leq 0.e_{-1}e_{-2}\cdots e_{-n} + 10^{-n}.$$

Then, by the previous theorem, $0.e_{-1}e_{-2}\cdots$ is the *unique* decimal expansion of y.
Why uniqueness matters. We need to check whether there exists another expansion of y such that $y = f(m)$ for some $m \in \mathbb{N}$, since we are manipulating digits. For example, suppose $f(1) = 0.200\cdots$, $f(2) = 0.111\cdots$, and $f(3) = 0.09999\cdots$, and we choose $y = 0.100$ (since $1 \neq 2$, $0 \neq 1$, $0 \neq 9$). Then

$$y = 0.100 = 0.09999\cdots = f(3).$$

Thus we need to make sure that there is no $n \in \mathbb{N}$ with $f(n) = y$ by checking the uniqueness of the decimal expansion of y; then our construction works.
Then, since $e_{-j} \neq d_{-j}^{(j)}$, the j-th decimal place of y is different from the j-th decimal place of $f(j)$, for every $j \in \mathbb{N}$. Thus $y \neq f(j)$ for any $j \in \mathbb{N}$. Hence there exists $y \in (0,1]$ such that there is no $n \in \mathbb{N}$ with $f(n) = y$. Therefore f is not a bijection, a contradiction. The assumption that $(0,1]$ is countable is false, so $(0,1]$ is uncountable.

Corollary

$\mathbb{R}$ is uncountable.

This was "proved" under the previous statement that $(0,1]$ is uncountable.

7 Convergent Sequences of Real Numbers

7.1 Sequences and series: first definitions

Definition

A *sequence* of real numbers is a function $x : \mathbb{N} \to \mathbb{R}$. We denote $x(n)$ by x_n, and the sequence by $\{x_n\}_{n=1}^{\infty}$, or $\{x_n\}$, or $x_1, x_2, \ldots$.

Note

We are able to think of a sequence as a list of numbers because the domain is $\mathbb{N}$, which is countable. However, do not confuse the sequence with a set because of the notation $\{x_n\}$: a sequence is *not* a set.

Example

$1, 1, 1, \ldots$; that is, $x(n) = 1$ for all $n \in \mathbb{N}$.

Note

We do not think of sequences as functions. Similarly, we will not think of a function $f : X \to Y$, $x \mapsto y$, as a subset of a Cartesian product, namely a relation $R \subset X \times Y$ (the "function") such that $(x, y) \in R$ and $(x, z) \in R$ imply $y = z$ (one x maps onto a unique y). Rather, we think of a function simply as a mapping, and of a sequence as a list of numbers.

Example

$$\left\{\frac{1}{n}\right\}_{n=1}^{\infty} = \left\{\frac{1}{n}\right\} = 1, \ \frac{1}{2}, \ \frac{1}{3}, \ \frac{1}{4}, \ \ldots$$

7.2 Bounded sequences

Definition

A sequence $\{x_n\}$ is *bounded* if there exists $B \geq 0$ such that, for all $n \in \mathbb{N}$,

$$|x_n| \leq B.$$

Examples

The sequences $\{1\}$ and $\{1/n\}$ are bounded. The sequence

$$\{(-1)^n\} = -1, 1, -1, 1, \ldots$$

is bounded, since $|(-1)^n| = 1$.

Non-example

The sequence $\{n\} = 1, 2, 3, 4, \ldots$ is not bounded.

Remark: unbounded sequences

A sequence $\{x_n\}$ is *unbounded* if, for every $B \geq 0$, there exists $n \in \mathbb{N}$ such that $|x_n| > B$. When negating a statement, "for all" becomes "there exists", "there exists" becomes "for all", and the conditions are negated:

$$\begin{aligned} \exists B,\ P(B) \text{ is true} \quad &\longrightarrow \quad \forall B,\ P(B) \text{ negated}, \\ P(B):\ \forall n,\ P(n) \text{ is true} \quad &\longrightarrow \quad \exists n,\ P(n) \text{ negated}. \end{aligned}$$

Proof (that $\{n\}$ is unbounded)

Let $B \geq 0$. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $n \cdot 1 > B$. Thus $\{n\}$ is unbounded.

7.3 Convergence of a sequence

Definition

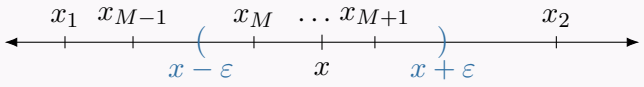
A sequence $\{x_n\}$ *converges* to $x \in \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that, for every $n \geq M$,

$$|x_n - x| < \varepsilon.$$

If a sequence converges to some $x \in \mathbb{R}$, we say it is *convergent*; otherwise, it is *divergent*.

Remark

Since $|x_n - x| < \varepsilon$ means $x - \varepsilon < x_n < x + \varepsilon$, if $\{x_n\}$ is convergent and converges to x, then, once we go out far enough, all the x_n should be within distance less than ε of the limit x.



Thus, for every ε (a "tolerance"), we can always find an $M \in \mathbb{N}$ describing how far we need to go along $x_1, x_2, \ldots, x_M, \ldots$ The definition shows that x_n gets arbitrarily close to the limit x if we go far enough in $\{x_n\}$.

Theorem

If $x, y \in \mathbb{R}$ and $|x - y| \leq \varepsilon$ for every $\varepsilon > 0$, then $x = y$.

Proof

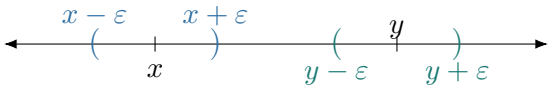
Suppose $x \neq y$. Then, by property 1) of the absolute value (positive definiteness), $|x - y| > 0$. Let

$$\varepsilon = \frac{|x - y|}{2} > 0.$$

Then $|x - y| > \frac{|x-y|}{2} = \varepsilon$, a contradiction. Thus $x = y$.

Theorem: uniqueness of limits

If $\{x_n\}$ converges to x and to y, then $x = y$.



Proof

Let $\varepsilon > 0$. Since $x_n \to x$, there exists $M_1 \in \mathbb{N}$ such that, for every $n \geq M_1$, $|x_n - x| < \varepsilon/2$. Since $x_n \to y$, there exists $M_2 \in \mathbb{N}$ such that, for every $n \geq M_2$, $|x_n - y| < \varepsilon/2$, and thus $|y - x_n| = |x_n - y| < \varepsilon/2$. Since $M_1 + M_2 \geq M_1$ and $M_1 + M_2 \geq M_2$, for every $n \geq M_1 + M_2$,

$$|x_n - x| < \frac{\varepsilon}{2} \qquad \text{and} \qquad |y - x_n| < \frac{\varepsilon}{2}.$$

By the triangle inequality,

$$|(x_n - x) + (y - x_n)| \leq |x_n - x| + |y - x_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $|y - x| < \varepsilon$. By the previous theorem, $y = x$.

Remark

Besides $M_1 + M_2$, we can choose $M = \max\{M_1, M_2\}$, which also works.

Definition

If $\{x_n\}$ converges to x, we call x the *limit* of $\{x_n\}$ and write

$$x = \lim_{n\to\infty} x_n \qquad \text{or} \qquad x_n \to x \quad (\text{as } n \to \infty).$$

Example

Let $x_n = 1$ for every n, that is, the sequence $1, 1, \ldots$. Show that $\lim_{n\to\infty} x_n = 1$.

Proof

Let $\varepsilon > 0$. Let $M = 1$. Then, for every $n \geq M$, $x_n = 1$. Thus

$$|x_n - x| = |1 - 1| = 0 < \varepsilon.$$

Example

Prove that $\lim_{n\to\infty} \frac{1}{n} = 0$.

Proof

Let $\varepsilon > 0$. We want to show that $|1/n| < \varepsilon$. Choose $M \in \mathbb{N}$ such that $M \cdot 1 > 1/\varepsilon$, and thus $1/M < \varepsilon$. Such an M exists by the Archimedean property. Since $n \in \mathbb{N}$, we have $|1/n| = 1/n$.

Hence, for every $n \geq M$,

$$\left|\frac{1}{n}\right| = \frac{1}{n} \leq \frac{1}{M} < \varepsilon.$$

A typical proof of $\lim_{n\to\infty} x_n = x$

Let $\varepsilon > 0$. ... Choose M such that ...; for this M, for every $n \geq M$,

$$|x_n - x| < \cdots < \varepsilon.$$

How to find an M?

Manipulate $|x_n - x|$ around until one eventually arrives at a simple enough expression that contains n and is bigger than $|x_n - x|$, say $f(n)$. Then, as long as $f(n) < \varepsilon$ for the M and every $n \geq M$, we have $|x_n - x| < \varepsilon$. Most of the time we can remove some terms that we do not want. For example,

$$\left|\frac{1}{n^2+2n}\right| = \frac{1}{n^2+2n} < \frac{1}{n^2} \leq \frac{1}{n},$$

and then we let $M > 1/\varepsilon$.

7.4 Proving convergence and divergence

Definition (recall)

$\{x_n\}$ converges to x if, for every $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that, for every $n \geq M$,

$$|x_n - x| < \varepsilon.$$

Negation of the definition

$\{x_n\}$ *does not converge to* x if there exists $\varepsilon_0 > 0$ such that, for every $M \in \mathbb{N}$, there exists $n \geq M$ such that

$$|x_n - x| \geq \varepsilon_0.$$

(Note that this is not the same statement as "$\{x_n\}$ is not convergent".)

Example

$$\lim_{n\to\infty} \frac{1}{n^2+30n+1} = 0.$$

Scratch work

We want M such that, if $n \geq M$, then $\frac{1}{n^2+30n+1} < \varepsilon$. Now

$$\frac{1}{n^2+30n+1} < \frac{1}{n^2+30n} \leq \frac{1}{30n} < \frac{1}{n},$$

so if $\frac{1}{n} < \varepsilon$, then $\frac{1}{n^2+30n+1} < \varepsilon$. Then we only need M such that $n \geq M$ implies $\frac{1}{n} < \varepsilon$.

Proof

Let $\varepsilon > 0$. Choose $M \in \mathbb{N}$ such that $M > 1/\varepsilon$; thus $1/M < \varepsilon$. For all $n \geq M$,

$$\left|\frac{1}{n^2+30n+1} - 0\right| = \frac{1}{n^2+30n+1} < \frac{1}{30n} < \frac{1}{n} \leq \frac{1}{M} < \varepsilon.$$

Example

$\{(-1)^n\}$ does not converge.

Proof (by contradiction)

Suppose there exists $x \in \mathbb{R}$ such that $(-1)^n \to x$. We want to show that there exists $\varepsilon_0 > 0$ such that, for every $M \in \mathbb{N}$, there exists $n \geq M$ with $|x_n - x| \geq \varepsilon_0$. Choose $\varepsilon_0 = \frac{1}{2}$. Since $\{(-1)^n\} = -1, 1, -1, 1, \ldots$, for every $m \in \mathbb{N}$,

$$|(-1)^m - (-1)^{m+1}| = 2.$$

Then, for every $M \in \mathbb{N}$,

$$2 = |(-1)^{M+1} - (-1)^M| \leq |(-1)^{M+1} - x| + |x - (-1)^M| = |(-1)^{M+1} - x| + |(-1)^M - x|.$$

Then either $|(-1)^{M+1} - x| \geq 1 > \frac{1}{2}$ or $|(-1)^M - x| \geq 1 > \frac{1}{2}$. Thus there exists $n \geq M$ such that $|x_n - x| \geq \varepsilon_0$, so $x_n \not\to x$.

Theorem: convergent sequences are bounded

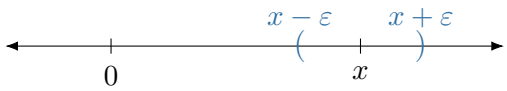
If $\{x_n\}$ converges, then $\{x_n\}$ is bounded.

Remark

The converse of the theorem is not true. Consider the sequence $\{(-1)^n\}$.

Proof

We want to show that there exists $B \geq 0$ such that, for every $n \in \mathbb{N}$, $|x_n| \leq B$.



Suppose $x_n \to x$. Then there exists $M \in \mathbb{N}$ such that, for every $n \geq M$, $|x_n - x| < 1$ (choosing $\varepsilon = 1$). Then, for every $n \geq M$,

$$|x_n| = |(x_n - x) + x| \leq |x_n - x| + |x| < 1 + |x|.$$

Define

$$B = |x_1| + |x_2| + \cdots + |x_{M-1}| + (1 + |x|),$$

a sum of finitely many finite terms. Then, for every $n \in \mathbb{N}$, $|x_n| \leq B$.

7.5 Monotone sequences

Definition

A sequence $\{x_n\}$ is *monotone increasing* if, for every $n \in \mathbb{N}$, $x_n \leq x_{n+1}$; that is, $x_1 \leq x_2 \leq x_3 \leq \cdots$. A sequence is *monotone decreasing* if, for every $n \in \mathbb{N}$, $x_n \geq x_{n+1}$; that is, $x_1 \geq x_2 \geq x_3 \geq \cdots$. If $\{x_n\}$ is monotone increasing or monotone decreasing, we say $\{x_n\}$ is *monotonic*.

Examples

- $\{1/n\}: \ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ is monotone decreasing.
- $\{-1/n\}: \ -1, -\frac{1}{2}, -\frac{1}{3}, \ldots$ is monotone increasing.
- $\{(-1)^n\}: \ -1, 1, -1, 1, \ldots$ is not monotone.

Theorem: monotone convergence

A monotonic sequence is convergent if and only if it is bounded.

Proof

Suppose $\{x_n\}$ is a monotone increasing sequence.
($\Rightarrow$) This follows from the previous theorem.
($\Leftarrow$) Suppose $\{x_n\}$ is bounded. Look at the set of entries of the sequence, $\{x_n \mid n \in \mathbb{N}\} \subset \mathbb{R}$. Since there exists $B \geq 0$ such that $|x_n| \leq B$ for every $n \in \mathbb{N}$ (the sequence is bounded), we have $x_n \leq B$ and $x_n \geq -B$. Thus the set $\{x_n \mid n \in \mathbb{N}\}$ is bounded above and below. Then, by the least upper bound property of $\mathbb{R}$, $\sup\{x_n \mid n \in \mathbb{N}\}$ exists in $\mathbb{R}$. Let

$$x = \sup\{x_n \mid n \in \mathbb{N}\}.$$

We want to show that $\lim_{n\to\infty} x_n = x$. Let $\varepsilon > 0$. Since $x = \sup\{x_n \mid n \in \mathbb{N}\}$, by the theorem on page 26, there exists $M \in \mathbb{N}$ such that

$$x - \varepsilon < x_M \leq x.$$

Since $\{x_n\}$ is monotone increasing, $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$, and since $x = \sup\{x_n \mid n \in \mathbb{N}\}$, we have $x_n \leq x$ for every n. Thus, for every $n \geq M$,

$$x - \varepsilon < x_M \leq x_n \leq x < x + \varepsilon.$$

Therefore $|x_n - x| < \varepsilon$. Thus $\lim_{n\to\infty} x_n = x$, and $\{x_n\}$ is convergent.

Remark

The same proof works for monotone decreasing sequences; then $\lim_{n\to\infty} x_n = \inf\{x_n \mid n \in \mathbb{N}\}$.

Theorem: geometric sequences

1) If $c \in \mathbb{R}$ and $c > 1$, then $\{c^n\}$ is unbounded (and hence c^n diverges).

2) If $c \in (0, 1)$, then $\lim_{n\to\infty} c^n = 0$.

Proof

1) We want to show that, for every $B \geq 0$, there exists $n \in \mathbb{N}$ such that $c^n > B$. Let $B \geq 0$. *Recall* the theorem on page 4: for every $n \in \mathbb{N}$, if $x \geq -1$, then $(1+x)^n \geq 1 + nx$. Hence

$$c^n = \big(1 + (c-1)\big)^n \geq 1 + n(c-1) \geq n(c-1).$$

Choose $n \in \mathbb{N}$ such that $n > \frac{B}{c-1}$ (it exists by the Archimedean property). Then

$$c^n = \big[1 + (c-1)\big]^n \geq 1 + n(c-1) \geq n(c-1) > \frac{B}{c-1}(c-1) = B.$$

Thus, for every $B \geq 0$, there exists n such that $c^n > B$; that is, $\{c^n\}$ is unbounded.
2) Since $c \in (0,1)$, we have $0 < c^n$. Since $c < 1$, for every $n \in \mathbb{N}$, $c^{n+1} < c^n$. Therefore $\{c^n\}$ is monotone decreasing and bounded, since $0 < c^n < c$. Then, by the previous theorem, $\{c^n\}$ is convergent. Let $\lim_{n\to\infty} c^n = L$. We want to show that $L = 0$. Equivalently, we will show that, for every $\varepsilon > 0$, $|L| = |L - 0| < \varepsilon$.
Let $\varepsilon > 0$. Then there exists $M \in \mathbb{N}$ such that, for every $n \geq M$,

$$|c^n - L| < \frac{(1-c)\varepsilon}{2}.$$

Then

$$\begin{aligned}(1-c)|L| = |L - cL| &= |L - c^{M+1} + c^{M+1} - cL| \leq |L - c^{M+1}| + |c^{M+1} - cL| \\ &= |L - c^{M+1}| + c\,|c^M - L| \\ &< \frac{(1-c)\varepsilon}{2} + c\,\frac{(1-c)\varepsilon}{2}.\end{aligned}$$

Since $c \in (0,1)$, we have $1 + c < 2$ and $1 - c > 0$. Thus

$$(1-c)|L| < \frac{(1-c)\varepsilon}{2} \cdot 2 = (1-c)\varepsilon,$$

and hence $|L| < \varepsilon$.

Note

We can choose our "ε" to be $\frac{(1-c)}{2}\varepsilon$, since this allows us to prove $|L| < \varepsilon$. If you would like, we can state "let $\varepsilon_0 = \frac{(1-c)\varepsilon}{2} > 0$." However, this is unnecessary.

7.6 Subsequences

Definition

Let $\{x_n\}$ be a sequence, and let $\{n_k\}_{k=1}^{\infty}$ be a sequence of natural numbers such that $n_1 < n_2 < n_3 < \cdots$. Then the sequence $\{x_{n_k}\}_{k=1}^{\infty}$ is called a *subsequence* of $\{x_n\}$.

Remark

What this definition is saying is: to generate a subsequence, pick an entry, then go to the right, pick another entry, and so on.

Example

Let $\{x_n\}: 1, 2, 3, 4, \ldots$ $(x_n = n)$. Then

$$\begin{aligned} \{x_{n_k}\} &= 1, 3, 5, 7, 9, \ldots && (n_k = 2k - 1), \\ \{x_{n_k}\} &= 2, 4, 6, 8, 10, \ldots && (n_k = 2k), \\ \{x_{n_k}\} &= 2, 3, 5, 7, 11, \ldots && (n_k = k\text{th prime}) \end{aligned}$$

are subsequences of $\{x_n\}$.

Non-example

Let $\{x_n\} = 1, 2, 3, 4, \ldots$ $(x_n = n)$. Then $\{x_{n_k}\} = 1, 1, 1, \ldots$ is *not* a subsequence of $\{x_n\}$: here $n_k = 1$ for every k, so $n_1 \not< n_2 \not< n_3 \not< \cdots$. Likewise, $\{x_{n_k}\} = 1, 1, 3, 3, 5, 5, \ldots$ is not a subsequence.

Example

Let $\{(-1)^n\}: -1, 1, -1, 1, -1, \ldots$. Then

$$\begin{aligned} \{x_{n_k}\} &: -1, -1, -1, -1, \ldots && n_k = 2k - 1, \\ \{x_{n_k}\} &: 1, 1, 1, \ldots && n_k = 2k \end{aligned}$$

are subsequences of $\{(-1)^n\}$.

Theorem: subsequences of a convergent sequence

If $\{x_n\}$ converges to x and $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$, then

$$\lim_{k\to\infty} x_{n_k} = x.$$

Proof

Claim. Since $1 \le n_1 < n_2 < n_3 < \cdots$, we have $n_k \ge k$ for every $k \in \mathbb{N}$.
Base case. For $k = 1$: since $n_1 \in \mathbb{N}$, $n_1 \ge 1$.
Inductive step. Suppose $n_m \ge m$. Since $n_{m+1} > n_m$ and $n_{m+1}, n_m \in \mathbb{N}$, we have $n_{m+1} \ge n_m + 1$. Since $n_m \ge m$, therefore $n_{m+1} \ge m + 1$.
By induction, $n_k \ge k$ for every $k \in \mathbb{N}$.
We want to show that, for every $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that, for every $k \ge M$, $|x_{n_k} - x| < \varepsilon$. Let $\varepsilon > 0$. Since $\lim_{n\to\infty} x_n = x$, there exists $M_0 \in \mathbb{N}$ such that, for every $n \ge M_0$, $|x_n - x| < \varepsilon$. Choose $M = M_0$. Then $k \ge M = M_0$ implies $n_k \ge M_0$. Thus $|x_{n_k} - x| < \varepsilon$.

8 The Squeeze Theorem and Operations Involving Limits

8.1 Operations involving limits

Theorem

$$\lim_{n\to\infty} x_n = x \iff \lim_{n\to\infty} |x_n - x| = 0.$$

Proof

Let $\varepsilon > 0$.
($\Rightarrow$) Since $x_n \to x$, there exists $M \in \mathbb{N}$ such that, for every $n \geq M$, $|x_n - x| < \varepsilon$. Thus, for every $n \geq M$,

$$\big|\,|x_n - x| - 0\,\big| = |x_n - x| < \varepsilon.$$

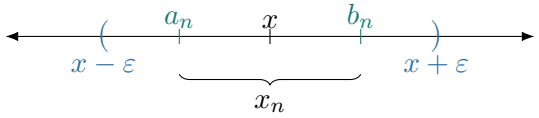
Thus $\lim_{n\to\infty} |x_n - x| = 0$.
($\Leftarrow$) Since $\lim_{n\to\infty} |x_n - x| = 0$, there exists $M \in \mathbb{N}$ such that, for every $n \geq M$, $\big|\,|x_n - x| - 0\,\big| < \varepsilon$. Thus $|x_n - x| < \varepsilon$, and so $x_n \to x$.

Theorem (Squeeze theorem)

Let $\{a_n\}$, $\{b_n\}$, $\{x_n\}$ be sequences such that, for every $n \in \mathbb{N}$,

$$a_n \leq x_n \leq b_n,$$

and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = x$. Then $\lim_{n\to\infty} x_n = x$.



Proof

Let $\varepsilon > 0$. Since $b_n \to x$, there exists $M_1 \in \mathbb{N}$ such that, for every $n \geq M_1$,

$$|b_n - x| < \varepsilon \implies -\varepsilon < b_n - x < \varepsilon \implies (x - \varepsilon <)\ b_n < x + \varepsilon.$$

Since $a_n \to x$, there exists $M_2 \in \mathbb{N}$ such that, for every $n \geq M_2$,

$$|a_n - x| < \varepsilon \implies -\varepsilon < a_n - x < \varepsilon \implies x - \varepsilon < a_n\ (< x + \varepsilon).$$

Thus, for every $n \geq M_1 + M_2$,

$$x - \varepsilon < a_n \leq x_n \leq b_n < x + \varepsilon \implies -\varepsilon < x_n - x < \varepsilon \implies |x_n - x| < \varepsilon.$$

Therefore $x_n \to x$.

Example

$$\lim_{n\to\infty} \frac{n^2}{n^2+n+1} = 1.$$

Proof

We want to show that

$$\lim_{n\to\infty} \left| \frac{n^2}{n^2+n+1} - 1 \right| = 0.$$

Let $\varepsilon > 0$. We have

$$\left| \frac{n^2}{n^2+n+1} - 1 \right| = \left| \frac{n^2-(n^2+n+1)}{n^2+n+1} \right| = \left| \frac{-n-1}{n^2+n+1} \right| = \left| \frac{n+1}{n^2+n+1} \right|$$
$$= \frac{n+1}{n^2+n+1} \leq \frac{n+1}{n^2+n} = \frac{1}{n}.$$

Thus

$$0 \leq \left| \frac{n^2}{n^2+n+1} - 1 \right| \leq \frac{1}{n}.$$

Since $\lim_{n\to\infty} 0 = 0$ and $\lim_{n\to\infty} \frac{1}{n} = 0$ (the example on page 37), the squeeze theorem gives $\left| \frac{n^2}{n^2+n+1} - 1 \right| \to 0$, and then, by the previous theorem, $\frac{n^2}{n^2+n+1} \to 1$.

8.2 Limits and the order of the real numbers

Question

How does the limit interact with the order of $\mathbb{R}$? (See page 15 and page 22.)

Theorem: limits preserve inequalities

1) If $\{x_n\}$ and $\{y_n\}$ are convergent sequences and $x_n \leq y_n$ for every $n \in \mathbb{N}$, then $\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n$.

2) If $\{x_n\}$ is a convergent sequence and $a \leq x_n \leq b$ for every $n \in \mathbb{N}$, then $a \leq \lim_{n\to\infty} x_n \leq b$.

Remark

We may lose the strict inequality "<"! In other words,

$$\forall n \in \mathbb{N},\ a < x_n < b \quad \not\Longrightarrow \quad a < \lim_{n\to\infty} x_n < b.$$

Counterexample. For every $n \in \mathbb{N}$, $1 - \frac{1}{n} < 1$; however,

$$\lim_{n\to\infty} 1 = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right) = 1,$$

so $\lim_{n\to\infty}(1 - \frac{1}{n}) \not< \lim_{n\to\infty} 1$.

Proof

2) is obvious from 1).
1) Let $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$. We want to show that $x \leq y$. We argue by contradiction: suppose $y < x$. Let

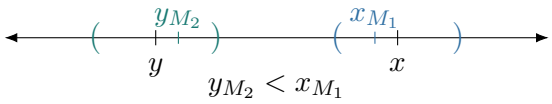
$$\varepsilon = \frac{x-y}{2} > 0.$$

Since $x_n \to x$, there exists $M_1 \in \mathbb{N}$ such that, for every $n \geq M_1$, $|x_n - x| < \varepsilon$. Since $y_n \to y$, there exists $M_2 \in \mathbb{N}$ such that, for every $n \geq M_2$, $|y_n - y| < \varepsilon$. Then, for every $n \geq M = M_1 + M_2$,

$$|x_n - x| < \varepsilon \implies x_n > x - \varepsilon = \frac{x+y}{2},$$
$$|y_n - y| < \varepsilon \implies y_n < y + \varepsilon = \frac{x+y}{2}.$$

Thus, for every $n \geq M$,

$$y_n < \frac{x+y}{2} < x_n.$$



Since $x_n \leq y_n$ for every $n \in \mathbb{N}$, this is a contradiction. Thus the assumption that $x > y$ is false, and $x \leq y$.

8.3 Limits and algebraic operations

Question

How does the limit interact with algebraic operations?

Theorem: algebraic operations and limits

Suppose $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$ (that is, both sequences converge). Then:

1) $\lim_{n\to\infty} (x_n + y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n = x + y$;
2) for every $c \in \mathbb{R}$, $\lim_{n\to\infty} (cx_n) = c \lim_{n\to\infty} x_n = cx$;
3) $\lim_{n\to\infty} (x_n y_n) = \left(\lim_{n\to\infty} x_n\right)\left(\lim_{n\to\infty} y_n\right) = xy$;
4) if $y_n \neq 0$ for every $n \in \mathbb{N}$ and $y \neq 0$, then $\lim_{n\to\infty} \left(\frac{x_n}{y_n}\right) = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n} = \frac{x}{y}$.

Proof

1) Let $\varepsilon > 0$. Since $x_n \to x$, there exists $M_1 \in \mathbb{N}$ such that, for every $n \geq M_1$, $|x_n - x| < \varepsilon/2$. Since $y_n \to y$, there exists $M_2 \in \mathbb{N}$ such that, for every $n \geq M_2$, $|y_n - y| < \varepsilon/2$. Then, for every

$n \geq M = M_1 + M_2$,

$$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $x_n + y_n \to x + y$.
2) Let $\varepsilon > 0$. Since $x_n \to x$, there exists $M_1 \in \mathbb{N}$ such that, for every $n \geq M_1$,

$$|x_n - x| < \frac{\varepsilon}{|c| + 1}.$$

Then

$$|c|\,|x_n - x| < \frac{\varepsilon}{|c| + 1} \cdot |c| \quad \Longrightarrow \quad |cx_n - cx| < \varepsilon \cdot \frac{|c|}{|c| + 1} < \varepsilon \cdot 1 = \varepsilon.$$

We choose $|c| + 1$ instead of $|c|$ because we do not know whether $c = 0$!
3) (By the squeeze theorem.) Since $y_n \to y$, $\{y_n\}$ is bounded. Thus there exists $B \geq 0$ such that $|y_n| \leq B$ for every $n \in \mathbb{N}$. Then

$$\begin{aligned} |x_n y_n - xy| &= |(x_n - x)y_n + (y_n - y)x| \\ &\leq |x_n - x|\,|y_n| + |y_n - y|\,|x| \\ &\leq |x_n - x|\,B + |x|\,|y_n - y|. \end{aligned}$$

(We are going to use the squeeze theorem and parts 1) and 2), which we just proved.) Thus

$$0 \leq |x_n y_n - xy| \leq B|x_n - x| + |x|\,|y_n - y|.$$

Since $x_n \to x$, we have $|x_n - x| \to 0$; similarly $|y_n - y| \to 0$. By 2), $|x_n - x|B \to 0$ and $|x|\,|y_n - y| \to 0$, and then, by 1), $B|x_n - x| + |x|\,|y_n - y| \to 0$. Thus, by the squeeze theorem, since $0 \to 0$ and $B|x_n - x| + |x|\,|y_n - y| \to 0$, we get $|x_n y_n - xy| \to 0$, that is, $x_n y_n \to xy$.
4) We want to show that $\frac{1}{y_n} \to \frac{1}{y}$. (Then $(\frac{1}{y_n})x_n \to x(\frac{1}{y}) = \frac{x}{y}$ by 3).)
Claim. There exists $b > 0$ such that $|y_n| \geq b$ for every $n \in \mathbb{N}$.
(Proof by contradiction.) Suppose that, for every $b > 0$, there exists $n \in \mathbb{N}$ such that $|y_n| < b$. Then, by the theorem on page 36, $|y_n - 0| \leq b$ implies $y_n = 0$. Since $y_n \neq 0$ for every n, this is a contradiction.
Now let $\varepsilon > 0$. Since $y_n \to y$, there exists $M_0 \in \mathbb{N}$ such that, for every $n \geq M_0$,

$$|y_n - y| < \varepsilon\,|y|\,b.$$

Choose $M = M_0$. Then, for every $n \geq M_0$,

$$\left|\frac{1}{y_n} - \frac{1}{y}\right| = \left|\frac{y - y_n}{y_n y}\right| = \frac{|y_n - y|}{|y_n|\,|y|} \leq \frac{|y_n - y|}{b\,|y|} < \frac{\varepsilon\,|y|\,b}{b\,|y|} = \varepsilon.$$

Thus $\frac{1}{y_n} \to \frac{1}{y}$, and hence $\frac{x_n}{y_n} \to \frac{x}{y}$ by 3).

Note

Part 4) can be proved through the squeeze theorem again. For the existence of b, one can simply let

$$b = \inf\Big\{|y_1|, \ldots, |y_{M-1}|, \frac{|y|}{2}\Big\},$$

and one can show that $b > 0$.

8.4 Square roots and absolute values

Theorem: limits and square roots

If $x_n \geq 0$ for every $n \in \mathbb{N}$ and $\lim_{n\to\infty} x_n = x$, then

$$\lim_{n\to\infty} \sqrt{x_n} = \sqrt{x}.$$

Proof

Case 1: $x = 0$. Let $\varepsilon > 0$. Since $x_n \to 0$, there exists $M_0 \in \mathbb{N}$ such that, if $n \geq M_0$, then $|x_n| = x_n < \varepsilon^2$. Choose $M = M_0$. Then, for every $n \geq M$,

$$|\sqrt{x_n} - 0| = \sqrt{x_n} < \sqrt{\varepsilon^2} = \varepsilon.$$

Thus $\sqrt{x_n} \to \sqrt{x} = 0$.
Case 2: $x \neq 0$. Then

$$|\sqrt{x_n} - \sqrt{x}| = \left|(\sqrt{x_n} - \sqrt{x})\,\frac{\sqrt{x_n} + \sqrt{x}}{\sqrt{x_n} + \sqrt{x}}\right| = \left|\frac{x_n - x}{\sqrt{x_n} + \sqrt{x}}\right| = \frac{|x_n - x|}{\sqrt{x_n} + \sqrt{x}} \leq \frac{1}{\sqrt{x}}\,|x_n - x|.$$

Thus, by the squeeze theorem again,

$$0 \leq |\sqrt{x_n} - \sqrt{x}| \leq \frac{1}{\sqrt{x}}\,|x_n - x|,$$

where $0 \to 0$ and $\frac{1}{\sqrt{x}}|x_n - x| \to 0$. Therefore, by the squeeze theorem, $|\sqrt{x_n} - \sqrt{x}| \to 0$. Thus $\sqrt{x_n} \to \sqrt{x}$.

Remark

Similarly, one can prove the following statement.

Theorem: limits and kth roots

If $x_n \geq 0$ for every $n \in \mathbb{N}$ and $x_n \to x$, then, for every $k \in \mathbb{N}$,

$$\lim_{n\to\infty} (x_n)^{1/k} = x^{1/k}.$$

The proof will not be given.

Theorem: limits and absolute values

If $\{x_n\}$ is a convergent sequence and $\lim_{n\to\infty} x_n = x$, then $\{|x_n|\}$ is convergent, with $|x_n| \to |x|$.

Proof

By the definition of the absolute value, $\sqrt{x^2} = |x|$ for every $x \in \mathbb{R}$. Thus, by the previous theorem,

$$\lim_{n\to\infty} \sqrt{x_n^2} = \lim_{n\to\infty} |x_n| = \sqrt{x^2} = |x|.$$

Note

The proof can also be done through the reverse triangle inequality (page 30):

$$0 \le \big|\, |x_n| - |x| \,\big| \le |x_n - x|,$$

and then by the squeeze theorem.

Remark

The converse of this theorem does not hold; that is,

$$|x_n| \to |x| \quad \not\Longrightarrow \quad x_n \to x.$$

Counterexample. Consider $\{(-1)^n\}$. Then $|(-1)^n| \to 1$, yet $\{(-1)^n\}$ diverges.

9 Limsup, Liminf, and the Bolzano–Weierstrass Theorem

9.1 Recollections and some standard limits

Recall

- **Theorem.** $\lim_{n\to\infty} x_n = x$ if and only if $\lim_{n\to\infty} |x_n - x| = 0$ (page 42).
- **Theorem (squeeze theorem).** If $a_n \le x_n \le b_n$ for every n, then

  $$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = L \quad \Longrightarrow \quad \lim_{n\to\infty} x_n = L$$

  (page 42).
- **Theorem.** If $|c| < 1$, then $\lim_{n\to\infty} c^n = 0$ (page 40).

Theorem (binomial theorem)

For all $n \in \mathbb{N}$ and all $x, y \in \mathbb{R}$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k, \qquad \text{where} \qquad \binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

The proof can be done via induction.

Theorem

1) If $p > 0$, then $\lim_{n\to\infty} \frac{1}{n^p} = 0$.

2) If $p > 0$, then $\lim_{n\to\infty} p^{1/n} = 1$.

3) $\lim_{n\to\infty} n^{1/n} = 1$.

Proof

1) Let $\varepsilon > 0$. Choose $M \in \mathbb{N}$ such that $M > (1/\varepsilon)^{1/p}$; such an M exists by the Archimedean property. Then, for all $n \geq M$,

$$\left| \frac{1}{n^p} - 0 \right| = \frac{1}{n^p} \leq \frac{1}{M^p} < \varepsilon.$$

2) If $p = 1$, the claim is a trivial fact.
If $p > 1$, then $p > 1$ gives $|p^{1/n} - 1| = p^{1/n} - 1$, so we want to show that $p^{1/n} - 1 \to 0$. Recall that, for every $x \geq -1$, $(1 + x)^n \geq 1 + nx$ (page 4). Then

$$p = \left[1 + (p^{1/n} - 1)\right]^n \geq 1 + n\,(p^{1/n} - 1),$$

and thus

$$0 \leq p^{1/n} - 1 \leq \frac{p - 1}{n}.$$

Since $0 \to 0$ and $(p - 1)/n \to 0$ as $n \to \infty$, the squeeze theorem gives

$$\lim_{n\to\infty} |p^{1/n} - 1| = \lim_{n\to\infty} (p^{1/n} - 1) = 0.$$

Thus $p^{1/n} \to 1$.
If $p < 1$, then

$$\lim_{n\to\infty} \frac{1}{(1/p)^{1/n}} = \frac{1}{1} = 1.$$

3) Let $x_n = n^{1/n} - 1 \geq 0$. We want to show that $\lim_{n\to\infty} x_n = 0$, where $\{x_n\}$ is a sequence. Since $x_n = n^{1/n} - 1$, we have that, for every $n \in \mathbb{N}$,

$$n = (1 + x_n)^n = \sum_{k=0}^{n} \binom{n}{k} x_n^k \geq \binom{n}{2} x_n^2 = \frac{n!}{2!\,(n-2)!}\, x_n^2 = \frac{n(n-1)}{2}\, x_n^2,$$

since all the terms are non-negative. That is,

$$n \geq \frac{n(n-1)}{2}\, x_n^2.$$

Then, for all $n > 1$, $n \in \mathbb{N}$,

$$x_n^2 \leq \frac{2}{n-1} \quad \Longrightarrow \quad x_n \leq \sqrt{\frac{2}{n-1}}.$$

Since $x_n = n^{1/n} - 1 \geq 0$, for every $n \in \mathbb{N} \setminus \{1\}$,

$$0 \leq x_n \leq \sqrt{\frac{2}{n-1}}.$$

Then, by the squeeze theorem (and the fact that limits preserve operations and square roots, as long as the sequences converge),

$$\sqrt{\frac{2}{n-1}} \to 0 \quad \Longrightarrow \quad x_n \to 0.$$

Thus $n^{1/n} \to 1$.

9.2 Limsup and liminf

Example (introduction)

If we look at $\{(-1)^n\}$, it is bounded and it has subsequences that are convergent, namely $\{x_{n_k}\}$ where $n_k = 2k$ or $n_k = 2k-1$ for all $k \in \mathbb{N}$.
However, if we consider $\{n\} = 1, 2, 3, \ldots$, it is unbounded, and it has no subsequence that is convergent. Thus a question is:

Q: Does every bounded sequence have a convergent subsequence?

A: Yes. (Bolzano–Weierstrass theorem.)

Definition

Let $\{x_n\}$ be a bounded sequence. We define (if they exist)

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \big(\sup\{x_k \mid k \geq n\}\big), \qquad \liminf_{n\to\infty} x_n = \lim_{n\to\infty} \big(\inf\{x_k \mid k \geq n\}\big),$$

namely the *limit inferior* and the *limit superior* of $\{x_n\}$.

Note

The expression $\{x_k \mid k \geq n\}$ is a set contained in $\mathbb{R}$, not a subsequence. We did not define what it means to take the "sup" or the "inf" of a sequence.

Theorem

Let $\{x_n\}$ be a bounded sequence, and let

$$a_n = \sup\{x_k \mid k \geq n\}, \qquad b_n = \inf\{x_k \mid k \geq n\},$$

so that $\limsup_{n\to\infty} x_n := \lim_{n\to\infty} a_n$ and $\liminf_{n\to\infty} x_n := \lim_{n\to\infty} b_n$. Then:

1) $\{a_n\}$ is monotone decreasing and bounded, and $\{b_n\}$ is monotone increasing and bounded. (Thus $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist.)

2) $\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n$.

We will prove a theorem first, then prove this one.

Theorem

If $A, B \subset \mathbb{R}$ are non-empty and bounded, and $A \subset B$, then

$$\inf B \leq \inf A \leq \sup A \leq \sup B.$$

Proof

Since $\sup B$ is an upper bound for B and $A \subset B$, $\sup B$ is an upper bound for A. Since $\sup A$ is the least upper bound for A, and $\sup B$ is an upper bound for A, we get $\sup A \leq \sup B$.
Since $A \neq \emptyset$, $\sup A$ is an upper bound for A and $\inf A$ is a lower bound for A, so there exists $x \in A$ such that $x \leq \sup A$ and $x \geq \inf A$. Thus

$$\inf A \leq x \leq \sup A \quad \Longrightarrow \quad \inf A \leq \sup A.$$

The inequality $\inf A \geq \inf B$ works exactly the same as the proof of $\sup A \leq \sup B$.

Proof of the theorem on limsup and liminf

1) Since $\{x_k \mid k \geq n+1\} \subset \{x_k \mid k \geq n\}$, the previous theorem gives

$$a_{n+1} = \sup\{x_k \mid k \geq n+1\} \leq \sup\{x_k \mid k \geq n\} = a_n.$$

Hence $a_{n+1} \leq a_n$ for every $n \in \mathbb{N}$, and $\{a_n\}$ is monotone decreasing. Similarly, $b_n \leq b_{n+1}$ for every $n \in \mathbb{N}$, so $\{b_n\}$ is monotone increasing.
Since $\{x_n\}$ is bounded, there is $B \geq 0$ such that $-B \leq x_n \leq B$ for every $n \in \mathbb{N}$. Therefore

$$\begin{aligned} & -B \leq \inf\{x_k \mid k \geq n\} \leq \sup\{x_k \mid k \geq n\} \leq B \\ \Longrightarrow \quad & -B \leq b_n \leq a_n \leq B \\ \Longrightarrow \quad & |b_n| \leq B \ \text{ and } \ |a_n| \leq B. \end{aligned}$$

Thus $\{a_n\}$ is a sequence that is bounded and monotone decreasing, and therefore $\lim_{n\to\infty} a_n$ exists. Similarly, $\lim_{n\to\infty} b_n$ exists.
2) Since $-B \leq b_n \leq a_n \leq B$, and limits preserve (loose) inequalities,

$$\lim_{n\to\infty} b_n \leq \lim_{n\to\infty} a_n, \qquad \text{that is,} \qquad \liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Example

Let $\{x_n\} = \{(-1)^n\}$. *Claim:* $\limsup_{n\to\infty} x_n = 1$ and $\liminf_{n\to\infty} x_n = -1$.
For every $n \in \mathbb{N}$, $\{(-1)^k \mid k \geq n\} = \{-1, 1\}$. Thus

$$\sup\{(-1)^k \mid k \geq n\} = \sup\{1, -1\} = 1 \qquad \text{for every } n \in \mathbb{N}.$$

Therefore

$$\begin{aligned} \limsup_{n\to\infty} (-1)^n &= \lim_{n\to\infty} \sup\{(-1)^k \mid k \geq n\} = \lim_{n\to\infty} 1 = 1, \\ \liminf_{n\to\infty} (-1)^n &= \lim_{n\to\infty} \inf\{(-1)^k \mid k \geq n\} = \lim_{n\to\infty} \inf\{1, -1\} = \lim_{n\to\infty} (-1) = -1. \end{aligned}$$

Example

Let $x_n = 1/n$. *Claim:* $\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = 0$.
For $n \in \mathbb{N}$, the set $\{x_k \mid k \geq n\}$ will be

$$\left\{\frac{1}{k} \;\middle|\; k \geq n\right\} = \left\{\frac{1}{n}, \frac{1}{n+1}, \frac{1}{n+2}, \frac{1}{n+3}, \ldots\right\}.$$

Again, this is a set, not a sequence: sup and inf apply to sets, while lim sup and lim inf apply to sequences. Then

$$\sup\left\{\frac{1}{k} \;\middle|\; k \geq n\right\} = \sup\left\{\frac{1}{n}, \frac{1}{n+1}, \frac{1}{n+2}, \ldots\right\} = \frac{1}{n}$$

(this can be proven). Thus

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \left(\sup\{x_k \mid k \geq n\}\right) = \lim_{n\to\infty} \frac{1}{n} = 0.$$

Also

$$\inf\left\{\frac{1}{k} \;\middle|\; k \geq n\right\} = \inf\left\{\frac{1}{n}, \frac{1}{n+1}, \frac{1}{n+2}, \ldots\right\} = 0 \qquad \text{for every } n \in \mathbb{N}.$$

Thus

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \left(\inf\{x_k \mid k \geq n\}\right) = \lim_{n\to\infty} 0 = 0.$$

The lim inf part can be shown either through the definition of inf or by the squeeze theorem: $0 \leq \inf\{x_k \mid k \geq n\} \leq 1/n$.

Remarks

1) Later we will see the fact that we can already see from these two examples: lim sup = lim inf for $\{x_n\}$ if and only if the sequence is convergent.

2) Be careful that the sequences obtained by taking inf and sup, namely

$$\{a_n\} = \{\sup\{x_k \mid k \geq n\}\}_{n=1}^{\infty} \qquad \text{and} \qquad \{b_n\} = \{\inf\{x_k \mid k \geq n\}\}_{n=1}^{\infty},$$

are **not necessarily subsequences** of the original sequence $\{x_n\}$! (The inner braces denote a set; the outer braces denote a sequence.)

9.3 The Bolzano–Weierstrass theorem

Theorem

Let $\{x_n\}$ be a bounded sequence. Then there exist subsequences $\{x_{n_k}\}$ and $\{x_{m_k}\}$ such that

$$\lim_{k\to\infty} x_{n_k} = \limsup_{n\to\infty} x_n \qquad \text{and} \qquad \lim_{k\to\infty} x_{m_k} = \liminf_{n\to\infty} x_n.$$

This immediately gives the Bolzano–Weierstrass theorem, which is essentially emphasizing and repeating what this theorem says.

Theorem (Bolzano–Weierstrass)

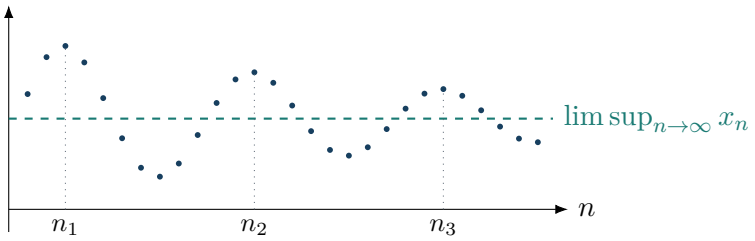
Every bounded sequence has a convergent subsequence.

Note

This holds only in finite-dimensional Euclidean space $\mathbb{R}^n$!

Proof of the previous theorem

For lim sup. Let $a_n = \sup\{x_k \mid k \geq n\}$, where $n \in \mathbb{N}$. An idea of the proof:



Recall the theorem on page 26: let $A \subset \mathbb{R}$, $A \neq \emptyset$; then $a = \sup A$ if and only if for every $\varepsilon > 0$ there exists $x \in A$ such that $a - \varepsilon < x \leq a$.
Since $a_1 = \sup\{x_k \mid k \geq 1\}$, there exists $n_1 \in \mathbb{N}$ (with $n_1 \geq 1$, in order that $x_{n_1} \in \{x_k \mid k \geq 1\}$) such that, for $\varepsilon = 1$,

$$a_1 - 1 < x_{n_1} \leq a_1.$$

Then, since $a_{n_1+1} = \sup\{x_k \mid k \geq n_1 + 1\}$, again by the theorem on page 26, choosing $\varepsilon = \frac{1}{2}$, there exists $n_2 \in \mathbb{N}$ such that $n_2 > n_1$ and

$$a_{n_1+1} - \tfrac{1}{2} < x_{n_2} \leq a_{n_1+1}.$$

Again, $n_2 > n_1$ means $x_{n_2} \in \{x_k \mid k \geq n_1 + 1\}$; otherwise the theorem fails, since x_{n_2} would not belong to this set.
Reiterating this process, we obtain that, for every $k \in \mathbb{N}$,

$$a_{n_k+1} - \frac{1}{k+1} < x_{n_{k+1}} \leq a_{n_k+1} \qquad \Big(\text{choosing } \varepsilon = \frac{1}{k+1}\Big),$$

where $n_1 < n_2 < \cdots$, which makes $\{x_{n_k}\}_{k=1}^{\infty}$ a subsequence of $\{x_n\}$.
Then $\{a_{n_k+1} - \frac{1}{k+1}\}_{k=1}^{\infty}$ converges to

$$\lim_{k\to\infty} a_{n_k+1} = \lim_{n\to\infty} a_n = \limsup_{n\to\infty} x_n.$$

Here $\lim_{k\to\infty} a_{n_k+1} = \lim_{n\to\infty} a_n$, since subsequences converge to the same limit if the original sequence converges (theorem on page 42). Also, $\{a_{n_k+1}\}$ converges to $\lim_{k\to\infty} a_{n_k+1} = \lim_{n\to\infty} a_n = \limsup_{n\to\infty} x_n$.
Thus, by the squeeze theorem, $x_{n_{k+1}} \to \limsup_{n\to\infty} x_n$ as $k \to \infty$. Therefore $\{x_{n_k}\}_{k=1}^{\infty}$ is a subsequence of $\{x_n\}$ converging to $\limsup_{n\to\infty} x_n$.
For lim inf. We are going to use the second part of the theorem on page 26. Let $b_n = \inf\{x_k \mid k \geq n\}$. Since $b_1 = \inf\{x_k \mid k \geq 1\}$, by the theorem on page 26 there exists $m_1 \in \mathbb{N}$ (with $m_1 \geq 1$, so that $x_{m_1} \in \{x_k \mid k \geq 1\}$) such that

$$b_1 \leq x_{m_1} < b_1 + 1 \qquad (\text{choosing } \varepsilon = 1).$$

Since $b_{m_1+1} = \inf\{x_k \mid k \geq m_1 + 1\}$, there exists $m_2 \in \mathbb{N}$ such that $m_2 > m_1$ and

$$b_{m_1+1} \leq x_{m_2} < b_{m_1+1} + \tfrac{1}{2} \qquad (\varepsilon = \tfrac{1}{2}).$$

Reiterating this process, for every $k \in \mathbb{N}$ there exists $m_{k+1} \in \mathbb{N}$ such that $m_{k+1} > m_k$ and

$$b_{m_k+1} \leq x_{m_{k+1}} < b_{m_k+1} + \frac{1}{k+1} \qquad \left(\varepsilon = \frac{1}{k+1}\right).$$

Thus we found a sequence of natural numbers with $m_1 < m_2 < m_3 < \cdots$, and hence $\{x_{m_k}\}$ is a subsequence of $\{x_n\}$ by definition.
Since $\{b_{m_k+1}\}_{k=1}^{\infty}$ converges to $\liminf_{n\to\infty} x_n$, and $\{b_{m_k+1} + \frac{1}{k+1}\}_{k=1}^{\infty}$ converges to $\liminf_{n\to\infty} x_n$ (a subsequence converges to the limit of the sequence if the sequence converges, and this limit L exists by the previous theorem on the existence of lim inf for bounded sequences), the squeeze theorem shows that the subsequence $\{x_{m_k}\}$ of $\{x_n\}$ converges to $\liminf_{n\to\infty} x_n$.

Remark

This theorem immediately gives the Bolzano–Weierstrass theorem, which is an emphasis that bounded sequences have convergent subsequences.

Theorem

Let $\{x_n\}$ be a bounded sequence. Then

$$\{x_n\} \text{ is convergent} \iff \limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n.$$

Moreover, if $\{x_n\}$ is convergent, then

$$\lim_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n.$$

Proof

($\Longleftarrow$) Suppose $L = \limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n$. Since, for every $n \in \mathbb{N}$,

$$\inf\{x_k \mid k \geq n\} \leq x_n \leq \sup\{x_k \mid k \geq n\},$$

the sequence on the left, $\{b_n\} = \{\inf\{x_k \mid k \geq n\}\}$, and the sequence on the right, $\{a_n\} = \{\sup\{x_k \mid k \geq n\}\}$, both converge to L. Then, by the squeeze theorem,

$$\lim_{n\to\infty} x_n = L = \limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n.$$

($\Longrightarrow$) Let $x = \lim_{n\to\infty} x_n$. Then, by the theorem on page 42, every subsequence of $\{x_n\}$ converges to x. By the previous theorem, there exist subsequences that converge to $\limsup_{n\to\infty} x_n$ and to $\liminf_{n\to\infty} x_n$. Thus $\limsup_{n\to\infty} x_n = x$ and $\liminf_{n\to\infty} x_n = x$.

10 The Completeness of the Real Numbers and Basic Properties of Infinite Series

10.1 Cauchy sequences

Definition

A sequence $\{x_n\}$ is *Cauchy* if for every $\varepsilon > 0$ there exists $M \in \mathbb{N}$ such that, for all $n \geq M$ and all $k \geq M$ with $n, k \in \mathbb{N}$, we have

$$|x_n - x_k| < \varepsilon.$$

Example

Let $x_n = 1/n$. *Claim:* $\{1/n\}$ is Cauchy.

Proof

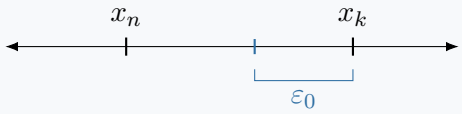
Let $\varepsilon > 0$. Choose $M \in \mathbb{N}$ such that $M > 2/\varepsilon$. Then, for all $n \geq M$ and all $k \geq M$,

$$\left|\frac{1}{n} - \frac{1}{k}\right| \leq \left|\frac{1}{n}\right| + \left|-\frac{1}{k}\right| = \frac{1}{n} + \frac{1}{k} \leq \frac{1}{M} + \frac{1}{M} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Negation of the definition

A real sequence $\{x_n\}$ is *not Cauchy* if there exists $\varepsilon_0 > 0$ such that for every $M \in \mathbb{N}$ there exist $n, k \in \mathbb{N}$ with $n \geq M$, $k \geq M$ and

$$|x_n - x_k| \geq \varepsilon_0.$$



Non-example

$\{(-1)^n\}$ is not Cauchy.

Proof

Choose $\varepsilon_0 = 2$. Let $M \in \mathbb{N}$. Choose $n = M$ and $k = M + 1$. Then

$$|(-1)^n - (-1)^k| = |\pm 2| = 2 \geq \varepsilon_0.$$

Note: "let" versus "choose"

When doing proofs, if we want to show something for all ..., we say "let"; for example, let $\varepsilon > 0$, let $n \in \mathbb{N}$. If we want to show that there exists something, we "choose"; for example, choose $M_0 = \cdots$, choose $\varepsilon_0 = \cdots > 0$.

Theorem

If $\{x_n\}$ is Cauchy, then $\{x_n\}$ is bounded.

Proof

Since $\{x_n\}$ is Cauchy, there exists $M \in \mathbb{N}$ such that, for all $n \geq M$ and all $k \geq M$, $|x_n - x_k| < 1$. Thus, for all $n \geq M$, $|x_n - x_M| < 1$. Then, for all $n \geq M$,

$$|x_n| = |x_n - x_M + x_M| \leq |x_n - x_M| + |x_M| < 1 + |x_M|.$$

Let

$$B = |x_1| + |x_2| + \cdots + |x_{M-1}| + |x_M| + 1.$$

Then $|x_n| < B$ for every $n \in \mathbb{N}$, and thus $\{x_n\}$ is bounded.

Theorem

If $\{x_n\}$ is Cauchy and there exists a subsequence $\{x_{n_k}\}$ that is convergent, then $\{x_n\}$ converges.

Proof

Suppose $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$ such that $x_{n_k} \to x$. We want to show that $x_n \to x$. Let $\varepsilon > 0$. Since $x_{n_k} \to x$, there exists $M_0 \in \mathbb{N}$ such that, for all $k \geq M_0$,

$$|x_{n_k} - x| < \frac{\varepsilon}{2}.$$

Since $\{x_n\}$ is Cauchy, there exists $M_1 \in \mathbb{N}$ such that, for all $n \geq M_1$ and all $k \geq M_1$,

$$|x_n - x_k| < \frac{\varepsilon}{2}.$$

Now choose $M = M_0 + M_1$. Then, for all $n \geq M$, we have $n_M \geq M > M_0$ and $n \geq M_1$. Therefore

$$|x_n - x| \leq |x_n - x_{n_M}| + |x_{n_M} - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Theorem

If $\{x_n\}$ is a sequence in $\mathbb{R}$, then

$$\{x_n\} \text{ is Cauchy} \iff \{x_n\} \text{ is convergent.}$$

Proof

($\Longrightarrow$) Suppose $\{x_n\}$ is Cauchy; then $\{x_n\}$ is bounded. By the Bolzano–Weierstrass theorem, $\{x_n\}$ has a convergent subsequence. Since $\{x_n\}$ is Cauchy and one of its subsequences converges, $\{x_n\}$ is convergent (by the previous theorem).
($\Longleftarrow$) Suppose $\{x_n\}$ is convergent. Let $\varepsilon > 0$. Then there exists $M_0 \in \mathbb{N}$ such that, for all $n \geq M_0$, $|x_n - x| < \varepsilon/2$. Choose $M = M_0$. Then, for all $n \geq M$ and all $k \geq M$, $|x_n - x| < \varepsilon/2$ and $|x_k - x| < \varepsilon/2$. Thus

$$|x_n - x_k| \leq |x_n - x| + |x_k - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $\{x_n\}$ is Cauchy.

Remark

If we consider sequences within $\mathbb{Q}$, things get more complicated because of the absence of the least upper bound property.

Example

Take $\{x_n\}$ such that $x_n \in \mathbb{Q}$ and $x_n \to \sqrt{2}$ (which is in $\mathbb{R}$ but not in $\mathbb{Q}$). Thus, in $\mathbb{Q}$:

$$\text{convergence} \implies \text{Cauchy}, \qquad \text{yet} \qquad \text{Cauchy} \not\Rightarrow \text{convergent}.$$

Questions

Q: What did we use precisely from $\mathbb{R}$ to achieve this completeness?

A: The part that monotone, bounded sequences always converge in $\mathbb{R}$ (not in $\mathbb{Q}$); see page 39. Recall that the proof used the existence of $\sup\{x_n \mid n \in \mathbb{N}\}$ and $\inf\{x_n \mid n \in \mathbb{N}\}$, which may not exist in $\mathbb{Q}$ (there, $\lim_{n\to\infty} x_n = \sup\{x_n \mid n \in \mathbb{N}\}$). Further, the Bolzano–Weierstrass theorem is true in $\mathbb{R}$, yet false in $\mathbb{Q}$. In other words, the lim sup and lim inf do not always exist in $\mathbb{Q}$.

Q: If, in $\mathbb{R}$, convergence $\iff$ Cauchy, then why bother defining Cauchy sequences?

A: Convergent sequences need a candidate for the limit. Yet for a Cauchy sequence we do not need to find one: we can still prove that it is Cauchy, and thus convergent.

10.2 Series

Definition

Given a real sequence $\{x_n\}$, the symbol

$$\sum_{n=1}^{\infty} x_n \qquad \text{or} \qquad \sum_{n} x_n$$

is the *series* associated to $\{x_n\}$. We say that $\sum x_n$ *converges* if the sequence

$$\left\{S_m = \sum_{n=1}^{m} x_n = x_1 + x_2 + \cdots + x_m\right\}_{m=1}^{\infty}$$

converges. Each entry $S_m = \sum_{n=1}^{m} x_n$ is called the *m-th partial sum* of $\{x_n\}$. If $S = \lim_{m\to\infty} S_m$, then we write $S = \sum x_n$ and treat $\sum_n x_n$ as a real number. Otherwise, i.e. if $\lim_{m\to\infty} S_m$ does not exist, then we say that this series $\sum_n x_n$ *diverges*.

Remark

The index of a series does not necessarily have to start from 1. We can have $\sum_{n=0}^{\infty} x_n$ or $\sum_{n=3}^{\infty} y_n$ (see the theorem on page 59).
Just like for bounded sequences, we do not really care about $x_1, x_2, \ldots, x_{M-1}$, but we look at what happens if we go far enough out: $x_M, x_{M+1}, \ldots$. This is simply because $x_1, \ldots, x_M$ must be finite.

Example

$\sum_{n=1}^{\infty} \frac{1}{(n+1)n}$ converges.

Proof

$$\begin{aligned} S_m = \sum_{n=1}^{m} \frac{1}{(n+1)n} &= \sum_{n=1}^{m} \left(\frac{1}{n} - \frac{1}{n+1}\right) = \sum_{n=1}^{m} \frac{1}{n} - \sum_{n=1}^{m} \frac{1}{n+1} \\ &= \left(1 + \frac{1}{2} + \cdots + \frac{1}{m}\right) - \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{m+1}\right) \\ &= 1 - \frac{1}{m+1}. \end{aligned}$$

Thus $\lim_{m\to\infty} S_m = 1$.

Example

$\sum_{n=1}^{\infty} (-1)^n$ diverges.

Proof

$S_m = (-1) + 1 + (-1) + \cdots + (-1)^m$. Thus

$$S_m = \begin{cases} -1, & \text{if } m \text{ is odd,} \\ 0, & \text{if } m \text{ is even.} \end{cases}$$

Then $\{S_m\} = -1, 0, -1, 0, -1, \ldots$. We can show that $\{S_m\}$ diverges simply using an ε–M proof.

Theorem (geometric series)

If $|r| < 1$, then $\sum_{n=0}^{\infty} r^n$ converges and

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Note that the index starts at 0.

Proof

We compute the partial sum by induction.
Claim. For all $m \in \mathbb{N}$,

$$\sum_{n=0}^{m} r^n = \frac{1 - r^{m+1}}{1-r}.$$

Base case. For $m = 1$,

$$\sum_{n=0}^{1} r^n = r + 1 = \frac{1 - r^2}{1 - r}.$$

Inductive step. Assume that $P(m)$ is true for m. Then

$$\sum_{n=0}^{m+1} r^n = r^{m+1} + \sum_{n=0}^{m} r^n = r^{m+1} + \frac{1 - r^{m+1}}{1 - r} = \frac{1 - r^{m+1} + r^{m+1} - r^{m+2}}{1 - r},$$

so

$$\sum_{n=0}^{m+1} r^n = \frac{1 - r^{m+2}}{1 - r}.$$

Thus $P(m+1)$ is true. By induction, $P(m)$ is true for all $m \in \mathbb{N}$.
From the theorem we proved on page 40, we can easily show that, for every $c \in \mathbb{R}$ with $|c| < 1$, $\lim_{m\to\infty} c^m = 0$; in particular, $r^m \to 0$. Then

$$\lim_{m\to\infty} S_m = \lim_{m\to\infty} \sum_{n=0}^{m} r^n = \frac{1 - r \cdot 0}{1 - r} = \frac{1}{1 - r}.$$

Remark

If $|r| \geq 1$, there are three cases: $r = -1$, $r = 1$, and $|r| > 1$. It can be proven separately in each case that $\lim_{m\to\infty} S_m$ diverges if $|r| \geq 1$.

Theorem

Let $\{x_n\}$ be a sequence and let $M \in \mathbb{N}$. Then

$$\sum_{n=0}^{\infty} x_n \text{ converges} \iff \sum_{n=M}^{\infty} x_n \text{ converges.}$$

Proof

The partial sums satisfy, for every $m \in \mathbb{N}$,

$$\sum_{n=0}^{m} x_n = \sum_{n=M}^{m} x_n + \sum_{n=0}^{M-1} x_n.$$

($\Longrightarrow$) Suppose $\{\sum_{n=0}^{m} x_n\}_m$ converges to L. Then $\{\sum_{n=M}^{m} x_n\}_m$ converges to $L - \sum_{n=0}^{M-1} x_n$.
($\Longleftarrow$) Suppose $\{\sum_{n=M}^{m} x_n\}_m$ converges to L. Then $\{\sum_{n=0}^{m} x_n\}_m$ converges to $L + \sum_{n=0}^{M-1} x_n$.

Definition

We say that the series $\sum_n x_n$ is *Cauchy* if $\{S_m = \sum_{n=1}^{m} x_n\}_{m=1}^{\infty}$, the sequence of partial sums, is a Cauchy sequence.

Theorem

The series $\sum_n x_n$ is Cauchy if and only if $\sum x_n$ is convergent.

Proof

This is direct from the theorem on page 56.

Theorem

$\sum x_n$ is Cauchy if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{N}$ such that, for all $m \geq M$ and all $\ell > m$,

$$\left| \sum_{n=m+1}^{\ell} x_n \right| < \varepsilon.$$

Proof

($\Longrightarrow$) Since $\sum x_n$ is Cauchy, $\{S_m\}_{m=1}^{\infty}$ is Cauchy. Let $\varepsilon > 0$. Then there exists $M_0 \in \mathbb{N}$ such that, for all $m \geq M_0$ and all $\ell \geq M_0$, $|S_\ell - S_m| < \varepsilon$. Choosing $M = M_0$, for all $m \geq M$ and all $\ell > m \geq M$,

$$|S_\ell - S_m| = \left| \sum_{n=1}^{\ell} x_n - \sum_{n=1}^{m} x_n \right| = \left| \sum_{n=m+1}^{\ell} x_n \right| < \varepsilon.$$

($\Longleftarrow$) Let $\varepsilon > 0$. Choose $M_0 = M$, where M is such that, for all $m \geq M$ and all $\ell > m$, $\left|\sum_{n=m+1}^{\ell} x_n\right| < \varepsilon$. If $\ell = m$, then $|S_m - S_\ell| = 0 < \varepsilon$. If $\ell > m$, then, since for all $m \geq M$ and all $\ell > m$ we have $\left|\sum_{n=m+1}^{\ell} x_n\right| < \varepsilon$, it follows that $|S_\ell - S_m| < \varepsilon$. Without loss of generality, the case $\ell < m$ is the same as the case $\ell > m$.

Theorem

If $\sum x_n$ converges, then $\lim_{n\to\infty} x_n = 0$.

Remark: the n-th term test

The negative converse (contrapositive) of this statement gives the *n-th term test*: if $\lim_{n\to\infty} x_n \neq 0$, then the series $\sum x_n$ diverges.

Proof

Suppose $\sum x_n$ converges. Then $\sum x_n$ is Cauchy. Let $\varepsilon > 0$. Then there exists $M_0 \in \mathbb{N}$ such that, for all $\ell > m \geq M_0$,

$$\left| \sum_{n=m+1}^{\ell} x_n \right| < \varepsilon.$$

Choose $M = M_0 + 1$. Then, for all $m \geq M$, we have $m - 1 \geq M_0$ and $m > m - 1$, so

$$\left| \sum_{n=(m-1)+1}^{m} x_n \right| = \left| \sum_{n=m}^{m} x_n \right| = |x_m| = |x_m - 0| < \varepsilon.$$

Thus, for every $\varepsilon > 0$ there exists $M = M_0 + 1 \in \mathbb{N}$ such that $|x_m| < \varepsilon$ for all $m \geq M$, which means $\lim_{n\to\infty} x_n = 0$.

Theorem

If $|r| \geq 1$, then $\sum_{n=0}^{\infty} r^n$ diverges.

Proof

Again, by part 1) of the theorem on page 40, for every $r \in \mathbb{R}$ with $|r| > 1$, $\lim_{n\to\infty} r^n$ does not exist. Thus $\sum_{n=0}^{\infty} r^n$ diverges by the previous theorem.
If $r = 1$, then $\{r^n\} = 1, 1, 1, 1, \ldots$, and $\lim_{n\to\infty} r^n = 1 \neq 0$. Thus, by the previous theorem, $\sum_{n=0}^{\infty} r^n$ diverges.
If $r = -1$, then $\{r^n\} = \{(-1)^n\}$. This sequence diverges. Thus $r^n \not\to 0$, and the series $\sum x_n$ diverges.

Theorem (corollary)

For $\alpha \in \mathbb{R}$,

$$\sum_{n=0}^{\infty} \alpha r^n \text{ converges} \quad \iff \quad |r| < 1.$$

11 Absolute Convergence and the Comparison Test

11.1 The harmonic series and the converse of the nth-term test

Recall: series

If $\sum x_n$ converges, then $\lim_{n\to\infty} x_n = 0$.

Question

Does the converse hold? That is, does

$$\lim_{n\to\infty} x_n = 0 \quad \stackrel{?}{\Longrightarrow} \quad \sum x_n \text{ converges}$$

hold? The answer is no. A counterexample is the harmonic series.

Theorem

The series $\sum_{n=1}^{\infty} \frac{1}{n}$ does not converge.

Remark

The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is called the *harmonic series*.

Proof

Idea. We will show that there exists a subsequence of $\{S_m\}_{m=1}^{\infty}$, the sequence of partial sums, which diverges. Since a convergent sequence has the property that all of its subsequences converge, the existence of a divergent subsequence shows that $\{S_m\}$ diverges. In fact, we are going to prove a stronger statement: this subsequence will be unbounded, which implies that $\{S_m\}_{m=1}^{\infty}$ is unbounded. Since convergence implies boundedness, unboundedness implies

divergence. Thus an unbounded subsequence implies that $\{S_m\}_{m=1}^{\infty}$ diverges, which implies that $\sum_{n=1}^{\infty} x_n$ does not converge.
We want to show that a subsequence

$$S_{m_k} = \sum_{n=1}^{m_k} \frac{1}{n}$$

is unbounded. Consider, for $\ell \in \mathbb{N}$, the following subsequence:

$$S_{2^\ell} = \sum_{n=1}^{2^\ell} \frac{1}{n}.$$

Then

$$\begin{aligned} S_{2^\ell} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{\ell-1}+1} + \cdots + \frac{1}{2^\ell}\right) \\ &= 1 + \sum_{n=2^0+1}^{2^1} \frac{1}{n} + \sum_{n=2^1+1}^{2^2} \frac{1}{n} + \sum_{n=2^2+1}^{2^3} \frac{1}{n} + \cdots + \sum_{n=2^{\ell-1}+1}^{2^\ell} \frac{1}{n} \\ &= 1 + \sum_{i=1}^{\ell} \sum_{n=2^{i-1}+1}^{2^i} \frac{1}{n}. \end{aligned}$$

Since, for every n such that $2^{i-1}+1 \le n \le 2^i$, we have $n \le 2^i$, and hence $\frac{1}{n} \ge \frac{1}{2^i}$, it follows that

$$\sum_{n=2^{i-1}+1}^{2^i} \frac{1}{n} \ge \sum_{n=2^{i-1}+1}^{2^i} \frac{1}{2^i}.$$

Since the summand $\frac{1}{2^i}$ no longer depends on the index n,

$$\sum_{n=2^{i-1}+1}^{2^i} \frac{1}{2^i} = \frac{1}{2^i} \sum_{n=2^{i-1}+1}^{2^i} 1 = \frac{1}{2^i}(2^i - (2^{i-1}+1) + 1) = \frac{1}{2^i}\, 2^{i-1} = \frac{1}{2},$$

where we are counting the indices from $2^{i-1}+1$ to 2^i. Thus

$$S_{2^\ell} = 1 + \sum_{i=1}^{\ell} \sum_{n=2^{i-1}+1}^{2^i} \frac{1}{n} \ge \sum_{i=1}^{\ell} \frac{1}{2} = \frac{\ell}{2}.$$

Let $B > 0$. Choose ℓ such that $\frac{\ell}{2} > B$; such an ℓ exists by the Archimedean property. Then $S_{2^\ell} \ge \frac{\ell}{2} > B$. Since this holds for every $B > 0$, the subsequence $\{S_{2^\ell}\}_{\ell=1}^{\infty}$ is unbounded. Therefore $\{S_m\}_{m=1}^{\infty}$ diverges.

Remark

Although $\mathbb{R}$ does not satisfy the converse, there are some fields in which

$$\lim_{n\to\infty} x_n = 0 \quad \Longrightarrow \quad \sum_{n=1}^{\infty} x_n \text{ converges.}$$

An example is the p-adic numbers.

11.2 Linearity and series with non-negative terms

Theorem: linearity of convergent series

Let $\alpha \in \mathbb{R}$, and let $\sum x_n$ and $\sum y_n$ be convergent series. Then $\sum_n (\alpha x_n + y_n)$ converges, and

$$\sum (\alpha x_n + y_n) = \alpha \sum x_n + \sum y_n.$$

Proof

Since the partial sums satisfy

$$\sum_{n=1}^{m} (\alpha x_n + y_n) = \alpha \sum_{n=1}^{m} x_n + \sum_{n=1}^{m} y_n,$$

we have

$$\lim_{m\to\infty} \sum_{n=1}^{m} (\alpha x_n + y_n) = \lim_{m\to\infty} \Big(\alpha \sum_{n=1}^{m} x_n + \sum_{n=1}^{m} y_n \Big).$$

By the linear properties of limits, since $\{\sum_{n=1}^{m} x_n\}_{m=1}^{\infty}$ and $\{\sum_{n=1}^{m} y_n\}_{m=1}^{\infty}$ converge,

$$\begin{aligned} \lim_{m\to\infty} \sum_{n=1}^{m} (\alpha x_n + y_n) &= \alpha \lim_{m\to\infty} \sum_{n=1}^{m} x_n + \lim_{m\to\infty} \sum_{n=1}^{m} y_n \\ &= \alpha \sum_{n=1}^{\infty} x_n + \sum_{n=1}^{\infty} y_n. \end{aligned}$$

Remark

Series with non-negative terms are easier to work with. This is because

$$\sum_{n=1}^{m+1} x_n = \sum_{n=1}^{m} x_n + x_{m+1},$$

so, since $x_{m+1} \geq 0$,

$$\sum_{n=1}^{m+1} x_n \;\geq\; \sum_{n=1}^{m} x_n.$$

In other words, the sequence of partial sums $\{S_m\}_{m=1}^{\infty}$ is monotone increasing.

Theorem: series with non-negative terms

If $x_n \geq 0$ for every $n \in \mathbb{N}$, then

$$\sum x_n \text{ converges} \quad \Longleftrightarrow \quad \{S_m\} \text{ is bounded.}$$

Proof

Suppose $x_n \geq 0$ for every $n \in \mathbb{N}$. Then, for every $m \in \mathbb{N}$,

$$S_{m+1} = \sum_{n=1}^{m+1} x_n = x_{m+1} + \sum_{n=1}^{m} x_n \;\geq\; \sum_{n=1}^{m} x_n = S_m.$$

Thus $\{S_m\}_{m=1}^{\infty}$ is monotone increasing. Therefore, by the theorem on page 39,

$$\sum_{n=1}^{\infty} x_n \text{ converges} \quad \Longleftrightarrow \quad \{S_m\}_{m=1}^{\infty} \text{ converges} \quad \Longleftrightarrow \quad \{S_m\} \text{ is bounded},$$

where the last equivalence uses that $\{S_m\}$ is monotone increasing.

11.3 Absolute convergence

Definition

The series $\sum x_n$ *converges absolutely* if $\sum |x_n|$ converges.

Theorem

If $\sum x_n$ converges absolutely, then $\sum x_n$ converges.

Lemma

If $m \geq 2$ and $x_1, x_2, \ldots, x_m \in \mathbb{R}$, then

$$\Big|\sum_{n=1}^{m} x_n\Big| \leq \sum_{n=1}^{m} |x_n|.$$

Proof (by induction)

Base case $m = 2$: $|x_1 + x_2| \leq |x_1| + |x_2|$ by the triangle inequality.
Inductive step. Assume the statement is true for m, that is,

$$\Big|\sum_{n=1}^{m} x_n\Big| \leq \sum_{n=1}^{m} |x_n|.$$

Then

$$\Big|\sum_{n=1}^{m+1} x_n\Big| = \Big|\sum_{n=1}^{m} x_n + x_{m+1}\Big| \leq \Big|\sum_{n=1}^{m} x_n\Big| + |x_{m+1}| \leq \sum_{n=1}^{m+1} |x_n|.$$

So the statement is true for $m+1$. By induction, $\left|\sum_{n=1}^{m} x_n\right| \leq \sum_{n=1}^{m} |x_n|$ for every $m \in \mathbb{N}$.

Proof of the theorem

We want to show that $\sum x_n$ is Cauchy; then, by the theorem on page 59, $\sum x_n$ converges. Let $\varepsilon > 0$. Since $\sum |x_n|$ converges, $\sum |x_n|$ is Cauchy by the theorem on page 59. Hence there exists

$M_0 \in \mathbb{N}$ such that, for every $m \geq M_0$ and every $\ell > m$,

$$\Big| \sum_{n=m+1}^{\ell} |x_n| \Big| = \sum_{n=m+1}^{\ell} |x_n| < \varepsilon.$$

Choose $M = M_0$. Then, for all ℓ, m such that $\ell > m \geq M$,

$$\Big| \sum_{n=m+1}^{\ell} x_n \Big| \leq \sum_{n=m+1}^{\ell} |x_n| < \varepsilon.$$

Thus, by the theorem on page 59, $\sum x_n$ is Cauchy, and therefore $\sum x_n$ converges.

Remark: two kinds of series

There are essentially two cases of series:

1) absolutely convergent series;

2) conditionally convergent series.

Ways to determine convergence:

- the comparison test;
- the p-series test;
- the ratio test;
- the root test;
- the alternating series test;
- the nth-term test (for divergence).

These tests use that absolute convergence implies convergence.

Remark

The converse of the previous theorem does not hold: convergence does not imply absolute convergence.

Example: the alternating harmonic series

The alternating harmonic series satisfies

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\ln 2.$$

It is not absolutely convergent, since $\sum \frac{1}{n}$ (the harmonic series) diverges.

Remark: absolutely convergent series are stable

Absolutely convergent series are more "stable" than conditionally convergent ones: one can change the order of summing (rearrange the terms), and $\sum x_n$ converges to the same number, if $\sum x_n$ is absolutely convergent. This is false for conditionally convergent series!

Example: a rearrangement with a different sum

$$\sum_{n=1}^{\infty} \frac{-(-1)^n}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2.$$

Yet the rearrangement

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots = \frac{3}{2} \ln 2.$$

11.4 Convergence tests

Theorem (comparison test)

Suppose $0 \leq x_n \leq y_n$ for every $n \in \mathbb{N}$. Then

1) if $\sum y_n$ converges, then $\sum x_n$ converges;

2) if $\sum x_n$ diverges, then $\sum y_n$ diverges.

Proof

1) Suppose $\sum y_n$ converges, with $y_n \geq 0$ for every $n \in \mathbb{N}$. Then $\{\sum_{n=1}^{m} y_n\}_{m=1}^{\infty}$ is bounded, by the theorem on page 63. So there exists $B \geq 0$ such that, for every $m \in \mathbb{N}$,

$$\left|\sum_{n=1}^{m} y_n\right| \leq B.$$

Since $y_n \geq 0$, this says $\sum_{n=1}^{m} y_n \leq B$. Since $0 \leq x_n \leq y_n$, for every $m \in \mathbb{N}$,

$$\sum_{n=1}^{m} x_n \leq \sum_{n=1}^{m} y_n \leq B.$$

Thus $\{\sum_{n=1}^{m} x_n\}_{m=1}^{\infty}$ is bounded. Since $x_n \geq 0$ for every $n \in \mathbb{N}$, the boundedness of $\{\sum_{n=1}^{m} x_n\}_{m=1}^{\infty}$ implies that $\sum_{n=1}^{\infty} x_n$ converges, by the theorem on page 63.
2) Since $0 \leq x_n \leq y_n$ for every $n \in \mathbb{N}$ and $\sum x_n$ diverges, the negative converse (contrapositive) of the theorem on page 63 shows that $\{\sum_{n=1}^{m} x_n\}_{m=1}^{\infty}$ is unbounded. Then, for every $B \geq 0$, there exists $m \in \mathbb{N}$ such that

$$\sum_{n=1}^{m} x_n > B.$$

Since $y_n \geq x_n \geq 0$, we have $\sum_{n=1}^{m} x_n \leq \sum_{n=1}^{m} y_n$. Hence, for every $B \geq 0$, there exists $m \in \mathbb{N}$ such that

$$\sum_{n=1}^{m} y_n > B.$$

Then $\{\sum_{n=1}^{m} y_n\}_{m=1}^{\infty}$ is unbounded. Since convergence implies boundedness, unboundedness implies divergence. Therefore $\sum_{n=1}^{\infty} y_n$ diverges.

Theorem (p-series test)

For $p \in \mathbb{R}$, the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

Proof

($\Longrightarrow$) Suppose $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges. Assume $p \leq 1$. Then $n^p \leq n$, so $\frac{1}{n^p} \geq \frac{1}{n}$. Since $\frac{1}{n^p} \geq \frac{1}{n} > 0$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, the comparison test shows that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges, a contradiction. Thus the assumption $p \leq 1$ is false, and $p > 1$.
($\Longleftarrow$) Suppose $p > 1$.
We want to show that the sequence of partial sums is bounded. Then, since $\frac{1}{n^p} \geq 0$, it follows that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges. We will first show that there exists a subsequence of $\{S_m\}$ that is bounded. This is similar to how we showed that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.
First claim: the subsequence $\{S_{2^k}\}$ is bounded. More precisely, for every $k \in \mathbb{N}$,

$$S_{2^k} = \sum_{n=1}^{2^k} \frac{1}{n^p} \leq 1 + \frac{1}{1 - 2^{-p+1}}.$$

Indeed,

$$\begin{aligned} S_{2^k} &= 1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \cdots + \frac{1}{(2^k)^p} \\ &= 1 + \left(\frac{1}{2^p}\right) + \left(\frac{1}{3^p} + \frac{1}{4^p}\right) + \cdots + \left[\frac{1}{(2^{k-1}+1)^p} + \cdots + \frac{1}{(2^k-1)^p} + \frac{1}{(2^k)^p}\right] \\ &= 1 + \sum_{i=1}^{k} \sum_{n=2^{i-1}+1}^{2^i} \frac{1}{n^p}. \end{aligned}$$

Since, for every n such that $2^{i-1} + 1 \leq n \leq 2^i$, we have $n \geq 2^{i-1} + 1$, it follows that

$$\frac{1}{n^p} \leq \frac{1}{(2^{i-1}+1)^p} \qquad \text{for every such } n.$$

Since $2^{i-1} + 1 > 2^{i-1}$, we have $\dfrac{1}{(2^{i-1}+1)^p} < \dfrac{1}{(2^{i-1})^p}$. Then

$$\begin{aligned} S_{2^k} = 1 + \sum_{i=1}^{k} \sum_{n=2^{i-1}+1}^{2^i} \frac{1}{n^p} &\leq 1 + \sum_{i=1}^{k} \sum_{n=2^{i-1}+1}^{2^i} \frac{1}{(2^{i-1})^p} \\ &= 1 + \sum_{i=1}^{k} \frac{2^{i-1}}{(2^{i-1})^p} \\ &= 1 + \sum_{i=1}^{k} 2^{(i-1)(1-p)} \end{aligned}$$

$$= 1 + \sum_{i=0}^{k-1} (2^{1-p})^i \qquad \text{(shift of index)}$$
$$\leq 1 + \sum_{i=0}^{\infty} (2^{1-p})^i.$$

Since $p > 1$, we have $p - 1 > 0$, so $2^{p-1} > 1^{p-1} = 1$. Thus

$$\frac{1}{2^{p-1}} = 2^{1-p} < 1.$$

Then $\sum_{i=0}^{\infty} (2^{1-p})^i$ is a convergent geometric series, because $|r| < 1$ (see the theorem on page 57). Thus

$$S_{2^k} \leq 1 + \frac{1}{1-r} = 1 + \frac{1}{1-2^{1-p}},$$

so $\{S_{2^k}\}_{k=1}^{\infty}$ is bounded.
Second claim: $\{S_m\}$ is bounded. Let $m \in \mathbb{N}$. By induction we can prove that $2^m > m$. Then

$$S_m = \sum_{n=1}^{m} \frac{1}{n^p} \leq \sum_{n=1}^{2^m} \frac{1}{n^p} \leq 1 + \frac{1}{1-2^{1-p}}.$$

Here $2^m > m$ implies that S_{2^m} has more positive terms than S_m, so $S_{2^m} \geq S_m$. Thus $\{S_m\}_{m=1}^{\infty}$ is bounded. Since $\frac{1}{n^p} \geq 0$ for every $n \in \mathbb{N}$, the theorem on page 63 shows that $\{S_m\}_{m=1}^{\infty}$ converges. Hence $\sum \frac{1}{n^p}$ converges when $p > 1$.

12 The Ratio, Root, and Alternating Series Tests

12.1 Using the comparison and p-series tests

Recall: comparison test

If $0 \leq x_n \leq y_n$, then

1) $\sum y_n$ converges $\implies \sum x_n$ converges;

2) $\sum x_n$ diverges $\implies \sum y_n$ diverges.

Recall: p-series test

For every $p \in \mathbb{R}$, $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

Example

Consider $\sum_{n=1}^{\infty} \frac{1}{n^2+2020n}$. We have

$$0 \leq \frac{1}{n^2+2020n} \leq \frac{1}{n^2}.$$

By the p-series test, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. Then, by the comparison test, $\sum_{n=1}^{\infty} \frac{1}{n^2+2020n}$ converges.

Note: the comparison test only goes one way

Although the following is true,

$$\frac{1}{n^2+2020n} \leq \frac{1}{2020n} \leq \frac{1}{n} \qquad \text{for every } n \in \mathbb{N},$$

we can *not* conclude that

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges} \quad \Longrightarrow \quad \sum_{n=1}^{\infty} \frac{1}{n^2+2020n} \text{ diverges.}$$

This is NOT RIGHT, because the comparison test states that if $0 \leq x_n \leq y_n$, then $\sum x_n$ diverges implies $\sum y_n$ diverges; here $x_n = \frac{1}{n^2+2020n}$ plays the role of the smaller term and $y_n = \frac{1}{n}$ the larger one.

12.2 The ratio test

Theorem (ratio test)

Suppose $x_n \neq 0$ for every $n \in \mathbb{N}$, and that the limit

$$L = \lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|}$$

exists. Then

1) $L > 1 \Longrightarrow \sum_{n=1}^{\infty} x_n$ diverges;

2) $L < 1 \Longrightarrow \sum_{n=1}^{\infty} x_n$ converges absolutely.

Remark

There is nothing we can conclude if $L = 1$.

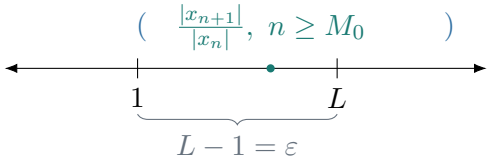
Example: $L = 1$ gives no information

Consider $\{x_n = \frac{1}{n}\}_{n=1}^{\infty}$. Then $L = 1$, but $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. Consider $\{x_n = \frac{1}{n^2}\}_{n=1}^{\infty}$. Then $L = 1$, but $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Proof

1) Suppose $L > 1$. Since $\frac{|x_{n+1}|}{|x_n|} \to L$, there exists $M_0 \in \mathbb{N}$ such that, for every $n \geq M_0$,

$$\left|\frac{|x_{n+1}|}{|x_n|} - L\right| < L - 1 \qquad \text{(choosing } \varepsilon = L - 1\text{)}.$$



Thus, for every $n \geq M_0$,

$$\frac{|x_{n+1}|}{|x_n|} > L - \varepsilon = L - (L - 1) = 1.$$

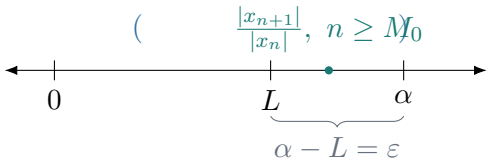
Then, for every $n \geq M_0$, $|x_{n+1}| > |x_n|$; that is,

$$|x_{M_0}| < |x_{M_0+1}| < \cdots,$$

so $\{|x_n|\}_{n=1}^{\infty}$ is monotone increasing (from M_0 on). Assume $|x_n| \to 0$. Then $|x_n| \leq 0$ (the sequence is bounded by 0). Since $|x_n| \geq 0$, this forces $|x_n| = 0$, a contradiction, since $|x_n| \neq 0$ for every n. Thus $|x_n| \not\to 0$, and the series diverges.

2) Suppose $L < 1$. Let $L < \alpha < 1$. Then, since $\frac{|x_{n+1}|}{|x_n|} \to L$ and $L < \alpha$, there exists $M_0 \in \mathbb{N}$ such that, for every $n \geq M_0$,

$$\left|\frac{|x_{n+1}|}{|x_n|} - L\right| < \alpha - L = \varepsilon \qquad (\alpha > L \implies \varepsilon > 0).$$



Then

$$\frac{|x_{n+1}|}{|x_n|} < L + (\alpha - L) = \alpha,$$

so, for every $n \geq M_0$, $|x_{n+1}| < \alpha|x_n|$. Then, for every $n \geq M_0$,

$$|x_n| < \alpha|x_{n-1}| < \alpha^2|x_{n-2}| < \cdots < \alpha^{n-M_0}|x_{M_0}|.$$

We want to show that $\{\sum_{n=1}^{m} |x_n|\}_{m=1}^{\infty}$ is bounded. Then, since $|x_n| \geq 0$, $\sum_{n=1}^{\infty} |x_n|$ converges, so $\sum x_n$ converges absolutely.

Let $m \in \mathbb{N}$. Then

$$\begin{aligned}
\sum_{n=1}^{m} |x_n| &= \sum_{n=1}^{M_0-1} |x_n| + \sum_{n=M_0}^{m} |x_n| \\
&\leq \sum_{n=1}^{M_0-1} |x_n| + |x_{M_0}| \sum_{n=M_0}^{m} \alpha^{n-M_0} \\
&\leq \sum_{n=1}^{M_0-1} |x_n| + |x_{M_0}| \sum_{n=0}^{\infty} \alpha^{n},
\end{aligned}$$

where in the last step we are adding more non-negative terms. Since $0 \le L < \alpha < 1$,

$$\sum_{n=1}^{m} |x_n| \le \sum_{n=1}^{M_0-1} |x_n| + |x_{M_0}| \frac{1}{1-\alpha}.$$

The first sum on the right has finitely many terms, and the right-hand side is a fixed number, independent of m. Since $|x_n| \ge 0$ and $\{\sum_{n=1}^{m} |x_n|\}_{m=1}^{\infty}$ is bounded, $\sum_{n=1}^{\infty} |x_n|$ converges, so $\sum_{n=1}^{\infty} x_n$ converges absolutely.

Example: the exponential series

For every $x \in \mathbb{R}$, $\sum \frac{x^n}{n!}$ converges absolutely (with $0! = 1$). By the ratio test,

$$L = \lim_{n\to\infty} \frac{\left|\frac{x^{n+1}}{(n+1)!}\right|}{\left|\frac{x^n}{n!}\right|} = \lim_{n\to\infty} \frac{|x|}{n+1} = |x| \lim_{n\to\infty} \frac{1}{n+1} = 0.$$

Thus $L < 1$, so $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges absolutely.

12.3 The root test

Theorem (root test)

Let $\sum_{n=1}^{\infty} x_n$ be a series and suppose the limit

$$L = \lim_{n\to\infty} |x_n|^{1/n}$$

exists. Then

1) $L < 1 \implies \sum x_n$ converges absolutely;

2) $L > 1 \implies \sum x_n$ diverges.

Remark

The case $L = 1$ gives no information.

Proof

1) Again, we will show that $\sum_{n=1}^{\infty} |x_n|$ is bounded by a convergent geometric series. Suppose $L < 1$, and let $L < \alpha < 1$. Since $|x_n|^{1/n} \to L$, there exists $M \in \mathbb{N}$ such that, for every $n \ge M$,

$$\left||x_n|^{1/n} - L\right| < \alpha - L \ (= \varepsilon).$$

Hence, for every $n \ge M$, $|x_n|^{1/n} < L + \alpha - L = \alpha$, and therefore $|x_n| < \alpha^n$ for every $n \ge M$. Then, for every $m \in \mathbb{N}$, the partial sum satisfies

$$\sum_{n=1}^{m} |x_n| = \sum_{n=1}^{M-1} |x_n| + \sum_{n=M}^{m} |x_n|$$

$$\leq \sum_{n=1}^{M-1} |x_n| + \sum_{n=M}^{m} \alpha^n$$
$$\leq \sum_{n=1}^{M-1} |x_n| + \sum_{n=0}^{\infty} \alpha^n = \sum_{n=1}^{M-1} |x_n| + \frac{1}{1-\alpha},$$

using $\alpha \in (0,1)$; the right-hand side is independent of m. Thus, for every $m \in \mathbb{N}$,

$$\sum_{n=1}^{m} |x_n| \leq \sum_{n=1}^{M-1} |x_n| + \frac{1}{1-\alpha}, \qquad M \in \mathbb{N},\ \alpha \in (L,1),$$

so $\{\sum_{n=1}^{m} |x_n|\}_{m=1}^{\infty}$ is bounded. Since $|x_n| \geq 0$ for every n, the theorem on page 63 shows that $\sum_{n=1}^{\infty} |x_n|$ converges; that is, $\sum_{n=1}^{\infty} x_n$ converges absolutely.
2) Suppose $L > 1$. Since $|x_n|^{1/n} \to L > 1$, there exists $M \in \mathbb{N}$ such that, for every $n \geq M$, $|x_n|^{1/n} > 1$. Thus, for every $n \geq M$, $|x_n| > 1^n = 1$. In other words, there exists $\varepsilon = 1$ such that, for every $M_0 \in \mathbb{N}$, there exists $n \geq M_0$ with $|x_n| > 1$. Therefore $x_n \not\to 0$, and $\sum_{n=1}^{\infty} x_n$ diverges.

12.4 The alternating series test

Theorem (alternating series test)

Let $\{x_n\}$ be a monotone decreasing sequence such that $x_n \to 0$ (therefore $x_n \geq 0$ for every $n \in \mathbb{N}$). Then the series

$$\sum_{n=1}^{\infty} (-1)^n x_n$$

converges.

Proof

Let $S_m = \sum_{n=1}^{m} (-1)^n x_n$.
We want to show: we will show that a subsequence of $\{S_m\}_m$ is monotone decreasing and bounded below. Then we will prove that $\{S_m\}_m$ converges to the same limit. (This is similar to the proof for p-series.)
Consider

$$S_{2k} = \sum_{n=1}^{2k} (-1)^n x_n = (x_2 - x_1) + (x_4 - x_3) + \cdots + (x_{2k} - x_{2k-1}).$$

Since $\{x_n\}$ is monotone decreasing, $x_{2k+2} - x_{2k+1} \leq 0$. Thus

$$S_{2k} \geq (x_2 - x_1) + \cdots + (x_{2k} - x_{2k-1}) + (x_{2k+2} - x_{2k+1}),$$

so $S_{2k} \geq S_{2(k+1)}$ for every $k \in \mathbb{N}$. Thus $\{S_{2k}\}_{k=1}^{\infty}$ is *monotone decreasing.*
Next we will show that it is bounded:

$$S_{2k} = -x_1 + (x_2 - x_3) + \cdots + (x_{2k-2} - x_{2k-1}) + x_{2k} \geq -x_1 + 0 = -x_1.$$

Thus $\{S_{2k}\}_{k=1}^{\infty}$ is bounded below by $-x_1$. Therefore $\{S_{2k}\}_{k=1}^{\infty}$ is a sequence that is bounded below and monotone decreasing, so $\{S_{2k}\}_k$ converges.

Let $s = \lim_{k\to\infty} S_{2k}$. Now we want to show that $S_m \to s$.
Let $\varepsilon > 0$. Since $S_{2k} \to s$, there exists $M_0 \in \mathbb{N}$ such that, for all $k \geq M_0$,

$$|S_{2k} - s| < \frac{\varepsilon}{2}.$$

Since $x_n \to 0$, there exists $M_1 \in \mathbb{N}$ such that, for every $n \geq M_1$,

$$|x_n - 0| = |x_n| < \frac{\varepsilon}{2}.$$

Now choose $M = \max\{2M_0 + 1,\ M_1\}$, and let $m \geq M$.
Case 1: m is even. Then $\frac{m}{2} \in \mathbb{N}$, and $\frac{m}{2} \geq M_0 + \frac{1}{2} > M_0$. Thus

$$|S_{2\cdot\frac{m}{2}} - s| = |S_m - s| < \frac{\varepsilon}{2} < \varepsilon.$$

Case 2: m is odd. Then $\frac{m-1}{2} \in \mathbb{N}$. Since $m \geq 2M_0 + 1$, we have $\frac{m-1}{2} \geq M_0$, so

$$|S_{2\cdot\frac{m-1}{2}} - s| = |S_{m-1} - s| < \frac{\varepsilon}{2}.$$

Since $m \geq M$, we have $|x_m| < \frac{\varepsilon}{2}$. Then, because $S_m = S_{m-1} + (-1)^m x_m$,

$$|S_m - s| \leq |S_{m-1} - s| + |x_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $|S_m - s| < \varepsilon$ if m is odd.
Thus, for every $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that, for every $m \geq M$, $|S_m - s| < \varepsilon$. Hence $\{S_m\} \to s$, and $\sum_{n=1}^{\infty}(-1)^n x_n$ converges.

12.5 Rearrangements of absolutely convergent series

Theorem: rearranging an absolutely convergent series

Suppose $\sum_{n=1}^{\infty} x_n$ converges absolutely and $\sum_{n=1}^{\infty} x_n = x$. Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection. Then $\sum_{n=1}^{\infty} x_{\sigma(n)}$ is absolutely convergent, and

$$\sum_{n=1}^{\infty} x_{\sigma(n)} = x.$$

In other words, given a series that is absolutely convergent, we can rearrange the terms, and the new series is still absolutely convergent and converges to the same thing as the original series $\sum x_n$.

Proof

We will show that $\{\sum_{n=1}^{m} |x_{\sigma(n)}|\}_{m=1}^{\infty}$ is bounded, which gives convergence. Then we will show that $\sum_{n=1}^{\infty} x_{\sigma(n)}$ converges to the same limit $\sum_{n=1}^{\infty} x_n$.
Since $\sum_{n=1}^{\infty} |x_n|$ converges, $\{\sum_{n=1}^{m} |x_n|\}_{m=1}^{\infty}$ converges, so $\{\sum_{n=1}^{m} |x_n|\}_{m=1}^{\infty}$ is bounded. Hence

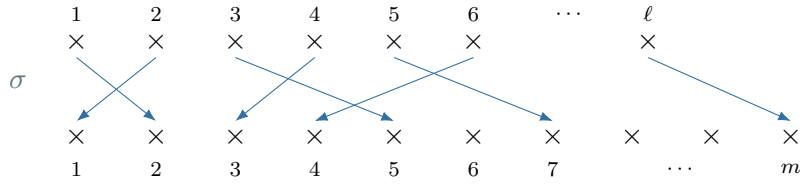
there exists $B \geq 0$ such that, for every $m \in \mathbb{N}$,

$$\sum_{n=1}^{m} |x_n| \leq B.$$

Let $\ell \in \mathbb{N}$. Then $\sigma(\{1, 2, \ldots, \ell\})$ is a *finite* subset of $\mathbb{N}$. Thus there exists $m \in \mathbb{N}$ such that

$$\sigma(\{1, 2, \ldots, \ell\}) \subset \{1, 2, 3, \ldots, m\}.$$

This is because there exists a supremum (a largest element) of the set $\sigma(\{1, \ldots, \ell\})$, which we can take to be m; note that m should be at least as big as ℓ.



Thus

$$\sum_{n=1}^{\ell} |x_{\sigma(n)}| = \sum_{n \in \sigma(\{1,\ldots,\ell\})} |x_n| \leq \sum_{n \in \{1,\ldots,m\}} |x_n| = \sum_{n=1}^{m} |x_n| \leq B.$$

Therefore $\left\{\sum_{n=1}^{m} |x_{\sigma(n)}|\right\}_{m=1}^{\infty}$ is bounded. Since $|x_{\sigma(n)}| \geq 0$, it follows that $\sum_{n=1}^{\infty} |x_{\sigma(n)}|$ converges, so $\sum_{n=1}^{\infty} x_{\sigma(n)}$ converges absolutely.
Now we want to show that

$$\sum_{n=1}^{m} x_{\sigma(n)} \longrightarrow \sum_{n=1}^{\infty} x_n \qquad \text{as } m \to \infty.$$

① Let $x = \sum_{n=1}^{\infty} x_n$. Let $\varepsilon > 0$. Then there exists $M_0 \in \mathbb{N}$ such that, for every $m \geq M_0$,

$$\left|\sum_{n=1}^{m} x_n - x\right| < \frac{\varepsilon}{2}.$$

② Since $\sum_{n=1}^{\infty} |x_n|$ converges, the theorem on page 59 shows that $\sum_{n=1}^{\infty} |x_n|$ is Cauchy, and hence, by the theorem on page 59, there exists $M_1 \in \mathbb{N}$ such that, for every $\ell > m \geq M_1$,

$$\left|\sum_{n=m+1}^{\ell} |x_n|\right| = \sum_{n=m+1}^{\ell} |x_n| < \frac{\varepsilon}{2}.$$

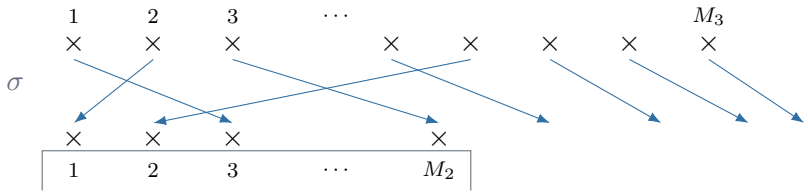
Let $M_2 = \max\{M_0, M_1\}$. Since $\{1, 2, \ldots, M_2\}$ is a finite subset of $\mathbb{N}$ and σ is a bijection, there exists $M_3 \in \mathbb{N}$ such that

$$\{1, 2, \ldots, M_2\} \subset \sigma(\{1, 2, 3, \ldots, M_3\}).$$

We want to show: there exists $M \in \mathbb{N}$ such that, for every $m \geq M$,

$$\left|\sum_{n=1}^{m} x_{\sigma(n)} - x\right| < \varepsilon.$$

Let $M = M_3$. This enables us to shift from $\sigma(n)$ to n.



Since $\{1,\dots,M_2\} \subset \sigma(\{1,\dots,M_3\})$, we get $M_3 \geq M_2$. Hence $M = M_3 \geq M_1$ and $M \geq M_0$.

Note: why we need M_3

This property of M_3 allows us to shift the index $\sigma(n)$ while preserving ① and ②, because $m \geq M_3$ implies $m \geq M_2$, which implies $m \geq M_0$ and $m \geq M_1$. We need M_3 because M_3 allows us to shift the index from $\sigma(n)$ back to n! This is because $\{1,\dots,M_2\} \subset \sigma(\{1,\dots,M_3\})$. We cannot use M_2 directly.

Thus, for every $m \geq M$,

$$\Big|\sum_{n=1}^{m} x_{\sigma(n)} - x\Big| = \Big|\sum_{n\in\sigma(\{1,\dots,m\})} x_n - x\Big|.$$

Since $m \geq M$ means $m \geq M_3$, we have $\sigma(\{1,\dots,M_3\}) \subset \sigma(\{1,\dots,m\})$, and therefore $\{1,\dots,M_2\} \subset \sigma(\{1,\dots,m\})$. Then

$$\begin{aligned}\sum_{n\in\sigma(\{1,\dots,m\})} x_n &= \sum_{n=1}^{M_2} x_n + \sum_{n\in\sigma(\{1,\dots,m\})\setminus\{1,2,\dots,M_2\}} x_n \\ &\leq \sum_{n=1}^{M_2} x_n + \sum_{n=M_2+1}^{\max\sigma(\{1,\dots,m\})\setminus\{1,2,\dots,M_2\}} x_n \\ &= \sum_{n=1}^{M_2} x_n + \sum_{n=M_2+1}^{\max\sigma(\{1,\dots,m\})} x_n.\end{aligned}$$

Since $M_2 \geq M_0$, and

$$\max\sigma(\{1,\dots,m\}) \;\geq\; m \;\geq\; M_3 \;\geq\; M_2 = \max\{M_0, M_1\} \;\geq\; M_1$$

(the first inequality is proved by contradiction: otherwise we would get $m < m$), we obtain, for every $m \geq M$,

$$\begin{aligned}\Big|\sum_{n=1}^{m} x_{\sigma(n)} - x\Big| &= \Big|\sum_{n\in\sigma(\{1,\dots,m\})} x_n - x\Big| \\ &\leq \Big|\sum_{n=1}^{M_2} x_n - x\Big| + \sum_{n=M_2+1}^{\max\sigma(\{1,\dots,m\})} |x_n| \qquad \text{(triangle inequality)} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,\end{aligned}$$

where the first term is estimated by ① and the second by ②. The existence of M_3, and the way we chose it, are given above.

Thus, for every $\varepsilon > 0$, there exists $M = M_3 \in \mathbb{N}$ such that, for every $m \geq M$,

$$\left|\sum_{n=1}^{m} x_{\sigma(n)} - x\right| < \varepsilon.$$

Hence $\sum_{n=1}^{\infty} x_{\sigma(n)} = x$.

Continuation

The counterpart of this theorem for conditionally convergent series, the Riemann series theorem, is stated and proved at the end of these notes, on page 164.

13 Introduction to Topology in Metric Spaces

About this section

This section is extra material: preparation for 18.101 and 18.102.

13.1 Countability theorems

Theorem

Every infinite subset of a countable set A is countable.

Proof

Suppose $E \subset A$ and E is infinite. Let $\{x_n\}$ be the elements of A. We arrange the elements of E in the following way.

- Let $n_1 \in \mathbb{N}$ be the least integer such that $x_{n_1} \in E$.
- Let $n_2 \in \mathbb{N}$ be the second least integer (that is, $n_2 > n_1$) such that $x_{n_2} \in E$.

Reiterating this process, we construct a sequence

$$\{x_{n_k}\}_{k=1}^{\infty}$$

in which n_k is the smallest $n_k \in \mathbb{N}$ such that $n_k > n_{k-1}$ and $x_{n_k} \in E$, for every $k \in \mathbb{N}$.
Recall that, by definition, a sequence is a function from $\mathbb{N}$ to $\mathbb{R}$ (see page 34). Thus let $f : \mathbb{N} \to E$ be given by $f(k) = x_{n_k}$. Apparently f is a bijection, by our way of choosing the n_k. Thus the subset E is countable.

Remark

We used the fact that A is countable when we listed the elements of A as a sequence $\{x_n\}_{n=1}^{\infty}$. If A were uncountable, we would not be able to do so.

Theorem

Let $\{E_n\}_{n=1}^{\infty}$ be a sequence of countable sets (thus each is infinite), and let

$$S = \bigcup_{n=1}^{\infty} E_n.$$

Then S is countable.

Proof

We rearrange the elements of the E_n into a sequence. Let every set E_n, $n \in \mathbb{N}$, be arranged in a sequence $\{x_{nk}\}$, $k = 1, 2, \ldots$:

$$\begin{array}{llllll} x_{11} & x_{12} & x_{13} & x_{14} & \cdots & \text{(elements in } E_1\text{)} \\ x_{21} & x_{22} & x_{23} & x_{24} & \cdots & \text{(elements in } E_2\text{)} \\ x_{31} & x_{32} & x_{33} & x_{34} & \cdots & \text{(elements in } E_3\text{)} \\ \vdots & \vdots & \vdots & \vdots & & \end{array}$$

Following the array, we arrange the $x_{nk} \in S$ in the order

$$x_{11},\ x_{21},\ x_{12},\ x_{31},\ x_{22},\ x_{13},\ x_{41},\ x_{32},\ x_{23},\ x_{14},\ \ldots$$

The subscripts make up an *infinite subset of* $\mathbb{N}$. Let it be J; that is,

$$J = \{11, 21, 12, 31, 22, 13, 41, \ldots\}.$$

Note. If there are elements in common between the E_n, then these will appear more than once in J. For example, $x_{12} = x_{21}$. Then we can simply remove one of these and obtain a subset $J' \subset J$.
Thus there exists a bijection

$$f : J \longrightarrow S = \bigcup_{n=1}^{\infty} E_n.$$

Since $J \subset \mathbb{N}$, by the previous theorem J is countable. Then, since $f : J \to S$ is a bijection, S is countable.

Theorem

Let A be a countable set, and let B_n be the set of all n-tuples $(a_1, \ldots, a_n)$, where $a_k \in A$ for $k = 1, \ldots, n$. The elements $a_1, \ldots, a_n$ need not be distinct. Then B_n is countable.

Proof (by induction)

Base case ($n = 1$): $B_1 = A$.
Inductive step: suppose B_{n-1} is countable. We now want to show that B_n is countable.

Consider the elements of B_n of the form (b, a), where $b \in B_{n-1}$ and $a \in A$. Let

$$A_b := \{(b, a) \mid a \in A\},$$

where $b \in B_{n-1}$ is given. Then $|A_b| = |A|$ (there is a bijection $f : A_b \to A$, so they have the same cardinality). Thus A_b is countable.
Therefore, we can write B_n as the union of all the A_b:

$$B_n = \bigcup_{b \in B_{n-1}} A_b.$$

Since B_{n-1} is countable, B_n is the union of a countable collection of countable sets (namely the sets A_b, $b \in B_{n-1}$). Thus, by the previous theorem, B_n is countable. Therefore, by induction, B_n is countable for every $n \in \mathbb{N}$.

Note: a hidden condition in the previous theorem

The union $\bigcup_{n=1}^{\infty} E_n$ in the previous theorem implies that we are taking a countable union, indexed by $n = 1, 2, \ldots$. Thus, if B_{n-1} were uncountable, we would not be able to use the previous theorem any more.

Remark

This theorem gives us another way to prove that $|\mathbb{Q}| = |\mathbb{N}|$, that is, that the rationals are countable. All rationals can be fixed by a pair of coprime integers (a, b) such that $a \in \mathbb{Z}$, $b \in \mathbb{N}$, and $r = a/b$. Then $|\mathbb{Z}| = |\mathbb{N}|$ and $|\mathbb{N}| = |\mathbb{N}|$, so $\mathbb{Q}$ is countable by this theorem.

13.2 Metric spaces

Definition

A set X, whose elements can be called *points*, is said to be a *metric space* if for all $p, q \in X$ there is an associated real number, denoted by $d(p, q) \in \mathbb{R}_{\geq 0}$ and called the *distance* between p and q, such that for all $p, q, r \in X$:

1) $d(p, q) \geq 0$, and $d(p, q) = 0 \iff p = q$;
2) $d(p, q) = d(q, p)$;
3) $d(p, q) \leq d(p, r) + d(r, q)$.

Example

One can verify that the distance in the Euclidean space $\mathbb{R}^k$,

$$d(\alpha, \beta) = |\alpha - \beta| \qquad \forall \alpha, \beta \in \mathbb{R}^k,$$

is a metric on $\mathbb{R}^k$.

Definition

Let X be a metric space. Let $p, q \in X$, $r \in \mathbb{R}_{>0}$, and $E \subset X$.

1) A *neighbourhood* of p is the set

$$N_r(p) := \{q \in X \mid d(p,q) < r\}.$$

Here r is the *radius* of $N_r(p)$.

2) The point p is a *limit point*, or *cluster point*, of E if $p \in X$ is such that, for every $r \in \mathbb{R}_{>0}$,

$$N_r(p) \cap E \setminus \{p\} \neq \emptyset.$$

In other words, every neighbourhood of p contains a point $q \neq p$ such that $q \in E$.

3) p is an *isolated point* of the set E if $p \in E$ and p is not a limit point of E.

4) E is *closed* if: p is a limit point of $E \implies p \in E$. That is, every limit point of E is a point of E.

5) p is an *interior point* of E if there exists a neighbourhood N of p such that $N \subset E$.

6) E is *open* if: $p \in E \implies p$ is an interior point. That is, every point of E is an interior point of E.

7) The *complement* of E, denoted by E^c, is the set of all q such that $q \in X \setminus E$.

8) E is *bounded* if there exist $M \in \mathbb{R}$ and $q \in X$ such that

$$d(p,q) < M \qquad \forall p \in E.$$

9) E is *dense in* X if every point of X is a limit point of E, or a point of E, or both.

Remark

A set can be both open and closed.

Example: density of $\mathbb{Q}$ in $\mathbb{R}$

We will see that the theorem on page 25 does agree with statement 9). One can take $y = x + r$, where r is the radius of the neighbourhood. Then every neighbourhood of x contains a rational. Thus every real number is a limit point of $\mathbb{Q}$.

Theorem

A set E is open if and only if E^c (its complement) is closed.

Proof

($\Longleftarrow$) Suppose E^c is closed. Let $x \in E$. Then

$$\implies x \notin E^c \implies x \text{ is not a limit point of } E^c$$
$$\implies \exists r \in \mathbb{R}_{>0} \text{ such that } E^c \setminus \{x\} \cap N_r(x) = E^c \cap N_r(x) = \emptyset$$

$$\implies \exists r \in \mathbb{R}_{>0} \text{ such that } \forall y \in N_r(x),\ y \notin E^c \implies y \in E.$$

Thus there exists $r \in \mathbb{R}_{>0}$ such that $y \in E$ for every $y \in N_r(x)$, so

$$N_r(x) \subset E \quad \implies \quad x \text{ is an interior point of } E,\ \forall x \in E \quad \implies \quad E \text{ is open.}$$

($\implies$) Suppose E is open. We want to show: x is a limit point of $E^c \implies x \in E^c$.
Let x be a limit point of E^c. Then, for every $r > 0$, there exists $y \in X$ such that $y \in E^c \setminus \{x\} \cap N_r(x)$. Thus, in every neighbourhood, there exists $y \in N_r(x)$ such that $y \in E^c$, so $y \notin E$. Hence

$$N_r(x) \not\subset E \qquad \forall r > 0,$$

and x cannot be an interior point of E. Since E is open, $x \notin E$, so $x \in E^c$. Thus E^c is closed.

Corollary

A set F is closed if and only if F^c is open.

Proof

Set $E^c = F$. The theorem is then proved by the previous one.

Theorem

1) For any collection $\{G_\alpha\}$ of open sets, their union $\bigcup_\alpha G_\alpha$ is open.

2) For any collection $\{F_\alpha\}$ of closed sets, their intersection $\bigcap_\alpha F_\alpha$ is closed.

3) For any finite collection $G_1, \ldots, G_n$ of open sets, their intersection $\bigcap_{i=1}^n G_i$ is open.

4) For any finite collection $F_1, \ldots, F_n$ of closed sets, their union $\bigcup_{i=1}^n F_i$ is closed.

Proof

1) Let $G = \bigcup_\alpha G_\alpha$. We want to show that x is an interior point of G. If $x \in G$, then $x \in G_\alpha$ for some α. Since G_α is open and $x \in G_\alpha$, x is an interior point of G_α: there exists $r > 0$ such that

$$N_r(x) \subset G_\alpha \quad \implies \quad N_r(x) \subset \bigcup_\alpha G_\alpha.$$

Thus x is also an interior point of $\bigcup_\alpha G_\alpha$, so $\bigcup_\alpha G_\alpha$ is open.
2) We want to show that $\bigcap_\alpha F_\alpha$ is closed. We can easily prove that

$$\left(\bigcap_\alpha F_\alpha\right)^c = \bigcup_\alpha (F_\alpha^c).$$

Then F_α closed implies F_α^c open, and by 1) $\bigcup_\alpha (F_\alpha^c)$ is open. Thus its complement is closed:

$$\left(\bigcup_\alpha (F_\alpha^c)\right)^c = \left(\left(\bigcap_\alpha F_\alpha\right)^c\right)^c = \bigcap_\alpha F_\alpha \quad \text{is closed.}$$

3) Let $H = \bigcap_{i=1}^n G_i$. Since each G_i is open, for every $x \in H$ there exists $r_i > 0$ such that

$$N_{r_i}(x) \subset G_i \qquad \forall i \in \{1, \ldots, n\}.$$

Thus choose $r = \min\{r_1, r_2, \ldots, r_n\}$. Then

$$\begin{aligned} & N_r(x) \subset N_{r_i}(x) \qquad \forall i \in \{1, \ldots, n\} \\ \implies \quad & N_r(x) \subset G_i \qquad \forall i \in \{1, \ldots, n\} \\ \implies \quad & N_r(x) \subset H = \bigcap_{i=1}^{n} G_i. \end{aligned}$$

Thus x is an interior point of H, so $H = \bigcap_{i=1}^{n} G_i$ is open.
4) Again, 4) follows from 3) by using

$$\Big(\bigcup_{i=1}^{n} F_i\Big)^c = \bigcap_{i=1}^{n} (F_i^c).$$

13.3 Compact sets

Definition

A collection $\{G_\alpha\}$ of open subsets of X is an *open cover* of a set $E \subset X$ in a metric space X if

$$E \subset \bigcup_{\alpha \in A} G_\alpha.$$

Here A is an *index set*. A collection $\{G_\beta\}$ is said to be a *subcover* of $\{G_\alpha \mid \alpha \in A\}$ if $\{G_\beta \mid \beta \in B\}$ is an open cover of E and the index set satisfies $B \subset A$. In other words,

$$\{G_\beta \mid \beta \in B\} \subset \{G_\alpha \mid \alpha \in A\}.$$

Definition

Let X be a metric space. A subset K of X is *compact* if every open cover $\{G_\alpha\}$ of K has a *finite subcover*; that is, there exists a finite $B \subset A$, $B = \{\beta_1, \ldots, \beta_n\}$, such that

$$K \subset \bigcup_{i=1}^{n} G_{\beta_i}.$$

Note that "finite" does not mean that G_{β_i} contains finitely many elements of X; rather, the subcover $\{G_{\beta_i} \mid i = 1, \ldots, n\}$, with $B = \{\beta_1, \ldots, \beta_n\} \subset A$, contains finitely many members (indicated by n), each of which is an open subset of X.

Remark

There is an alternative definition of "compactness".

Definition

A subset $A \subset \mathbb{R}^n$ is *(sequentially) compact* if for every sequence $\{x_n\}_{n=1}^{\infty}$ in A there exists a subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ such that

$$\lim_{k \to \infty} x_{n_k} = x \in A.$$

That is, every sequence has a subsequence converging in A.

Note

Sequentially compact $\iff$ compact, if A is a metric space.

Example

$[a, b] \subset \mathbb{R}$ is compact for all $a, b \in \mathbb{R}$ with $a \leq b$.

Non-example

The set $E = \{r \in \mathbb{Q} \mid r > 0 \text{ and } r^2 < 2\}$ is not compact.

Negation of the definition

A set $K \subset X$ is *not* compact if either

1) there exists an open cover $\{G_\alpha \mid \alpha \in A\}$ of K such that, for every subcover $\{G_\beta \mid \beta \in B\}$, the index set B is an infinite subset of A; or

2) there exists $\{x_n\}_{n=1}^{\infty}$ in K such that, for every subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ of $\{x_n\}_{n=1}^{\infty}$, either $\lim_{k\to\infty} x_{n_k} = x \notin K$ or $\{x_{n_k}\}_k$ diverges.

Proof of the non-example, using 1)

Claim. The collection $\{S_n\}$, $n \in \mathbb{N}$, where

$$S_n = \left\{x \in X \;\middle|\; \frac{1}{n} < x < \sqrt{2} - \frac{1}{n}\right\}, \qquad X = \mathbb{Q}$$

($X = \mathbb{R}$ also works), is an open cover of the set E, and every subcover $\{S_{n_k}\}$, $k \in \mathbb{N}$, of $\{S_n\}$ has to be infinite.
First, we want to show that

$$E = \{r \in \mathbb{Q} \mid r > 0 \text{ and } r^2 < 2\} \subset S_1 \cup S_2 \cup \cdots = \bigcup_{n=1}^{\infty} S_n.$$

Let $r \in E$. Thus $r \in \mathbb{Q}_{>0}$ and $r^2 < 2$. Therefore

$$(\sqrt{2} - r)(\sqrt{2} + r) > 0 \quad \Longrightarrow \quad \sqrt{2} - r > 0.$$

Let $\sqrt{2} - r = p \in \mathbb{R}$. By the Archimedean property, there exists $n_1 \in \mathbb{N}$ such that $n_1 p > 1$. Then

$$\sqrt{2} - r = p > \frac{1}{n_1} \quad \Longrightarrow \quad r < \sqrt{2} - \frac{1}{n_1}.$$

By the Archimedean property, there exists $n_2 \in \mathbb{N}$ such that $n_2 r > 1$, so $r > 1/n_2$. Choose $n = \max\{n_1, n_2\}$. Thus

$$\frac{1}{n} \leq \frac{1}{n_2} < r < \sqrt{2} - \frac{1}{n_1} \leq \sqrt{2} - \frac{1}{n} \quad \Longrightarrow \quad r \in S_n.$$

Therefore, for every $r \in E$ there exists $n \in \mathbb{N}$ such that $r \in S_n$, so $E \subset \bigcup_{n=1}^{\infty} S_n$.

Second, we want to show that S_n is open for every $n \in \mathbb{N}$. This is trivial to prove. Then $\{S_n\}$, $n \in \mathbb{N}$, is an open cover of the set E.
Third, we want to show that every subcover of $\{S_n\}$ has to be infinite. We argue by contradiction. Suppose there exists a subcover $\{S_{n_1}, S_{n_2}, \ldots, S_{n_k}\}$ which is finite.
Idea: we will show that there exists $r \in E$ such that $r \notin \bigcup_{i=1}^{k} S_{n_i}$. Then $\{S_{n_1}, \ldots, S_{n_k}\}$ is not an open cover of E, hence not a subcover of $\{S_n\}$.
Since $\{S_{n_1}, \ldots, S_{n_k}\}$ is a subcover of $\{S_n\}$, $n \in \mathbb{N}$, we have $n_1, \ldots, n_k \in \mathbb{N}$. Let

$$r = \frac{1}{2 \max\{n_1, \ldots, n_k\}}.$$

Then

$$0 < r \leq \frac{1}{2n_i} < \frac{1}{n_i}, \qquad i = 1, \ldots, k.$$

Since $n_i \in \mathbb{N}$ for $i = 1, \ldots, k$, we have $r \in \mathbb{Q}$. And

$$r^2 \leq \frac{1}{(2n_i)^2} \leq \frac{1}{2^2} = \frac{1}{4} < 2.$$

Therefore $r \in E$ and $r \notin S_{n_i}$ for $i \in \{1, \ldots, k\}$. Hence $r \in E$ and $r \notin \bigcup_{i=1}^{k} S_{n_i}$, so $\{S_{n_1}, \ldots, S_{n_k}\}$ is not an open cover of E, and therefore $\{S_{n_1}, \ldots, S_{n_k}\}$ is not a subcover of E. This is a contradiction.
Thus any subcover of $\{S_n\}$, $n \in \mathbb{N}$, has to be infinite. By the negation of the definition (page 81), E is not compact.

Theorem (Heine–Borel)

Let A be a subset of a Euclidean space. Then

$$A \text{ is compact} \iff A \text{ is closed and bounded.}$$

14 Limits of Functions

14.1 Continuous functions

Limits of functions

Definition

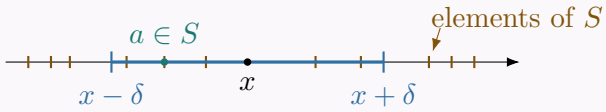
Let $S \subset \mathbb{R}$ and $x \in \mathbb{R}$. We say x is a *cluster point of the set* S if, for every $\delta > 0$,

$$(x - \delta, x + \delta) \cap S \setminus \{x\} \neq \emptyset \quad \text{(non-empty)}.$$

Remark

In other words, for every $\delta > 0$ there exists $a \in S$ such that $a \neq x$ and $a \in (x - \delta, x + \delta)$; or, for every $\delta > 0$ there exists $a \in S$ such that $0 < |a - x| < \delta$.
In terms of Rudin's definitions, a neighbourhood of p is the set $N_r(p)$ consisting of all q in the metric space such that $d(p, q) < r$. Thus q is a cluster (limit) point of a set S if and only if

every neighbourhood of q in the metric space contains a point p such that $p \neq q$ and $p \in S$.



Example

Let $S = \{1/n \mid n \in \mathbb{N}\}$. Then $0 \in \mathbb{R}$ is a cluster point of S. Let $\delta > 0$. Choose $n \in \mathbb{N}$ such that $n > 1/\delta$ (by the Archimedean property). Then

$$\frac{1}{n} \in (0 - \delta, 0 + \delta) \cap S \setminus \{0\} \quad \Longrightarrow \quad (-\delta, \delta) \cap S \setminus \{0\} \neq \emptyset.$$

Example

Let $S = (0, 1)$. The set of cluster points is $[0, 1]$.

Example

Let $S = \mathbb{Q}$. The set of cluster points of $\mathbb{Q}$ is $\mathbb{R}$ (see the example on page 79). This follows from the theorem on page 25 (density of $\mathbb{Q}$), or equivalently from Rudin's version on page 79.

Example

Let $S = \{0\}$. Then S has no cluster points.

Negation of the definition

x is not a cluster point of S if there exists $\delta_0 > 0$ such that

$$(x - \delta_0, x + \delta_0) \cap S \setminus \{x\} = \emptyset.$$

Proof (that $S = \{0\}$ has no cluster points)

Let $x > 0$. Choose $\delta_0 = x/2 > 0$. Then

$$(x - \delta_0, x + \delta_0) \cap S \setminus \{x\} = \left(\frac{x}{2}, \frac{3x}{2}\right) \cap S \setminus \{x\}.$$

Since $0 \notin (\frac{x}{2}, \frac{3x}{2})$, we get $(\frac{x}{2}, \frac{3x}{2}) \cap S \setminus \{x\} = \emptyset$. The same proof goes for $x < 0$. If $x = 0$, then $S \setminus \{0\} = \emptyset$.

Example

Let $S = \mathbb{Z}$. Then S has no cluster points.

Note

A cluster point of a set S does not have to be in S.

Theorem

Let $S \subset \mathbb{R}$. Then x is a cluster point of S if and only if there exists a sequence $\{x_n\}$ with $x_n \in S \setminus \{x\}$ such that $x_n \to x$.

Proof

($\Longrightarrow$) Suppose x is a cluster point of S. Let $\delta > 0$. Then $(x-\delta, x+\delta) \cap S \setminus \{x\} \neq \emptyset$, so there exists $y \in S \setminus \{x\}$ such that $|y - x| < \delta$.
Let $\delta_n = 1/n$ for every $n \in \mathbb{N}$. We construct a sequence $\{x_n\}$ using the δ_n:

- $n = 1$: there exists $y \in S \setminus \{x\}$ such that $|y - x| < 1$. Let $x_1 = y$.
- $n = 2$: there exists $y \in S \setminus \{x\}$ such that $|y - x| < 1/2$. Let $x_2 = y$.

$\vdots$

- $n = k$: there exists $y \in S \setminus \{x\}$ such that $|y - x| < 1/k$. Let $x_k = y$.

Thus we obtained a sequence $\{x_n\}_{n=1}^{\infty}$.
Let $\varepsilon > 0$. By the Archimedean property, there exists $M_0 \in \mathbb{N}$ such that $M_0 > 1/\varepsilon$. Choose $M = M_0$. Then, for every $n \geq M$, by our construction of $\{x_n\}_n$ we have

$$|x_n - x| < \frac{1}{n} \leq \frac{1}{M} = \frac{1}{M_0} < \varepsilon.$$

Thus $x_n \to x$, where $x_n \in S \setminus \{x\}$. Or one can use the squeeze theorem:

$$x - \frac{1}{n} < x_n < x + \frac{1}{n} \qquad \forall n \in \mathbb{N},$$

where both $x - \frac{1}{n} \to x$ and $x + \frac{1}{n} \to x$.
($\Longleftarrow$) Suppose $\{x_n\}$ is a sequence with $x_n \in S \setminus \{x\}$ and $x_n \to x$. Then, for every $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n \geq M$.
Let $\delta > 0$. Choose $\varepsilon = \delta$. Then there exists M such that, for all $n \geq M$,

$$x_n \in S \setminus \{x\} \quad \text{and} \quad |x_n - x| < \varepsilon = \delta.$$

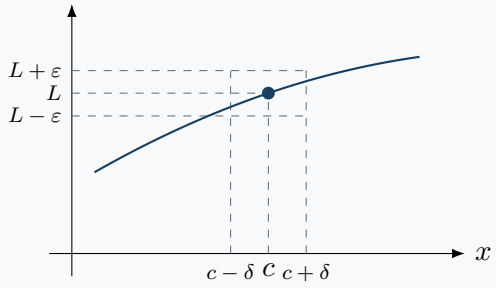
Hence $(x-\delta, x+\delta) \cap S \setminus \{x\} \neq \emptyset$. Thus x is a cluster point of S.

Definition

Let $S \subset \mathbb{R}$ and let c be a cluster point of S. Let $f : S \to \mathbb{R}$. We say $f(x)$ *converges to* L *as* x *approaches* c (written $x \to c$) if for every $\varepsilon > 0$ there exists $\delta > 0$ such that, if $x \in S$ and $0 < |x - c| < \delta$, then

$$|f(x) - L| < \varepsilon.$$

If x is near c (but $x \neq c$), then $f(x)$ is near L.



Notation

If $f(x)$ does converge to L as $x \to c$, then we write

$$f(x) \to L \quad \text{as} \quad x \to c \qquad \text{or} \qquad \lim_{x \to c} f(x) = L.$$

Theorem (uniqueness of the limit)

Let c be a cluster point of $S \subset \mathbb{R}$, and let $f : S \to \mathbb{R}$. If

$$\lim_{x \to c} f(x) = L_1 \qquad \text{and} \qquad \lim_{x \to c} f(x) = L_2,$$

then $L_1 = L_2$.

Proof

We want to show that $|L_1 - L_2| < \varepsilon$ for every $\varepsilon > 0$.
Let $\varepsilon > 0$. Since $f(x) \to L_1$ as $x \to c$, there exists $\delta_1 > 0$ such that

$$0 < |x - c| < \delta_1 \quad \Longrightarrow \quad |f(x) - L_1| < \frac{\varepsilon}{2}.$$

Similarly, there exists $\delta_2 > 0$ such that

$$0 < |x - c| < \delta_2 \quad \Longrightarrow \quad |f(x) - L_2| < \frac{\varepsilon}{2}.$$

Choose $\delta = \min\{\delta_1, \delta_2\} > 0$, so that $0 < |x - c| < \delta_1$ and $0 < |x - c| < \delta_2$. Thus

$$|f(x) - L_1| < \frac{\varepsilon}{2}, \qquad |f(x) - L_2| < \frac{\varepsilon}{2}.$$

By the triangle inequality,

$$|L_1 - L_2| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $L_1 = L_2$.

Example

$\lim_{x \to c}(ax + b) = ac + b$, with $S = \mathbb{R}$ and $c \in \mathbb{R}$.

Proof

Let $\varepsilon > 0$.
Scratch work. We want to find δ such that $0 < |x - c| < \delta \Longrightarrow |f(x) - (ac + b)| < \varepsilon$. Now

$$|f(x) - (ac + b)| = |a|\,|x - c| < |a|\,\delta.$$

Thus choose $\delta = \dfrac{\varepsilon}{|a| + 1}$ (note that $|a| + 1 \neq 0$).
Choose $\delta = \dfrac{\varepsilon}{1 + |a|}$. Then, for every $x \in \mathbb{R}$ such that $0 < |x - c| < \delta$,

$$|f(x) - (ac + b)| = |a|\,|x - c| < |a|\,\delta = \frac{|a|}{|a| + 1}\,\varepsilon < \varepsilon.$$

Thus $\lim_{x \to c} f(x) = ac + b$.

Example

Let $f(x) = \sqrt{x}$. Then, for every $c > 0$, $\lim_{x \to c} f(x) = \sqrt{c}$ (here $S = [0, \infty)$ and $c > 0$).

Proof

Scratch work: how to find δ? We want to show $|\sqrt{x} - \sqrt{c}| < \varepsilon$. Now

$$|\sqrt{x} - \sqrt{c}| = \left|(\sqrt{x} - \sqrt{c})\frac{\sqrt{x} + \sqrt{c}}{\sqrt{x} + \sqrt{c}}\right| = \left|\frac{x - c}{\sqrt{x} + \sqrt{c}}\right| < \frac{|x - c|}{\sqrt{c}} < \varepsilon.$$

Thus let $|x - c| < \delta = \sqrt{c}\,\varepsilon$.
Let $\varepsilon > 0$. Choose $\delta = \varepsilon\sqrt{c}$. Then, for $x > 0$ with $0 < |x - c| < \delta$,

$$|f(x) - \sqrt{c}| = \frac{|x - c|}{\sqrt{x} + \sqrt{c}} \leq \frac{|x - c|}{\sqrt{c}} < \frac{\delta}{\sqrt{c}} = \varepsilon.$$

Thus $\lim_{x \to c} f(x) = \sqrt{c}$.

Example

Let

$$f(x) = \begin{cases} 1, & x \neq 0, \\ 2, & x = 0. \end{cases}$$

Then $\lim_{x \to 0} f(x) = 1$.

Proof

Let $\varepsilon > 0$. Choose $\delta = 1$. Then $0 < |x - 0| < 1$ and $x \in \mathbb{R}$ imply

$$x \neq 0 \quad \Longrightarrow \quad f(x) = 1 \quad \Longrightarrow \quad |f(x) - L| = |1 - 1| = 0 < \varepsilon.$$

Note

$\lim_{x\to 0} f(x) \neq f(0)$! We do not care about what happens at c; we look at what happens to f near c.

Theorem

Let $S \subset \mathbb{R}$, let c be a cluster point of S, and let $f : S \to \mathbb{R}$. Then

$$\lim_{x\to c} f(x) = L$$

if and only if, for every sequence $\{x_n\}_{n=1}^{\infty}$ in $S \setminus \{c\}$ such that $x_n \to c$, the sequence $\{f(x_n)\}_{n=1}^{\infty}$ converges to L.

Proof

($\Longrightarrow$) Suppose $\lim_{x\to c} f(x) = L$. Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in $S \setminus \{c\}$ such that $x_n \to c$. We want to show that $\{f(x_n)\}_{n=1}^{\infty}$ converges to L.
Let $\varepsilon > 0$. Since $\lim_{x\to c} f(x) = L$, there exists $\delta > 0$ such that

$$x \in S \text{ and } 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon.$$

Since $x_n \to c$, there exists $M_0 \in \mathbb{N}$ such that $0 < |x_m - c| < \delta$ for all $m \geq M_0$. Choose $M = M_0$. Then, for all $n \geq M$, since $0 < |x_n - c| < \delta$, we have $|f(x_n) - L| < \varepsilon$. Thus $f(x_n) \to L$ as $n \to \infty$.
($\Longleftarrow$) Suppose that, for every sequence $\{x_n\}$ in $S \setminus \{c\}$ with $x_n \to c$, we have $f(x_n) \to L$. We argue by contradiction: assume $\lim_{x\to c} f(x) \neq L$. Then there exists $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there exists $x \in S$ with

$$0 < |x - c| < \delta \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.$$

We will show that there exists a sequence $\{x_n\}$ in $S \setminus \{c\}$ such that $f(x_n) \not\to L$. Choose $\delta = 1/n$. Then, for every $n \in \mathbb{N}$, there exists $x \in S$ such that

$$0 < |x - c| < \frac{1}{n} \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.$$

We denote this x by x_n for each $n \in \mathbb{N}$. Thus

$$0 < |x_n - c| < \frac{1}{n} \qquad \forall n \in \mathbb{N}.$$

By the squeeze theorem, since in $0 < |x_n - c| < 1/n$ both $0 \to 0$ and $1/n \to 0$, we get $|x_n - c| \to 0$. Note that $\{x_n\}$ is a sequence, so $|x_n - c| \to 0$ implies $x_n \to c$.
We found a sequence $\{x_n\}$ converging to c; now we show $f(x_n) \not\to L$. Since $\{x_n\}$ is in $S \setminus \{c\}$ and $x_n \to c$, by assumption $f(x_n) \to L$, so $|f(x_n) - L| \to 0$ (since $\{f(x_n)\}_n$ is a sequence).
Since for every $n \in \mathbb{N}$ there is $x_n \in S$ with $|f(x_n) - L| \geq \varepsilon_0$, and since limits of sequences preserve inequalities,

$$\lim_{n\to\infty} |f(x_n) - L| \geq \lim_{n\to\infty} \varepsilon_0 = \varepsilon_0.$$

Since $|f(x_n) - L| \to 0$, we have that $0 \geq \varepsilon_0 > 0$, so $0 > 0$, a contradiction. Therefore our assumption that $f(x) \not\to L$ as $x \to c$ is false.

15 Limits of Functions in Terms of Sequences and Continuity

Definition (recall)

Let $S \subset \mathbb{R}$, let c be a cluster point of S, and let $f : S \to \mathbb{R}$. Then

$$\lim_{x \to c} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $x \in S$ with $0 < |x - c| < \delta$, we have $|f(x) - L| < \varepsilon$.

Theorem: limits of functions in terms of sequences

Let $S \subset \mathbb{R}$, let c be a cluster point of S, and let $f : S \to \mathbb{R}$. Then

$$\lim_{x \to c} f(x) = L \iff \text{for every sequence } \{x_n\}_{n=1}^{\infty} \text{ in } S \setminus \{c\} \text{ such that } x_n \to c, \text{ we have that } \{f(x_n)\}_{n=1}^{\infty} \text{ converges to } L.$$

Remark: why do we need this theorem?

Question. Why do we need this theorem?
Answer. Because, using this theorem, we get to easily prove the analogous statements for limits of functions from what we have proven for limits of sequences. For example: the squeeze theorem, how limits interact with operations, absolute values, inequalities, n-th roots, basically.

Theorem

For every $c \in \mathbb{R}$, $\lim_{x \to c} x^2 = c^2$.

Proof

We could use the ε–δ definition, but we will use the theorem on page 45 for limits of sequences to prove this one.
We want to show: for every sequence $\{x_n\}$ in $\mathbb{R} \setminus \{c\}$ such that $x_n \to c$, we have $x_n^2 \to c^2$.
Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in $\mathbb{R} \setminus \{c\}$ such that $x_n \to c$. Then, by the theorem on page 45 (the limit of convergent sequences preserves operations),

$$\lim_{n \to \infty} x_n^2 = \left(\lim_{n \to \infty} x_n \right)^2 = c^2.$$

Therefore, by the previous theorem, $\lim_{x \to c} f(x) = c^2$.

Theorem

Let $S = \mathbb{R} \setminus \{0\}$.

1) $\lim_{x \to 0} \sin \frac{1}{x}$ does not exist.

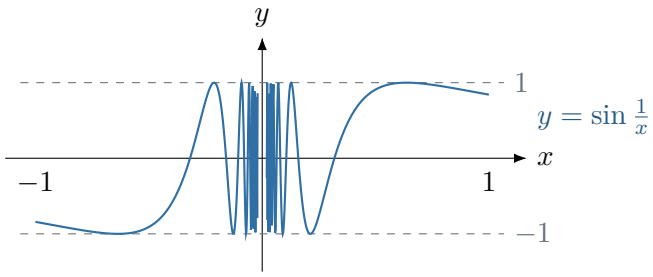
2) $\lim_{x \to 0} x \sin \frac{1}{x} = 0$.

Note

Obviously, for this limit we cannot evaluate by plugging in.

Proof of 1)

Idea of the proof. The idea of the proof is the fact that $f(x) = \sin\frac{1}{x}$ is highly oscillating between 1 and -1 near 0. Thus we do not know what is happening near 0. And if we take $\{x_n\}_{n=1}^{\infty}$ that converges to 0, then $f(x_n)$ should also oscillate between positive and negative values. Even better, we can find an $\{x_n\}$ such that $f(x_n)$ exactly goes like $+1, -1, +1, -1, \ldots$



By the theorem above, one can easily show that

$$\lim_{x\to 0} \sin\frac{1}{x} \text{ does not exist} \iff \text{there exists a sequence } \{x_n\} \text{ in } S\setminus\{c\} \text{ such that } x_n \to c, \text{ for which } f(x_n) \text{ does not converge.}$$

Thus we want to show: there exists $\{x_n\}$ with $x_n \in S\setminus\{c\}$ such that $f(x_n)$ does not converge. Let

$$x_n = \frac{2}{(2n-1)\pi}.$$

Then $x_n \to 0$, and $x_n \in \mathbb{R}\setminus\{0\}$ for every $n \in \mathbb{N}$. Now we will show that $f(x_n)$ diverges. We have

$$f(x_n) = \sin\Big(\frac{1}{x_n}\Big) = \sin\Big(\frac{2n-1}{2}\pi\Big).$$

If n is even,

$$\sin\Big(\frac{2n-1}{2}\pi\Big) = \sin\Big(n\pi - \frac{1}{2}\pi\Big) = \sin\Big(-\frac{1}{2}\pi\Big) = -1.$$

If n is odd,

$$\sin\Big(\frac{2n-1}{2}\pi\Big) = \sin\Big(n\pi - \frac{1}{2}\pi\Big) = \sin\Big(\frac{1}{2}\pi\Big) = 1.$$

Thus

$$\{f(x_n)\}_{n=1}^{\infty} = \{(-1)^{n+1}\}_{n=1}^{\infty} = 1, -1, 1, -1, 1, \ldots$$

Since $\{f(x_n)\}_n$ does not converge, $\lim_{x\to 0}\sin\frac{1}{x}$ does not exist.

Proof of 2)

Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in $\mathbb{R}\setminus\{0\}$ such that $x_n \to 0$. Then

$$0 \le \Big|x_n \sin\frac{1}{x_n}\Big| \le |x_n|.$$

By the squeeze theorem,

$$\lim_{n\to\infty}\left|x_n \sin\frac{1}{x_n}\right| = 0 \implies \left\{x_n \sin\frac{1}{x_n}\right\}_{n=1}^{\infty} \text{ converges to } 0 \implies \lim_{x\to 0} x\sin\frac{1}{x} = 0.$$

Theorem: limits of functions preserve inequalities

Let $S \subset \mathbb{R}$, let c be a cluster point of S, and let $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$. If $f(x) \leq g(x)$ for every $x \in S$, and $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ exist, then

$$\lim_{x\to c} f(x) \leq \lim_{x\to c} g(x).$$

Proof

Let $L_1 = \lim_{x\to c} f(x)$ and $L_2 = \lim_{x\to c} g(x)$. Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in $S \setminus \{c\}$ such that $x_n \to c$. Then, since $x_n \in S \setminus \{c\}$, we have $f(x_n) \leq g(x_n)$ for every $n \in \mathbb{N}$. Therefore, by the theorem on page 44, we have that

$$\lim_{n\to\infty} f(x_n) \leq \lim_{n\to\infty} g(x_n),$$

that is,

$$L_1 = \lim_{x\to c} f(x) \leq \lim_{x\to c} g(x) = L_2.$$

Remark: more consequences of the sequence characterization

By using the theorem which gives the analogous statement between limits of sequences and limits of functions, we can similarly show the following.

- *Squeeze theorem.* If c is a cluster point of S, $f, g, h : S \to \mathbb{R}$, $f(x) \leq g(x) \leq h(x)$ for every $x \in S$, and $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} h(x) = L$, then $\lim_{x\to c} g(x) = L$.
- Algebraic operations of limits.
- Absolute value of f.

15.1 Left and Right Limits of Functions

Definition: left limit

Let $S \subset \mathbb{R}$, and let c be a cluster point of the set $S \cap (-\infty, c)$. Then we say $f(x)$ converges to L as $x \to c^-$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, if $x \in S$ and $c - \delta < x < c$, then $|f(x) - L| < \varepsilon$.
Notation. This is denoted by

$$L = \lim_{x\to c^-} f(x),$$

called the *left limit* of f as x approaches c (from the left).

Definition: right limit

Let $S \subset \mathbb{R}$, and let c be a cluster point of $S \cap (c, +\infty)$. Then we say $f(x)$ converges to L as $x \to c^+$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, if $x \in S$ and $c < x < c + \delta$, then $|f(x) - L| < \varepsilon$.

Notation. This is denoted by

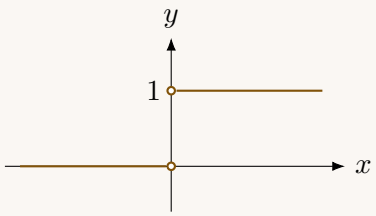
$$L = \lim_{x \to c^+} f(x),$$

called the *right limit* of f as x approaches c (from the right).

Example

Let

$$f(x) = \begin{cases} 0, & \text{if } x < 0, \\ 1, & \text{if } x > 0. \end{cases}$$



Then

$$\lim_{x \to 0^-} f(x) = 0, \qquad \lim_{x \to 0^+} f(x) = 1,$$

and $f(0)$ is not defined.

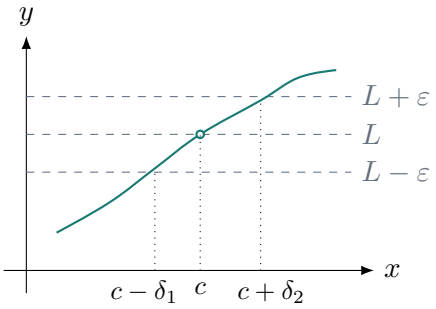
Theorem: two-sided limits from one-sided limits

Let $S \subset \mathbb{R}$, $f : S \to \mathbb{R}$, and let c be a cluster point of both $S \cap (-\infty, c)$ and $S \cap (c, +\infty)$. (Thus c is a cluster point of S.) Then

$$\lim_{x \to c} f(x) = L \iff \lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L.$$

Proof

Obvious from the definition of limits: take $\delta = \min\{\delta_1, \delta_2\}$.



15.2 Continuity

Definition

Let $S \subset \mathbb{R}$ and $c \in S$. We say f is *continuous at* c if for every $\varepsilon > 0$, there exists $\delta > 0$ such that, if $x \in S$ and $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.
We say f is *continuous on* $A \subset S$ if f is continuous at every point in A.

Note: limits versus continuity

The difference between the definition of limit and continuity is the fact that now we allow $x = c$ by $|x - c| < \delta$, without letting it be bigger than 0. Then $f(x) = f(c)$. This connects the limit of the function and the function evaluated at the point.

Example

$f(x) = ax + b$ is continuous, that is, it is continuous over its domain $\mathbb{R}$.

Proof

Let $c \in \mathbb{R}$. We want to show that f is continuous at c. Let $\varepsilon > 0$. Choose

$$\delta = \frac{\varepsilon}{1 + |a|}.$$

Then $|x - c| < \delta$ implies

$$|f(x) - f(c)| = |ax + b - (ac + b)| = |a|\,|x - c| < |a|\,\delta = \frac{|a|}{|a| + 1}\,\varepsilon < \varepsilon.$$

Thus, for every $c \in \mathbb{R}$, $f(x)$ is continuous at c.

Non-example

Let

$$f(x) = \begin{cases} 1, & x \neq 0, \\ 2, & x = 0. \end{cases}$$

Then f is not continuous at $c = 0$.

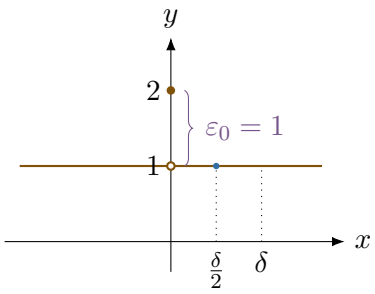
Negation of the definition

f is *not continuous at* c if there exists $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there exists $x \in S$ such that $|x - c| < \delta$ and $|f(x) - f(c)| \geq \varepsilon_0$.

Proof (of the non-example)

Choose $\varepsilon_0 = 1$. Let $\delta > 0$. Then, if $x = \frac{\delta}{2}$, we have

$$|x - 0| = \frac{\delta}{2} < \delta, \qquad \text{and} \qquad |f(x) - f(0)| = |2 - 1| \geq 1 = \varepsilon_0.$$



Question

Let $f : S \to \mathbb{R}$. Does there have to exist a $c \in S$ such that f is continuous at c? In other words, does there exist a function that is continuous nowhere over its domain?
Answer. No, there does not have to be such a point: this is shown by Dirichlet's function.

16 The Continuity of Sine and Cosine, and the Discontinuity of Dirichlet's Function

Recall

The definition of continuity connects the limits of a function at a point to how the function behaves at c.

Definition (recall)

Let $S \subset \mathbb{R}$, let $c \in S$ (c does not have to be a cluster point), and let $f : S \to \mathbb{R}$. Then f is continuous at c if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, if $x \in S$ and $|x - c| < \delta$, then

$$|f(x) - f(c)| < \varepsilon.$$

Remark: where f must be defined

f need not be defined at c (c has to be a cluster point of the set S, though) when we discuss the notion of limit. Yet f has to be defined at c (c does not need to be a cluster point of S) if we are discussing the notion of continuity at c, since the definition uses $f(c)$.

Theorem: characterizations of continuity

Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \to \mathbb{R}$. Then:

1) If c is not a cluster point of S, then f is continuous at c.
2) If c is a cluster point of S, then
$$f \text{ is continuous at } c \iff \lim_{x \to c} f(x) = f(c).$$
3) f is continuous at c if and only if, for every sequence $\{x_n\}_{n=1}^{\infty}$ in S such that $x_n \to c$, we have $f(x_n) \to f(c)$.

Proof of 1)

Let $\varepsilon > 0$. Since c is not a cluster point of S, there exists $\delta_0 > 0$ such that

$$(c - \delta_0, c + \delta_0) \cap S \setminus \{c\} = \emptyset.$$

Choose $\delta = \delta_0$. Then

$$x \in S \text{ and } |x - c| < \delta \quad \Longrightarrow \quad x \in S \cap (c - \delta, c + \delta) \quad \Longrightarrow \quad x = c.$$

Thus $|f(x) - f(c)| = |f(c) - f(c)| = 0 < \varepsilon$.

Proof of 2)

($\Longrightarrow$) Suppose f is continuous at c, a cluster point of S. Let $\varepsilon > 0$. Then there exists $\delta > 0$ such that

$$x \in S \text{ and } |x - c| < \delta \quad \Longrightarrow \quad |f(x) - f(c)| < \varepsilon.$$

Thus, for every $x \in S$ such that $0 < |x - c| < \delta$, we have $|f(x) - f(c)| < \varepsilon$. By the definition of limit, $f(x) \to f(c)$ as $x \to c$.
($\Longleftarrow$) Suppose $\lim_{x \to c} f(x) = f(c)$. Then, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $x \in S$ with $0 < |x - c| < \delta$, we have $|f(x) - f(c)| < \varepsilon$. Let $|x - c| = 0$, that is, $x = c$. Then

$$|f(x) - f(c)| = |f(c) - f(c)| = 0 < \varepsilon.$$

Thus, for every $x \in S$ with $|x - c| < \delta$, we have $|f(x) - f(c)| < \varepsilon$. Thus $f(x)$ is continuous at c.

Proof of 3)

($\Longrightarrow$) Suppose f is continuous at c. Let $\varepsilon > 0$. Let $\{x_n\}$ be a sequence in S such that $x_n \to c$. Thus there exists $M_0 \in \mathbb{N}$ such that $|x_n - c| < \delta$ for every $n \geq M_0$ (for any given $\delta > 0$). Since f is continuous at c, for every $\varepsilon > 0$ there exists $\delta_0 > 0$ such that

$$x \in S \text{ and } |x - c| < \delta_0 \quad \Longrightarrow \quad |f(x) - f(c)| < \varepsilon.$$

Choose $\delta = \delta_0$. Then $|x_n - c| < \delta_0$ for every $n \geq M_0$. Then choose $M = M_0$. Then, for every $n \geq M = M_0$, we have $x_n \in S$ and $|x_n - c| < \delta_0$, so $|f(x_n) - f(c)| < \varepsilon$. Therefore $\{f(x_n)\}_{n=1}^{\infty}$ converges to $f(c)$.
($\Longleftarrow$) Suppose that, for every sequence $\{x_n\}_{n=1}^{\infty}$ in S such that $x_n \to c$, the sequence $\{f(x_n)\}_{n=1}^{\infty}$ converges to $f(c)$. We argue by contradiction: suppose f is not continuous at c. Then there exists $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there exists $x \in S$ such that

$$|x - c| < \delta \qquad \text{and} \qquad |f(x) - f(c)| \geq \varepsilon_0.$$

We will denote this x by x_n: choose $\delta = \frac{1}{n}$. Then (for every $\delta = \frac{1}{n}$, that is) for every $n \in \mathbb{N}$, there exists $x = x_n \in S$ such that

$$|x_n - c| < \delta = \frac{1}{n} \qquad \text{and} \qquad |f(x_n) - f(c)| \geq \varepsilon_0.$$

Explicitly:

$$\begin{aligned} n=1,\ \delta=1: &\quad \exists x_1 \in S \text{ such that } |x_1-c|<1 \text{ and } |f(x_1)-f(c)|\geq \varepsilon_0,\\ n=2,\ \delta=\tfrac{1}{2}: &\quad \exists x_2 \in S \text{ such that } |x_2-c|<\tfrac{1}{2} \text{ and } |f(x_2)-f(c)|\geq \varepsilon_0,\\ &\quad\ \vdots \end{aligned}$$

We have constructed a sequence $\{x_n\}_{n=1}^{\infty}$ by our assumption that f is not continuous at c. Therefore, by taking all $n \in \mathbb{N}$, we obtained a sequence $\{x_n\}_{n=1}^{\infty}$ such that, for every $n \in \mathbb{N}$,

$$|x_n - c| < \frac{1}{n} \qquad \text{and} \qquad |f(x_n)-f(c)| \geq \varepsilon_0.$$

Then, by the squeeze theorem, $|x_n - c| \to 0$, so $x_n \to c$. By our assumption, $\{x_n\}_{n=1}^{\infty}$ is a sequence in S with $x_n \to c$, so $f(x_n) \to f(c)$. Then

$$\lim_{n\to\infty} |f(x_n)-f(c)| = 0.$$

Yet, by our construction of $\{x_n\}_{n=1}^{\infty}$, for every $n \in \mathbb{N}$, $|f(x_n)-f(c)| \geq \varepsilon_0$. Since limits preserve inequalities,

$$\lim_{n\to\infty} |f(x_n)-f(c)| \geq \varepsilon_0.$$

Thus $0 \geq \varepsilon_0 > 0$, that is, $0 > 0$, a contradiction. Hence the assumption that f is not continuous at c is false.

Note: another proof of 3)

We can also prove 3) by cases.
Case 1. c is not a cluster point of S. Then, by 1), f is continuous at c.
Case 2. c is a cluster point of S. Then we can go through the same proof as we did on pages 87–88 for the theorem and show that $\lim_{x\to c} f(x) = f(c)$. Then, by 2), f is continuous at c.

Theorem: continuity of sine and cosine

$f(x) = \sin x$ and $g(x) = \cos x$ are continuous functions (on $\mathbb{R}$).

Proof of 1): $\sin x$ ***is continuous***

We show that $f(x) = \sin x$ is continuous at c, for every $c \in \mathbb{R}$. Let $\varepsilon > 0$. Choose $\delta = \varepsilon$. Then $|x-c| < \delta$ implies

$$|\sin x - \sin c| = \left|2\cos\frac{x+c}{2}\,\sin\frac{x-c}{2}\right| \leq \left|2\sin\frac{x-c}{2}\right|,$$

since $|\cos y| \leq 1$ for every $y \in \mathbb{R}$. Since $|\sin x| \leq |x|$ for every $x \in \mathbb{R}$,

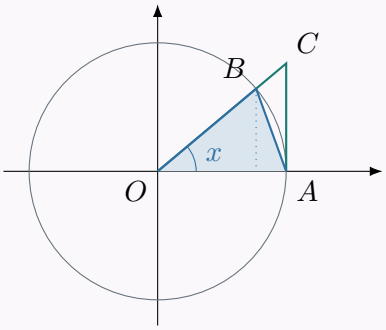
$$|\sin x - \sin c| \leq \left|2\sin\frac{x-c}{2}\right| \leq \left|2\cdot\frac{x-c}{2}\right| = |x-c| < \delta = \varepsilon.$$

Thus $\sin x = f(x)$ is continuous at c, for every $c \in \mathbb{R}$.

Note: why $|\sin x| \leq |x|$

Comparing areas in the unit circle (with x the angle $\angle AOB$),

$$\text{Area}_{\triangle AOB} < \text{Area}_{\text{sector } ABO} < \text{Area}_{\triangle ACO} \quad \Longrightarrow \quad |\sin x| \leq |x| \quad \text{for every } x \in \mathbb{R}.$$



Proof of 2):* $\cos x$ *is continuous

Now we will use the previous theorem to show that g is continuous; that is, we will show that, for every sequence $\{x_n\}_{n=1}^{\infty}$ in S such that $x_n \to c$, we have $g(x_n) \to g(c)$.
Let $c \in \mathbb{R}$. Suppose $\{x_n\}$ is a sequence such that $x_n \to c$. Then

$$x_n + \frac{\pi}{2} \longrightarrow c + \frac{\pi}{2}.$$

Since $\sin(x)$ is continuous, for every sequence in $\mathbb{R}$ such that $y_n \to c$, we have $\sin(y_n) \to \sin(c)$. Therefore

$$\sin\Big(x_n + \frac{\pi}{2}\Big) \longrightarrow \sin\Big(c + \frac{\pi}{2}\Big), \qquad \text{that is,} \qquad \cos x_n \longrightarrow \cos c.$$

And so $g(x) = \cos x$ is continuous on $\mathbb{R}$.

The following theorem will answer our question at the beginning.

Theorem (the Dirichlet function)

The function

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}, \end{cases}$$

is nowhere continuous; in other words, $f(x)$ is not continuous at c, for every $c \in \mathbb{R}$.

Lemma: density of the irrationals in $\mathbb{R}$

For every $x, y \in \mathbb{R}$ such that $x < y$, there exists $i \in \mathbb{Q}^c = \mathbb{R} \setminus \mathbb{Q}$ such that $x < i < y$.

Proof

We want to show: there exists $i \in \mathbb{Q}^c$ such that $x < i < y$. We will find a candidate for i first. Since $x, y \in \mathbb{R}$ and $x < y$, by the density of $\mathbb{Q}$ there exists $r \in \mathbb{Q}$ such that $x < r < y$. By the

Archimedean property, since $y - r > 0$, there exists $n \in \mathbb{N}$ such that $n(y-r) > 1$. Choose

$$i = r + \frac{1}{n\sqrt{2}}.$$

Since $n(y-r) > 1$, we get $y - r > \frac{1}{n}$. By the theorem on page 17, $\sqrt{2} > 1$. Thus

$$y - r > \frac{1}{n\sqrt{2}} \quad \Longrightarrow \quad y > r + \frac{1}{n\sqrt{2}} \quad \Longrightarrow \quad y > i > x.$$

Now we want to show: $i \in \mathbb{Q}^c$. Suppose, for contradiction, that $i = r + \frac{1}{n\sqrt{2}} \in \mathbb{Q}$. Since $\mathbb{Q}$ is a field,

$$\frac{1}{n\sqrt{2}} \in \mathbb{Q} \quad \Longrightarrow \quad \frac{1}{n\sqrt{2}} \cdot n = \frac{1}{\sqrt{2}} \in \mathbb{Q} \quad \Longrightarrow \quad \left(\frac{1}{\sqrt{2}}\right)^{-1} = \sqrt{2} \in \mathbb{Q}.$$

Then, by the theorem on page 19, $\sqrt{2} \notin \mathbb{Q}$ and $\sqrt{2} \in \mathbb{Q}$, a contradiction.
Thus, for every $x, y \in \mathbb{R}$ with $x < y$, there exists $i \in \mathbb{Q}^c$ such that $x < i < y$.

Proof of the theorem

Let $c \in \mathbb{R}$. Then $c \in \mathbb{Q}$ or $c \notin \mathbb{Q}$.
We want to show the following: by the theorem on page 94, there exists a sequence $\{x_n\}$ such that $x_n \to c$ and $\{f(x_n)\}_{n=1}^{\infty}$ does not converge to $f(c)$, which is equivalent to f not being continuous at c. Thus this is what we will show.
Case 1: $c \in \mathbb{Q}$. For every $n \in \mathbb{N}$, since $c \in \mathbb{R}$, $c + \frac{1}{n} \in \mathbb{R}$, and $c < c + \frac{1}{n}$, there exists $x_n \in \mathbb{Q}^c$ (the "i" of the lemma) such that

$$c < x_n < c + \frac{1}{n},$$

by the lemma. Thus we obtained an irrational sequence $\{x_n\}_{n=1}^{\infty}$ such that $x_n \to c$ (by the squeeze theorem). Yet $f(x_n) = 0$, since $x_n \notin \mathbb{Q}$. Thus

$$0 = \lim_{n\to\infty} f(x_n) \neq f(c) = 1 \qquad (c \in \mathbb{Q}).$$

Therefore there exists $\{x_n\}_n$ in $\mathbb{R}$ such that $x_n \to c$ and $f(x_n) \not\to f(c)$, so f is not continuous at c ($c \in \mathbb{Q}$).
Case 2: $c \in \mathbb{Q}^c$. By the density of $\mathbb{Q}$, for every $c \in \mathbb{Q}^c$ there exists $r = x_n \in \mathbb{Q}$ such that

$$c < x_n < c + \frac{1}{n}, \qquad \text{for every } n \in \mathbb{N}.$$

Thus, by the squeeze theorem, $x_n \to c$. Since $x_n \in \mathbb{Q}$, $f(x_n) = 1$. Therefore,

$$1 = \lim_{n\to\infty} f(x_n) \neq f(c) = 0 \qquad (c \in \mathbb{Q}^c).$$

There exists a sequence $\{x_n\}_{n=1}^{\infty}$ such that $x_n \to c$ and $f(x_n) \not\to f(c)$, so f is not continuous at c.

Theorem: polynomials are continuous

$f(x) = a_d x^d + \cdots + a_1 x + a_0$ is continuous (on $\mathbb{R}$).

Proof

Let $c \in \mathbb{R}$. Let $\{x_n\}_{n=1}^{\infty}$ be a sequence such that $x_n \to c$. Then, by the theorems for limits of sequences,

$$\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} \sum_{i=0}^{d} a_i x_n^i = \sum_{i=0}^{d} a_i \Big(\lim_{n\to\infty} x_n\Big)^i = \sum_{i=0}^{d} a_i c^i = f(c).$$

Thus, for every sequence $\{x_n\}$ in $\mathbb{R}$ such that $x_n \to c$, we have $f(x_n) \to f(c)$, so f is continuous at c.

Remark

After proving the following theorem, we will have another way to prove the previous theorem.

Theorem: algebraic operations preserve continuity

If $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ are continuous at $c \in S$, then

1) $f + g$ is continuous at c;
2) $f \cdot g$ is continuous at c;
3) if $g(x) \neq 0$ for every $x \in S$, then $\dfrac{f}{g}$ is continuous at c.

Proof

1) Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in S such that $x_n \to c$. Then $f(x_n) \to f(c)$ and $g(x_n) \to g(c)$. Thus, by the theorem for limits of sequences,

$$f(x_n) + g(x_n) \longrightarrow f(c) + g(c), \qquad \text{that is,} \qquad (f+g)(x_n) \longrightarrow (f+g)(c).$$

Hence $f + g$ is continuous at c.
2) and 3) have exactly the same proof as 1).

Theorem: composition preserves continuity

Let $A, B \subset \mathbb{R}$, $f : B \to \mathbb{R}$, and $g : A \to B$. If g is continuous at $c \in A$ and f is continuous at $g(c)$, then $f \circ g$ is continuous at c.

Proof

Suppose $\{x_n\}_{n=1}^{\infty}$ in A converges to c. Then $g(x_n) \to g(c)$, and hence

$$f(g(x_n)) \longrightarrow f(g(c)), \qquad \text{that is,} \qquad (f \circ g)(x_n) \longrightarrow (f \circ g)(c).$$

Hence $f \circ g$ is continuous at c.

Examples

1) For every $n \in \mathbb{N}$, $f(x) = x^n$ is continuous. *Proof.* By induction and the previous theorems.
2) $\dfrac{1}{x^2}$ is continuous on $(0, +\infty)$. *Proof.* $g(x) = x^2$ is continuous on $(0, +\infty)$ and $g(x) \neq 0$.

3) $f(x) = \sum_{i=0}^{d} a_i x^i$ is continuous on $\mathbb{R}$ (by the previous two theorems).

4) $f(x) = \left[\frac{1}{\cos(x^3)}\right]^2$ is continuous on $(0, +\infty)$.

5) $f(x) = \frac{1}{3 + (\sin x)^5}$ is continuous on $\mathbb{R}$.

17 The Min/Max Theorem and Bolzano's Intermediate Value Theorem

Recall

Theorem. Let $S \subset \mathbb{R}$, $c \in S$, and $f : S \to \mathbb{R}$. Then f is continuous at c if and only if, for every sequence $\{x_n\}_{n=1}^{\infty}$ in S such that $x_n \to c$, we have $f(x_n) \to f(c)$.
Theorem (Bolzano–Weierstrass). Every bounded sequence has a convergent subsequence.

Note

We will see that if $f : [a, b] \to \mathbb{R}$ is continuous, then it is well-behaved.

17.1 Bounded functions

Definition

A function $f : S \to \mathbb{R}$ is said to be *bounded* if there exists $B \geq 0$ such that, for every $x \in S$,

$$|f(x)| \leq B.$$

Examples

1) $f(x) = 3x + 1$ on $S = [0, 1]$: f is bounded.

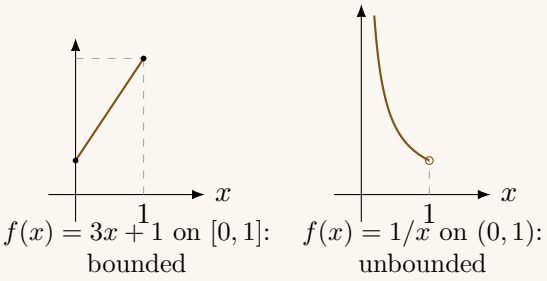
2) $f(x) = 3x + 1$ on $S = \mathbb{R}$: f is unbounded.

Negation of the definition

A function $f : S \to \mathbb{R}$ is *unbounded* if, for every $B \geq 0$, there exists $x \in S$ such that $|f(x)| > B$.

Example

$f : (0, 1) \to \mathbb{R}$, $f(x) = \frac{1}{x}$, is unbounded.



$f(x) = 3x + 1$ on $[0, 1]$: bounded

$f(x) = 1/x$ on $(0, 1)$: unbounded

Theorem: continuous functions on $[a, b]$ are bounded

If $f : [a, b] \to \mathbb{R}$ is continuous, then f is bounded.

Proof (by contradiction)

Suppose $f : [a, b] \to \mathbb{R}$ is continuous and unbounded. Then, for every $n \in \mathbb{N}$, there exists $x_n \in [a, b]$ such that $|f(x_n)| > n$. In this way we obtain a sequence $\{x_n\}$. Since $x_n \in [a, b]$,

$$|x_n| \leq \max\{|a|, |b|\},$$

so $\{x_n\}_{n=1}^{\infty}$ is bounded. By the Bolzano–Weierstrass theorem, there exist a subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ of $\{x_n\}_{n=1}^{\infty}$ and $x \in \mathbb{R}$ such that $x_{n_k} \to x$. Since $a \leq x_{n_k} \leq b$ for every $k \in \mathbb{N}$,

$$a \leq \lim_{k\to\infty} x_{n_k} = x \leq b.$$

Since f is continuous at $x \in [a, b]$, by the theorem on page 94,

$$f(x) = \lim_{k\to\infty} f(x_{n_k}) \quad \Longrightarrow \quad |f(x)| = \lim_{k\to\infty} |f(x_{n_k})|, \qquad x \in [a, b],$$

where the implication uses the theorem on page 47. Since $\{|f(x_{n_k})|\}_{k=1}^{\infty}$ converges to $|f(x)|$, the sequence $\{|f(x_{n_k})|\}_{k=1}^{\infty}$ is bounded.
By our construction of $\{x_n\}_n$, for every $n \in \mathbb{N}$,

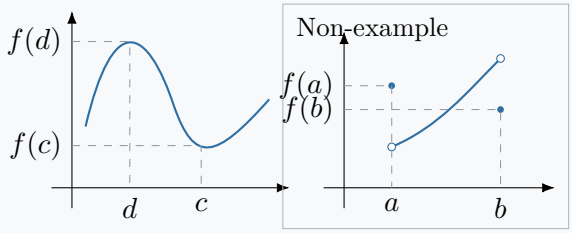
$$|f(x_n)| > n.$$

Thus $|f(x_{n_k})| > n_k$ for every $k \in \mathbb{N}$. Since $\{|f(x_{n_k})|\}_k$ is bounded, $\{n_k\}_k$ is bounded. Yet $k \leq n_k$ for every $k \in \mathbb{N}$, so $\{n_k\}_k$ is unbounded, a contradiction. Thus f has to be continuous and bounded.

17.2 The extreme value theorem

Definition

Let $f : S \to \mathbb{R}$, where $S \subset \mathbb{R}$ and $S \neq \emptyset$. Then f *achieves an absolute minimum* at $c \in S$ if, for every $x \in S$, $f(x) \geq f(c)$. Similarly, f *achieves an absolute maximum* at d if, for every $x \in S$, $f(x) \leq f(d)$.



Theorem (Extreme Value Theorem, E.V.T., or Min–Max Theorem)

Let $f : [a, b] \to \mathbb{R}$. If f is continuous, then f achieves an absolute maximum and an absolute minimum.

Remark

In other words, there exist $c, d \in [a, b]$ such that

$$f([a, b]) \subset [f(c), f(d)].$$

Proof

We will prove this first for the absolute maximum. Suppose $f : [a, b] \to \mathbb{R}$ is continuous. Then, by the previous theorem, f is bounded. Then the set

$$E = \{f(x) \mid x \in [a, b]\}$$

is bounded above. By the least upper bound property of $\mathbb{R}$, $\sup E$ exists in $\mathbb{R}$. Let $L = \sup E$.
Claim. There exists a sequence $\{x_n\}_{n=1}^{\infty}$ in $[a, b]$ such that $f(x_n) \to L$.
Since $L = \sup E$, for every $\varepsilon > 0$ there exists $x \in [a, b]$ such that

$$f(x) - \varepsilon < \sup E \leq f(x).$$

Choose $\varepsilon_1 = 1$: there exists $x_1 \in [a, b]$ such that $f(x_1) - 1 < \sup E \leq f(x_1)$. Choose $\varepsilon_2 = \frac{1}{2}$: there exists $x_2 \in [a, b]$ such that $f(x_2) - \frac{1}{2} < \sup E \leq f(x_2)$, and so on. In general, let $\varepsilon_n = \frac{1}{n}$; for every $n \in \mathbb{N}$ there exists $x_n \in [a, b]$ such that

$$f(x_n) - \frac{1}{n} < \sup E \leq f(x_n).$$

Thus, by the squeeze theorem, we have obtained a sequence $\{x_n\}_{n=1}^{\infty}$ such that

$$f(x_n) \to L = \sup E.$$

This proves the claim.
By the Bolzano–Weierstrass theorem, there exists a subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ of $\{x_n\}_{n=1}^{\infty}$ such that

$$\lim_{k \to \infty} x_{n_k} = d.$$

Since $a \leq x_{n_k} \leq b$, we get $a \leq \lim_{k\to\infty} x_{n_k} = d \leq b$. Then, since f is continuous at d,

$$f(d) = \lim_{k \to \infty} f(x_{n_k})$$

by the theorem we recalled on page 100. Note that $\{f(x_{n_k})\}_{k=1}^{\infty}$ becomes a subsequence of $\{f(x_n)\}_{n=1}^{\infty}$, as we picked $\{x_{n_k}\}_k$ to be a subsequence of $\{x_n\}$. Since our choice of $\{x_n\}$ gives that $f(x_n) \to L$, the subsequence $\{f(x_{n_k})\}_{k=1}^{\infty}$ should also converge to L. Thus

$$f(d) = \lim_{k\to\infty} f(x_{n_k}) = \lim_{n\to\infty} f(x_n) = L.$$

That is, we have shown that there exists $d \in [a,b]$ such that, for every $x \in [a,b]$, $f(x) \le f(d)$, since $L = \sup E$.

Absolute minimum. Suppose f is continuous. Then f is bounded, so the set

$$E = \{f(x) \mid x \in [a,b]\}$$

is bounded. By the greatest lower bound property of $\mathbb{R}$ (the theorem on page 23), $\inf E$ exists in $\mathbb{R}$. Let $\ell = \inf E$. Since ℓ is $\inf E$, for every $n \in \mathbb{N}$, with $\varepsilon_n = \frac{1}{n}$, there exists $x_n \in [a,b]$ such that

$$\ell \le f(x_n) < \ell + \varepsilon_n.$$

Then, by the squeeze theorem, $\{f(x_n)\}_{n=1}^{\infty}$ converges to ℓ. Since $x_n \in [a,b]$, the sequence is bounded, so by Bolzano–Weierstrass there exists a subsequence $\{x_{m_k}\}_{k=1}^{\infty}$ such that $x_{m_k} \to c$ for some $c \in [a,b]$. By our choice of $\{x_{m_k}\}_{k=1}^{\infty}$, we have that $\{f(x_{m_k})\}_{k=1}^{\infty}$ is a subsequence of $\{f(x_n)\}_{n=1}^{\infty}$. Since $f(x_n) \to \ell$, its subsequence satisfies $f(x_{m_k}) \to \ell$. Since f is continuous, by the theorem on page 94 (recalled on page 100),

$$f(c) = \lim_{k\to\infty} f(x_{m_k}).$$

Therefore

$$f(c) = \lim_{k\to\infty} f(x_{m_k}) = \lim_{n\to\infty} f(x_n) = \ell.$$

That is, there exists $c \in [a,b]$ such that $f(c) = \ell = \inf E$, so f achieves an absolute minimum at c.

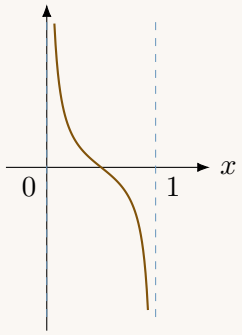
Question

Could we drop the assumption that $f : S \to \mathbb{R}$ is defined particularly with $S = [a,b]$? That is, does S have to be a closed, bounded interval?

Answer. No, the assumption cannot be dropped.

Non-example

$f(x) = \dfrac{1}{x} - \dfrac{1}{1-x}$ on $(0,1)$.



Thus our assumptions are all in fact necessary in this theorem:

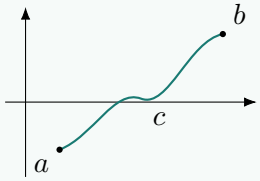
1) a closed and bounded set;

2) a continuous function.

17.3 The bisection method and the intermediate value theorem

Theorem (Bisection Method)

Let $f : [a, b] \to \mathbb{R}$ be continuous. If $f(a) < 0$ and $f(b) > 0$, then there exists $c \in (a, b)$ such that $f(c) = 0$.



Idea of the proof

We will construct sequences $\{a_n\}_{n=1}^{\infty}$, $\{b_n\}_{n=1}^{\infty}$ such that a_n and b_n converge to the same limit, and $f(a_n) \leq 0$, $f(b_n) \geq 0$. Then

$$\lim_{n\to\infty} f(a_n) = \lim_{n\to\infty} f(b_n) = f(c) = 0,$$

since f is continuous (page 94, recalled on page 100).

Proof

Let $a_1 = a$, $b_1 = b$. Then $f(a_1) < 0$, $f(b_1) > 0$. If $f\left(\frac{a_1+b_1}{2}\right) \geq 0$, let $a_2 = a_1$, $b_2 = \frac{a_1+b_1}{2}$. Otherwise $f\left(\frac{a_1+b_1}{2}\right) < 0$, and we let $a_2 = \frac{a_1+b_1}{2}$, $b_2 = b_1$.
Generally, for every $n \in \mathbb{N}$:

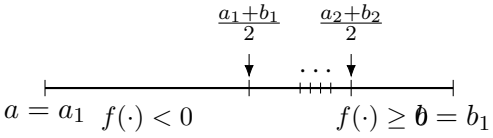
- if $f\left(\frac{a_n+b_n}{2}\right) \geq 0$, choose $b_{n+1} = \frac{a_n+b_n}{2}$ and $a_{n+1} = a_n$;
- if $f\left(\frac{a_n+b_n}{2}\right) < 0$, choose $b_{n+1} = b_n$ and $a_{n+1} = \frac{a_n+b_n}{2}$.

By this way of picking a_n, b_n, we have that, for every $n \in \mathbb{N}$,

1) $a \le a_n \le a_{n+1} \le b_{n+1} \le b_n \le b$;

2) $b_{n+1} - a_{n+1} = \dfrac{b_n - a_n}{2}$;

3) $f(a_n) < 0$, $f(b_n) \ge 0$.



Thus, by 1), $\{a_n\}_n$ is monotone increasing and $\{b_n\}_n$ is monotone decreasing; $\{a_n\}$ is bounded by b and $\{b_n\}$ is bounded by a. Therefore there exist $c, d \in [a, b]$ such that $a_n \to c$ and $b_n \to d$. By 2), $b_n - a_n = \dfrac{1}{2^{n-1}}(b - a)$. Since $\{a_n\}$ and $\{b_n\}$ converge,

$$\lim_{n\to\infty}(b_n - a_n) = \lim_{n\to\infty} b_n - \lim_{n\to\infty} a_n = d - c = \lim_{n\to\infty} \frac{1}{2^{n-1}}(b - a) = 0.$$

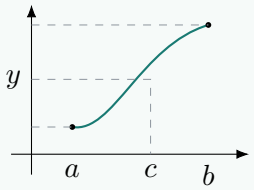
Thus $d - c = 0$, so $d = c$, and hence $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = c = d$. Therefore $c \ne a, b$, so $c \in (a, b)$.
By 3), and since f is continuous, using the theorem on page 94 (recalled on page 100),

$$f(c) = \lim_{n\to\infty} f(a_n) \le 0, \qquad f(c) = \lim_{n\to\infty} f(b_n) \ge 0.$$

Hence $f(c) = 0$ with $c \in (a, b)$.

Theorem (Bolzano's Intermediate Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be continuous. If $f(a) < y < f(b)$ or $f(b) < y < f(a)$, then there exists $c \in (a, b)$ such that $f(c) = y$.



Proof

Consider the case $f(a) < y < f(b)$. Let $g(x) = f(x) - y$. Then $g : [a, b] \to \mathbb{R}$ is continuous (page 99). Thus

$$g(a) = f(a) - y < 0 \qquad \text{and} \qquad g(b) = f(b) - y > 0.$$

By the previous theorem, there exists $c \in (a, b)$ such that $g(c) = 0$. Then $g(c) = f(c) - y = 0$, so there exists $c \in (a, b)$ such that $f(c) = y$.

Theorem: the image of $[a, b]$

Let $f : [a, b] \to \mathbb{R}$ be continuous. Let $c \in [a, b]$ be where f achieves an absolute minimum, and $d \in [a, b]$ be where f achieves an absolute maximum. Then

$$f([a, b]) = [f(c), f(d)].$$

That is, every value between the absolute minimum and the absolute maximum is achieved.

Proof

By the extreme value theorem, since f is continuous, $f([a, b]) \subset [f(c), f(d)]$. We want to show that $[f(c), f(d)] \subset f([a, b])$. By the intermediate value theorem,

$$[f(c), f(d)] \subset f([c, d]) \subset f([a, b]).$$

Therefore $[f(c), f(d)] = f([a, b])$. If $d < c$, then $[f(c), f(d)] \subset f([d, c]) \subset f([a, b])$.

Example

The polynomial $f(x) = x^{2021} + x^{2020} + 9x + 1$ has at least one real root.

Proof

Trivial.

Remark

Another way to state the last theorem is: the image of a continuous function on a closed, bounded interval is a closed, bounded interval.

18 Uniform Continuity and the Definition of the Derivative

18.1 Uniform continuity

Recall

$f : S \to \mathbb{R}$ is continuous at $c \in S$ if, for every $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon, c) > 0$ such that

$$x \in S \text{ and } |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon.$$

Writing $\delta(\varepsilon, c) > 0$ indicates that we are choosing δ in terms of c; see the example below for detail. So δ *can* depend on ε and c.

Example

$f(x) = \dfrac{1}{x}$ is continuous on $(0, 1)$.

Proof

Let $\varepsilon > 0$. Choose $\delta = \min\left\{\frac{c}{2}, \frac{c^2}{2}\varepsilon\right\}$. Then

$$|x-c| < \delta \implies |x-c| < \frac{c}{2} \implies \frac{c}{2} < c - |x-c| \leq |c - |x-c|| \leq |c + (x-c)| = |x|,$$

where the last two inequalities are a property of the absolute value and the reverse triangle inequality. Thus $\frac{1}{|x|} < \frac{2}{c}$. Therefore, for every $c \in (0,1)$,

$$\left|\frac{1}{x} - \frac{1}{c}\right| = \frac{|x-c|}{|x|\,|c|} < \frac{1}{|x|} \cdot \frac{\delta}{|c|} < \frac{2}{c} \cdot \frac{\delta}{c} \leq \frac{2c^2\varepsilon}{2c^2} = \varepsilon.$$

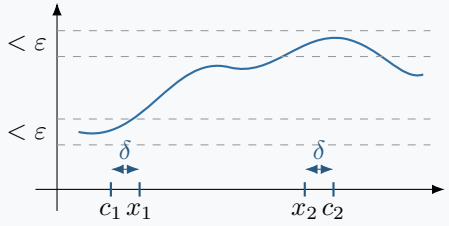
Remark

As shown, for some $f(x)$, if we want to show that f is continuous at c, then we have to choose δ in terms of c, not only ε, to have that $|x-c| < \delta \implies |f(x) - f(c)| < \varepsilon$. For $f(x) = \frac{1}{x}$ on $(0,1)$, δ depends on *both* ε and c.

Definition

Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$. Then f is *uniformly continuous* on S if, for every $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that, for every $c \in S$,

$$x \in S \text{ and } |x-c| < \delta \quad \implies \quad |f(x) - f(c)| < \varepsilon.$$



Remark

Thus, for the definition of uniform continuity, δ only depends on ε.

Theorem

If a function $f : S \to \mathbb{R}$ is uniformly continuous on S, then f is continuous on S.

Proof

Trivial.

Example

$f(x) = x^2$ is uniformly continuous on $[0,1)$.

Proof

Let $\varepsilon > 0$. Choose $\delta = \varepsilon/2$. Then, for $x, c \in [0, 1)$,

$$|x - c| < \delta \quad \Longrightarrow \quad |x - c|\,|x + c| \leq |x - c|(|x| + |c|) < [1 + 1]\delta = 2\delta = \varepsilon.$$

Thus f is uniformly continuous on $[0, 1)$.

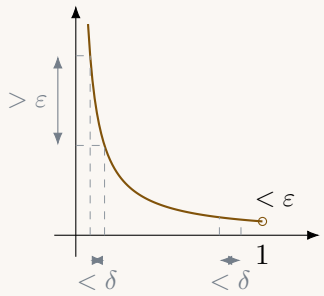
Negation of the definition

$f : S \to \mathbb{R}$ is *not uniformly continuous* on S if there exists $\varepsilon_0 > 0$ such that, for every $\delta > 0$, there exist $x, c \in S$ such that

$$|x - c| < \delta \qquad \text{and} \qquad |f(x) - f(c)| \geq \varepsilon_0.$$

Example

$f(x) = \dfrac{1}{x}$ is not uniformly continuous on $(0, 1)$.



Proof

Choose $\varepsilon_0 = 2$. Let $\delta > 0$. Choose $c = \min\{\delta, \frac{1}{2}\} > 0$ and $x = \dfrac{c}{2}$. Then

$$|x - c| = \frac{c}{2} \leq \frac{\delta}{2} < \delta \qquad \text{and} \qquad \left|\frac{1}{x} - \frac{1}{c}\right| = \left|\frac{2}{c} - \frac{1}{c}\right| = \frac{1}{c} \geq 2.$$

We have shown that there exists $\varepsilon_0 = 2 > 0$ such that, for every $\delta > 0$, there exist $c = \min\{\delta, \frac{1}{2}\}$ and $x = \frac{c}{2}$ such that $|x - c| < \delta$ and $\left|\frac{1}{x} - \frac{1}{c}\right| \geq \varepsilon_0$.

Remark: ε_0 cannot be chosen in terms of δ

Note that we can*not* choose ε_0 in terms of δ!
Counter-example. Consider $f(x) = x$. One can show that it is uniformly continuous on $\mathbb{R}$. Yet suppose we choose $\varepsilon_0 = \frac{\delta}{4}$. Let $\delta > 0$. Then choose $c = 0$, $x = \frac{\delta}{2}$. Then

$$|x - c| = \frac{\delta}{2} < \delta \qquad \text{and} \qquad |f(x) - f(c)| = \frac{\delta}{2} > \frac{\delta}{4} = \varepsilon_0.$$

Then $f(x)$ would *not* be uniformly continuous on $\mathbb{R}$, a contradiction.
Similarly, in the negated ε–M, negated ε–δ, and negated continuity proofs, we can *not* choose the bad ε_0 in terms of M, δ, or anything that comes after it: there exists an ε_0 such that, for

any $M \in \mathbb{N}$, there exists $n \in \mathbb{N}$ such that $n \geq M$ and $|x_n - x| \geq \varepsilon_0$. Because there M comes after our choice of ε_0, we cannot choose ε_0 depending on M.

Example

$f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Proof

Choose $\varepsilon_0 = 1$. Let $\delta > 0$.
How to choose x, c? We want to show that $|x^2 - c^2| \geq 1$ and $|x - c| < \delta$. Suppose $x > c > 0$, say $x = c + \frac{\delta}{2}$. Then

$$|x^2 - c^2| = \left|\left(c + \frac{\delta}{2}\right)^2 - c^2\right| = \left|c^2 + c\delta + \frac{\delta^2}{4} - c^2\right| = c\delta + \frac{\delta^2}{4} \geq c\delta.$$

Thus, as long as $c\delta > 1$, our choice works. The more variables we have, the more space we have to play with the inequalities.
By the Archimedean property, since $\delta > 0$, there exists $M \in \mathbb{N}$ such that $M\delta > 1$. Choose $c = M$ and $x = c + \frac{\delta}{2}$. Then $|x - c| = \frac{\delta}{2} < \delta$ and

$$|x^2 - c^2| = \left|\left(c + \frac{\delta}{2}\right)^2 - c^2\right| = \left|c\delta + \frac{\delta^2}{4}\right| > c\delta > 1 = \varepsilon_0.$$

Thus $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$.

Remark: compact domains

As you may wonder: yes, a finite union of closed bounded intervals is compact. And the extreme value theorem and the intermediate value theorem do not necessarily have to be about a function $f : [a, b] \to \mathbb{R}$; they can be about a continuous $f : S \to \mathbb{R}$ where S is compact (the next theorem as well).
However, note that S being compact does *not* imply that S is a union of closed, bounded intervals. Counter-examples include $S = \{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$, the Cantor set, etc., which are compact but contain no proper intervals.

Theorem: continuity on $[a, b]$ is uniform

Let $f : [a, b] \to \mathbb{R}$. Then f is continuous on $[a, b]$ if and only if f is uniformly continuous on $[a, b]$.

Proof

($\Longleftarrow$) Trivial.
($\Longrightarrow$) By contradiction. Suppose f is continuous and *not* uniformly continuous.
Idea. We will again construct two bounded sequences $\{x_n\}_{n=1}^{\infty}$ and $\{c_n\}_{n=1}^{\infty}$ and use the Bolzano–Weierstrass theorem. (This trick is valid using the definition of limits, continuity, and the theorem about inf and sup.)
Since f is not uniformly continuous, there exists $\varepsilon_0 > 0$ such that, letting $\delta_n = \frac{1}{n}$ for $n \in \mathbb{N}$, for

every $n \in \mathbb{N}$ there exist $x_n, c_n \in [a,b]$ such that

$$|x_n - c_n| < \frac{1}{n} \qquad \text{and} \qquad |f(x_n) - f(c_n)| \geq \varepsilon_0.$$

Then $\{x_n\}_{n=1}^{\infty}$ and $\{c_n\}_{n=1}^{\infty}$ are bounded. By Bolzano–Weierstrass, there exists a subsequence $\{x_{n_k}\}_k$ of $\{x_n\}$ such that

$$\lim_{k\to\infty} x_{n_k} = x \qquad \text{for some } x \in [a,b].$$

Then, by picking the same indices, the subsequence $\{c_{n_k}\}_k$ of $\{c_n\}$ is still bounded by $[a,b]$. Then, by Bolzano–Weierstrass, there exists a subsequence $\{c_{n_{k_j}}\}_j$ such that

$$\lim_{j\to\infty} c_{n_{k_j}} = c \qquad \text{for some } c \in [a,b].$$

Since $x_{n_k} \to x$, the subsequence $\{x_{n_{k_j}}\}_j$ of $\{x_{n_k}\}$ (obtained by picking the same entries as $\{c_{n_{k_j}}\}_j$) also converges to x:

$$\lim_{j\to\infty} x_{n_{k_j}} = x.$$

By our choice of $\{x_n\}$, $\{c_n\}$, since $n_{k_j} \in \mathbb{N}$, we have

$$0 \leq |x_{n_{k_j}} - c_{n_{k_j}}| \leq \frac{1}{n_{k_j}} \leq \frac{1}{k_j} \leq \frac{1}{j}.$$

By the squeeze theorem, as $j \to \infty$, $|x_{n_{k_j}} - c_{n_{k_j}}| \to 0$. By the algebraic operations of limits of sequences,

$$\lim_{j\to\infty} |x_{n_{k_j}} - c_{n_{k_j}}| = \left|\lim_{j\to\infty} x_{n_{k_j}} - \lim_{j\to\infty} c_{n_{k_j}}\right| = |x - c| = 0.$$

Thus $x = c$; that is, $x_{n_{k_j}} \to c$ and $c_{n_{k_j}} \to c$.
Then, by the theorem on page 94 (recalled on page 100), since f is continuous,

$$f(x_{n_{k_j}}) \to f(c), \qquad f(c_{n_{k_j}}) \to f(c).$$

Thus

$$\lim_{j\to\infty} |f(x_{n_{k_j}}) - f(c_{n_{k_j}})| = |f(c) - f(c)| = 0.$$

By our choice of $\{x_n\}$, $\{c_n\}$, we have $|f(x_n) - f(c_n)| \geq \varepsilon_0$ for every $n \in \mathbb{N}$. Since $n_{k_j} \in \mathbb{N}$, and limits respect inequalities,

$$0 = \lim_{j\to\infty} |f(x_{n_{k_j}}) - f(c_{n_{k_j}})| \geq \varepsilon_0 > 0 \quad \Longrightarrow \quad 0 > 0,$$

a contradiction.

18.2 The derivative

Definition

Let I be an interval (it does not have to be closed or bounded), let $f : I \to \mathbb{R}$, and let $c \in I$; thus c is a cluster point of $I \setminus \{c\}$. We say f is *differentiable at* c if the limit

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. If f is differentiable at c, we put

$$f'(c) := \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

If f is differentiable at c for every $c \in I$, then $f'(x)$ gives a *new function* from I to $\mathbb{R}$, which we denote by f' or $\dfrac{df}{dx}$.

Example

Let $\alpha \in \mathbb{R}$ and $n \in \mathbb{N} \cup \{0\}$. Then $f(x) = \alpha x^n$ is differentiable, and for every $c \in \mathbb{R}$, $f'(c) = n\alpha c^{n-1}$.

Proof

Consider

$$(x - c) \sum_{j=0}^{n-1} x^{n-1-j} c^j = \sum_{j=0}^{n-1} x^{n-j} c^j - \sum_{j=0}^{n-1} x^{n-(1+j)} c^{j+1}.$$

Choosing the dummy variable $l = j + 1$,

$$\begin{aligned}(x - c) \sum_{j=0}^{n-1} x^{n-1-j} c^j &= \sum_{j=0}^{n-1} x^{n-j} c^j - \sum_{l=1}^{n} x^{n-l} c^l \\ &= x^{n-0} c^0 - x^{n-n} c^n \\ &= x^n - c^n.\end{aligned}$$

Then

$$\begin{aligned}f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = \lim_{x \to c} \frac{\alpha x^n - \alpha c^n}{x - c} &= \alpha \lim_{x \to c} \sum_{j=0}^{n-1} x^{n-1-j} c^j \\ = \alpha \sum_{j=0}^{n-1} c^{n-1-j} c^j = \alpha c^{n-1} \sum_{j=0}^{n-1} 1 &= \alpha n c^{n-1}.\end{aligned}$$

Example

$f(x) = ax + b$ is differentiable on $\mathbb{R}$, and for every $c \in \mathbb{R}$, $f'(c) = a$.

Proof

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}=\lim_{x\to c}\frac{(ax+b)-(ac+b)}{x-c}=a\lim_{x\to c}\frac{x-c}{x-c}=a, \qquad \text{for every } c\in\mathbb{R}.$$

19 Weierstrass's Example of a Continuous and Nowhere Differentiable Function

Remark: the derivative in terms of sequences

By the theorems on page 87–page 88,

$$L=\lim_{x\to c}\frac{f(x)-f(c)}{x-c}$$

if and only if, for every sequence $\{x_n\}$ in $I\setminus\{c\}$ such that $x_n\to c$, we have

$$L=\lim_{n\to\infty}\frac{f(x_n)-f(c)}{x_n-c}.$$

Theorem: differentiability implies continuity

If $f:I\to\mathbb{R}$ is differentiable at $c\in I$, then f is continuous at c. In other words,

$$\lim_{x\to c}f(x)=f(c).$$

(Since c has to be a cluster point of I for f to be differentiable at c, this is continuity at c by the theorem on page 94.)

Proof

We want to show that $\lim_{x\to c}f(x)=f(c)$. Since f is differentiable at c, the limit

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}$$

exists, and $c\in I$ is a cluster (limit) point of I. Then

$$\begin{aligned}\lim_{x\to c}f(x)&=\lim_{x\to c}[f(x)-f(c)+f(c)]=\lim_{x\to c}\left\{\left[\frac{f(x)-f(c)}{x-c}\right](x-c)+f(c)\right\}\\&=\left[\lim_{x\to c}\frac{f(x)-f(c)}{x-c}\right]\lim_{x\to c}(x-c)+\lim_{x\to c}f(c)=0\cdot f'(c)+f(c)=f(c).\end{aligned}$$

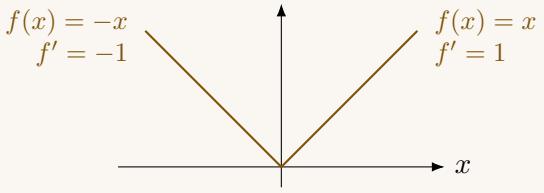
Thus f is continuous at c.

Question: does the converse hold?

Does continuity imply differentiability? No.

Example: $|x|$ is continuous but not differentiable at 0

The function $f(x) = |x|$ is continuous, but it is not differentiable at $c = 0$.



Proof

We will show that

1) $f(x) = |x|$ is continuous at 0;

2) $f(x) = |x|$ is not differentiable at 0.

1) Let $\varepsilon > 0$. Choose $\delta = \varepsilon$. Then

$$|x - 0| < \delta \implies |x| < \delta \implies |f(x) - f(0)| = ||x| - |0|| = |x| < \delta = \varepsilon.$$

2) We will show that there exists a sequence $\{x_n\}_n$ in $I \setminus \{0\}$ such that $x_n \to 0$ and

$$\left\{\frac{f(x_n) - f(c)}{x_n - c}\right\}_{n=1}^{\infty}$$

does not converge. Choose $x_n = \dfrac{(-1)^n}{n}$. Thus, for every $n \in \mathbb{N}$, $x_n \neq 0$, and $x_n \to 0$. Moreover,

$$\frac{f(x_n) - f(0)}{x_n - 0} = \frac{\left|\frac{(-1)^n}{n}\right| - |0|}{\frac{(-1)^n}{n}} = (-1)^n.$$

Therefore

$$\left\{\frac{f(x_n) - f(c)}{x_n - c}\right\}_{n=1}^{\infty} \quad \text{does not converge.}$$

Note

One can also show this by showing that

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} \neq \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Then, by the theorem on page 91, the limit $\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$ does not exist.

Question: must a continuous function be differentiable somewhere?

If $f : \mathbb{R} \to \mathbb{R}$ is continuous, does there have to exist $c \in \mathbb{R}$ such that f is differentiable at c? No (Weierstrass).
Goal. Construct a continuous $f : \mathbb{R} \to \mathbb{R}$ which is nowhere differentiable.

Theorem I

1) For every $x, y \in \mathbb{R}$, $|\cos x - \cos y| \leq |x - y|$.

2) Let $c \in \mathbb{R}$. Then, for every $K \in \mathbb{N}$, there exists $y \in \left(c + \frac{\pi}{K}, c + \frac{3\pi}{K}\right)$ such that

$$|\cos(Kc) - \cos(Ky)| \geq 1.$$

Proof

1) Recall that on page 96 we showed that $|\sin x - \sin y| \leq |x - y|$. Then

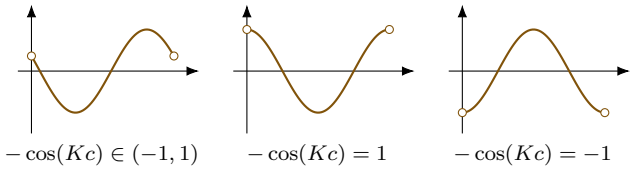
$$|\cos x - \cos y| = \left|\sin(x + \tfrac{\pi}{2}) - \sin(y + \tfrac{\pi}{2})\right| \leq \left|(x + \tfrac{\pi}{2}) - (y + \tfrac{\pi}{2})\right| = |x - y|.$$

2) The function $f(x) = \cos(Kx)$ is $\frac{2\pi}{K}$-periodic. Thus we have

$$[-1, 1] \subset f\left(\left(c + \tfrac{\pi}{K}, c + \tfrac{3\pi}{K}\right)\right) \qquad (*)$$

if

$$\cos(K \cdot (c + \tfrac{\pi}{K})) = \cos\left(K\left(c + \tfrac{3\pi}{K}\right)\right) = -\cos(Kc) \neq \pm 1.$$



$-\cos(Kc) \in (-1, 1)$ $\quad -\cos(Kc) = 1$ $\quad -\cos(Kc) = -1$

And $(-1, 1]$ or $[-1, 1)$ is contained in $f\left(\left(c + \frac{\pi}{K}, c + \frac{3\pi}{K}\right)\right)$ if $-\cos(Kc) = -1$ or 1, respectively $(*)$. In each case the required y exists by $(*)$:

- if $\cos(Kc) = \pm 1$, choose y such that $\cos(Ky) = 0$;
- if $\cos(Kc) \in [0, 1)$, choose y such that $\cos(Ky) = -1$;
- if $\cos(Kc) \in (-1, 0)$, choose y such that $\cos(Ky) = 1$.

Hence, for every $c \in \mathbb{R}$ and every $K \in \mathbb{N}$, there exists $y \in \left(c + \frac{\pi}{K}, c + \frac{3\pi}{K}\right)$ such that

$$|\cos(Kc) - \cos(Ky)| \geq 1.$$

Theorem II

For every $a, b, c \in \mathbb{R}$,

$$|a + b + c| \geq |a| - |b| - |c|.$$

Proof

We have

$$|a| = |a + b + c - (b + c)| \leq |a + b + c| + |b + c| \leq |a + b + c| + |b| + |c|.$$

Hence $|a| - |b| - |c| \leq |a + b + c|$.

Theorem III

1) For every $x \in \mathbb{R}$, the series
$$\sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}$$
is *absolutely convergent*. (Then, since for every $x \in \mathbb{R}$ the series $\sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}$ gives a real number, we can define a function from it in 2).)

2) The function $f : \mathbb{R} \to \mathbb{R}$, where
$$f(x) = \sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k},$$
is bounded and continuous.

Proof

1) For every $k \in \mathbb{N}$ and every $x \in \mathbb{R}$, $|\cos(160^k x)| \leq 1$, so
$$\left|\frac{\cos(160^k x)}{4^k}\right| \leq \left(\frac{1}{4}\right)^k.$$
The series $\sum_{k=0}^{\infty} \left(\frac{1}{4}\right)^k$ converges by the convergence of geometric series. By the comparison test, since for every $k \in \mathbb{N}$
$$0 \leq \left|\frac{\cos(160^k x)}{4^k}\right| \leq \left(\frac{1}{4}\right)^k,$$
the series $\sum_{k=0}^{\infty} \left|\frac{\cos(160^k x)}{4^k}\right|$ converges.
2) For every $x \in \mathbb{R}$,
$$|f(x)| = \left|\sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}\right| \leq \sum_{k=0}^{\infty} \frac{|\cos(160^k x)|}{4^k} \leq \sum_{k=0}^{\infty} 4^{-k} = \frac{1}{1-r} = \frac{4}{3}.$$
Thus f is bounded (see the definition on page 100).
Now we want to show that f is continuous over $\mathbb{R}$ (using the theorems on page 100 and page 94). Suppose $c \in \mathbb{R}$ and the sequence $\{x_n\}$ in $\mathbb{R}$ converges to c. Since, for every $n \in \mathbb{N}$,
$$|f(x_n) - f(c)| \leq |f(x_n)| + |f(c)| \leq \frac{4}{3} + \frac{4}{3} = \frac{8}{3},$$
the sequence $\{|f(x_n) - f(c)|\}_{n=1}^{\infty}$ is bounded. We want to show that $f(x_n) \to f(c)$. By the theorem on page 53, since $\{|f(x_n) - f(c)|\}_n$ is bounded and $|f(x_n) - f(c)| \geq 0$, we have
$$\lim_{n\to\infty} |f(x_n) - f(c)| = 0 \iff \limsup_{n\to\infty} |f(x_n) - f(c)| = 0.$$
(Indeed, $\liminf \leq \limsup$. If $\limsup = 0$, then, since $|f(x_n) - f(c)| \geq 0$, we get $\liminf \geq \inf \geq 0$, so $0 \leq \liminf \leq \limsup = 0$, and hence $\liminf = \limsup = 0$. Then, by the theorem on page 53, the limit is 0.)
Claim. For every $\varepsilon > 0$, $\limsup_{n\to\infty} |f(x_n) - f(c)| < \varepsilon$.

Let $\varepsilon > 0$. Since $\sum_{k=0}^{\infty} 4^{-k}$ converges, the series $\sum_{k=0}^{\infty} 4^{-k}$ is Cauchy, so for every $\varepsilon > 0$ there exists M_0 such that

$$\left| \sum_{k=M_0+1}^{\infty} 4^{-k} \right| < \frac{\varepsilon}{2}.$$

Then

$$\begin{aligned}
\limsup_{n\to\infty} |f(x_n) - f(c)| &= \limsup_{n\to\infty} \left| \sum_{k=0}^{\infty} \frac{\cos(160^k x_n)}{4^k} - \sum_{k=0}^{\infty} \frac{\cos(160^k c)}{4^k} \right| \\
&= \limsup_{n\to\infty} \Bigg| \sum_{k=0}^{M_0} \frac{\cos(160^k x_n)}{4^k} - \sum_{k=0}^{M_0} \frac{\cos(160^k c)}{4^k} \\
&\qquad + \sum_{k=M_0+1}^{\infty} \frac{\cos(160^k x_n)}{4^k} - \sum_{k=M_0+1}^{\infty} \frac{\cos(160^k c)}{4^k} \Bigg| \\
&\le \limsup_{n\to\infty} \sum_{k=0}^{M_0} \frac{|\cos(160^k x_n) - \cos(160^k c)|}{4^k} \\
&\qquad + \limsup_{n\to\infty} \sum_{k=M_0+1}^{\infty} \frac{|\cos(160^k x_n)| + |\cos(160^k c)|}{4^k} \\
&\le \limsup_{n\to\infty} \sum_{k=0}^{M_0} 4^{-k} \, |160^k x_n - 160^k c| + \limsup_{n\to\infty} \sum_{k=M_0+1}^{\infty} \frac{2}{4^k} \\
&< \limsup_{n\to\infty} \sum_{k=0}^{M_0} \left[(4^{-1})(160)\right]^k |x_n - c| + \frac{\varepsilon}{2} \cdot 2 \\
&= \left(\sum_{k=0}^{M_0} 40^k \right) \limsup_{n\to\infty} |x_n - c| + \varepsilon.
\end{aligned}$$

In the third inequality, the first sum was estimated by Theorem I, 1), and the second by the fact that $|\cos|$ is bounded by 1. The factor $\sum_{k=0}^{M_0} 40^k$ is a fixed number, as the sum is finite. Since our $\{x_n\}_{n=1}^{\infty}$ converges to c, $\lim_{n\to\infty} |x_n - c| = 0$, so by the theorem on page 53,

$$\limsup_{n\to\infty} |x_n - c| = \liminf_{n\to\infty} |x_n - c| = 0.$$

Thus

$$\limsup_{n\to\infty} |f(x_n) - f(c)| < \left(\sum_{k=0}^{M_0} 40^k \right) \limsup_{n\to\infty} |x_n - c| + \varepsilon = 0 + \varepsilon = \varepsilon.$$

Then $|f(x_n) - f(c)| \to 0$, so $f(x_n) \to f(c)$, and therefore f is continuous at c.

Theorem (Weierstrass)

The function

$$f(x) = \sum_{k=0}^{\infty} \frac{\cos(160^k x)}{4^k}$$

is nowhere differentiable.

Proof

Let $c \in \mathbb{R}$.
Idea. Like what we did to show that $|x|$ is not differentiable at 0, we will construct $\{x_n\}_{n=1}^{\infty}$ in $\mathbb{R} \setminus \{c\}$ such that $x_n \to c$ and $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_{n=1}^{\infty}$ does not converge. We will even show that $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_{n=1}^{\infty}$ is unbounded.
By Theorem I, 2), choosing $K = 160^n$ with $n \in \mathbb{N}$, we get: for every $n \in \mathbb{N}$ (with $K = 160^n$) there exists

$$x = x_n \in \left(\frac{\pi}{160^n} + c,\ \frac{3\pi}{160^n} + c\right) \quad \text{such that} \quad |\cos(160^n c) - \cos(160^n x_n)| \geq 1.$$

Since $c + \frac{\pi}{160^n} < x < c + \frac{3\pi}{160^n}$, we have $|x - c| < \frac{3\pi}{160^n}$, so by the squeeze theorem $|x_n - c| \to 0$, that is, $x_n \to c$. Now we want to show that $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_n$ is unbounded.
Let

$$f_k(x) = \frac{\cos(160^k x)}{4^k}$$

(for simplification); then $f(x) = \sum_{k=0}^{\infty} f_k(x)$.
Let $n \in \mathbb{N}$. We will split $f(c) - f(x_n)$ into three parts, and then use Theorem II:

$$\begin{aligned} f(c) - f(x_n) &= \sum_{k=0}^{\infty} f_k(c) - \sum_{k=0}^{\infty} f_k(x_n) \\ &= \left[\sum_{k=0}^{n-1} f_k(c) - \sum_{k=0}^{n-1} f_k(x_n)\right] + [f_n(c) - f_n(x_n)] + \left[\sum_{k=n+1}^{\infty} f_k(c) - \sum_{k=n+1}^{\infty} f_k(x_n)\right] \\ &= \sum_{k=0}^{n-1} [f_k(c) - f_k(x_n)] + [f_n(c) - f_n(x_n)] + \sum_{k=n+1}^{\infty} [f_k(c) - f_k(x_n)]. \end{aligned}$$

Let

$$a_n = f_n(c) - f_n(x_n), \qquad b_n = \sum_{k=0}^{n-1} [f_k(c) - f_k(x_n)], \qquad c_n = \sum_{k=n+1}^{\infty} [f_k(c) - f_k(x_n)].$$

Then, by Theorem II,

$$|f(c) - f(x_n)| = |a_n + b_n + c_n| \geq |a_n| - |b_n| - |c_n| \qquad \text{for every } n \in \mathbb{N}.$$

Now we will show that $|a_n| - |b_n| - |c_n|$ is unbounded.
The term a_n. By our choice of $\{x_n\}_{n=1}^{\infty}$,

$$|a_n| = |f_n(c) - f_n(x_n)| = \left|\frac{\cos(160^n c)}{4^n} - \frac{\cos(160^n x_n)}{4^n}\right| = 4^{-n}|\cos(160^n c) - \cos(160^n x_n)| \geq 4^{-n}.$$

The term b_n. We have

$$\begin{aligned} |b_n| &= \left|\sum_{k=0}^{n-1} [f_k(c) - f_k(x_n)]\right| \leq \sum_{k=0}^{n-1} |f_k(c) - f_k(x_n)| \\ &\leq \sum_{k=0}^{n-1} \left|\frac{\cos(160^k c) - \cos(160^k x_n)}{4^k}\right| = \sum_{k=0}^{n-1} 4^{-k}|\cos(160^k c) - \cos(160^k x_n)| \\ &\leq \sum_{k=0}^{n-1} 4^{-k}|160^k c - 160^k x_n| = |x_n - c| \cdot \sum_{k=0}^{n-1} 40^k, \end{aligned}$$

where the last inequality is Theorem I, 1). Since $|x_n - c| \le \frac{3\pi}{160^n}$, and using the geometric-sum formula shown by induction on page 57 and page 59,

$$\begin{aligned}|b_n| &\le \sum_{k=0}^{n-1} \frac{3\pi}{160^n} \cdot 40^k = \frac{3\pi}{160^n} \cdot \frac{1 - 40^{(n-1)+1}}{1 - 40} = \frac{3\pi}{160^n} \cdot \frac{40^n - 1}{39} \\ &\le \frac{3 \cdot 4 \cdot 40^n}{160^n \cdot 39} = \frac{4^{-n+1}}{13}.\end{aligned}$$

The term c_n. We have

$$\begin{aligned}|c_n| &= \left| \sum_{k=n+1}^{\infty} [f_k(c) - f_k(x_n)] \right| = \left| \sum_{k=n+1}^{\infty} \frac{\cos(160^k c) - \cos(160^k x_n)}{4^k} \right| \\ &\le \sum_{k=n+1}^{\infty} 4^{-k} |\cos(160^k c) - \cos(160^k x_n)|.\end{aligned}$$

Continuing,

$$\begin{aligned}|c_n| &\le \sum_{k=n+1}^{\infty} 4^{-k} \Big[|\cos(160^k c)| + |\cos(160^k x_n)| \Big] \\ &\le 2 \sum_{k=n+1}^{\infty} 4^{-k} = 2 \cdot 4^{-n-1} \cdot \frac{4}{3} \le 4^{-n} \cdot \frac{2}{3}.\end{aligned}$$

Thus

$$|f(c) - f(x_n)| \ge |a_n| - |b_n| - |c_n| \ge 4^{-n} \left(1 - \frac{4}{13} - \frac{2}{3} \right) = \frac{4^{-n}}{39}.$$

Therefore

$$\frac{|f(c) - f(x_n)|}{|c - x_n|} \ge \frac{160^n}{3\pi} \cdot \frac{4^{-n}}{39} = \frac{40^n}{117\pi},$$

which is unbounded. Thus $\left\{ \frac{f(x_n) - f(c)}{x_n - c} \right\}_{n=1}^{\infty}$ is unbounded, and hence it does not converge.

Remark: the Weierstrass functions

The class of functions given by Weierstrass,

$$f(x) = \sum_{k=0}^{\infty} \frac{\cos(a^k x)}{b^k} \qquad \text{where } \frac{a}{b} > 1,$$

is continuous on $\mathbb{R}$ yet nowhere differentiable.

20 Differentiation Rules, Rolle's Theorem, and the Mean Value Theorem

Recall: the derivative

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

Theorem: linearity, product rule, and quotient rule

Let $f, g : I \to \mathbb{R}$ be differentiable at $c \in I$. Then:

1) (Linearity) For every $\alpha \in \mathbb{R}$, $\alpha f + g : I \to \mathbb{R}$ is differentiable at c, and

$$(\alpha f + g)'(c) = \alpha f'(c) + g'(c).$$

2) (Product rule) $fg : I \to \mathbb{R}$ is differentiable at c, and

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

3) (Quotient rule) If $g(x) \neq 0$ for every $x \in I$, then $\frac{f}{g} : I \to \mathbb{R}$ is differentiable at c, and

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}.$$

Proof

1) We compute

$$\lim_{x \to c} \frac{(\alpha f + g)(x) - (\alpha f + g)(c)}{x - c} = \lim_{x \to c} \left[\alpha \frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c}\right] = \alpha f'(c) + g'(c).$$

2) Since g is differentiable at c, it is continuous at c: $\lim_{x \to c} g(x) = g(c)$. Then

$$\begin{aligned}
\lim_{x \to c} \frac{f(x)g(x) - f(c)g(c)}{x - c} &= \lim_{x \to c} \frac{f(x)g(x) - f(c)g(x) + f(c)g(x) - f(c)g(c)}{x - c} \\
&= \lim_{x \to c} \left[\frac{f(x) - f(c)}{x - c} g(x) + \frac{g(x) - g(c)}{x - c} \cdot f(c)\right] \\
&= f'(c)g(c) + g'(c)f(c).
\end{aligned}$$

3) *Claim.*

$$\left[\frac{1}{g(x)}\right]' = \frac{-g'(x)}{[g(x)]^2}.$$

Indeed,

$$\lim_{x \to c} \frac{\frac{1}{g(x)} - \frac{1}{g(c)}}{x - c} = \lim_{x \to c} \frac{g(c) - g(x)}{g(x)\, g(c)\, (x - c)} = \frac{-g'(c)}{[g(c)]^2}.$$

Then, by 2),

$$\left(\frac{f}{g}\right)'(c) = \left(f \cdot \frac{1}{g}\right)'(c) = \left[f' \cdot \frac{1}{g} + f \cdot \left(\frac{1}{g}\right)'\right](c)$$

$$= \frac{f'(c)}{g(c)} + f(c) \cdot \frac{-g'(c)}{[g(c)]^2}$$
$$= \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}.$$

Theorem (Chain rule)

Let I_1, I_2 be two intervals, let $g : I_1 \to I_2$ be differentiable at $c \in I_1$, and let $f : I_2 \to \mathbb{R}$ be differentiable at $g(c)$. Then $f \circ g : I_1 \to \mathbb{R}$ is differentiable at c, and

$$(f \circ g)'(c) = f'(g(c))\, g'(c).$$

Idea of the proof

$$\frac{f(g(x)) - f(g(c))}{g(x) - g(c)} \cdot \frac{g(x) - g(c)}{x - c} = \frac{f(g(x)) - f(g(c))}{x - c}.$$

The only problem is that we have to be careful when $x = c$.

Proof

Let $h(x) = f(g(x))$ and $d = g(c)$. We want to show that $h'(c) = f'(d)\, g'(c)$. Let

$$u(y) = \begin{cases} \dfrac{f(y) - f(d)}{y - d}, & y \neq d, \\ f'(d), & y = d, \end{cases} \qquad v(x) = \begin{cases} \dfrac{g(x) - g(c)}{x - c}, & x \neq c, \\ g'(c), & x = c. \end{cases}$$

Then

$$f(y) - f(d) = u(y)\,(y - d) \quad \text{for every } y \in I_2, \qquad \text{and}$$
$$g(x) - g(c) = v(x)\,(x - c) \quad \text{for every } x \in I_1.$$

(Note that $y = d$ gives $0 = u(y) \cdot 0$, and $x = c$ gives $0 = v(x) \cdot 0$.)
Note that u is continuous at d and v is continuous at c, by our construction of u and v; that is, $\lim_{y \to d} u(y) = u(d)$ and $\lim_{x \to c} v(x) = v(c)$. Indeed,

$$\lim_{y \to d} u(y) = \lim_{y \to d} \frac{f(y) - f(d)}{y - d} = f'(d) = u(d).$$

Similarly,

$$\lim_{x \to c} v(x) = \lim_{x \to c} \frac{g(x) - g(c)}{x - c} = g'(c) = v(c).$$

Thus u is continuous at d, and v is continuous at c. Then

$$\begin{aligned} h(x) - h(c) &= f(g(x)) - f(g(c)) \\ &= u(g(x))\,[g(x) - g(c)] \\ &= u(g(x))\, v(x)\, (x - c), \end{aligned}$$

so

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \lim_{x \to c} u(g(x))\, v(x).$$

Since $\lim_{y\to d} u(y) = u(d)$ and $\lim_{x\to c} v(x) = v(c)$,

$$\lim_{x\to c} \frac{h(x)-h(c)}{x-c} = \lim_{x\to c} u(g(x))\, v(x) = u(g(c))\, v(c) = f'(g(c))\, g'(c),$$

by our definition of $u(y)$ and $v(x)$.

20.1 Mean Value Theorem

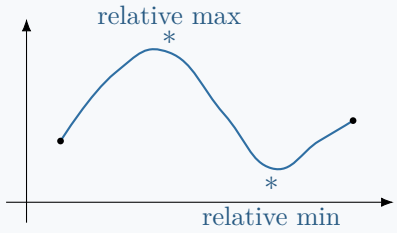
Definition (relative minimum and maximum)

Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$. Then f has a *relative maximum* at $c \in S$ if there exists $\delta > 0$ such that, for every $x \in S$,

$$|x-c| < \delta \quad \Longrightarrow \quad f(x) \le f(c).$$

The function f has a *relative minimum* at $c \in S$ if there exists $\delta > 0$ such that, for every $x \in S$,

$$|x-c| < \delta \quad \Longrightarrow \quad f(x) \ge f(c).$$



Theorem: the derivative vanishes at an interior extremum

If $f : [a,b] \to \mathbb{R}$ has a relative maximum or a relative minimum at $c \in (a,b)$, and f is differentiable at c, then

$$f'(c) = 0.$$

Proof

Relative maximum. Since f has a relative maximum at c, there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subset (a,b)$ and, for every $x \in (c-\delta, c+\delta)$, $f(x) \le f(c)$. We will construct two sequences as follows. Let

$$x_n = c - \frac{\delta}{2n} \in (c-\delta, c), \qquad n = 1, 2, \ldots;$$

then $x_n \to c$ as $n \to \infty$, and since f is differentiable at c,

$$f'(c) = \lim_{n\to\infty} \frac{f(x_n)-f(c)}{x_n - c} \ge 0.$$

Let

$$y_n = c + \frac{\delta}{2n} \in (c, c+\delta);$$

then $y_n \to c$, so

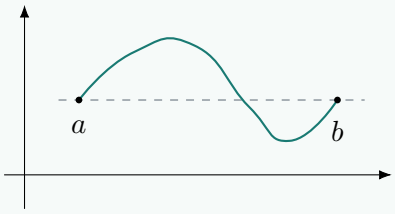
$$f'(c) = \lim_{n\to\infty} \frac{f(y_n)-f(c)}{y_n - c} \le 0.$$

Therefore $f'(c) = 0$.
For a relative minimum, simply invert the inequalities.

Theorem (Rolle)

Let $f : [a, b] \to \mathbb{R}$ be continuous and differentiable on (a, b), with $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$



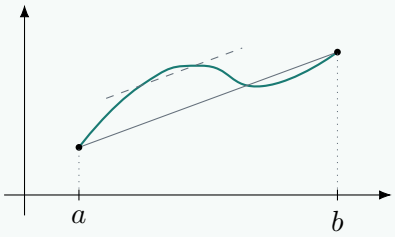
Proof

Let $K = f(a) = f(b)$. Since f is continuous on $[a, b]$, by the extreme value theorem (page 101) there exist $c_1, c_2 \in [a, b]$ such that f achieves an absolute maximum and an absolute minimum at c_1 and c_2, respectively.
Then $f(c_1) > K$ implies $c_1 \in (a, b)$, and hence $f'(c_1) = 0$ by the previous theorem. Similarly, $f(c_2) < K$ implies $c_2 \in (a, b)$, and hence $f'(c_2) = 0$ by the previous theorem.
If $f(c_1) \leq K \leq f(c_2)$, then $f(x) = K$ for every $x \in [a, b]$, so $f'(c) = 0$ for every $c \in (a, b)$.

Theorem (Mean Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be continuous, and let f be differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)\,(b - a).$$



Proof

Let $g : [a, b] \to \mathbb{R}$ be

$$g(x) = f(x) - f(b) + \frac{f(b) - f(a)}{b - a}\,(b - x).$$

Then $g(a) = g(b) = 0$. Note that g is continuous on $[a, b]$ and g is differentiable on (a, b). By Rolle's theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$, so

$$0 = f'(c) - \frac{f(b) - f(a)}{b - a} \quad \Longrightarrow \quad f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem: zero derivative means constant

If $f : I \to \mathbb{R}$ is differentiable and $f'(x) = 0$ for every $x \in I$, then f is constant; that is, $f(x) = C$ for some $C \in \mathbb{R}$.

Proof

Let $a, b \in I$ with $a < b$. Then f is continuous on $[a, b]$ and differentiable on (a, b). Therefore there exists $c \in (a, b)$ such that

$$f(b) - f(a) = (b - a)\, f'(c) = 0.$$

Hence $f(b) = f(a)$ for all $a, b \in I$ with $a < b$.

Theorem: monotonicity and the sign of the derivative

Let $f : I \to \mathbb{R}$ be differentiable. Then:

1) f is increasing if and only if $f'(x) \geq 0$ for every $x \in I$;

2) f is decreasing if and only if $f'(x) \leq 0$ for every $x \in I$.

Proof

1) ($\Longleftarrow$) Suppose $f'(x) \geq 0$ for every $x \in I$. Let $a, b \in I$ with $a < b$. By the Mean Value Theorem, there exists $c \in (a, b)$ such that

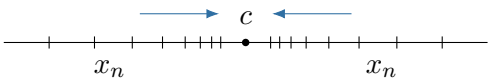
$$f(b) - f(a) = (b - a)\, f'(c) \geq 0 \quad \Longrightarrow \quad f(a) \leq f(b).$$

($\Longrightarrow$) Suppose f is increasing. Let $c \in I$, and let $\{x_n\}$ be a sequence in I such that $x_n \to c$ and $x_n < c$ for every $n \in \mathbb{N}$. Then, for every n, $f(x_n) - f(c) \leq 0$, so

$$\frac{f(x_n) - f(c)}{x_n - c} \geq 0 \quad \Longrightarrow \quad f'(c) = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} \geq 0.$$

Next, let $\{x_n\}$ be a sequence in I such that $x_n > c$ for every n. Then, for every n, $f(x_n) - f(c) \geq 0$, so

$$\frac{f(x_n) - f(c)}{x_n - c} \geq 0 \quad \Longrightarrow \quad f'(c) = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} \geq 0.$$



Hence, in either case, we have that $f'(c) \geq 0$ for every $c \in I$.
2) f is decreasing if and only if $-f$ is increasing, which, by 1), holds if and only if $(-f)'(x) = -f'(x) \geq 0$ for every $x \in I$, that is, $f'(x) \leq 0$ for every $x \in I$.

20.2 Important Remark on Continuity and Differentiability

Remark: a false idea

The following idea is false!

$$f \text{ continuous at } c \iff \lim_{x\to c} f(x) = f(c) \iff \lim_{x\to c^+} f(x) = \lim_{x\to c^-} f(x) = f(c);$$

$$f \text{ differentiable at } c \iff \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \text{ exists} \iff \lim_{x\to c^+} \frac{f(x)-f(c)}{x-c} = \lim_{x\to c^-} \frac{f(x)-f(c)}{x-c}.$$

Do **not** ignore the assumptions of those theorems!

Remark: one-sided limits need cluster points on both sides

For $\lim_{x\to c^-}$ or $\lim_{x\to c^+}$ to exist, we need c to be a cluster point of both $S \cap (c, +\infty)$ and $S \cap (-\infty, c)$, yet our definition of the limit and of continuity/differentiability used at most the fact that c is a cluster point of $S \setminus \{c\}$. Continuity only needs f to be defined at c!

Remark: when the statements above are true

The only condition under which the statements above are true is that c is a cluster point of both $S \cap (c, +\infty)$ and $S \cap (-\infty, c)$. Therefore we absolutely can say that $f : [0, 1] \to \mathbb{R}$, $f(x) = x$, is differentiable and continuous at $x = 0$ and $x = 1$.

Remark: the lie from calculus

This is the lie from calculus: that a limit exists if and only if $\lim_{x\to c^+}$ and $\lim_{x\to c^-}$ exist and are equal. (This is true only if c is a cluster point of both $S \cap (c, +\infty)$ and $S \cap (-\infty, c)$.)

Remark: continuity at an isolated point

Similarly, we now consider f continuous at c if c is not a cluster point (a single point).

21 Taylor's Theorem and the Definition of Riemann Sums

21.1 Taylor's theorem

Remark

Taylor's theorem is essentially the mean value theorem for higher order derivatives.

Definition (n-times differentiable)

We say $f : I \to \mathbb{R}$ is *n-times differentiable* on $J \subset I$ if, for every $c \in J$,

$$f'(c),\ f''(c),\ \ldots,\ f^{(n)}(c)$$

exist.

Theorem (Taylor)

Suppose $f : [a, b] \to \mathbb{R}$ is continuous and is n-times continuously differentiable, i.e. $f', f'', \ldots, f^{(n)}$ are continuous on $[a, b]$. Assume $f^{(n+1)}$ exists on (a, b). Given $x_0, x \in [a, b]$, there exists $c \in (x_0, x)$ such that

$$f(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}.$$

Denote

$$P_n(x) := \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k,$$

the *n-th order Taylor polynomial* of f at x_0, and

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1},$$

the *n-th order remainder term*.

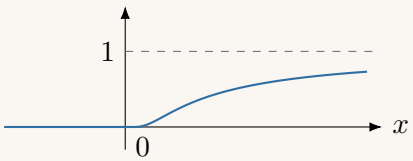
Remark

$R_n(x)$ is *not* necessarily so small that it can be ignored! In some cases, $R_n(x)$ can be large.

Example

Let

$$f(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \le 0. \end{cases}$$



In fact, f is n-times differentiable for every $n \in \mathbb{N}$, and

$$\boxed{f^{(n)}(0) = 0} \qquad \text{for every } n.$$

Yet if we consider the n-th order Taylor polynomial and the n-th order remainder term of it at $x_0 = 0$:

$$\begin{aligned} f(x) &= \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}, \qquad c \text{ between } x_0 \text{ and } x, \\ &= \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(0)(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1} \\ &= \underbrace{0}_{P_n(x)} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}}_{R_n(x)}. \end{aligned}$$

In other words, when we choose $x_0 = 0$ for this $f(x)$, the value of $f(x)$ entirely relies on the remainder term and on the c between x_0 and x. In this case, we cannot throw $R_n(x)$ away and only use $P_n(x)$ to approximate $f(x)$.

Proof of Taylor's theorem

Let $x, x_0 \in [a, b]$. If $x = x_0$, then this is true for every c.
Suppose $x \neq x_0$. Let

$$M_{x,x_0} = \frac{f(x) - P_n(x)}{(x - x_0)^{n+1}}.$$

So

$$f(x) = P_n(x) + [f(x) - P_n(x)] = P_n(x) + M_{x,x_0}(x - x_0)^{n+1}.$$

Here $f(x)$ is fixed, not a function. We want to show that there exists $c \in (a, b)$ such that

$$M_{x,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!}.$$

The significance of $P_n(x)$ is that it agrees with $f(x)$ up to the n-th order derivative right at the point $x = x_0$! For $0 \leq k \leq n$, we have that

$$f^{(k)}(x_0) = P_n^{(k)}(x_0).$$

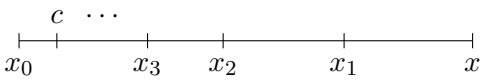
(This can be proven.)
Now we will construct a function to which we will apply the mean value theorem. Let

$$g(s) = f(s) - P_n(s) - M_{x,x_0}(s - x_0)^{n+1}.$$

Then

$$\begin{aligned}
g(x_0) &= f(x_0) - P_n(x_0) - M_{x,x_0}(x_0 - x_0)^{n+1} = 0, \\
g'(x_0) &= f'(x_0) - P_n'(x_0) - M_{x,x_0}(n+1)(x_0 - x_0)^n = 0, \\
&\vdots \\
g^{(n)}(x_0) &= f^{(n)}(x_0) - P_n^{(n)}(x_0) - M_{x,x_0}(n+1)!\,(x_0 - x_0) = 0.
\end{aligned}$$

Now note that $g(x) = 0$ by our definition of M_{x,x_0}. By the mean value theorem, there exists $x_1 \in (x_0, x)$ such that $g'(x_1) = 0$.



Since $g'(x_1) = 0$ and $g'(x_0) = 0$, by the mean value theorem there exists $x_2 \in (x_0, x_1)$ such that $g''(x_2) = 0$.
Reiterating this process, we eventually have an $x_n \in (x_0, x_{n-1})$ such that $g^{(n)}(x_n) = 0$. Then, since $g^{(n)}(x_0) = 0$, by the mean value theorem there exists $c \in (x_0, x_n)$ such that

$$g^{(n+1)}(c) = 0. \qquad (*)$$

Compute the following $(n+1)$-th order derivatives:

$$\frac{d^{n+1}}{ds^{n+1}} M_{x,x_0}(s-x_0)^{n+1} = M_{x,x_0}(n+1)! \cdot 1,$$

$$P_n^{(n+1)}(c) = \left[\sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(s-x_0)^k\right]^{(n+1)}(c) = 0+0+\cdots+0 = 0.$$

Therefore, plugging $s=c$ into $g^{(n+1)}(s)$ (we cannot do this with $f(x)$, because x is chosen to be fixed):

$$\begin{aligned} 0 \overset{(*)}{=} g^{(n+1)}(c) &= \left[f - P_n - M_{x,x_0}(s-x_0)^{n+1}\right]^{(n+1)}(c) \\ &= f^{(n+1)}(c) - P_n^{(n+1)}(c) - M_{x,x_0}(n+1)! \\ &= f^{(n+1)}(c) - 0 - M_{x,x_0}(n+1)!. \end{aligned}$$

Hence we have shown that there exists $c \in (x_0, x_n) \subset (x_0, x)$ such that

$$\frac{f^{(n+1)}(c)}{(n+1)!} = M_{x,x_0}.$$

That is, there exists $c \in (x_0, x)$ such that

$$\frac{f^{(n+1)}(c)}{(n+1)!} = \frac{f(x) - P_n(x)}{(x-x_0)^{n+1}} \quad \Longrightarrow \quad f(x) = P_n(x) + \frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}.$$

Remark

As we have seen in the example of

$$f(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \le 0, \end{cases}$$

the Taylor polynomial does give us a way to approximate a function near a point x_0, but the approximation is *not necessarily good*; i.e. $f(x)$ heavily relies on the remainder term $R_n(x)$.

21.2 The second derivative test and L'Hôpital's rule

Theorem (Second derivative test)

Suppose $f : (a,b) \to \mathbb{R}$ is twice continuously differentiable. If $x_0 \in (a,b)$, $f'(x_0) = 0$, and $f''(x_0) > 0$, then f has a *strict relative minimum* at x_0.

Proof

Since f'' is continuous at x_0,

$$\lim_{c \to x_0} f''(c) = f''(x_0) > 0.$$

Claim. There exists $\delta > 0$ such that, for every $c \in (x_0 - \delta, x_0 + \delta)$, $f''(c) > 0$.
(Proof by contradiction.) Suppose that for every $\delta > 0$ there exists $c \in (x_0 - \delta, x_0 + \delta)$ with $f''(c) \leq 0$. Choose $\delta_n = 1/n$, $n \in \mathbb{N}$. Then for every $n \in \mathbb{N}$ there exists $c_n \in (x_0 - \delta_n, x_0 + \delta_n)$ such that

$$0 < |c_n - x_0| < \frac{1}{n}, \qquad f''(c_n) \leq 0.$$

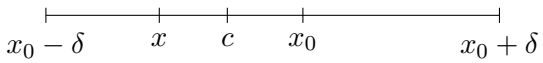
By the squeeze theorem, $\{c_n\}_{n=1}^{\infty}$ converges to x_0. Since f'' is continuous at x_0, by the theorem on page 94 (see also page 100),

$$\lim_{n\to\infty} f''(c_n) = f''(x_0).$$

Since limits of sequences preserve inequalities, $f''(x_0) \leq 0$, a contradiction. Thus our claim is true.
Now we want to show: for every $x \in (x_0-\delta, x_0+\delta)$, $f(x) > f(x_0)$ if $x \neq x_0$. Let $x \in (x_0-\delta, x_0+\delta)$. By Taylor's theorem, there exists c between x_0 and x such that

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2.$$

Since c is between x_0 and x, we have $c \in (x_0 - \delta, x_0 + \delta)$, and hence $f''(c) > 0$.



Since $f'(x_0) = 0$, then

$$\begin{aligned} f(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2 \\ &= f(x_0) + \frac{f''(c)}{2}(x - x_0)^2 > f(x_0) + 0 \qquad \text{if } x \neq x_0, \end{aligned}$$

and $f(x) = f(x_0)$ when $x = x_0$. Therefore x_0 is a strict relative minimum of f.

Theorem (L'Hôpital's rule for right-hand limits)

Let $-\infty \leq a < b \leq \infty$. Let $f, g : \mathbb{R} \to \mathbb{R}$ be differentiable on (a, b) such that $g'(x) \neq 0$ for every $x \in (a, b)$, and suppose that

$$\lim_{x\to a^+} f(x) = 0 = \lim_{x\to a^+} g(x) \qquad \text{or} \qquad \lim_{x\to a^+} g(x) = \pm\infty.$$

Then: if

$$\lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L \in \mathbb{R} \cup \{-\infty, \infty\}, \qquad \text{then} \qquad \lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

Lemma (Generalized mean value theorem)

If $f, g : \mathbb{R} \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b), then there exists a point $c \in (a, b)$ such that

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

Proof

Let

$$h(t) = \big[f(b) - f(a)\big]g(t) - \big[g(b) - g(a)\big]f(t), \qquad t \in [a, b].$$

Then we want to show that there exists $c \in (a, b)$ such that $h'(c) = 0$. Note that

$$h(a) = \big[f(b) - f(a)\big]g(a) - \big[g(b) - g(a)\big]f(a) = f(b)g(a) - g(b)f(a),$$
$$h(b) = \big[f(b) - f(a)\big]g(b) - \big[g(b) - g(a)\big]f(b) = f(b)g(a) - f(a)g(b).$$

Thus $h(a) = h(b)$. Note that $h(t)$ is continuous on $[a, b]$ and differentiable on (a, b). Therefore, by Rolle's theorem, there exists $c \in (a, b)$ such that $h'(c) = 0$, that is,

$$\big[f(b) - f(a)\big]g'(c) = \big[g(b) - g(a)\big]f'(c) \qquad \text{for some } c \in (a, b).$$

Proof of L'Hôpital's rule

If $a < \alpha < \beta < b$, then Rolle's theorem implies $g(\beta) \neq g(\alpha)$. By the generalized mean value theorem, there exists $u \in (\alpha, \beta)$ such that

$$\frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} = \frac{f'(u)}{g'(u)}.$$

Case 1.1: $L \in \mathbb{R}$, $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$.
Let $\varepsilon > 0$. Then there exists $c \in (a, b)$ such that

$$a < u < c \quad (\delta = c - a) \quad \Longrightarrow \quad \left|\frac{f'(u)}{g'(u)} - L\right| < \varepsilon. \tag{1}$$

Thus, for all α, β such that $a < \alpha < \beta \leq c$, since

$$\frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} = \frac{f'(u)}{g'(u)} \qquad \text{for some } a < u < c,$$

we have that

$$\left|\frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} - L\right| < \varepsilon.$$

We need $\beta \leq c$ because $u \in (\alpha, \beta)$: if $\beta > c$, then u could be bigger than c, and then (1) no longer holds, because u needs to be in (a, c).
If we take the limit $\alpha \to a^+$, then, since limits preserve loose inequalities, we have

$$\left|\frac{f(\beta)}{g(\beta)} - L\right| \leq \varepsilon, \qquad \beta \in (a, c],$$

that is,

$$L - \varepsilon \leq \frac{f(\beta)}{g(\beta)} \leq L + \varepsilon.$$

In summary, since $a < \alpha < \beta \leq c < b$, we have shown that there exists $c \in (a, b)$ such that, for every β with $a < \beta < c$,

$$\left|\frac{f(\beta)}{g(\beta)} - L\right| \leq \varepsilon.$$

Therefore $\dfrac{f(x)}{g(x)} \to L$ as $x \to a^+$.
Case 1.2: $L \in \{-\infty, \infty\}$, $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$.
$L = +\infty$: Let $M > 0$. Then there exists $c \in (a, b)$ such that

$$a < u < c \quad \Longrightarrow \quad \frac{f'(u)}{g'(u)} > M.$$

Thus again, for $a < \alpha < \beta \leq c$, we have that

$$\frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} = \frac{f'(u)}{g'(u)} > M.$$

As $\alpha \to a^+$, we have that there exists $c \in (a, b)$ such that, for every $\beta \in (a, c)$,

$$\frac{f(\beta)}{g(\beta)} \geq M.$$

Thus $\dfrac{f(\beta)}{g(\beta)} \to +\infty$.
Similarly, if $L = -\infty$, we can show $\dfrac{f(\beta)}{g(\beta)} \to -\infty$ by showing $\dfrac{f(\beta)}{g(\beta)} \leq -M$ for $\beta \in (a, c)$.
Case 2.1: $L \in \mathbb{R}$, $\lim_{x\to a^+} g(x) = \infty$. (The case $\lim_{x\to a^+} g(x) = -\infty$ is analogous.)
We will show this for $L > 0$; the cases $L = 0$ and $L < 0$ are similar. Suppose $L > 0$. Let $\varepsilon > 0$. Then there exists $c \in (a, b)$ such that

$$\left| \frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} - L \right| < \varepsilon \qquad \text{for } a < \alpha < \beta \leq c.$$

Since $g(x) \to \infty$ as $x \to a^+$, and by the fact that c is close to a, we may assume $g(c) > 0$. Choose $\beta = c$. Then we have that

$$L - \varepsilon < \frac{f(c) - f(\alpha)}{g(c) - g(\alpha)} < L + \varepsilon \qquad \text{for } \alpha \in (a, c). \tag{*}$$

Since $\lim_{\alpha\to a^+} \dfrac{g(c)}{g(\alpha)} = 0$, we may assume that

$$0 < \frac{g(c)}{g(\alpha)} < 1 \qquad \text{for every } \alpha \in (a, c).$$

We may assume this because c is chosen by ourselves: we can make c as close to a as we would like. Then

$$\frac{g(\alpha) - g(c)}{g(\alpha)} = 1 - \frac{g(c)}{g(\alpha)} > 0.$$

Multiplying this into $(*)$, we get

$$(L - \varepsilon)\left(1 - \frac{g(c)}{g(\alpha)}\right) < \frac{f(\alpha)}{g(\alpha)} - \frac{f(c)}{g(\alpha)} < (L + \varepsilon)\left(1 - \frac{g(c)}{g(\alpha)}\right).$$

Since $\lim_{\alpha\to a^+}\dfrac{g(c)}{g(\alpha)}=0$ and $\lim_{\alpha\to a^+}\dfrac{f(c)}{g(\alpha)}=0$, for every $\delta\in(0,1)$ there exists $d\in(a,c)$ such that, for every $\alpha\in(a,d)$,

$$\left|\frac{g(c)}{g(\alpha)}\right|<\delta \qquad \text{and} \qquad \left|\frac{f(c)}{g(\alpha)}\right|<\delta.$$

Since $g(c)>0$, we have that, for every $\alpha\in(a,d)$,

$$0<\frac{g(c)}{g(\alpha)}<\delta \qquad \text{and} \qquad \frac{|f(c)|}{g(\alpha)}<\delta.$$

Thus

$$(L-\varepsilon)(1-\delta)-\delta<\frac{f(\alpha)}{g(\alpha)}<(L+\varepsilon)+\delta.$$

Choose $\delta=\min\{1,\varepsilon,\varepsilon/(|L|+1)\}$; then $\delta\le 1$, $\delta\le\varepsilon$, and $\delta\le\varepsilon/(|L|+1)$. Thus

$$\begin{aligned}(L-\varepsilon)(1-\delta)-\delta &\ge L-L\delta-\varepsilon+\varepsilon\delta-\delta\ge L-\varepsilon-(L+1)\frac{\varepsilon}{|L|+1}\\ &\ge L-2\varepsilon,\end{aligned}$$

since we have assumed $L>0$. Similarly, we can eventually show that

$$L-2\varepsilon<\frac{f(\alpha)}{g(\alpha)}<L+2\varepsilon.$$

Thus, since ε is arbitrary, $\dfrac{f(\alpha)}{g(\alpha)}\to L$.

Case 2.2: $L\in\{-\infty,\infty\}$, $\lim_{x\to a^+}g(x)=\infty$.

Let $M>1$. Again there exists $c\in(a,b)$ such that

$$\frac{f'(u)}{g'(u)}>M \qquad \text{for every } u\in(a,c).$$

Then again

$$\frac{f(\beta)-f(\alpha)}{g(\beta)-g(\alpha)}>M \qquad \text{for } a<\alpha<\beta\le c.$$

Since $g(x)\to\infty$, we may assume $g(c)>0$,

$$\frac{|f(c)|}{g(\alpha)}<\frac{1}{2}, \qquad \text{and} \qquad 0<\frac{g(c)}{g(\alpha)}<\frac{1}{2} \qquad \text{for every } \alpha\in(a,c).$$

Choose $\beta=c$. Multiplying by $1-\dfrac{g(c)}{g(\alpha)}>\dfrac{1}{2}$, we get

$$\frac{f(\alpha)-f(c)}{g(\alpha)}>M\left(1-\frac{g(c)}{g(\alpha)}\right)>\frac{1}{2}M.$$

Thus

$$\frac{f(\alpha)}{g(\alpha)}>\frac{1}{2}M+\frac{f(c)}{g(\alpha)}>\frac{1}{2}(M-1) \qquad \text{for every } \alpha\in(a,c).$$

Since $M>1$ is arbitrary, $\dfrac{f(\alpha)}{g(\alpha)}\to\infty$. The argument is analogous for $L=-\infty$.

21.3 The Riemann integrals

Remark

Although it is interpreted to be the area underneath a curve, we did not define the Riemann integral via the area under a curve.

Remark

Lebesgue integration is some sort of a *completion* of Riemann integration.

Question: what functions do have Riemann integrals?

Answer: functions that are in some sense continuous everywhere. However, we need measure theory to rigorously define what it means to be "almost continuous everywhere". Therefore we will only define the Riemann integral for (completely) continuous functions for now.

Definition/Notation

Denote

$$C([a,b]) := \{f : [a,b] \to \mathbb{R} \mid f \text{ is continuous}\},$$

the set of all continuous functions on $[a,b]$.

Definition (partition, norm, tag)

A *partition* of $[a,b]$ is a finite set, denoted by $\underline{x}$, where

$$\underline{x} = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

(The elements of $\underline{x}$ are $x_0, x_1, \ldots, x_n$.) The *norm* of $\underline{x}$, denoted by $\|\underline{x}\|$, is the number

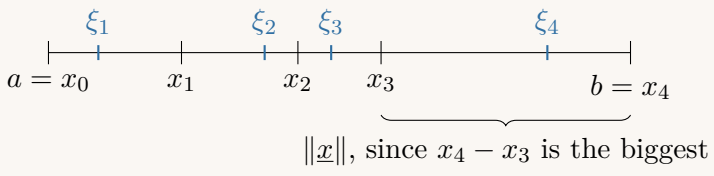
$$\|\underline{x}\| := \max\{x_1 - x_0,\ x_2 - x_1,\ \ldots,\ x_n - x_{n-1}\}.$$

A *tag* of the partition $\underline{x}$ is a finite set $\underline{\xi} := \{\xi_1, \ldots, \xi_n\}$, where

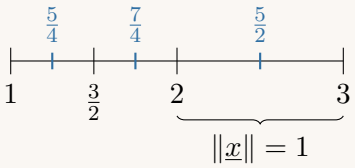
$$a = x_0 \le \xi_1 \le x_1 \le \xi_2 \le x_2 \le \cdots \le x_{n-1} \le \xi_n \le x_n = b.$$

If a partition $\underline{x}$ is tagged by $\underline{\xi}$, we say the pair $(\underline{x}, \underline{\xi})$ is a *tagged partition*.

Example ($n = 4$)



Example



Here

$$\underline{x} = \{1, \tfrac{3}{2}, 2, 3\}, \qquad \underline{\xi} = \{\tfrac{5}{4}, \tfrac{7}{4}, \tfrac{5}{2}\},$$

and

$$\|\underline{x}\| = \max\{\tfrac{3}{2} - 1,\ 2 - \tfrac{3}{2},\ 3 - 2\} = 1.$$

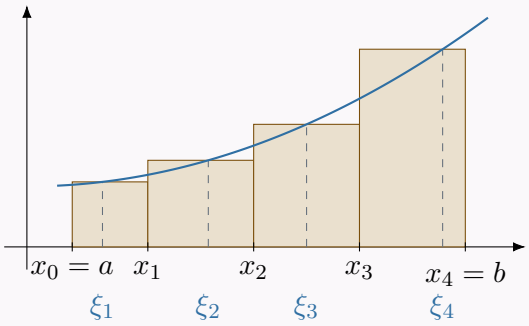
Definition (Riemann sum)

Let $f \in C([a,b])$, and let $(\underline{x}, \underline{\xi})$ be a tagged partition. The *Riemann sum* of f associated to $(\underline{x}, \underline{\xi})$ is the number

$$S_f(\underline{x}, \underline{\xi}) := \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}).$$

Interpretation

We may interpret the Riemann sum as an approximate area.



22 The Riemann Integral of a Continuous Function

22.1 Existence and uniqueness of the Riemann integral

Recall

- $\underline{x} = \{a = x_0 < x_1 < \cdots < x_n = b\}$ is a *partition*.
- $\|\underline{x}\| = \max\{x_1 - x_0,\ x_2 - x_1,\ \ldots,\ x_n - x_{n-1}\}$ is the *norm* of the partition.
- $\underline{\xi} = \{\xi_1, \ldots, \xi_n\}$ with $x_0 \le \xi_1 \le x_1 \le \xi_2 \le x_2 \le \cdots \le \xi_n \le x_n$ is a *tag*.
- $(\underline{x}, \underline{\xi})$ is a *tagged partition*.

- $S_f(\underline{x}, \underline{\xi}) = \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1})$ is the *Riemann sum* of f associated to $(\underline{x}, \underline{\xi})$.

Theorem (Riemann integral)

Let $f \in C([a,b])$. Then there exists a unique number, denoted by

$$\int_a^b f(x)\,dx \in \mathbb{R},$$

with the following property: for all sequences of tagged partitions $\{(\underline{x}^r, \underline{\xi}^r)\}_{r=1}^{\infty}$ such that $\|\underline{x}^r\| \to 0$ as $r \to \infty$, we have

$$\lim_{r\to\infty} S_f(\underline{x}^r, \underline{\xi}^r) = \int_a^b f(x)\,dx.$$

Remark

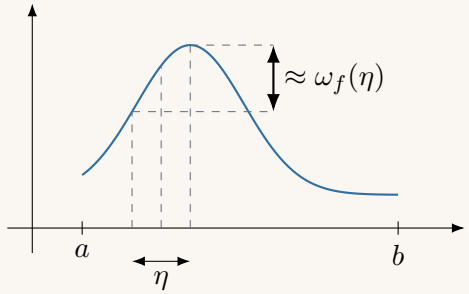
Uniqueness becomes obvious after the proof of existence, since the limit of a sequence is unique.

Definition (modulus of continuity)

For $f \in C([a,b])$ and $\eta > 0$, we define

$$\omega_f(\eta) = \sup\{|f(x) - f(y)| : x, y \text{ such that } |x - y| \le \eta\}.$$

Example



The arrow is approximately $\omega_f(\eta)$, if we choose η to be the length shown on the graph.

Example

Let $f(x) = ax + b$. *Claim:* $\omega_f(\eta) = |a|\eta$.
Then $|f(x) - f(y)| = |a|\,|x - y|$. If $|x - y| \le \eta$, then $|f(x) - f(y)| \le |a|\eta$. And since $|x - y| = \eta$ implies $|f(x) - f(y)| = |a|\eta$, we have

$$|a|\eta = \sup\{|f(x) - f(y)| : |x - y| \le \eta\} = \omega_f(\eta).$$

Theorem I

For every $f \in C([a,b])$, $\lim_{\eta\to 0} \omega_f(\eta) = 0$; i.e. for every $\varepsilon > 0$ there exists $\delta > 0$ such that, for every $\eta < \delta$, $\omega_f(\eta) < \varepsilon$.

Proof

Let $\varepsilon > 0$. Since $f \in C([a,b])$, by the theorem on page 109, f is uniformly continuous. Hence there exists $\delta_0 > 0$ such that, for every x, y with $|x-y| < \delta_0$, $|f(x) - f(y)| < \varepsilon/2$.
Then choose $\delta = \delta_0$. Let $\eta < \delta$. Then if $|x-y| \le \eta < \delta = \delta_0$, then $|f(x) - f(y)| < \varepsilon/2$. Therefore $\varepsilon/2$ is an upper bound for $\{|f(x) - f(y)| : |x-y| \le \eta\}$. By the least upper bound property of $\mathbb{R}$, the supremum $\omega_f(\eta)$ exists, and thus

$$\omega_f(\eta) \le \frac{\varepsilon}{2} < \varepsilon.$$

Remark

By the definition of the modulus of continuity, we have, for every $x, y \in [a,b]$,

$$|f(x) - f(y)| \le \omega_f(|x-y|).$$

Theorem II

If $(\underline{x}, \underline{\xi})$ and $(\underline{x}', \underline{\xi}')$ are tagged partitions such that $\underline{x} \subset \underline{x}'$, then, if $f \in C([a,b])$,

$$\left|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')\right| \le \omega_f(\|\underline{x}\|)(b-a).$$

If such, we say $\underline{x}'$ is a *refinement* of $\underline{x}$, which is obtained by adding more partition points.

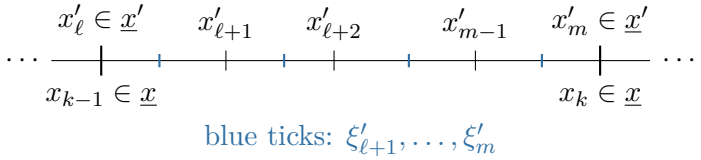
Proof

Suppose $\underline{x} = \{x_0, \ldots, x_n\}$ and $\underline{\xi} = \{\xi_1, \ldots, \xi_n\}$. For $k = 1, \ldots, n$, note that $\underline{x}'$ has more partition points than $\underline{x}$. Let

$$\underline{y}^{(k)} = \{x_{k-1} = x'_\ell < x'_{\ell+1} < \cdots < x'_m = x_k\},$$
$$\underline{\zeta}^{(k)} = \{\xi'_{\ell+1}, \xi'_{\ell+2}, \ldots, \xi'_m\}.$$

Therefore $\underline{y}^{(k)} \subset \underline{x}'$ and $\underline{\zeta}^{(k)} \subset \underline{\xi}'$. Then

$$(\underline{x}', \underline{\xi}') = \bigcup_{k=1}^{n} (\underline{y}^{(k)}, \underline{\zeta}^{(k)}).$$



blue ticks: $\xi'_{\ell+1}, \ldots, \xi'_m$

Hence

$$S_f(\underline{x}', \underline{\xi}') = \sum_{k=1}^{n} S_f(\underline{y}^{(k)}, \underline{\zeta}^{(k)}).$$

Then we compute, for $k = 1, \ldots, n$,

$$\begin{aligned}
&|f(\xi_k)(x_k - x_{k-1}) - S_f(\underline{y}^{(k)}, \underline{\zeta}^{(k)})| \qquad (*)\\
&= \left| f(\xi_k)(x_k - x_{k-1}) - \sum_{i=\ell+1}^{m} f(\xi_i')(x_i' - x_{i-1}') \right| \\
&= \left| f(\xi_k)(x_m' - x_\ell') - \sum_{i=\ell+1}^{m} f(\xi_i')(x_i' - x_{i-1}') \right| \\
&= \Big| f(\xi_k)(x_m' - x_{m-1}' + x_{m-1}' - x_{m-2}' + \cdots + x_{\ell+1}' - x_\ell') \\
&\qquad\qquad - \sum_{i=\ell+1}^{m} f(\xi_i')(x_i' - x_{i-1}') \Big| \\
&= \left| \sum_{i=\ell+1}^{m} [f(\xi_k) - f(\xi_i')](x_i' - x_{i-1}') \right| \\
&\le \sum_{i=\ell+1}^{m} |f(\xi_k) - f(\xi_i')|(x_i' - x_{i-1}') \\
&\le \sum_{i=\ell+1}^{m} \omega_f(|\xi_k - \xi_i'|)(x_i' - x_{i-1}').
\end{aligned}$$

By the definition of the modulus of continuity, since $\xi_k, \xi_i' \in [x_{k-1}, x_k]$, we have $|\xi_k - \xi_i'| \le |x_k - x_{k-1}|$, and hence $\omega_f(|\xi_k - \xi_i'|) \le \omega_f(|x_k - x_{k-1}|)$. Thus

$$\begin{aligned}
(*) &\le \sum_{i=\ell+1}^{m} \omega_f(|\xi_k - \xi_i'|)(x_i' - x_{i-1}') \\
&\le \sum_{i=\ell+1}^{m} \omega_f(|x_k - x_{k-1}|)(x_i' - x_{i-1}') \\
&= \omega_f(|x_k - x_{k-1}|)(x_k - x_{k-1}).
\end{aligned}$$

Again, since

$$|x_k - x_{k-1}| \le \|\underline{x}\| = \sup\{x_1 - x_0,\ \ldots,\ x_k - x_{k-1},\ \ldots,\ x_n - x_{n-1}\},$$

we have $\omega_f(|x_k - x_{k-1}|) \le \omega_f(\|\underline{x}\|)$. Thus

$$(*) \le \omega_f(\|\underline{x}\|)(x_k - x_{k-1}). \qquad (**)$$

Then

$$\begin{aligned}
|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| &= \left| \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) - \sum_{k=1}^{n} S_f(\underline{y}^{(k)}, \underline{\zeta}^{(k)}) \right| \\
&\le \sum_{k=1}^{n} |f(\xi_k)(x_k - x_{k-1}) - S_f(\underline{y}^{(k)}, \underline{\zeta}^{(k)})|.
\end{aligned}$$

By $(**)$, it follows that

$$|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| \le \sum_{k=1}^{n} \omega_f(\|\underline{x}\|)(x_k - x_{k-1}) = \omega_f(\|\underline{x}\|)(b - a).$$

Theorem III

For any two tagged partitions $(\underline{x}, \underline{\xi})$ and $(\underline{x}', \underline{\xi}')$ of $[a,b]$, and $f \in C([a,b])$, we have

$$|S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| \leq [\omega_f(\|\underline{x}\|) + \omega_f(\|\underline{x}'\|)](b-a).$$

Proof

Let $\underline{x}'' = \underline{x} \cup \underline{x}'$, and let $\underline{\xi}''$ be a tag of $\underline{x}''$. By Theorem II, since $\underline{x} \subset \underline{x}''$ and $\underline{x}' \subset \underline{x}''$,

$$\begin{aligned} |S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}', \underline{\xi}')| &\leq |S_f(\underline{x}, \underline{\xi}) - S_f(\underline{x}'', \underline{\xi}'')| \\ &\quad + |S_f(\underline{x}'', \underline{\xi}'') - S_f(\underline{x}', \underline{\xi}')| \\ &\leq \omega_f(\|\underline{x}\|)(b-a) + \omega_f(\|\underline{x}'\|)(b-a) \\ &= [\omega_f(\|\underline{x}\|) + \omega_f(\|\underline{x}'\|)](b-a). \end{aligned}$$

Proof of the Riemann integral theorem

We now prove the theorem stated on page 133. Let $\{(\underline{y}^{(r)}, \underline{\zeta}^{(r)})\}_{r=1}^{\infty}$ be a sequence of tagged partitions of $[a,b]$ such that $\|\underline{y}^{(r)}\| \to 0$ as $r \to \infty$.

Claim I. $\{S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)})\}_{r=1}^{\infty}$ is Cauchy. By the theorem on page 56, $\{S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)})\}_{r=1}^{\infty}$ is therefore convergent.

Proof of Claim I. Let $\varepsilon > 0$. By Theorem I, there exists $\delta > 0$ such that, for every $\eta < \delta$, $\omega_f(\eta) < \dfrac{\varepsilon}{2(b-a)}$. Since $\|\underline{y}^{(r)}\| \to 0$, there exists $M_0 \in \mathbb{N}$ such that, for every $r \geq M_0$, $\|\underline{y}^{(r)}\| < \delta$. Thus, for every $r \geq M_0$,

$$\omega_f(\|\underline{y}^{(r)}\|) < \frac{\varepsilon}{2(b-a)}. \qquad (*)$$

We want to show: for every $\varepsilon > 0$ there exists $M \in \mathbb{N}$ such that, for every $\ell, r \geq M$,

$$|S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)}) - S_f(\underline{y}^{(\ell)}, \underline{\zeta}^{(\ell)})| < \varepsilon$$

(the definition of a Cauchy sequence on page 54). Choose $M = M_0$. Then, for every $r, \ell \geq M_0$, by Theorem III,

$$\begin{aligned} |S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)}) - S_f(\underline{y}^{(\ell)}, \underline{\zeta}^{(\ell)})| &\leq [\omega_f(\|\underline{y}^{(r)}\|) + \omega_f(\|\underline{y}^{(\ell)}\|)](b-a) \\ &< \left[\frac{\varepsilon}{2(b-a)} + \frac{\varepsilon}{2(b-a)}\right](b-a) = \varepsilon. \end{aligned}$$

Thus $\{S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)})\}_{r=1}^{\infty}$ is Cauchy, and hence converges. This proves Claim I.

Let

$$L := \lim_{r \to \infty} S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)}).$$

Claim II. Let $\{(\underline{x}^{(r)}, \underline{\xi}^{(r)})\}_{r=1}^{\infty}$ be any sequence of tagged partitions with $\|\underline{x}^{(r)}\| \to 0$. Then

$$\lim_{r \to \infty} S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = L.$$

By the triangle inequality and Theorem III,

$$\begin{aligned} 0 \le |S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) - L| &\le |S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) - S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)})| + |S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)}) - L| \\ &\le \Big[\underbrace{\omega_f(\|\underline{x}^{(r)}\|)}_{\to 0 \text{ (Thm I)}} + \underbrace{\omega_f(\|\underline{y}^{(r)}\|)}_{\to 0 \text{ (Thm I)}}\Big](b-a) \\ &\quad + \underbrace{|S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)}) - L|}_{\to 0 \text{ (definition of } L)}. \end{aligned}$$

Therefore, by the squeeze theorem,

$$\lim_{r\to\infty} S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = L.$$

In other words, for any sequence of tagged partitions $(\underline{x}^{(r)}, \underline{\xi}^{(r)})$ such that $\|\underline{x}^{(r)}\| \to 0$, the Riemann sums converge to the same limit L, which we denote by the Riemann integral of f on $[a,b]$:

$$\int_a^b f(x)\,dx.$$

22.2 Properties of Riemann integrals

Notation

We will use the abbreviation $\int_a^b f$ for $\int_a^b f(x)\,dx$.

Theorem (Linearity)

If $f, g \in C([a,b])$ and $\alpha \in \mathbb{R}$, then

$$\int_a^b (\alpha f + g) = \alpha \int_a^b f + \int_a^b g.$$

Proof

Let $\{(\underline{x}^{(r)}, \underline{\xi}^{(r)})\}_r$ be a sequence of tagged partitions such that $\|\underline{x}^{(r)}\| \to 0$. Then, by the definition of Riemann sums,

$$S_{\alpha f+g}(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = \alpha S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) + S_g(\underline{x}^{(r)}, \underline{\xi}^{(r)}).$$

Hence

$$\begin{aligned} \int_a^b (\alpha f + g) &= \lim_{r\to\infty} S_{\alpha f+g}(\underline{x}^{(r)}, \underline{\xi}^{(r)}) \\ &= \lim_{r\to\infty} [\alpha S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) + S_g(\underline{x}^{(r)}, \underline{\xi}^{(r)})] \\ &= \alpha \int_a^b f + \int_a^b g. \end{aligned}$$

23 The Fundamental Theorem of Calculus, Integration by Parts, and the Change of Variable Formula

23.1 Basic properties of the Riemann integral

Recall: the Riemann integral

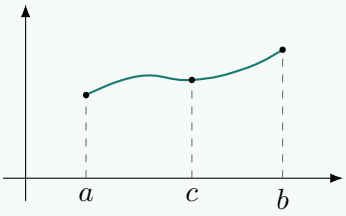
For every $f \in C([a,b])$ there exists a unique number $\int_a^b f$ with the property that, for every sequence of tagged partitions $\{(\underline{x}^{(r)}, \underline{\xi}^{(r)})\}_{r=1}^{\infty}$ with $\|\underline{x}^{(r)}\| \to 0$, we have

$$\lim_{r\to\infty} S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = \int_a^b f.$$

Theorem (additivity)

If $f \in C([a,b])$ and $a < c < b$, then

$$\int_a^b f = \int_a^c f + \int_c^b f.$$



Proof

Let $\{(\underline{y}^{(r)}, \underline{\zeta}^{(r)})\}_{r=1}^{\infty}$ and $\{(\underline{z}^{(r)}, \underline{\eta}^{(r)})\}_{r=1}^{\infty}$ be tagged partitions of $[a,c]$ and $[c,b]$ respectively, such that $\|\underline{y}^{(r)}\| \to 0$ and $\|\underline{z}^{(r)}\| \to 0$. Let

$$\underline{x}^{(r)} = \underline{y}^{(r)} \cup \underline{z}^{(r)}, \qquad \underline{\xi}^{(r)} = \underline{\zeta}^{(r)} \cup \underline{\eta}^{(r)}.$$

Thus $\{(\underline{x}^{(r)}, \underline{\xi}^{(r)})\}_{r=1}^{\infty}$ is a sequence of tagged partitions of $[a,b]$. Then

$$\|\underline{x}^{(r)}\| \le \|\underline{y}^{(r)}\| + \|\underline{z}^{(r)}\| \to 0.$$

Then

$$\begin{aligned}
\int_a^b f &= \lim_{r\to\infty} S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) \\
&= \lim_{r\to\infty} \left[S_f(\underline{y}^{(r)}, \underline{\zeta}^{(r)}) + S_f(\underline{z}^{(r)}, \underline{\eta}^{(r)}) \right] \\
&= \int_a^c f + \int_c^b f.
\end{aligned}$$

Theorem (bounds for the integral)

Let $f \in C([a,b])$, and let

$$m_f = \inf\{f(x) \mid x \in [a,b]\} = \min f \in \mathbb{R}, \qquad M_f = \sup\{f(x) \mid x \in [a,b]\} = \max f \in \mathbb{R}.$$

Here f achieves its minimum and maximum since f is continuous (see page 101). Then

$$m_f(b-a) \leq \int_a^b f \leq M_f(b-a).$$

Proof

Let $\{(\underline{x}^{(r)}, \underline{\xi}^{(r)})\}_{r=1}^{\infty}$ be a sequence of tagged partitions with $\|\underline{x}^{(r)}\| \to 0$. Then

$$S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = \sum_{k=1}^{n} f(\xi_k^{(r)})(x_k^{(r)} - x_{k-1}^{(r)}) \geq m_f \sum_{k=1}^{n} (x_k^{(r)} - x_{k-1}^{(r)}) = m_f(b-a).$$

Similarly,

$$S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = \sum_{k=1}^{n} f(\xi_k^{(r)})(x_k^{(r)} - x_{k-1}^{(r)}) \leq M_f \sum_{k=1}^{n} (x_k^{(r)} - x_{k-1}^{(r)}) = M_f(b-a).$$

Therefore, for every $r \in \mathbb{N}$,

$$m_f(b-a) \leq S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) \leq M_f(b-a).$$

Because limits of sequences preserve inequalities,

$$m_f(b-a) \leq \int_a^b f \leq M_f(b-a).$$

Theorem (monotonicity and the triangle inequality for integrals)

Suppose $f, g \in C([a,b])$.

1) If $f(x) \leq g(x)$ for every $x \in [a,b]$, then $\int_a^b f \leq \int_a^b g$.

2) (Triangle inequality for integrals) $\left|\int_a^b f\right| \leq \int_a^b |f|$.

Proof

1) Let $\{(\underline{x}^{(r)}, \underline{\xi}^{(r)})\}_{r=1}^{\infty}$ be such that $\|\underline{x}^{(r)}\| \to 0$. Then, for every $r \in \mathbb{N}$,

$$\begin{aligned} S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) &= \sum_{j=1}^{n} f(\xi_j^{(r)})(x_j^{(r)} - x_{j-1}^{(r)}) \\ &\leq \sum_{j=1}^{n} g(\xi_j^{(r)})(x_j^{(r)} - x_{j-1}^{(r)}) = S_g(\underline{x}^{(r)}, \underline{\xi}^{(r)}). \end{aligned}$$

As $r \to \infty$,

$$\lim_{r\to\infty} S_f(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = \int_a^b f \leq \int_a^b g = \lim_{r\to\infty} S_g(\underline{x}^{(r)}, \underline{\xi}^{(r)}).$$

2) Notice that $\pm f(x) \leq |f(x)|$ for every x. Thus, by 1),

$$\pm \int_a^b f \leq \int_a^b |f| \quad \Longrightarrow \quad -\int_a^b |f| \leq \int_a^b f \leq \int_a^b |f| \quad \Longrightarrow \quad \left|\int_a^b f\right| \leq \int_a^b |f|.$$

Remark

The limiting process of the Riemann integral *does* respect strict inequalities, i.e. if $f \in C([a,b])$ and $f > 0$, then $\int_a^b f > 0$.

Conventions

We define

$$\int_a^a f := 0, \qquad \int_a^b f := -\int_b^a f.$$

Theorem

$$\int_a^b 1 = b - a.$$

Proof

Let $\{(\underline{x}^{(r)}, \underline{\xi}^{(r)})\}_{r=1}^{\infty}$ be such that $\|\underline{x}^{(r)}\| \to 0$. Then

$$S_1(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = \sum_{j=1}^{n} (x_j^{(r)} - x_{j-1}^{(r)}) = x_{n(r)}^{(r)} - x_0^{(r)} = b - a.$$

Taking the limit as $r \to \infty$,

$$\int_a^b 1 = \lim_{r\to\infty} S_1(\underline{x}^{(r)}, \underline{\xi}^{(r)}) = b - a.$$

23.2 The Fundamental Theorem of Calculus

Theorem (Fundamental Theorem of Calculus)

Let $f \in C([a,b])$.

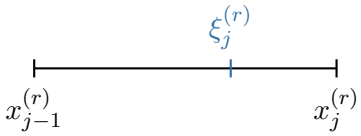
1) If $F : [a,b] \to \mathbb{R}$ is differentiable and $F' = f$, then
$$\int_a^b f = F(b) - F(a) = \int_a^b F'.$$

2) The function $G(x) = \int_a^x f$, $G : [a,b] \to \mathbb{R}$, is differentiable on $[a,b]$ and
$$\begin{cases} G' = f, \\ G(a) = 0. \end{cases}$$

Proof of 1)

Let $\{\underline{x}^{(r)}\}_r$ be a sequence of partitions with $\|\underline{x}^{(r)}\| \to 0$. We will come up with a set of tags. By the Mean Value Theorem (this is where we use that F is differentiable), for every r and every j there exists $\xi_j^{(r)} \in (x_{j-1}^{(r)}, x_j^{(r)})$ such that

$$F(x_j^{(r)}) - F(x_{j-1}^{(r)}) = F'(\xi_j^{(r)})(x_j^{(r)} - x_{j-1}^{(r)}).$$



Since $F' = f$,

$$F(x_j^{(r)}) - F(x_{j-1}^{(r)}) = F'(\xi_j^{(r)})(x_j^{(r)} - x_{j-1}^{(r)}) = f(\xi_j^{(r)})(x_j^{(r)} - x_{j-1}^{(r)}).$$

Choose $(\underline{x}^{(r)}, \underline{\xi}^{(r)})$ to be our tagged partition. Thus

$$\begin{aligned} \int_a^b f &= \lim_{r\to\infty} \sum_{j=1}^{n(r)} f(\xi_j^{(r)})(x_j^{(r)} - x_{j-1}^{(r)}) \\ &= \lim_{r\to\infty} \sum_{j=1}^{n(r)} (F(x_j^{(r)}) - F(x_{j-1}^{(r)})) \\ &= \lim_{r\to\infty} \big[F(b) - F(a)\big] = F(b) - F(a). \end{aligned}$$

Proof of 2)

Let $c \in [a,b]$. We want to show that

$$\lim_{x\to c} \frac{\int_a^x f - \int_a^c f}{x - c} = f(c).$$

Let $\varepsilon > 0$. Since f is continuous at c, there exists $\delta_0 > 0$ such that

$$|t - c| < \delta_0 \quad \Longrightarrow \quad |f(t) - f(c)| < \frac{\varepsilon}{2}.$$

Choose $\delta = \delta_0$. Suppose $0 < x - c < \delta$. If $t \in [c, x]$, then

$$|t - c| \leq |x - c| < \delta = \delta_0.$$

Thus, for $0 < x - c < \delta$ and every $t \in [c, x]$,

$$\begin{aligned}
\left| \frac{1}{x-c} \int_c^x f(t)\,dt - f(c) \right| &= \left| \frac{1}{x-c} \int_c^x f(t)\,dt - \frac{1}{x-c} \int_c^x f(c)\,dt \right| \\
&= \frac{1}{x-c} \left| \int_c^x [f(t) - f(c)]\,dt \right| \\
&\leq \frac{1}{x-c} \int_c^x |f(t) - f(c)|\,dt \qquad \text{(theorem on page 140)} \\
&\leq \frac{1}{x-c} \int_c^x \frac{\varepsilon}{2}\,dt = \frac{1}{x-c} \cdot \frac{\varepsilon}{2}\,(x-c) = \frac{\varepsilon}{2}.
\end{aligned}$$

The existence of this limit implies the differentiability of G. Similarly,

$$0 < c - x < \delta \quad \Longrightarrow \quad \left| \frac{1}{x-c} \int_c^x f(t)\,dt - f(c) \right| \leq \frac{\varepsilon}{2}.$$

Therefore

$$0 < |x - c| < \delta \quad \Longrightarrow \quad \left| \frac{\int_a^x f - \int_a^c f}{x-c} - f(c) \right| \leq \frac{\varepsilon}{2} < \varepsilon.$$

23.3 Integration by parts and change of variables

Theorem (Integration by Parts, IBP)

Suppose $f, g \in C([a, b])$ and $f', g' \in C([a, b])$. Then

$$\int_a^b f'g = [f(b)g(b) - f(a)g(a)] - \int_a^b fg'.$$

Proof

By the product rule of differentiation, $(fg)' = f'g + fg'$. Then, by linearity of the Riemann integral and the Fundamental Theorem of Calculus,

$$f(b)g(b) - f(a)g(a) = \int_a^b (fg)' = \int_a^b (f'g + fg') = \int_a^b f'g + \int_a^b fg'.$$

Theorem (Change of Variables)

Let $\varphi : [a, b] \to [c, d]$ be continuously differentiable such that $\varphi' > 0$ on $[a, b]$, and $\varphi(a) = c$, $\varphi(b) = d$. Then

$$\int_c^d f(u)\,du = \int_a^b f(\varphi(x))\,\varphi'(x)\,dx \qquad [\,u = \varphi(x)\,].$$

Proof

Let $F : [c, d] \to \mathbb{R}$ be such that $F' = f$. Then, by the Chain Rule of differentiation,

$$\frac{d}{dx} F[\varphi(x)] = F'(\varphi(x))\,\varphi'(x) = f(\varphi(x))\,\varphi'(x).$$

Thus, by the Fundamental Theorem of Calculus,

$$\begin{aligned}\int_a^b f(\varphi(x))\,\varphi'(x)\,dx &= \int_a^b \big[F(\varphi(x))\big]'\,dx \\ &= F(\varphi(b)) - F(\varphi(a)) \\ &= F(d) - F(c).\end{aligned}$$

By the Fundamental Theorem of Calculus,

$$\int_c^d f(u)\,du = \int_c^d \big[F(u)\big]'\,du = F(d) - F(c).$$

Therefore,

$$\int_c^d f(u)\,du = \int_a^b f(\varphi(x))\,\varphi'(x)\,dx.$$

24 Pointwise and Uniform Convergence of Sequences of Functions

Recall: Integration by Parts

If $f, g : [a, b] \to \mathbb{R}$ are continuously differentiable, then

$$\int_a^b f'g = f(b)g(b) - f(a)g(a) - \int_a^b fg'.$$

24.1 Fourier series: an application of integration by parts

Fourier series

Suppose $f : [-\pi, \pi] \to \mathbb{R}$ is 2π-periodic. Then (Fourier) f can be expressed as

$$f(x) = \sum_{n=0}^{\infty} [a_n \sin nx + b_n \cos nx].$$

Analogy to vectors

Suppose $\underline{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$. Take the standard basis $e_1, \ldots, e_n$ (any orthonormal basis would work). Suppose $\underline{x} = \sum_{i=1}^n a_i e_i$. How do we find a_i? Compute

$$\underline{x} \cdot e_j = \Big(\sum_{i=1}^n a_i e_i\Big) \cdot e_j = \sum_{i=1}^n a_i \delta_{ij} = a_j, \qquad \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

Back to $f(x)$:

$$\begin{aligned}\int_{-\pi}^{\pi} f(x)\sin \ell x\,dx &= \int_{-\pi}^{\pi}\sum_{n=0}^{\infty}\left[a_n \sin nx \sin \ell x + b_n \cos nx \sin \ell x\right] dx \\ &= \sum_{n=0}^{\infty}\Big[a_n \underbrace{\int_{-\pi}^{\pi} \sin nx \sin \ell x\,dx}_{=\pi\delta_{n\ell}} + b_n \underbrace{\int_{-\pi}^{\pi} \cos nx \sin \ell x\,dx}_{=0}\Big] \\ &= \sum_{n=0}^{\infty} \pi\delta_{n\ell}\, a_n = \pi a_\ell.\end{aligned}$$

Similarly,

$$\pi b_\ell = \int_{-\pi}^{\pi} f(x)\cos \ell x\,dx.$$

Definition (Fourier coefficients)

If $f : [-\pi, \pi] \to \mathbb{R}$ is continuous and 2π-periodic, then

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx, \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

are the *Fourier coefficients* of f.

Theorem (Riemann–Lebesgue Lemma)

If $f : [-\pi, \pi] \to \mathbb{R}$ is continuously differentiable, then

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = 0,$$

where a_n, b_n are the Fourier coefficients of f.

Remark

The Riemann–Lebesgue Lemma has a stronger version: f only needs to be continuous, and we have the same conclusion. (The assumption $f' \in C([-\pi, \pi])$ was unnecessary.)

Proof

We will show $\lim_{n\to\infty} b_n = 0$. Similar arguments work for a_n.

$$\begin{aligned}|b_n| &= \left|\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx\right| = \frac{1}{\pi}\left|\int_{-\pi}^{\pi}\Big[\frac{1}{n}\sin(nx)\Big]' f(x)\,dx\right| \\ &\le \left|\frac{1}{n}[f(\pi)\sin(n\pi) - f(-\pi)\sin(-n\pi)] - \frac{1}{n}\int_{-\pi}^{\pi}\sin(nx) f'(x)\,dx\right| \\ &\le |0| + \left|\frac{1}{n}\int_{-\pi}^{\pi}\sin(nx) f'(x)\,dx\right| \\ &\le \frac{1}{n}\int_{-\pi}^{\pi}|\sin nx|\,|f'(x)|\,dx \\ &\le \frac{1}{n}\underbrace{\int_{-\pi}^{\pi} 1\cdot|f'|}_{\text{fixed number}} \xrightarrow{n\to\infty} 0.\end{aligned}$$

Here the second line uses integration by parts together with $\frac{1}{\pi} < 1$, the fourth line uses a property of the Riemann integral (the triangle inequality for integrals), and the last line uses $|\sin(nx)| \leq 1$. Therefore, by the Squeeze Theorem, $|b_n| \to 0$, so $b_n \to 0$.

Remark

For the function $f(x) = \sum_{n=0}^{\infty} [a_n \sin nx + b_n \cos nx]$, it can be shown that the series converges uniformly on $\mathbb{R}$. (Harmonic Analysis.)

24.2 Sequences of functions and power series

Definition

A *power series* about x_0 is a series of the form

$$\sum_{m=0}^{\infty} a_m (x - x_0)^m.$$

Theorem

Suppose $R = \lim_{m\to\infty} |a_m|^{1/m}$ exists, and let

$$\rho = \begin{cases} \dfrac{1}{R}, & R > 0, \\ \infty, & R = 0. \end{cases}$$

Then $\sum_{m=0}^{\infty} a_m (x - x_0)^m$ converges absolutely for every x such that $|x - x_0| < \rho$, and diverges if $|x - x_0| > \rho$.

Proof

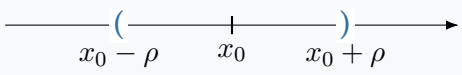
By the root test,

$$\lim_{m\to\infty} |a_m (x - x_0)^m|^{1/m} = R\,|x - x_0|.$$

Hence $\sum a_m (x - x_0)^m$ converges if $R|x - x_0| < 1$, that is, if $|x - x_0| < \rho$, and diverges if $R|x - x_0| > 1$, that is, if $|x - x_0| > \rho$.

Definition

ρ is the *radius of convergence* of $\sum_{n=0}^{\infty} a_n (x - x_0)^n$.



Remark

Thus we can now define $f : (x_0 - \rho, x_0 + \rho) \to \mathbb{R}$ by

$$f(x) = \sum_{j=0}^{\infty} a_j (x - x_0)^j.$$

Example

$$\sum_{j=0}^{\infty} x^j = \frac{1}{1-x}, \qquad \forall x \in (-1,1).$$

Example: the exponential function

Choosing $a_j = \dfrac{1}{j!}$ and $x_0 = 0$, the series $\sum_{j=0}^{\infty} \frac{1}{j!} x^j$ has radius of convergence $\rho = \infty$. We define e^x by

$$\boxed{\exp(x) = \sum_{j=0}^{\infty} \frac{1}{j!} x^j.}$$

Remark

Therefore, for every $x \in (x_0 - \rho, x_0 + \rho)$,

$$f(x) = \lim_{n\to\infty} f_n(x), \qquad \text{where} \qquad f_n(x) = \sum_{j=0}^{n} a_j (x - x_0)^j.$$

Questions

1) Is f continuous?

2) If 1) holds, is f differentiable? Does $f' = \lim_{n\to\infty} f_n'$?

3) If 1) holds, does $\displaystyle\int_a^b f = \lim_{n\to\infty} \int_a^b f_n$?

In other words, are these limiting processes interchangeable? (Note that the f_n are polynomials, which are differentiable, continuous, and integrable.)

Remark

In general, these questions apply not only to power series but to other sequences of functions as well. If they converge to some function, does their limiting behaviour respect other limits?

24.3 Pointwise and uniform convergence of sequences of functions

Definition (pointwise convergence)

For $n \in \mathbb{N}$, let $S \subset \mathbb{R}$, $S \neq \emptyset$, and let $f_n : S \to \mathbb{R}$. Let $f : S \to \mathbb{R}$. We say $\{f_n\}_n$ *converges pointwise* to f if, for every $x \in S$,

$$\lim_{n\to\infty} f_n(x) = f(x).$$

Example

Let $f(x) = \dfrac{1}{1-x}$ and $f_n(x) = \sum_{j=0}^{n} x^j$. Then, for every $x \in (-1,1)$, $\lim_{n\to\infty} f_n(x) = f(x)$.

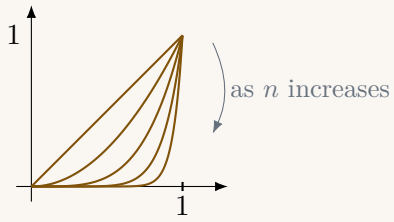
Example

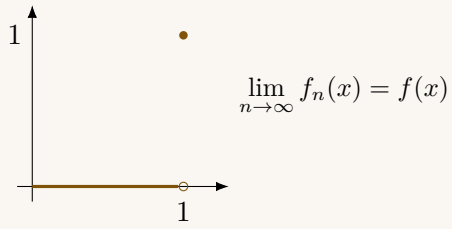
Let $f_n(x) = x^n$, $x \in [0,1]$. If $x = 1$, then $\lim_{n\to\infty} f_n(x) = 1$. If $x \in [0,1)$, then

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} x^n = 0 \qquad \text{(theorem on page 40)}.$$

Thus, for every $x \in [0,1]$,

$$\lim_{n\to\infty} f_n(x) = f(x) = \begin{cases} 0, & \text{if } x \in [0,1), \\ 1, & \text{if } x = 1. \end{cases}$$





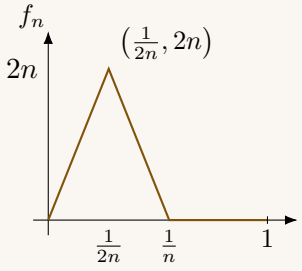
Remark

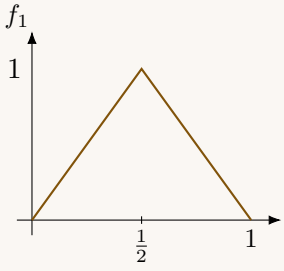
Note that we already have some answers to our questions: in this example, $f_n(x)$ is continuous for every $n \in \mathbb{N}$, yet $\lim_{n\to\infty} f_n(x)$ is no longer continuous at $x = 1$. Thus, if $f_n \to f$ pointwise, the answers to those questions are absolutely NO.

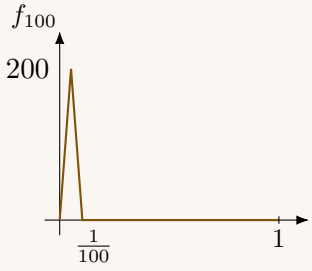
Example: a sequence of spikes

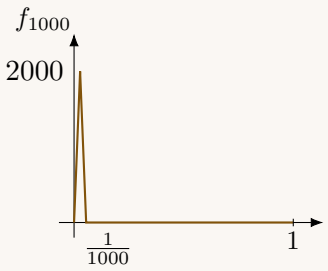
Let $f_n : [0,1] \to \mathbb{R}$ be defined by

$$f_n(x) = \begin{cases} 4n^2 x, & x \in [0, \frac{1}{2n}], \\ 4n - 4n^2 x, & x \in [\frac{1}{2n}, \frac{1}{n}], \\ 0, & x \in [\frac{1}{n}, 1]. \end{cases}$$







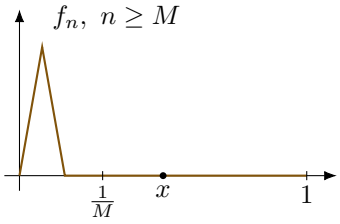


Claim

$\{f_n\}_{n=1}^{\infty}$ converges pointwise to $f(x) = 0$ on $[0, 1]$.

Proof

If $x = 0$, then $\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} 0 = 0$.
Suppose $x \in (0, 1]$. We want to show that $\lim_{n\to\infty} f_n(x) = 0$. Let $\varepsilon > 0$. We want to show that there exists $M \in \mathbb{N}$ such that, for every $n \geq M$, $|f_n(x) - 0| < \varepsilon$.
Let $x \in (0, 1]$. Choose $M \in \mathbb{N}$ such that $\frac{1}{M} < x$ (Archimedean property). Then $x > \frac{1}{M}$, and for every $n \geq M$, $x > \frac{1}{n}$. Hence $f_n(x) = 0$, since $x \in [\frac{1}{n}, 1]$, and so $|f_n(x)| < \varepsilon$.



Given $x \in (0, 1]$, the sequence $\{f_n(x)\}_{n=1}^{\infty}$ is

$$f_1(x),\ f_2(x),\ \ldots,\ f_{M-1}(x),\ \underbrace{0}_{f_M(x)},\ 0,\ 0,\ \ldots$$

Definition (uniform convergence)

For $n \in \mathbb{N}$, let $S \subset \mathbb{R}$, $S \neq \emptyset$, and let $f_n : S \to \mathbb{R}$. Let $f : S \to \mathbb{R}$. Then we say $\{f_n\}_{n=1}^{\infty}$ *converges uniformly* to f if, for every $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that, for every $n \geq M$ and every $x \in S$,

$$|f_n(x) - f(x)| < \varepsilon.$$

Remark

This is a much stronger statement than pointwise convergence... Yet, if we write out what it means to have $\lim_{n\to\infty} f_n(x) = f(x)$, the two definitions look suspiciously the same...?

The two definitions, side by side

The difference comes in when we state "$\forall x \in S$".

- Definition of pointwise convergence: $\forall x \in S,\ \forall \varepsilon > 0,\ \exists M = M(x, \varepsilon) \in \mathbb{N}$ such that

$\forall n \geq M,\ |f_n(x) - f(x)| < \varepsilon$.

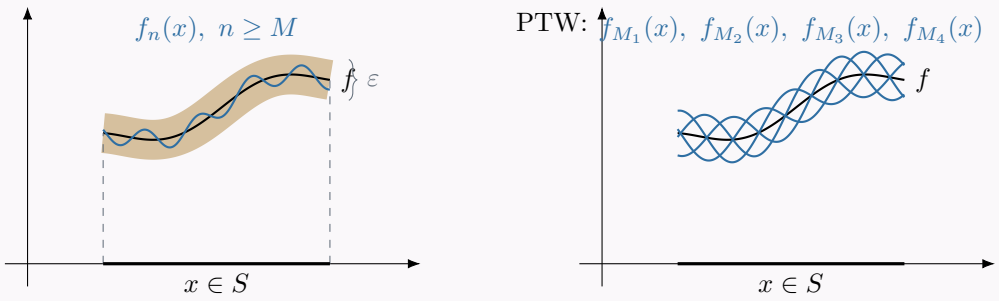
- Definition of uniform convergence: $\forall \varepsilon > 0,\ \exists M = M(\varepsilon) \in \mathbb{N}$ such that $\forall n \geq M,\ \forall x \in S,\ |f_n(x) - f(x)| < \varepsilon$.

In other words, our choice of M for uniform convergence has to be *independent* of x!

Analogy with uniform continuity

This is analogous to what we did for uniform continuity of functions: the choice of δ can no longer depend on $c \in S$, but only on ε: $\forall \varepsilon > 0,\ \exists \delta$ such that $\forall x,\ \forall c,\ \cdots \implies \cdots$.

Picture: uniform versus pointwise convergence



In the left picture, the shaded band is $\{(x, y) \in \mathbb{R}^2 : |y - f(x)| < \varepsilon\}$, and the blue curve is $f_n(x)$ for $n \geq M$. So, for a given ε, $f_n(x)$ with $n \geq M$ has to be less than ε away from $f(x)$ for every $x \in S$, and the choice of M is independent of x. In the pointwise picture on the right, the choice of M_1, M_2, M_3, M_4 only works for certain $x \in S$, but is not versatile.

Theorem

If $f_n : S \to \mathbb{R}$ and $f : S \to \mathbb{R}$, then

$$\{f_n\}_n \text{ converges to } f \text{ uniformly} \quad \implies \quad f_n \to f \text{ pointwise.}$$

Proof

Trivial from our discussion of the definitions of pointwise convergence and uniform convergence.

Theorem

Let $f_n(x) = x^n$ and

$$f(x) = \begin{cases} 0, & \text{if } x \in [0, 1), \\ 1, & \text{if } x = 1. \end{cases}$$

Then

1) for every $0 < b < 1$, $f_n \to f$ uniformly on $[0, b]$;

2) $\{f_n\}_n$ does not converge uniformly to f on $[0, 1]$.

Negation of the definition

$f_n : S \to \mathbb{R}$ *does not converge* to $f : S \to \mathbb{R}$ uniformly if there exists $\varepsilon_0 > 0$ such that, for every $M \in \mathbb{N}$, there exist $n \geq M$ and $x \in S$ such that

$$|f_n(x) - f(x)| \geq \varepsilon_0.$$

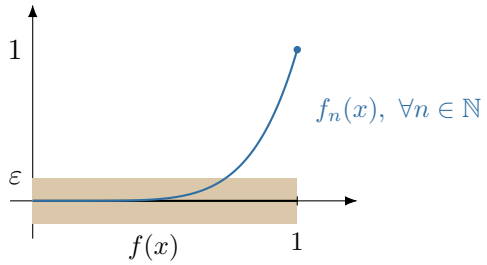
Proof

1) Let $\varepsilon > 0$ and $b \in (0, 1)$. Then $b^n \to 0$ (theorem on page 40). Therefore there exists $M_0 \in \mathbb{N}$ such that, for every $n \geq M_0$, $b^n < \varepsilon$. Choose $M = M_0$. Then, for every $n \geq M$ and every $x \in [0, b]$,

$$|f_n(x) - f(x)| = |f_n(x)| = x^n \leq b^n < \varepsilon.$$

2) Choose $\varepsilon_0 = \frac{1}{4}$. Let $M \in \mathbb{N}$. Choose $x = \left(\frac{1}{2}\right)^{1/M} \in (0, 1)$ and choose $n = M$. Then

$$|f_n(x) - f(x)| = |f_M(x)| = \left[\left(\tfrac{1}{2}\right)^{1/M}\right]^M = \frac{1}{2} > \frac{1}{4} = \varepsilon_0.$$



Intuitively, no matter what n we choose, f_n always leaves $(f(x) - \varepsilon, f(x) + \varepsilon)$ for some $x \in S$.

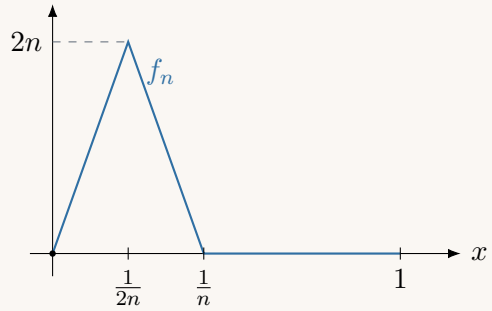
25 Uniform Convergence, the Weierstrass M-Test, and Interchanging Limits

We will continue looking at examples of sequences of functions that converge pointwise but not uniformly.

Example: a sequence of spikes

For $n \in \mathbb{N}$, define $f_n : [0, 1] \to \mathbb{R}$ by

$$f_n(x) = \begin{cases} 4n^2 x, & x \in \left[0, \dfrac{1}{2n}\right], \\ 4n - 4n^2 x, & x \in \left[\dfrac{1}{2n}, \dfrac{1}{n}\right], \\ 0, & x \in \left[\dfrac{1}{n}, 1\right]. \end{cases}$$



Claim. $f_n \to 0$ pointwise on $[0,1]$, but not uniformly on $[0,1]$.

Proof

Choose $\varepsilon_0 = 1$. Let $M \in \mathbb{N}$. Choose $n = M$ and $x = \frac{1}{2M}$. Then

$$|f_n(x) - f(x)| = |f_M(x)| = 2M \geq 1 = \varepsilon_0.$$

25.1 Interchange of limits

Remark

In general, limiting processes can not be interchanged.

Example: iterated limits

On the one hand,

$$\lim_{n\to\infty} \lim_{k\to\infty} \frac{\frac{n}{k}}{\frac{n}{k}+1} = \lim_{n\to\infty} \frac{0}{0+1} = 0.$$

On the other hand,

$$\lim_{k\to\infty} \lim_{n\to\infty} \frac{\frac{n}{k}}{\frac{n}{k}+1} = \lim_{k\to\infty} \lim_{n\to\infty} \frac{1}{1+\frac{k}{n}} = \lim_{k\to\infty} 1 = 1.$$

That is,

$$\lim_{k\to\infty} \lim_{n\to\infty} \frac{\frac{n}{k}}{\frac{n}{k}+1} \neq \lim_{n\to\infty} \lim_{k\to\infty} \frac{\frac{n}{k}}{\frac{n}{k}+1}.$$

Again, we now propose the three questions.

Questions

1) If $f_n : S \to \mathbb{R}$ is continuous and $f_n \to f$ uniformly (or pointwise), is f continuous?

2) If $f_n : [a,b] \to \mathbb{R}$ is differentiable and $f_n \to f$, $f_n' \to g$ uniformly (or pointwise), then is f differentiable with $g = f'$? In other words, is the derivative of the limit equal to the limit of the derivatives?

3) If $f_n : [a,b] \to \mathbb{R}$, with f_n and f continuous, are such that $f_n \to f$, does

$$\lim_{n\to\infty} \int_a^b f_n = \int_a^b f \ ?$$

Answer. Yes, if $f_n \to f$ uniformly. (For 2), we need only $f_n' \to g$ uniformly.) No, if $f_n \to f$ pointwise. The counterexamples are as follows.

Counterexamples under pointwise convergence

1) Let $f_n(x) = x^n$ on $[0,1]$, and let

$$f(x) = \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1. \end{cases}$$

Then $f_n \in C([0,1])$ for all n, $f_n \to f$ pointwise, and $f \notin C([0,1])$.

2) $f_n(x) = x^n$ is differentiable at 1 for all n, but

$$f = \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1, \end{cases}$$

is not differentiable at 1.

Or: use $f_n(x) = \dfrac{x^n}{n}$ on $[0,1]$. Then $f_n \to f = 0$ pointwise, and

$$f_n' \longrightarrow g(x) = \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1, \end{cases} \qquad \text{pointwise,}$$

but $g \neq f' = 0$.

3) Let

$$f_n(x) = \begin{cases} 4n^2 x, & x \in \left[0, \dfrac{1}{2n}\right], \\ 4n - 4n^2 x, & x \in \left[\dfrac{1}{2n}, \dfrac{1}{n}\right], \\ 0, & x \in \left[\dfrac{1}{n}, 1\right]. \end{cases}$$

Then

$$\int_0^1 f_n(x)\,dx = \text{area of the triangle} = \frac{1}{n} \cdot 2n \cdot \frac{1}{2} = 1 \qquad \text{for all } n.$$

Thus $\lim_{n\to\infty} \int_0^1 f_n = 1$. Yet $f_n \to f = 0$ pointwise, so

$$\int_0^1 f = 0 \neq \lim_{n\to\infty} \int_0^1 f_n.$$

Theorem: uniform limits of continuous functions are continuous

If $f_n : S \to \mathbb{R}$ is continuous for all n, $f : S \to \mathbb{R}$, and $f_n \to f$ uniformly, then f is continuous.

Proof

Let $c \in S$, and let $\varepsilon > 0$. Since $f_n \to f$ uniformly, there exists $M \in \mathbb{N}$ such that

$$\forall n \geq M,\ \forall y \in S: \quad |f_n(y) - f(y)| < \frac{\varepsilon}{3}. \tag{*}$$

Look at $n = M$ and $c \in S$; thus

$$|f_M(c) - f(c)| < \frac{\varepsilon}{3}. \tag{**}$$

Since $f_M : S \to \mathbb{R}$ is continuous, there exists $\delta_0 > 0$ such that

$$|x - c| < \delta_0 \quad \Longrightarrow \quad |f_M(x) - f_M(c)| < \frac{\varepsilon}{3}. \tag{***}$$

Choose $\delta = \delta_0$. If $|x - c| < \delta$ and $x \in S$, then

$$|f(x) - f(c)| = |f_M(c) - f_M(c) + f_M(x) - f_M(x) + f(x) - f(c)|$$

$$\leq |f(x) - f_M(x)| + |f_M(c) - f(c)| + |f_M(x) - f_M(c)|$$
$$< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,$$

where the three terms are estimated by $(*)$, $(**)$, and $(***)$ respectively.

Theorem: integrals of uniform limits

If $f_n : [a, b] \to \mathbb{R}$ is continuous for all n, $f : [a, b] \to \mathbb{R}$, and $f_n \to f$ uniformly, then

$$\lim_{n\to\infty} \int_a^b f_n = \int_a^b f.$$

Proof

Let $\varepsilon > 0$. Since $f_n \to f$ uniformly, there exists $M_0 \in \mathbb{N}$ such that, for all $n \geq M_0$ and all $x \in [a, b]$,

$$|f_n(x) - f(x)| < \frac{\varepsilon}{b - a}.$$

Then for all $n \geq M = M_0$, we have

$$\left| \int_a^b f_n - \int_a^b f \right| = \left| \int_a^b (f_n - f) \right| \leq \int_a^b |f_n - f| < \int_a^b \frac{\varepsilon}{b - a} = \varepsilon,$$

where the strict inequality holds because integration respects strict inequalities (page 140, page 141).

Remark

Notationally, this states that

$$\lim_{n\to\infty} \int_a^b f_n = \int_a^b \lim_{n\to\infty} f_n = \int_a^b f.$$

In other words, we are changing the order of two separate limiting processes.

Theorem: derivatives of limits

If $f_n : [a,b] \to \mathbb{R}$ is continuously differentiable for all n, $f : [a,b] \to \mathbb{R}$, $g : [a,b] \to \mathbb{R}$, $f_n \to f$ pointwise, and $f_n' \to g$ uniformly, then f is continuously differentiable and $f' = g$.

Proof

Let $x \in [a,b]$. Then, by part 1) of the fundamental theorem of calculus,

$$f_n(x) - f_n(a) = \int_a^x f_n'.$$

Hence

$$f(x) - f(a) = \lim_{n\to\infty} [f_n(x) - f_n(a)] = \lim_{n\to\infty} \int_a^x f_n'.$$

By the previous theorem,

$$f(x) - f(a) = \int_a^x g, \qquad \text{so} \qquad f(x) = f(a) + \int_a^x g.$$

By part 2) of the fundamental theorem of calculus, f is now differentiable and

$$f' = \Big(\int_a^x g\Big)' = g.$$

25.2 The Weierstrass M-test

Theorem (Weierstrass M-test)

Let $f_j : S \to \mathbb{R}$, and suppose there exists $\{M_j\}$ with $M_j > 0$ such that

a) for all $x \in S$, $|f_j(x)| \le M_j$;

b) $\sum_{j=1}^{\infty} M_j$ converges.

Then we have:

1) for all $x \in S$, $\sum_{j=1}^{\infty} f_j(x)$ converges absolutely;

2) let $f(x) = \sum_{j=1}^{\infty} f_j(x)$ for all $x \in S$, that is, $\sum_{j=1}^{n} f_j(x) \to f(x)$ pointwise on S. Then

$$\sum_{j=1}^{n} f_j \longrightarrow f \quad \text{uniformly on } S \text{ as } n \to \infty.$$

Remark: Weierstrass's example revisited

For the examples we had from Weierstrass, the class of continuous yet nowhere differentiable functions (page 112–page 118), we showed that they are continuous by showing

$$\limsup_{n\to\infty} |f(x_n) - f(c)| \le \varepsilon \quad \Longrightarrow \quad \lim_{n\to\infty} |f(x_n) - f(c)| = 0.$$

But now, with the theorems we have just shown, we can see that

$$\sum_{j=0}^{\infty} \frac{\cos(160^j x)}{2^j}$$

is continuous in a straightforward way. Since

$$\left|\frac{\cos(160^j x)}{2^j}\right| \leq \frac{1}{2^j} \qquad \text{and} \qquad \sum_{j=0}^{\infty} \frac{1}{2^j} \text{ converges absolutely,}$$

the M-test gives that

$$\left\{\sum_{j=0}^{n} \frac{\cos(160^j x)}{2^j}\right\}_{n=1}^{\infty} \quad \text{converges uniformly.}$$

Since each partial sum f_n is continuous by the properties of continuity, the theorem on page 153 therefore gives that

$$f = \sum_{j=0}^{\infty} \frac{\cos(160^j x)}{2^j}$$

is continuous on $\mathbb{R}$.

Proof

1) This follows directly from a), b), and the comparison test (page 66).

2) Let $\varepsilon > 0$. Since $\sum_{j=1}^{\infty} M_j$ converges, $\sum M_j$ is Cauchy, so there exists $N_0 \in \mathbb{N}$ such that, for all $n \geq N_0$,

$$\sum_{j=n+1}^{\infty} M_j = \left|\sum_{j=1}^{\infty} M_j - \sum_{j=1}^{n} M_j\right| < \varepsilon.$$

Choose $N = N_0$. Then for all $n \geq N = N_0$ and all $x \in S$,

$$\left|f(x) - \sum_{j=1}^{n} f_j(x)\right| = \left|\sum_{j=n+1}^{\infty} f_j(x)\right| \leq \sum_{j=n+1}^{\infty} |f_j(x)| \leq \sum_{j=n+1}^{\infty} M_j < \varepsilon.$$

26 Power Series and the Weierstrass Approximation Theorem

Recall: the Weierstrass M-test

Suppose $f_j : S \to \mathbb{R}$ are such that, for every $j \in \mathbb{N}$, there exists $M_j \geq 0$ with

a) $|f_j(x)| \leq M_j$ for all $x \in S$;

b) $\sum_{j=1}^{\infty} M_j$ convergent.

Then $\{\sum_{j=1}^{n} f_j\}_{n=1}^{\infty}$ converges uniformly.

Example

Let $f_j(x) = \dfrac{\cos(160^j x)}{4^j}$ for $x \in \mathbb{R}$. Then $|f_j(x)| \leq 4^{-j}$, and $\sum_{j=1}^{\infty} 4^{-j}$ converges, so

$$\sum_{j=1}^{\infty} \frac{\cos(160^j x)}{4^j}$$

converges uniformly on $\mathbb{R}$. Since a uniform limit of continuous functions is continuous (theorem on page 153), $\sum_{j=1}^{\infty} \frac{\cos(160^j x)}{4^j}$ is continuous on $\mathbb{R}$.

26.1 Back to power series

Recall

That $\sum_{j=1}^{\infty} f_j$ converges uniformly means that there exists $f : S \to \mathbb{R}$ such that $\sum_{j=1}^{n} f_j \to f$ uniformly on S.

Theorem: uniform convergence of power series

Let $\sum_{j=0}^{\infty} a_j (x - x_0)^j$ be a power series of radius of convergence

$$\rho = \left(\lim_{j \to \infty} |a_j|^{1/j} \right)^{-1} \in (0, \infty].$$

Then, for all $r \in [0, \rho)$, $\sum_{j=0}^{\infty} a_j (x - x_0)^j$ converges uniformly on $[x_0 - r, x_0 + r]$.

Proof

Let $r \in [0, \rho)$. Then, for all $j \in \mathbb{N} \cup \{0\}$ and all $x \in [x_0 - r, x_0 + r]$,

$$|a_j (x - x_0)^j| \leq |a_j| r^j =: M_j.$$

We have

$$\lim_{j \to \infty} M_j^{1/j} = r \left[\lim_{j \to \infty} |a_j|^{1/j} \right] = \begin{cases} \dfrac{r}{\rho}, & \text{if } \rho < \infty, \\ 0, & \text{if } \rho = \infty, \end{cases} \quad < 1,$$

so $\sum_{j=0}^{\infty} |a_j| r^j$ converges. By the Weierstrass M-test, $\sum_{j=0}^{\infty} a_j (x - x_0)^j$ then converges uniformly on $[x_0 - r, x_0 + r]$.

Theorem: differentiating and integrating power series

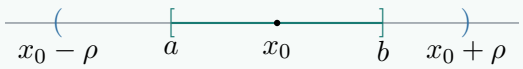
Let $\sum_{j=0}^{\infty} a_j (x - x_0)^j$ be a power series with a radius of convergence $\rho \in (0, \infty]$. Then:

1) for all $c \in (x_0 - \rho, x_0 + \rho)$, $\sum_{j=0}^{\infty} a_j (x - x_0)^j$ is differentiable at c, and

$$\frac{d}{dx} \sum_{j=0}^{\infty} a_j (x - x_0)^j \Big|_{x=c} = \sum_{j=0}^{\infty} j a_j (x - x_0)^{j-1} \Big|_{x=c};$$

2) for all a, b such that $[a, b] \subsetneq (x_0 - \rho, x_0 + \rho)$,

$$\int_a^b \sum_{j=0}^{\infty} a_j (x - x_0)^j \, dx = \sum_{j=0}^{\infty} a_j \left[\frac{(b - x_0)^{j+1} - (a - x_0)^{j+1}}{j + 1} \right].$$



Proof

Part 2) follows directly from the previous theorem and the theorem on page 154 that the limit of the integrals is the integral of the uniform limit (of f_n).
For part 1), we first prove a claim.
Claim. The series

$$\sum_{j=0}^{\infty} \frac{d}{dx} a_j (x - x_0)^j = \sum_{j=0}^{\infty} a_j \cdot j \, (x - x_0)^{j-1}$$

has the same radius of convergence, which is ρ.
We compute, using the root test (page 71) and the theorem on page 48,

$$\lim_{j \to \infty} |j a_j|^{1/j} = \lim_{j \to \infty} |a_j|^{1/j} \cdot 1 = \rho^{-1},$$

since $\sum_{j=0}^{\infty} a_j (x - x_0)^j$ has radius of convergence ρ. By the theorem on page 146, $\rho = (\rho^{-1})^{-1}$ is the radius of convergence of the power series $\sum_{j=0}^{\infty} j a_j (x - x_0)^{j-1}$. This completes the proof of the claim; therefore $\sum_{j=0}^{\infty} j a_j (x - x_0)^{j-1}$ is convergent at $x = c$ for every $c \in (x_0 - \rho, x_0 + \rho)$.

Remark: power series are infinitely differentiable

Iterating 1), we have, for all $x \in (x_0 - \rho, x_0 + \rho)$,

$$\frac{d^k}{dx^k} \sum_{j=0}^{\infty} a_j (x - x_0)^j = \sum_{j=0}^{\infty} \frac{d^k}{dx^k} a_j (x - x_0)^j \qquad \text{for all } k \in \mathbb{N}.$$

In other words, $\sum_{j=0}^{\infty} a_j (x - x_0)^j : (x_0 - \rho, x_0 + \rho) \to \mathbb{R}$ is infinitely differentiable. In particular, for $k \in \mathbb{N}$,

$$\begin{aligned}
\left[\frac{d^k}{dx^k} \sum_{j=0}^{\infty} a_j (x - x_0)^j \right]_{x=x_0} &= \left[\sum_{j=0}^{\infty} \frac{d^k}{dx^k} a_j (x - x_0)^j \right]_{x=x_0} \\
&= \Big[\frac{d^k}{dx^k} a_0 (x - x_0)^0 + \frac{d^k}{dx^k} a_1 (x - x_0)^1 + \cdots + \frac{d^k}{dx^k} a_{k-1} (x - x_0)^{k-1} \\
&\qquad + \frac{d^k}{dx^k} a_k (x - x_0)^k + \frac{d^k}{dx^k} a_{k+1} (x - x_0)^{k+1} + \cdots \Big]_{x=x_0} \\
&= \Big[\underbrace{0}_{j=0} + \underbrace{0}_{j=1} + \cdots + \underbrace{0}_{j=k-1} \\
&\qquad + k! \, a_k (x - x_0)^0 + (k+1)! \, a_{k+1} (x - x_0)^1 + \cdots \Big]_{x=x_0} \\
&= 0 + 0 + \cdots + 0 + k! \, a_k \cdot 1 + 0 + 0 + \cdots \\
&= k! \, a_k.
\end{aligned}$$

That is,

$$k!\, a_k = \Big[\frac{d^k}{dx^k} \sum_{j=0}^{\infty} a_j (x - x_0)^j\Big]_{x=x_0}.$$

26.2 The Weierstrass approximation theorem

Remark

The theorem states: "every continuous function on $[a, b]$ is almost a polynomial."

Theorem (Weierstrass approximation theorem)

If $f \in C([a, b])$, then there exists a sequence of polynomials $\{P_n(x)\}$ such that

$$P_n \longrightarrow f \quad \text{uniformly on } [a, b].$$

Remark

We will prove an easier statement, stated below.

Theorem (Weierstrass approximation theorem on $[0, 1]$)

If $f \in C([0, 1])$, then there exists a sequence of polynomials $\{P_n(x)\}$ such that $P_n \to f$ uniformly on $[0, 1]$. This proves the statement for $f \in C([a, b])$ as well.

Remark: reduction to the case $f(0) = f(1) = 0$

We only consider the case $f(0) = 0$, $f(1) = 0$. If we prove this case, then for every $\tilde{f} \in C([0, 1])$ there exists $\{P_n\}$ such that, uniformly,

$$P_n(x) \longrightarrow \tilde{f}(x) - \tilde{f}(0) - x[\tilde{f}(1) - \tilde{f}(0)],$$

since

$$\begin{aligned} \tilde{f}(0) - \tilde{f}(0) - 0 \cdot [\tilde{f}(1) - \tilde{f}(0)] &= 0 \qquad \text{(limit evaluated at 0)}, \\ \tilde{f}(1) - \tilde{f}(0) - 1 \cdot [\tilde{f}(1) - \tilde{f}(0)] &= 0 \qquad \text{(limit evaluated at 1)}. \end{aligned}$$

Furthermore, the new polynomial obtained,

$$\tilde{P}_n(x) = P_n(x) + \tilde{f}(0) + x[\tilde{f}(1) - \tilde{f}(0)] \longrightarrow \tilde{f}(x) \quad \text{uniformly.}$$

Thus our goal is to show the following.

Goal

If $f \in C([0, 1])$ with $f(0) = 0$ and $f(1) = 0$, then there exists a sequence of polynomials $\{P_n\}$ such that $P_n \to f$ uniformly on $[0, 1]$.

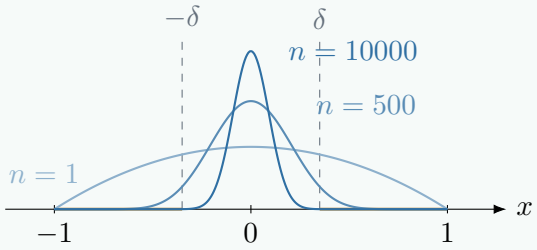
Theorem: the polynomials Q_n

Let

$$c_n := \left(\int_{-1}^{1}(1-x^2)^n\,dx\right)^{-1} \quad \text{for all } n \in \mathbb{N}, \qquad \text{and let} \qquad Q_n(x) := c_n(1-x^2)^n.$$

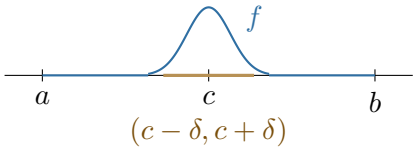
Then:

1) for all $n \in \mathbb{N}$, $Q_n(x) \geq 0$ on $[-1,1]$;
2) for all $n \in \mathbb{N}$, $\int_{-1}^{1} Q_n = 1$;
3) for all $\delta \in (0,1)$, $Q_n \to 0$ uniformly on $\{x : \delta \leq |x| \leq 1\}$.



Proof

Part 1). We first prove a claim.
Claim. If $f \in C([a,b])$ satisfies $f \geq 0$ for all $x \in [a,b]$, and there exists $c \in [a,b]$ where $f(c) > 0$, then $\int_a^b f > 0$.
Proof of the claim. In other words, we will show that the integral of a continuous, non-negative, and somewhere positive function is strictly positive. The idea is that, for every element in a neighbourhood of c, f evaluated there is strictly positive. (This neighbourhood exists by the continuity of f.)



Let $\varepsilon = f(c)$. Then, since f is continuous, there exists $\delta > 0$ such that

$$\begin{aligned} |x-c| < \delta \iff x \in (c-\delta, c+\delta) \quad &\implies \quad |f(x)-f(c)| < \varepsilon \\ &\implies \quad f(x) \in (f(c)-\varepsilon, f(c)+\varepsilon) = (0, 2f(c)). \end{aligned}$$

Thus, for all $x \in (c-\delta, c+\delta)$, $f(x) > 0$. Since integrals respect strict inequalities (page 140–page 141),

$$\int_{c-\delta/2}^{c+\delta/2} f > 0$$

(removing the endpoints $c \pm \delta$). Since $(c-\frac{\delta}{2}, c+\frac{\delta}{2}) \subset [a,b]$, by additivity,

$$\int_a^b f = \int_a^{c-\delta/2} f + \int_{c-\delta/2}^{c+\delta/2} f + \int_{c+\delta/2}^{b} f > 0 + 0 + 0 = 0.$$

Note. For $c = a$ or $c = b$ (endpoints), use $(c, c+\delta)$ or $(c-\delta, c)$ instead. This proves the claim. Thus

$$\int_{-1}^{1}(1-x^2)^n\,dx > 0 \quad \Longrightarrow \quad \Big(\int_{-1}^{1}(1-x^2)^n\,dx\Big)^{-1} = c_n > 0.$$

Since $(1-x^2)^n \geq 0$ for all $x \in [-1,1]$, thus

$$Q_n(x) = c_n(1-x^2)^n \geq 0 \qquad \text{for all } x \in [-1,1].$$

Part 2). We have

$$\int_{-1}^{1} Q_n = c_n \int_{-1}^{1}(1-x^2)^n\,dx = \frac{\int_{-1}^{1}(1-x^2)^n\,dx}{\int_{-1}^{1}(1-x^2)^n\,dx} = 1.$$

Part 3). Estimate of c_n. We have that, for all $n \in \mathbb{N}$ and all $x \in [-1,1]$,

$$(1-x^2)^n \geq 1 - nx^2.$$

Note. We have shown that, for all $c \geq -1$, $(1+c)^n \geq 1+nc$. Or, we can show this by showing that $g(x) = (1-x^2)^n - (1-nx^2)$ has $g(0) = 0$ and

$$g'(x) = n \cdot 2x\big[1-(1-x^2)^{n-1}\big] \geq 0,$$

and then using the theorem on page 122.
Now, noting that $(1-x^2)^n$ is even,

$$\begin{aligned}
\frac{1}{c_n} = \int_{-1}^{1}(1-x^2)^n\,dx &= 2\int_{0}^{1}(1-x^2)^n\,dx \\
&\geq 2\int_{0}^{1/\sqrt{n}}(1-x^2)^n\,dx \\
&\geq 2\int_{0}^{1/\sqrt{n}}(1-nx^2)\,dx \\
&= 2\Big(x - \frac{nx^3}{3}\Big)\Big|_0^{1/\sqrt{n}} \\
&= \frac{2}{\sqrt{n}} - \frac{2}{3\sqrt{n}} = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}}.
\end{aligned}$$

Therefore, for all $n \in \mathbb{N}$,

$$\frac{1}{c_n} > \frac{1}{\sqrt{n}} \quad \Longrightarrow \quad 0 < c_n < \sqrt{n}.$$

Let $\delta \in (0,1)$. We want to show that $Q_n \to 0$ uniformly at the x with $\delta \leq |x| \leq 1$, for all $\delta \in (0,1)$.
Claim. $\sqrt{n}\,(1-\delta^2)^n \to 0$ as $n \to \infty$.
Proof of the claim. We have

$$\lim_{n\to\infty}\big[\sqrt{n}\,(1-\delta^2)^n\big]^{1/n} = \lim_{n\to\infty}(n^{\frac{1}{n}\cdot\frac{1}{2}})(1-\delta^2) = 1^{\frac{1}{2}}\cdot(1-\delta^2) = 1-\delta^2 < 1.$$

Since $(1-\delta^2)^n \to 0$ (theorem on page 40) because $0 < 1-\delta^2 < 1$, thus

$$\lim_{n\to\infty}\sqrt{n}\,(1-\delta^2)^n = \lim_{n\to\infty}(1-\delta^2)^n = 0.$$

This proves the claim.
Then, let $\varepsilon > 0$. There exists $M_0 \in \mathbb{N}$ such that, for all $n \geq M_0$,

$$\sqrt{n}\,(1-\delta^2)^n < \varepsilon.$$

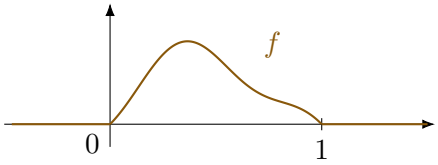
Choose $M = M_0$. Then, for all $n \geq M$ and all x with $\delta \leq |x| \leq 1$,

$$|Q_n(x)| = |c_n(1-x^2)^n| < \sqrt{n}\,(1-x^2)^n \leq \sqrt{n}\,(1-\delta^2)^n < \varepsilon.$$

Here the first inequality uses that c_n was shown to be bounded by $\sqrt{n}$, which is enough, and the second uses $\delta \leq |x| \leq 1$. Therefore Q_n converges to 0 uniformly on $[-1,-\delta] \cup [\delta, 1]$.

Proof of the Weierstrass approximation theorem

Suppose $f \in C([0,1])$ with $f(0) = f(1) = 0$.



We will extend the domain of f for the sake of convenience: let $f(x) = 0$ for all $x \in \mathbb{R} \setminus [0,1]$. Then we can show that $f \in C(\mathbb{R})$; that is, f is continuous on $\mathbb{R}$ now.
Now we will have our candidate polynomial. Define

$$P_n(x) = \int_0^1 f(t)\,Q_n(t-x)\,dt.$$

Here Q_n is evaluated at $t-x$; it is not $Q_n \cdot (t-x)$.
Claim 1. $P_n(x)$ is indeed a polynomial in terms of x.
Proof of Claim 1. We will use the binomial theorem twice. We have

$$\begin{aligned} P_n(x) &= \int_0^1 f(t)\,c_n\big[1-(x-t)^2\big]^n\,dt \\ &= \int_0^1 f(t)\,c_n\big[(x-t)^2-1\big]^n(-1)^n\,dt \\ &= \int_0^1 f(t)\,c_n\Big[\sum_{j=0}^{n}\binom{n}{j}(x-t)^{2j}(-1)^{n-j}(-1)^n\Big]dt \end{aligned}$$

$$= \int_0^1 f(t)\,c_n\Big[\sum_{j=0}^{n}\sum_{k=0}^{2j}\binom{n}{j}\binom{2j}{k}(-1)^{2n}(-1)^{-j}(-t)^k x^{2j-k}\Big]dt$$

$$= \int_0^1 f(t)\,c_n\Big[\sum_{j=0}^{n}\sum_{k=0}^{2j}\binom{n}{j}\binom{2j}{k}(-t)^k(-1)^j x^{2j-k}\Big]dt$$

$$= c_n\sum_{j=0}^{n}\sum_{k=0}^{2j}\Big\{\binom{n}{j}\binom{2j}{k}(-1)^j x^{2j-k}\int_0^1\big[f(t)(-t)^k\big]dt\Big\},$$

where we used $(-1)^{2n} = 1$ and $(\frac{1}{-1})^j = (-1)^j$. In the last expression, c_n is a constant, $\binom{n}{j}\binom{2j}{k}(-1)^j$ are constants, x^{2j-k} is the term in x, and $\int_0^1 f(t)(-t)^k\,dt$ is a fixed constant (the result of the integral). Thus we can see that $P_n(x)$ is indeed a polynomial. This proves Claim 1.

Claim 2. $P_n \to f$ uniformly on $[0,1]$.

Proof of Claim 2. Note that

$$\begin{aligned} P_n(x) &= \int_0^1 f(t)\,Q_n(t-x)\,dt \\ &= \int_{-x}^{1-x} f(x+u)\,Q_n(u)\,du \\ &= \int_{-x}^{1-x} f(x+t)\,Q_n(t)\,dt \\ &= \int_{-1}^{1} f(x+t)\,Q_n(t)\,dt. \end{aligned}$$

The second equality is the change of variables theorem on page 143, setting $u = t - x$; the last equality holds since, by our extension of f, $f(x+t) = 0$ for all $t \notin [-x, 1-x]$.

Let $\varepsilon > 0$. Since $f \in C([0,1])$, that is, f is continuous on a compact set, f is uniformly continuous. Then there exists $\delta > 0$ such that, for all $x, y \in [0,1]$ with $|x-y| < \delta$,

$$|f(x) - f(y)| < \frac{\varepsilon}{2}.$$

Now let $C = \sup\{f(x) : x \in [0,1]\}$, which exists by the min/max theorem, or extreme value theorem (page 101). Since $\sqrt{n}\,(1-\delta^2)^n \to 0$ by the previous theorem, there exists $M \in \mathbb{N}$ such that, for all $n \geq M$,

$$\sqrt{n}\,(1-\delta^2)^n < \frac{\varepsilon}{8C}.$$

Then, for all $n \geq M$ and all $x \in [0,1]$, we have

$$\begin{aligned} |P_n(x) - f(x)| &= \left| \int_{-1}^{1} f(x+t)\,Q_n(t)\,dt - f(x)\cdot 1 \right| \\ &= \left| \int_{-1}^{1} f(x+t)\,Q_n(t)\,dt - f(x)\int_{-1}^{1} Q_n(t)\,dt \right| \\ &= \left| \int_{-1}^{1} [f(x+t) - f(x)]\,Q_n(t)\,dt \right| \\ &\leq \int_{-1}^{1} |f(x+t) - f(x)|\,Q_n(t)\,dt \\ &= \int_{|t|\leq\delta} |f(x+t) - f(x)|\,Q_n(t)\,dt + \int_{\delta\leq|t|\leq 1} |f(x+t) - f(x)|\,Q_n(t)\,dt \\ &\leq \frac{\varepsilon}{2}\int_{|t|\leq\delta} Q_n(t)\,dt + \int_{\delta\leq|t|\leq 1} [|f(x+t)| + |f(x)|]\sqrt{n}\,(1-\delta^2)^n\,dt \\ &\leq \frac{\varepsilon}{2}\int_{-1}^{1} Q_n(t)\,dt + \sqrt{n}\,(1-\delta^2)^n \int_{\delta\leq|t|\leq 1} 2C \\ &\leq \frac{\varepsilon}{2}\cdot 1 + \sqrt{n}\,(1-\delta^2)^n \int_{-1}^{1} 2C \\ &= \frac{\varepsilon}{2} + 4C\sqrt{n}\,(1-\delta^2)^n \end{aligned}$$

$$< \frac{\varepsilon}{2} + 4C \cdot \frac{\varepsilon}{8C} = \varepsilon.$$

The steps are justified as follows.

- The second equality uses part 2) of the previous theorem, $\int_{-1}^{1} Q_n = 1$.
- The first inequality is a property of the integral.
- The integral over $[-1, 1]$ is split into the parts $|t| \leq \delta$ and $\delta \leq |t| \leq 1$:



- The second inequality uses the uniform continuity of f on the first integral, and the triangle inequality together with part 3) of the previous theorem ($Q_n(t) \leq \sqrt{n}\,(1-\delta^2)^n$ for $\delta \leq |t| \leq 1$) on the second.
- The third inequality uses the properties of the Riemann integral and $Q_n(x) \geq 0$ (part 1) of the previous theorem) on the first integral, and $C = \sup\{f(x) : x \in [0,1]\}$ on the second.
- The fourth inequality uses part 2) of the previous theorem on the first integral and $C \geq 0$ on the second.

Thus, for all $\varepsilon > 0$, there exists $M \in \mathbb{N}$ such that, for all $n \geq M$ and all $x \in [0,1]$,

$$|P_n(x) - f(x)| < \varepsilon.$$

Hence $P_n \to f$ uniformly.

27 The Riemann Series Theorem (continued)

This section continues the Riemann series theorem stated on page 72.

Theorem (Riemann series theorem)

Let $\sum_n a_n$ be a conditionally convergent series with real-valued terms. Let $x \leq y$, where $x, y \in [-\infty, +\infty]$. Then there exists a rearrangement $\sum_n b_n$ of $\sum_n a_n$ such that

$$\liminf_{n\to\infty} t_n = x \qquad \text{and} \qquad \limsup_{n\to\infty} t_n = y, \qquad \text{where } t_n = \sum_{k=1}^{n} b_k.$$

Remark

In other words, by picking $x = y \in [-\infty, +\infty]$, we will have that the rearrangement $\sum_n b_n$ can converge to whatever we would like.

Proof

For the purpose of convenience, we will omit the terms that are 0. Let P_n denote the n-th positive term of $\sum a_n$, and let $-q_n$ denote the n-th negative term of $\sum a_n$. Then $\sum P_n$ and $\sum q_n$ are divergent series with positive terms (otherwise $\sum a_n$ would be absolutely convergent). Now, since $\mathbb{R}$ is perfect, choose two sequences $\{x_n\}_n$ and $\{y_n\}_n$ such that

$$x_n \to x, \qquad y_n \to y, \qquad x_n < y_n, \qquad y_1 > 0.$$

Now, take enough (k_1) positive terms such that

$$P_1 + \cdots + P_{k_1 - 1} < y_1 \qquad \text{and} \qquad P_1 + \cdots + P_{k_1} > y_1,$$

that is, just enough to pass y_1. Then take just enough (r_1) negative terms from $\sum a_n$ such that

$$P_1 + \cdots + P_{k_1} - q_1 - \cdots - q_{r_1} < x_1.$$

Then take just enough $(k_2 - k_1)$ positive terms from $\sum a_n$ such that

$$P_1 + \cdots + P_{k_1} - q_1 - \cdots - q_{r_1} + P_{k_1+1} + \cdots + P_{k_2} > y_2.$$

Reiterating this process, we obtain a rearrangement $\sum_n b_n$. Now we claim that the partial sums $t_n = \sum_{k=1}^{n} b_k$ satisfy

$$\limsup_{n\to\infty} t_n = y \qquad \text{and} \qquad \liminf_{n\to\infty} t_n = x.$$

To sum up what we have shown, there exist subsequences $\{t_{k_n}\}_{n=1}^{\infty}$ and $\{t_{r_n}\}_{n=1}^{\infty}$ such that

$$t_{k_n} > y_n \qquad \text{and} \qquad t_{r_n} < x_n.$$

Now note that $P_n, q_n \to 0$ as $n \to \infty$ by the n-th term test; then showing $\limsup_{n\to\infty} t_n = y$ and $\liminf_{n\to\infty} t_n = x$ becomes trivial.